

Intro. to Basic Electronics

Instructors:

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General Information

- **Suggested pre-requisites:**
 - Basic course on applied electricity and linear algebra, Knowledge of electrical components will be an advantage.
- **Course Content:**
 - Introduction to Electronics and its Applications
 - Semiconductor Materials and Properties
 - Semiconductor Diodes
 - Semiconductor Diodes and Applications
 - Bipolar Junction Transistor
 - Transistor as an Amplifier
 - Operational Amplifier
 - Switching Theory and Logic Design
- **Grading :**
 - Cont. Ass. : 30% Final Exams 70%

Reference Books

- Electronic Principles by Albert Paul Malvino
(copies can be found in both Engineering and Main Lib.)
- Electronic Engineering by Sanjay Sharma PhD.
(copies can be found at Kingdom Books)
- etc

Lecture Notes

<https://sites.google.com/site/klogoclass/home>

Intro. to Electricity and Electronics



What is Electronics

Electronics is the branch of physics and technology concerned with the design of circuits using transistors and microchips, and with the behaviour and movement of electrons in a semiconductor, conductor, vacuum, or gas

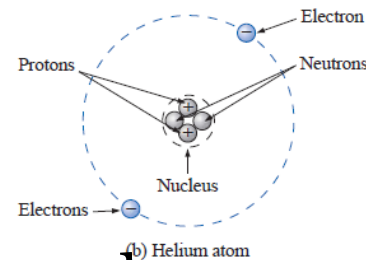
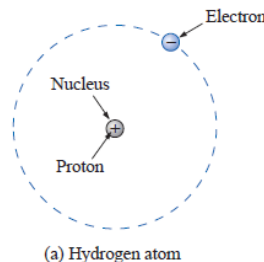
System of Units

Multiplication Factors	SI Prefix	SI Symbol
1 000 000 000 000 = 10^{12}	tera	T
1 000 000 000 = 10^9	giga	G
1 000 000 = 10^6	mega	M
1 000 = 10^3	kilo	k
0.001 = 10^{-3}	milli	m
0.000 001 = 10^{-6}	micro	μ
0.000 000 001 = 10^{-9}	nano	n
0.000 000 000 001 = 10^{-12}	pico	p

Applications

Atoms And Their Structure

- Everything is made of atoms
- The simplest of all atoms is the hydrogen atom.
- It is made up of two basic particles
 - the proton
 - the electron



- In all other elements the nucleus also contains **neutrons** which have no charge
- In every element the number of protons is equivalent to the number of electrons.

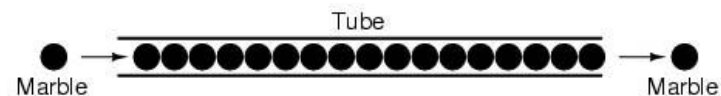
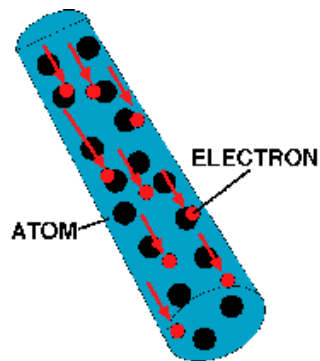
Coulomb's Law

$$F \text{ (attraction or repulsion)} = \frac{kQ_1Q_2}{r^2}$$

- Unlike charges attract; like charges repel.
- So there are forces of attraction acting in the atom between the **protons** in the nucleus and the **electrons** in the orbiting shells.
- This force is stronger when they are closer and weaker when they are far apart.
- Therefore it is easier to break away an electron that is distant from the nucleus.
- Also it is easier to break an electron from a shell that is incomplete and has fewer electrons.

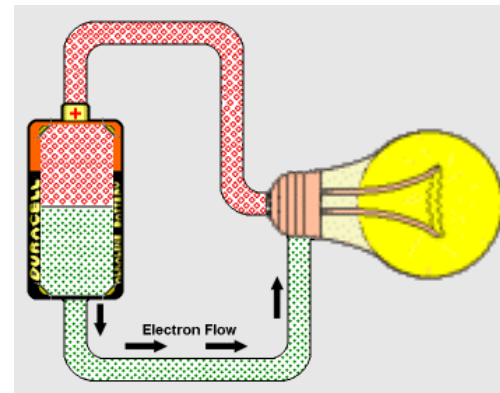
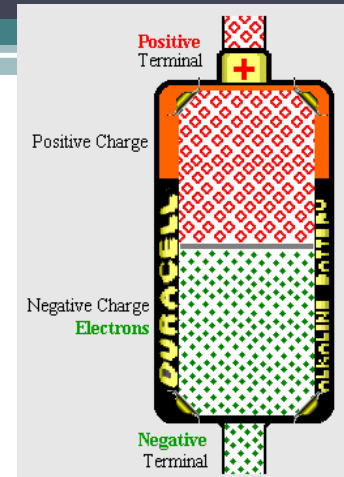
Electricity

- An electron that breaks away from its atom is known as a **Free Electron**.
- These free electrons are known as **charge carriers**.
- The movement of free electrons is known as **current of Electricity**.



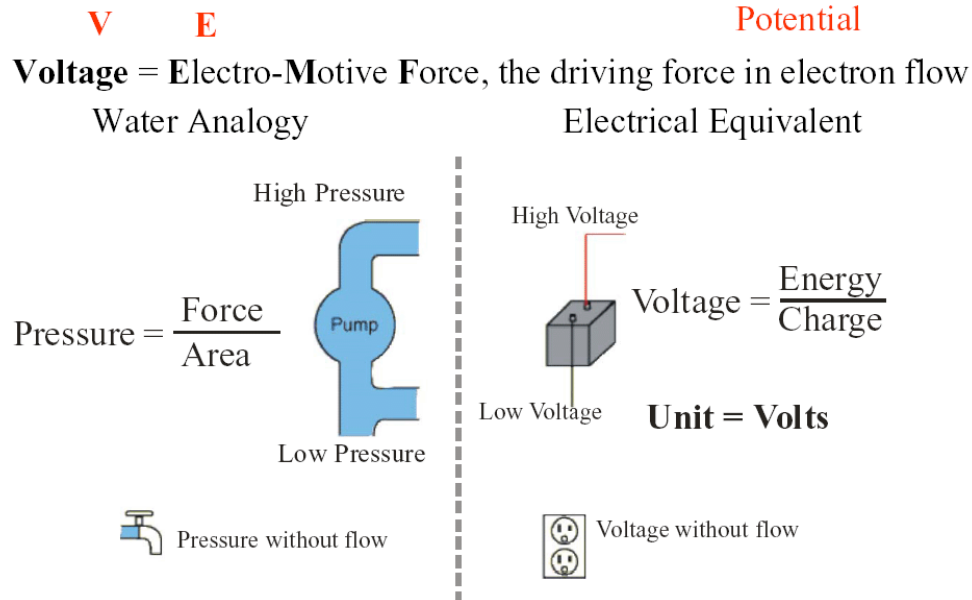
- Some materials have strong attraction and refuse to lose electrons (have less free electrons), these are called **insulators** (air, porcelain, oils, Bakelite, rubber, teflon, glass, mica)
- Some materials have weak attractions and allow electrons to be lost, these are called **conductors** (silver, copper, gold, aluminium, tungsten, nickel, iron)

- Surplus of electrons is called a negative charge (-). A shortage of electrons is called a positive charge (+).
- A battery provides a surplus of electrons by chemical reaction.
- By connecting a conductor from the positive terminal to negative terminal electrons will flow.



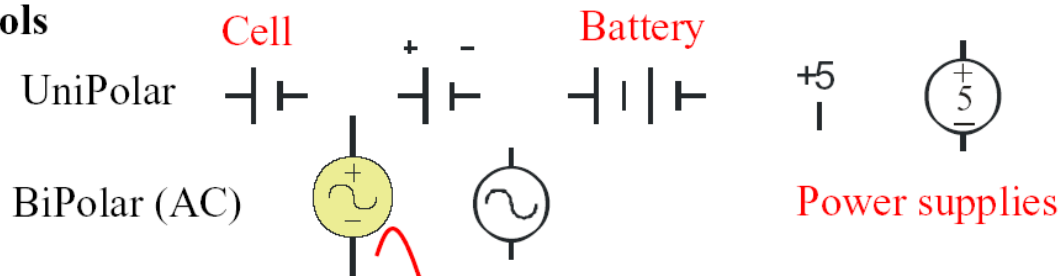
Potential Difference(Voltage)

- The applied potential difference (measured in volts) of a voltage source in an electric circuit is the “pressure” needed to set the system in motion and “cause” the flow of charge or current through the electrical system.
- Compare this pressure to the pressure from a water tap connected to a hose



Voltage Sources:

Symbols

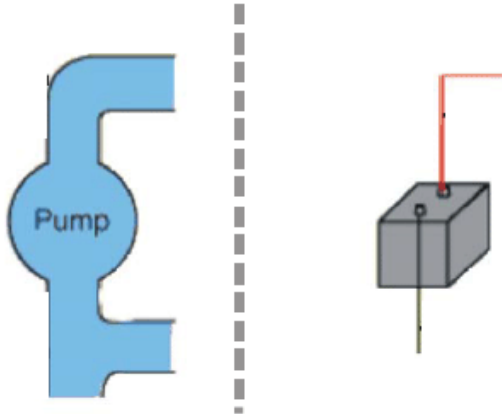


Properties

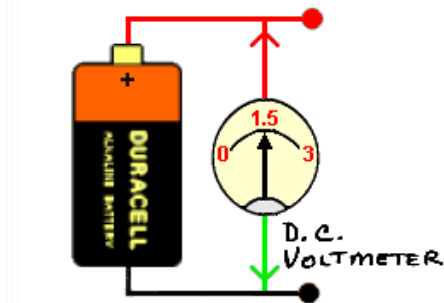
Constant Voltage, independent of the amount of current
Usually ideal

Examples

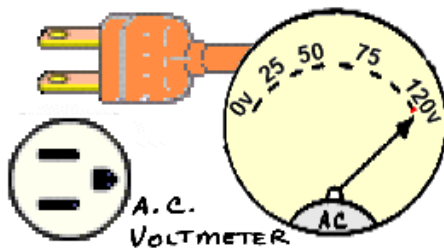
Batteries
Power Supplies
Signal Generators



- Voltage is like differential pressure, always measure between two points.

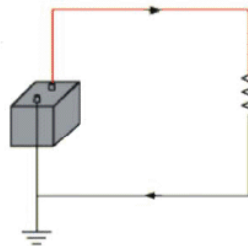
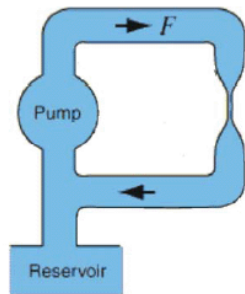


- Measure voltage between two points or across a component in a circuit.
- When measuring DC voltage make sure polarity of meter is correct, positive (+) red, negative (-) black.



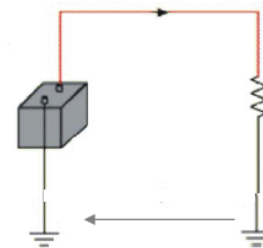
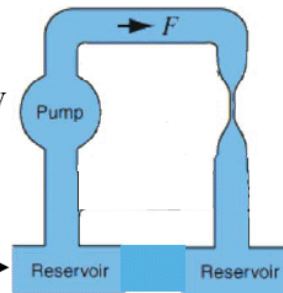
Ground

Provides a reference point



Purely a reference point
Does not participate in current flow

An integral path in the current flow



Symbols



Earth

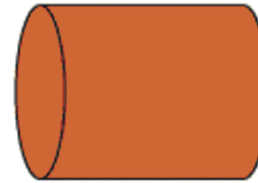


Analog Gnd

Current



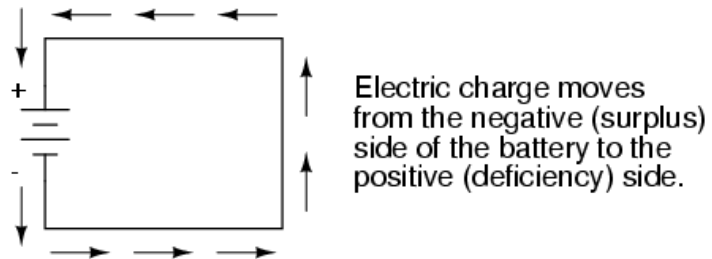
Flow of Water



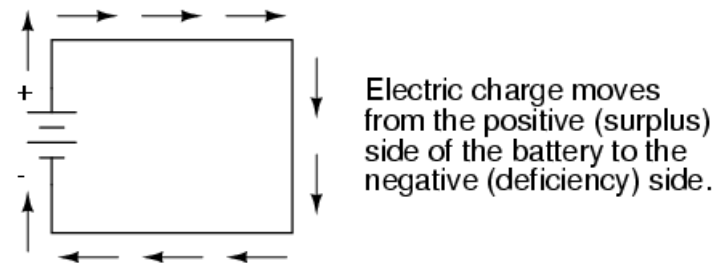
Flow of Charge

- Uniform flow of electrons thru a circuit is called *current*.

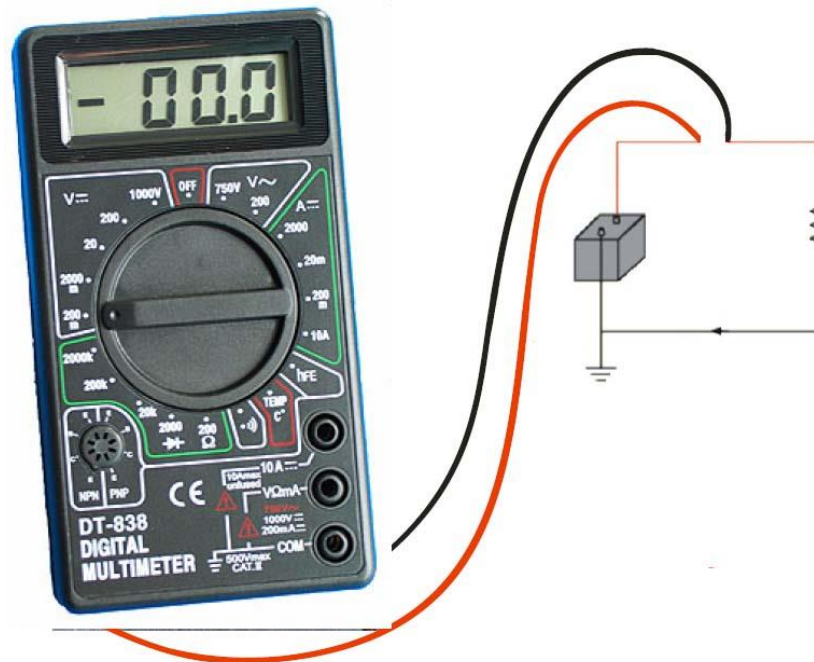
Electron flow notation



Conventional flow notation



**WILL USE CONVENTIONAL FLOW NOTATION
ON ALL SCHEMATICS**

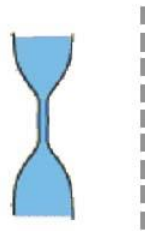


- To measure current, must break circuit and install meter in line.
- Measurement is imperfect because of voltage drop created by meter.

Resistance

Constriction
creates

Resistance to water flow

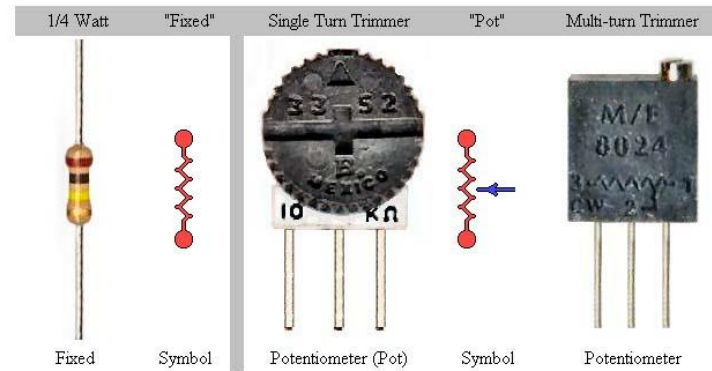


Resistor creates
Resistance to current
flow



- The resistance of a material is the opposing force that a flowing charge encounters
- All materials have a resistance that is dependent on cross-sectional area, material type and temperature.
- A resistor dissipates power in the form of heat

Various resistors types



Potentiometer

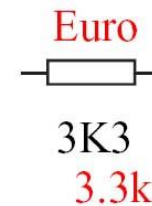
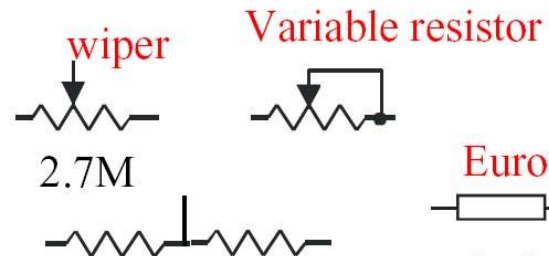
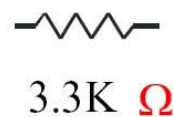


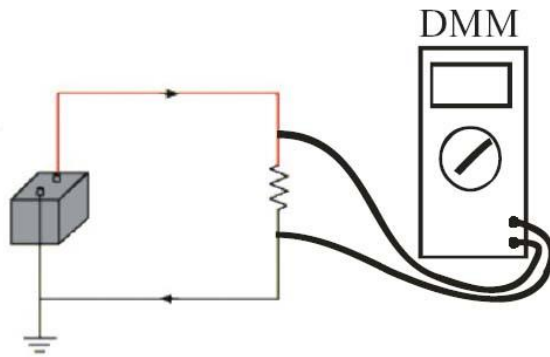
Potentiometer



Sliding Potentiometer

Symbols

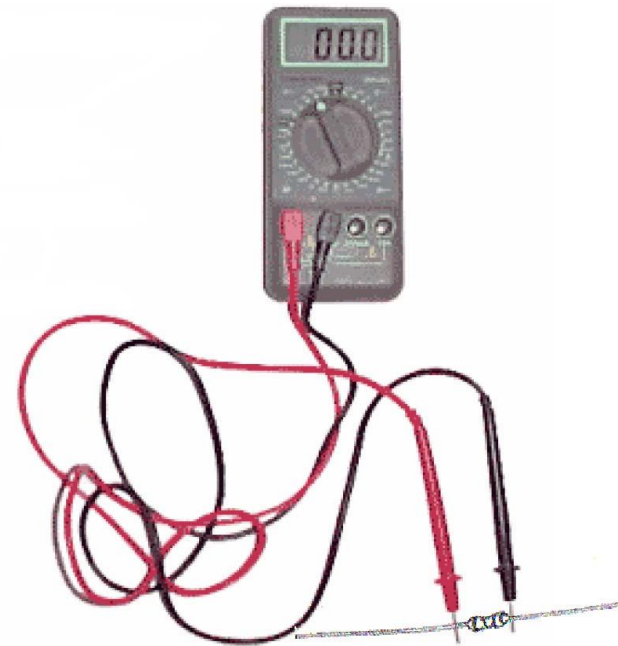




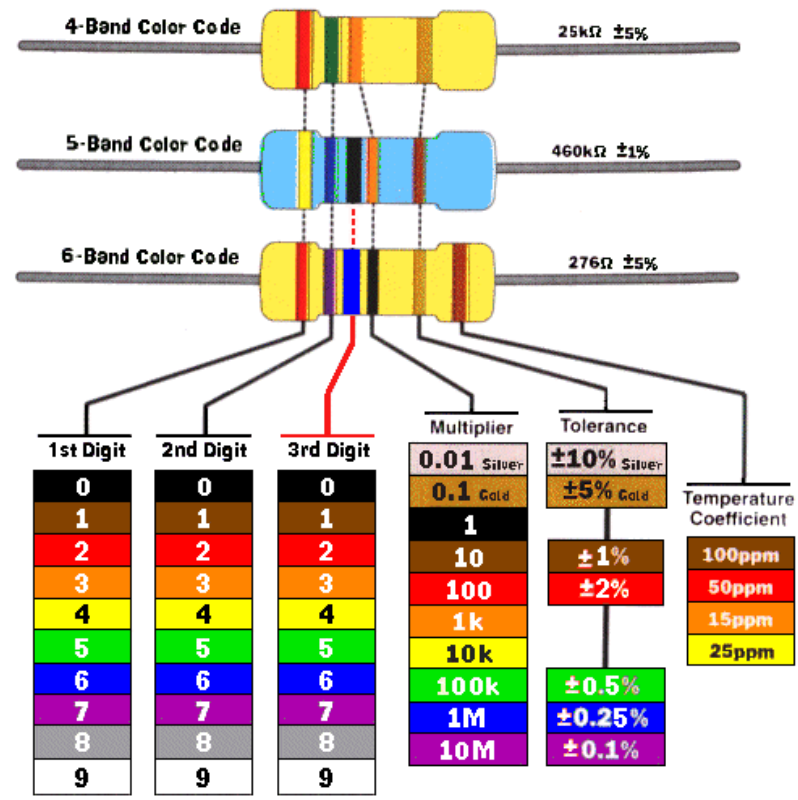
Will this work like we expect it to?

NO! don't do this!

When measuring resistance, remove component from the circuit.

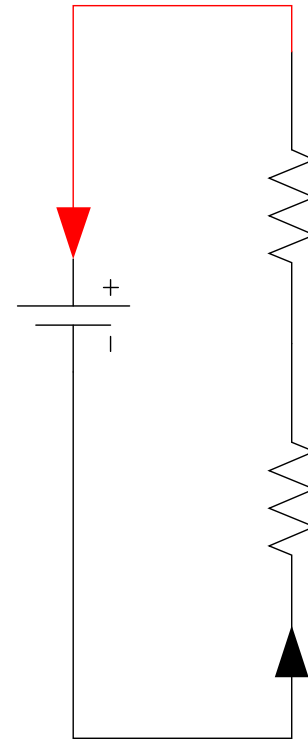


Resistor Color Code



Resistors in Circuits Series

- Looking at the current path, if there is only one path, the components are in series.



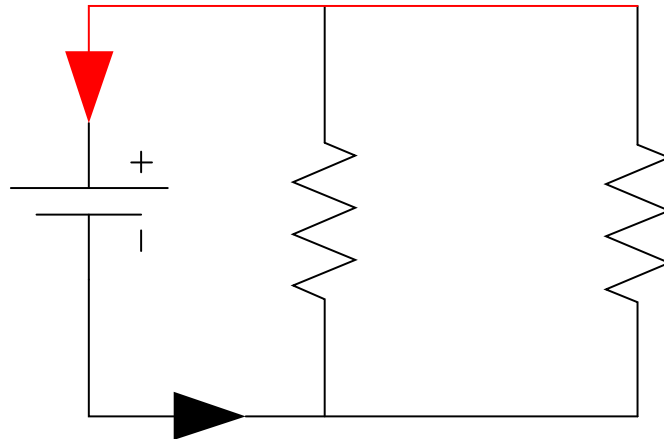
Resistors in Circuits Series

$$R_1 + R_2 + R_n$$

Resistors in Circuits

Parallel

- If there is more than one way for the current to complete its path, the circuit is parallel



Resistors in Circuits Parallel

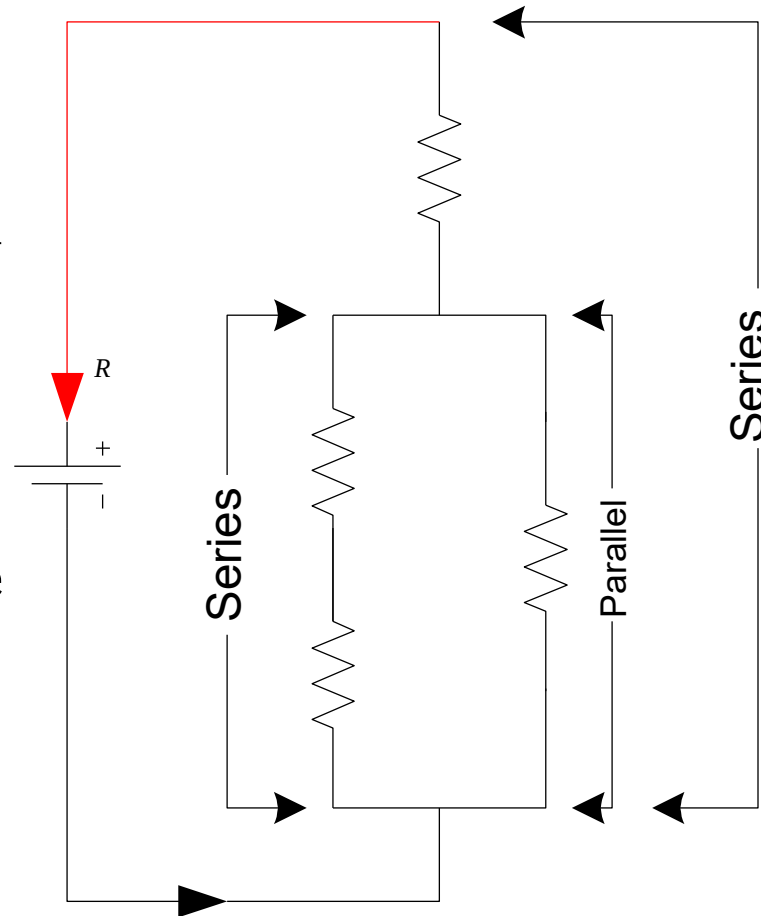
$$\frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_n}}$$

$$\frac{R_1 R_2}{R_1 + R_2}$$

Resistors in Circuits

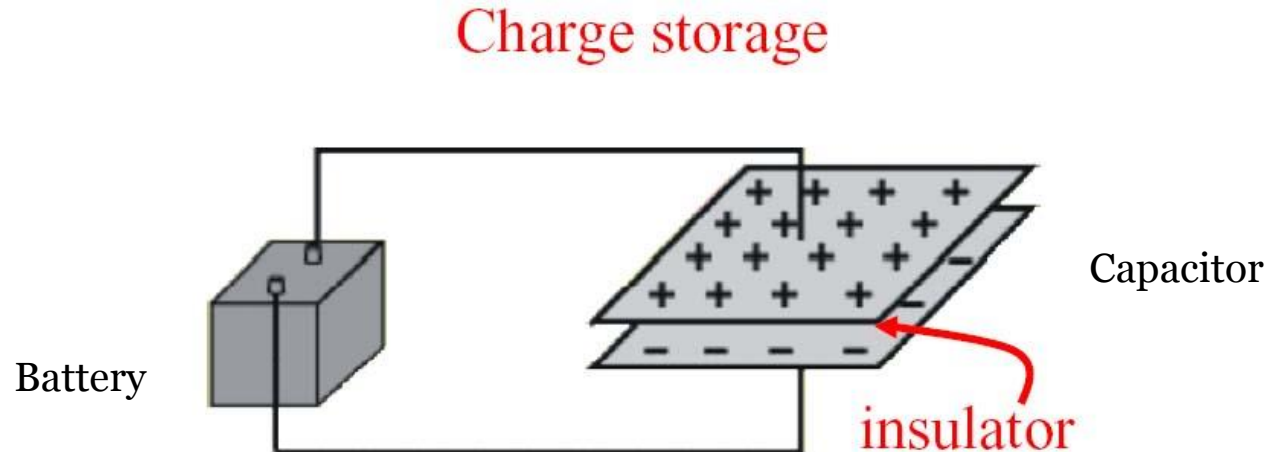
Mixed

- If the path for the current in a portion of the circuit is a single path, and in another portion of the circuit has multiple routes, the circuit is a mix of series and parallel.



Capacitance

A capacitor is used to store charge for a short amount of time



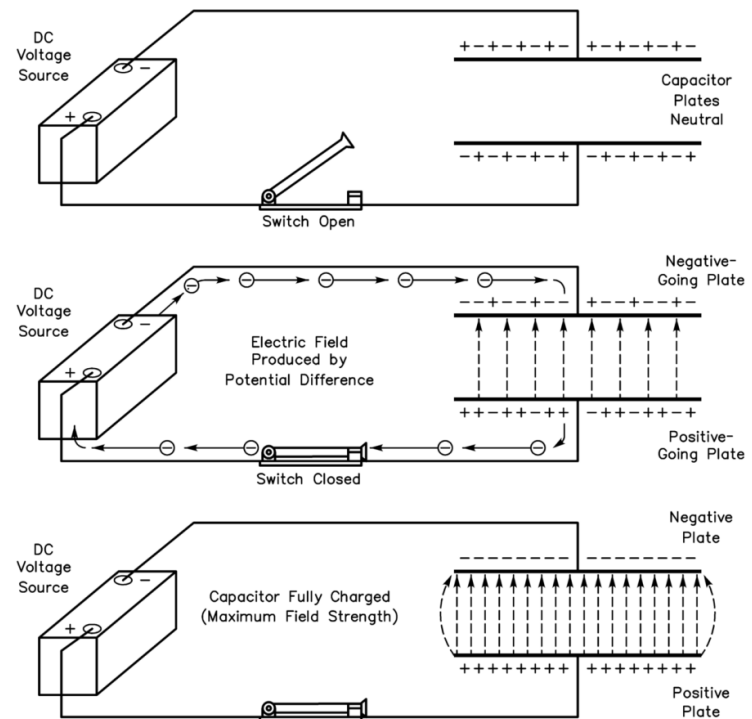
Unit = Farad

Pico Farad - $\text{pF} = 10^{-12}\text{F}$

Micro Farad - $\text{uF} = 10^{-6}\text{F}$

The Capacitor Defined

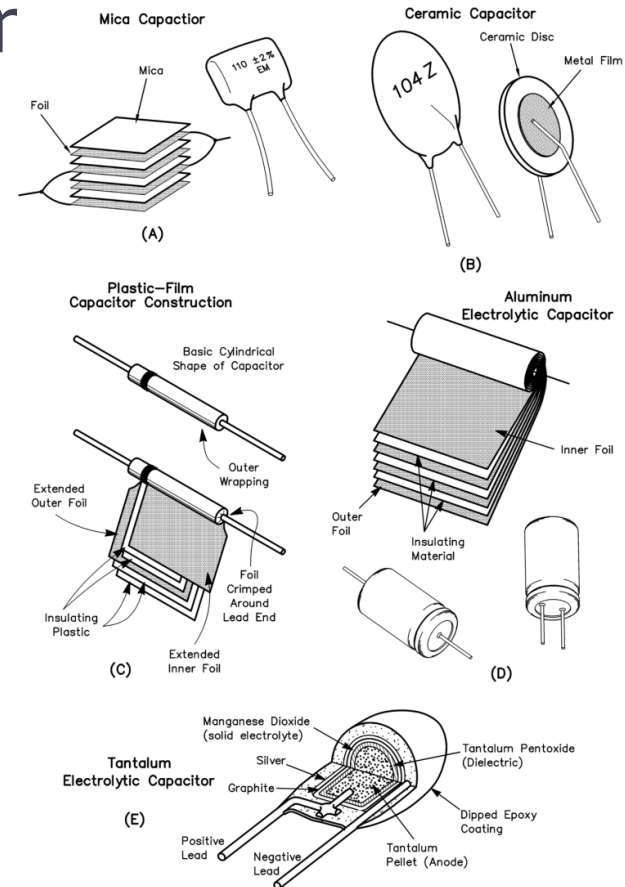
- A device that stores energy in electric field.
- Two conductive plates separated by a non conductive material.
- Electrons accumulate on one plate forcing electrons away from the other plate leaving a net positive charge.
- Think of a capacitor as very small, temporary storage battery.

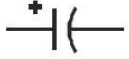


The Capacitor

Physical Construction

- Capacitors are rated by:
 - Amount of charge that can be held.
 - The voltage handling capabilities.
 - Insulating material between plates.



Symbols nonPolar  Polar  Euro 

Properties

Characteristic Equations: $I = C \frac{dV}{dT}$

$$V = \frac{1}{C} \int IdT \quad \text{Integrating Charge (storage)}$$

Markings

Polar vs Non-Polar

Values

Electrolytics mark (-)
Tantalums mark (+)
Longer lead

Examples



Mylar



Monolythic
Cermamic



Tantalum



Ceramic

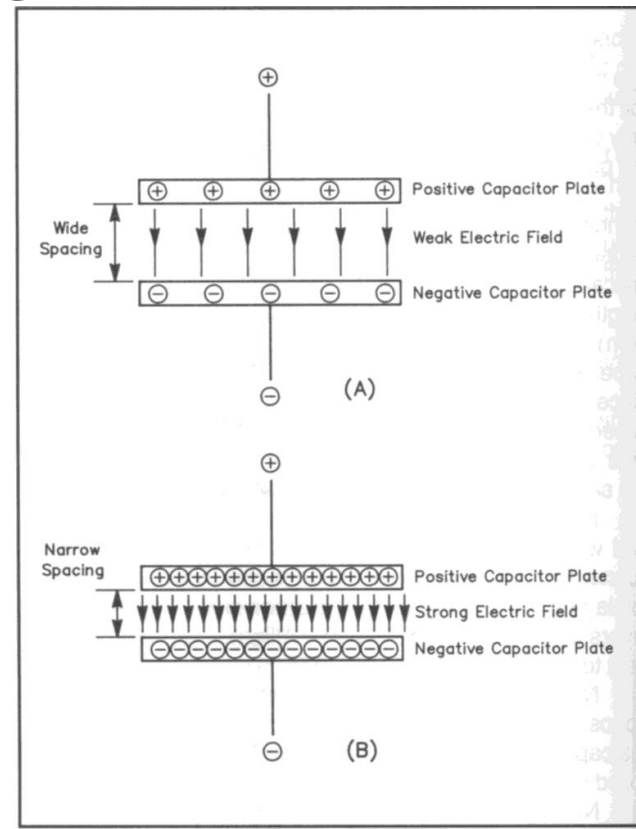


Electrolytic

The Capacitor

Ability to Hold a Charge

- Ability to hold a charge depends on:
 - Conductive plate surface area.
 - Space between plates.
 - Material between plates.



The Capacitor Behavior in DC

- When exposed to DC, the capacitor charges and holds the charge as long as the DC voltage is applied.
- The capacitor essentially blocks DC voltage from passing through.

The Capacitor Behavior in AC

- When AC current is applied, during one half of the cycle the capacitor accepts a charge in one direction.
- During the next half of the cycle, the capacitor is discharges then recharged in the reverse direction.
- During the next half cycle the pattern reverses.
- Essentially, it appears that AC current passes through a capacitor

The Capacitor Behavior

- A capacitor blocks the passage of DC
- A capacitor passes AC

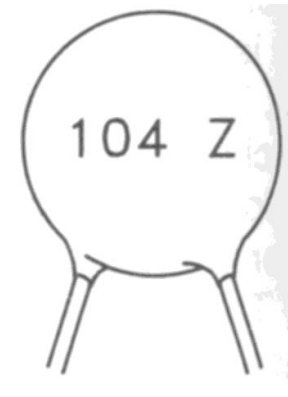
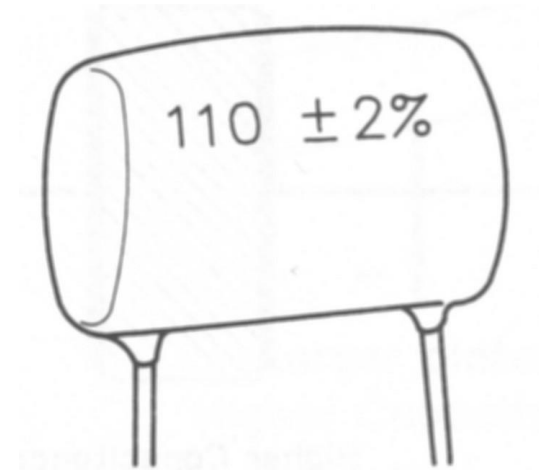
The Capacitor

Capacitance Value

- The unit of capacitance is the farad.
 - A single farad is a huge amount of capacitance.
 - Most electronic devices use capacitors that have a very tiny fraction of a farad.
- Common capacitance ranges are:
 - Micro μ - 10^{-6}
 - Nano n - 10^{-9}
 - Pico p - 10^{-12}

The Capacitor Capacitance Value

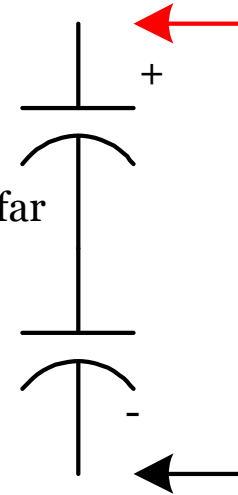
- Capacitor identification depends on the capacitor type.
- Could be color bands, dots, or numbers.
- Wise to keep capacitors organized and identified to prevent a lot of work trying to re-identify the values.



Capacitors in Circuits

- Two physical factors affect capacitance values.
 - Plate spacing
 - Plate surface area
- In series, plates are far apart making capacitance less

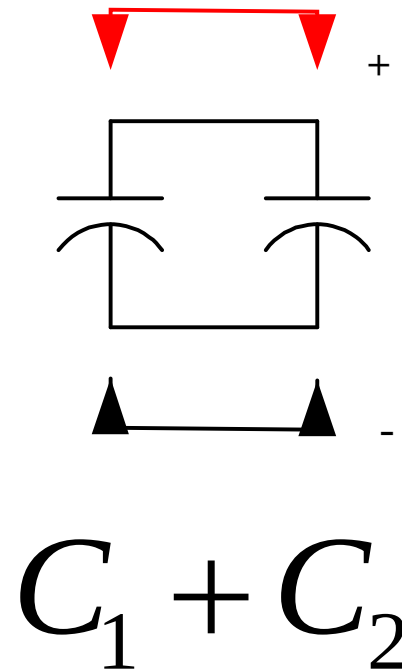
Charged plates far apart



$$\frac{C_1 C_2}{C_1 + C_2}$$

Capacitors in Circuits

- In parallel, the surface area of the plates add up to be greater, and close together.
- This makes the capacitance more the Capacitor

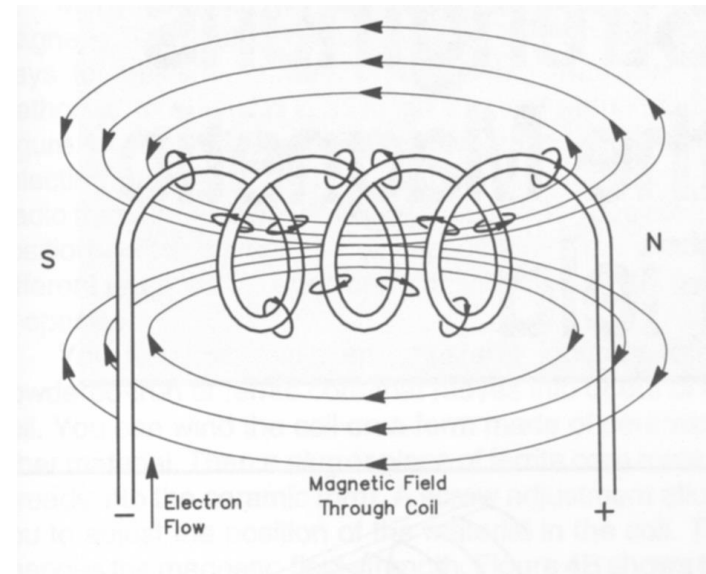


The Inductor

- There are two fundamental principles of electronics:
 1. Moving electrons create a magnetic field.
 2. Moving or changing magnetic fields cause electrons to move.
- An inductor is a coil of wire through which electrons move, and energy is stored in the resulting magnetic field.

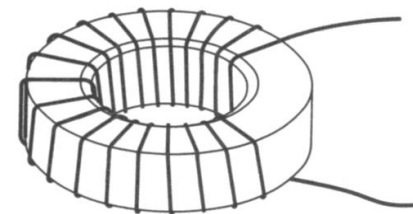
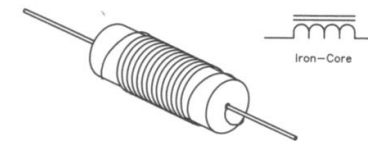
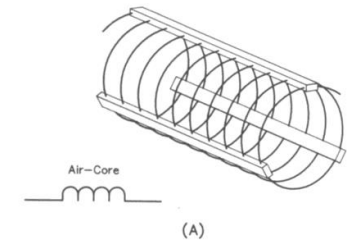
The Inductor

- Like capacitors, inductors temporarily store energy.
- Unlike capacitors:
 - Inductors store energy in a magnetic field, not an electric field.
 - When the source of electrons is removed, the magnetic field collapses immediately.



The Inductor

- Inductors are simply coils of wire.
 - Can be air wound (nothing in the middle of the coil)
 - Can be wound around a permeable material (material that concentrates magnetic fields)
 - Can be wound around a circular form (toroid)



The Inductor

- Inductance is measured in Henry(s).
- A Henry is a measure of the intensity of the magnetic field that is produced.
- Typical inductor values used in electronics are in the range of milli Henry ($1/1000$) and micro Henry ($1/1,000,000$)

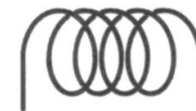
The Inductor

- The amount of inductance is influenced by a number of factors:

- Number of coil turns.
- Diameter of coil.
- Spacing between turns.
- Size of the wire used.
- Type of material inside the coil.



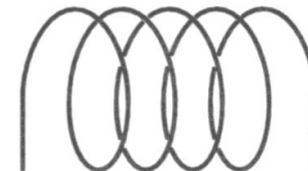
Wide Spacing Between Turns
Low Inductance



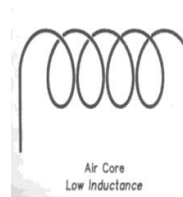
Close Spacing Between Turns
Higher Inductance



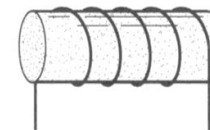
Small Diameter
Low Inductance



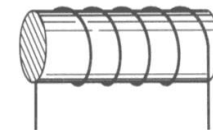
Larger Diameter
Higher Inductance



Air Core
Low Inductance



Powdered-iron Core
Higher Inductance



Soft-iron Core
Highest Inductance

Inductor Performance With DC Currents

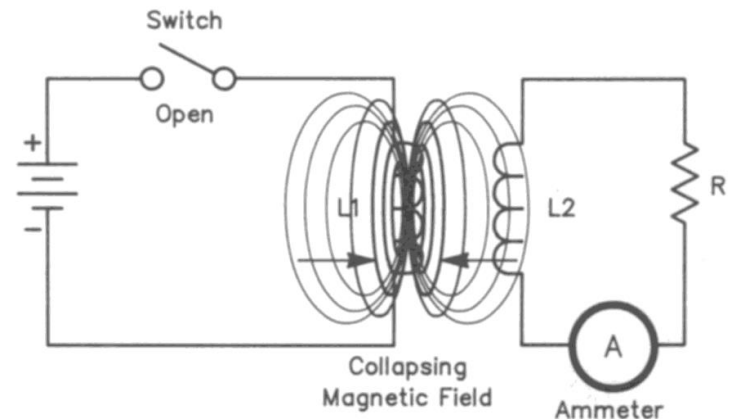
- When DC current is applied to an inductor, the wire in the inductor momentarily appears as a short circuit and maximum current flows.
- As the magnetic field builds (changes) there is a tendency for the current flow to slow down (due to an opposition cause the the changing magnetic field).
- Finally, the magnetic field is at its maximum and the current flows to maintain the field.
- As soon as the current source is removed, the magnetic field begins to collapse and creates a rush of current in the other direction, sometimes at very high voltages.

Inductor Performance With AC Currents

- When AC current is applied to an inductor, during the first half of the cycle, the magnetic field builds as if it were a DC voltage.
- During the next half of the cycle, the current is reversed and the magnetic field first has to decrease the reverse polarity in step with the changing current.
- Depending on the value of inductance, these forces can work against each other, making for a less than simple situation.

The Inductor

- Because the magnetic field surrounding an inductor can cut across another inductor in close proximity, the changing magnetic field in one can cause current to flow in the other ... the basis of transformers



Laws and Principles of Electricity and Electronics

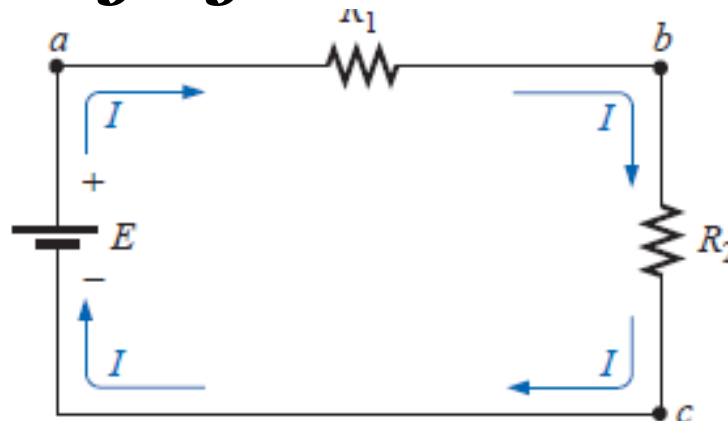


Laws and Rules

- Ohm's
- Kirchhoff's
- Voltage Divider
- Current Divider
- Thevenin's
- Norton's

Reminder: Elements in series

- Two elements are in series if
- **1. They have only one terminal in common (i.e., one lead of one is connected to only one lead of the other).**
- **2. The common point between the two elements is not connected to another current-carrying element.**



Reminder: Resistors in series

- The total resistance of a series circuit is the sum of the resistance levels.

$$R_T = R_1 + R_2 + R_3 + \cdots + R_N$$

Ohm's Law

- The mathematical relationship
 - $E=I \times R$
- Doing the math

Ohm's Law

- In 1827 George Ohm proved there was a direct relationship between Voltage (E), Current (I), and Resistance (R) in an electrical circuit. This relationship is known as Ohm's Law.
- Ohm's Law states that current in a circuit is proportional to the voltage and inversely proportional to the resistance.

Ohm's Law

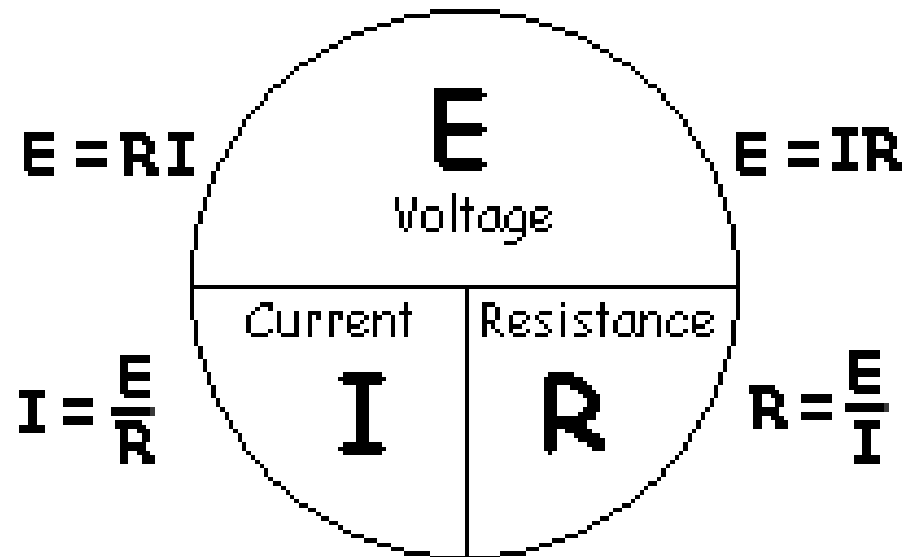
- There is a mathematical relationship between the three components of electricity. That relationship is **Ohm's Law**.
 - **E** = volts
 - **R** = resistance in ohms
 - **I** = current in amps

$$E = I * R$$

$$R = \frac{E}{I}$$

$$I = \frac{E}{R}$$

Ohm's Law



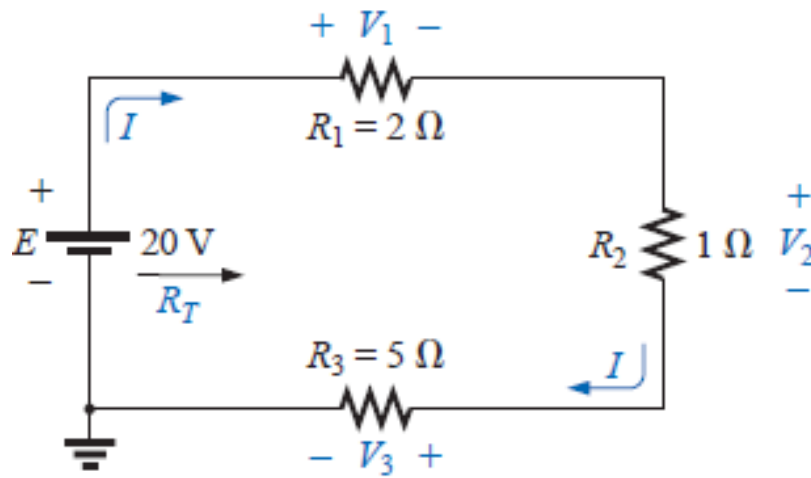
E = Voltage - Volts

I = Current - Amps

**R = Resistance or Reactance
(Impedance) - Ohms**

Question

- Find the total resistance for the series circuit of the figure below
- Calculate the source current I_s .
- Determine the voltages V_1 , V_2 , and V_3 .
- Calculate the power dissipated by R_1 , R_2 , and R_3 .
- Determine the power delivered by the source, and compare it to the sum of the power levels of part (d).



Power

- Transforming energy from one form to another is called **work**. The greater the energy transformed, the more work that is done.
- There are six basic forms of energy and they are light, heat, magnetic, chemical, electrical, and mechanical energy.
- The unit for measuring **work** is called the **Joule (J)**.

Power

- **Power (P)** is the rate at which **work** is performed and is measured by the unit called **Watt (W)**. **$Watts = Joules \text{ per second}$** .
- The output **Power, or power ratings** of electrical, electronic or mechanical devices can be expressed in **Watts (W)** and describes the number of **Joules** of energy converted every second.

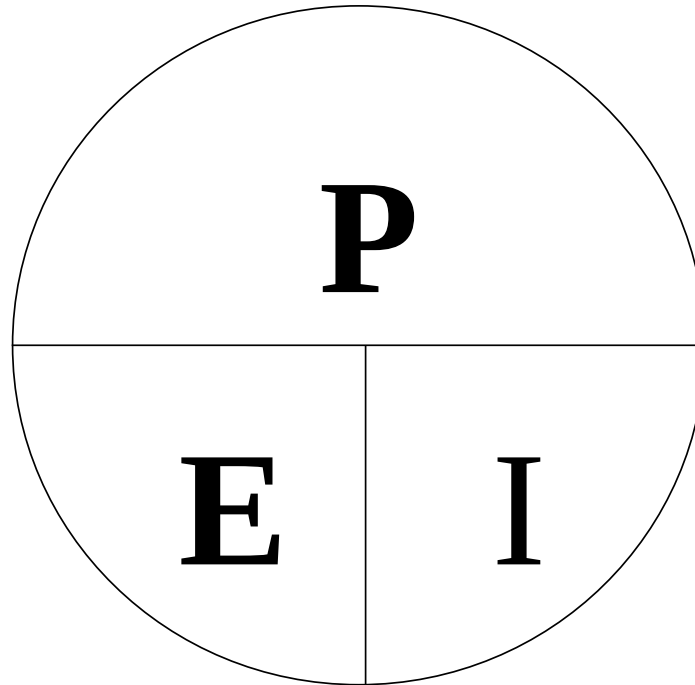
Power

- **Power** is the rate at which electric energy (W) is converted to some other form and can be expressed mathematically as $P = I \times V$.
- This formula states that the amount of power delivered to a device is dependent on the electrical pressure (or voltage applied across the device) and the current flowing through the device.

Power Formula

- The ***Power Formula*** is the relationship between Power (P), Voltage (E), and Current (I).

P = Power -Watts
E = Voltage - Volts
I = Current - Amps



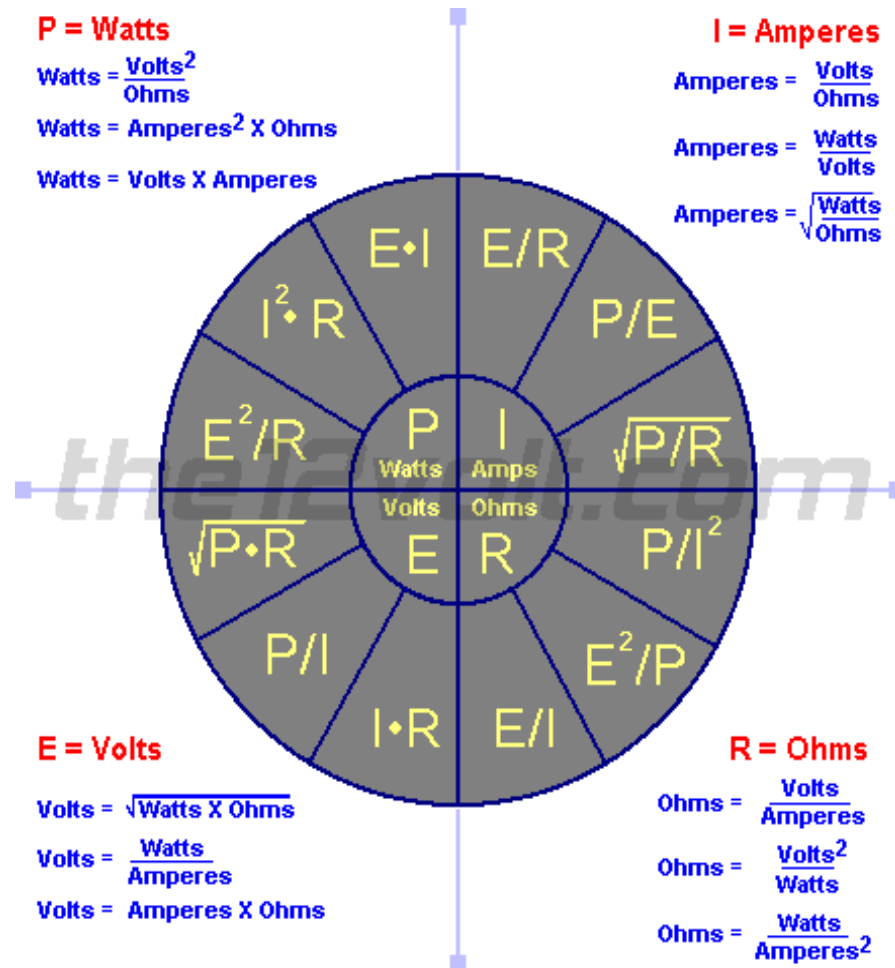
Power Formula

- The ***Power Formula*** states that if the voltage in a circuit changes, the current in the circuit also changes. The power required from a circuit changes any time loads are added (power increases) or removed (power decreases).
- The ***Power Formula*** is used when troubleshooting and to predict circuit characteristics before power is applied.

Combining Ohm's Law and Power Formula

- ***Ohm's Law*** and the ***Power Formula*** may be combined mathematically and written as any combination of Voltage (E), Current (I), Resistance (R), or Power (P).
- ***Ohm's Law*** and the ***Power Formula*** are limited to circuits in which electrical resistance is the only significant opposition to the flow of current. This limitation includes all DC circuits and AC circuits that do not contain a significant amount of inductance and/or capacitance – which we will learn about later.

Combining Ohm's Law and Power Formula



Kirchhoff's Voltage Law (KVL)

- The algebraic sum of the voltage that rises and drops around a closed loop is equal to zero
- $$E_T - V_1 - V_2 - V_3 - \dots - V_n = 0$$

$$\sum_{\text{C}} V_{\text{rises}} = \sum_{\text{C}} V_{\text{drops}}$$

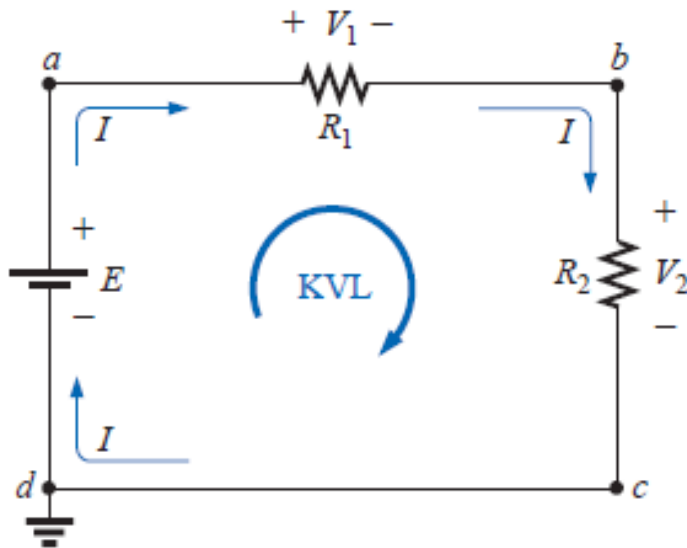
Kirchhoff's Voltage Law (KVL)

- Another way of stating KVL is:
 - Summation of voltage rises is equal to the summation of voltage drops around a closed loop

$$V_1 + V_2 + V_3 + \cdots + V_n = E_T$$

Kirchhoff's Voltage Law (KVL)

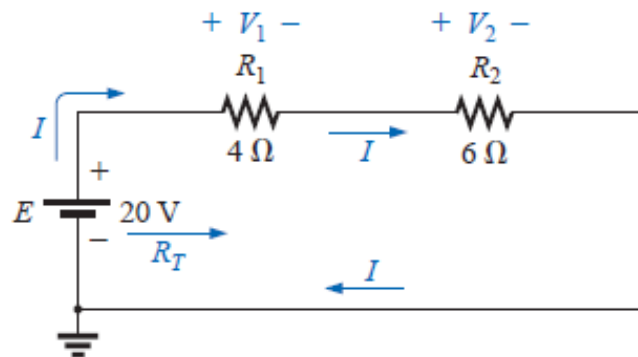
- the applied voltage of a series circuit equals the sum of the voltage drops across the series elements.*



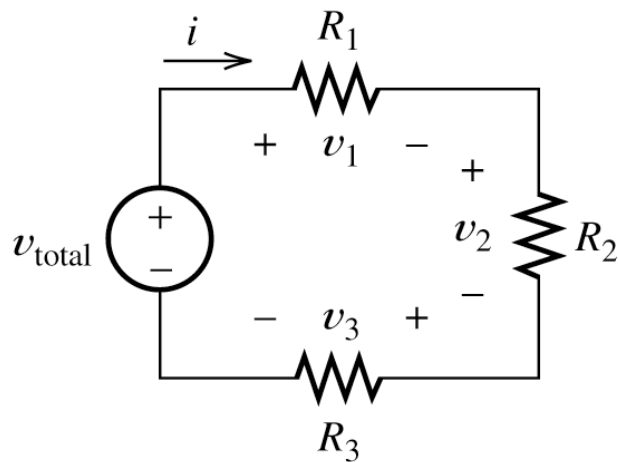
$$\begin{aligned}\sum_{\text{C}} V &= 0 \\ -E + V_2 + V_1 &= 0 \\ E &= V_1 + V_2\end{aligned}$$

Kirchhoff's Voltage Law (KVL)

- Find RT .
- Find I .
- Find V_1 and V_2 .
- Find the power to the 4- and 6- resistors.
- Find the power delivered by the battery, and compare it to that dissipated by the 4- and 6- resistors combined.
- Verify Kirchhoff's voltage law (clockwise direction).



Voltage Divider Rule



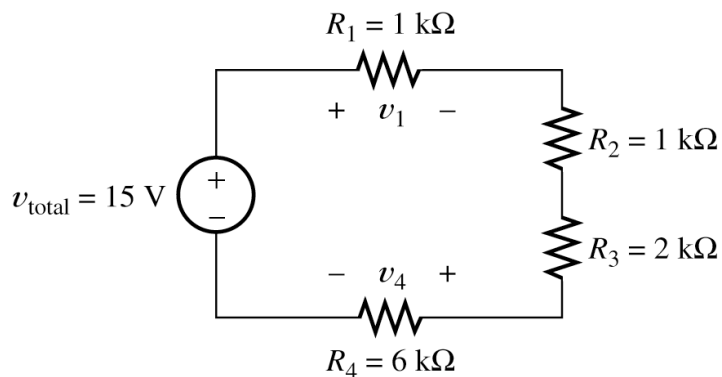
$$v_1 = R_1 i = \frac{R_1}{R_1 + R_2 + R_3} v_{\text{total}}$$

$$v_2 = R_2 i = \frac{R_2}{R_1 + R_2 + R_3} v_{\text{total}}$$

$$v_k = \frac{R_k}{R_1 + R_2 + \dots + R_N} v$$

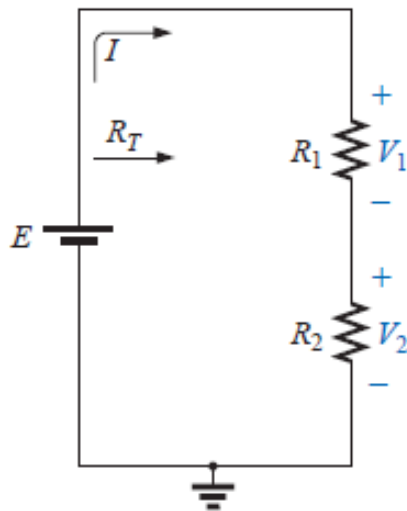
The voltage across the resistive elements will divide as the magnitude of the resistance levels.

Application of the Voltage-Division Principle



$$\begin{aligned} v_1 &= \frac{R_1}{R_1 + R_2 + R_3 + R_4} v_{\text{total}} \\ &= \frac{1000}{1000 + 1000 + 2000 + 6000} \times 15 \\ &= 1.5 \text{ V} \end{aligned}$$

Simplifying the Voltage Divider Rule



and

$$R_T = R_1 + R_2$$

$$I = \frac{E}{R_T}$$

Applying Ohm's law:

$$V_1 = IR_1 = \left(\frac{E}{R_T}\right)R_1 = \frac{R_1 E}{R_T}$$

with

$$V_2 = IR_2 = \left(\frac{E}{R_T}\right)R_2 = \frac{R_2 E}{R_T}$$

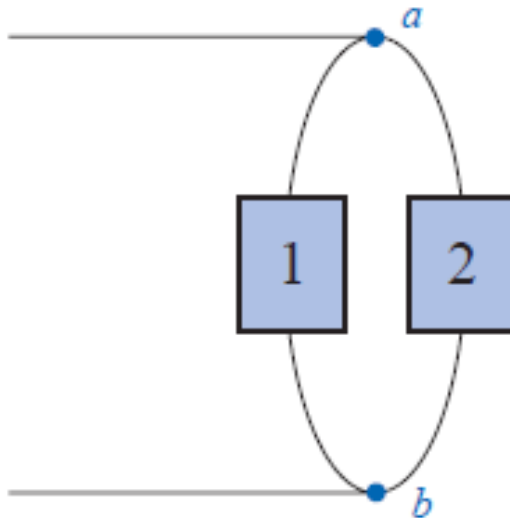
Note that the format for V_1 and V_2 is

$$V_x = \frac{R_x E}{R_T}$$

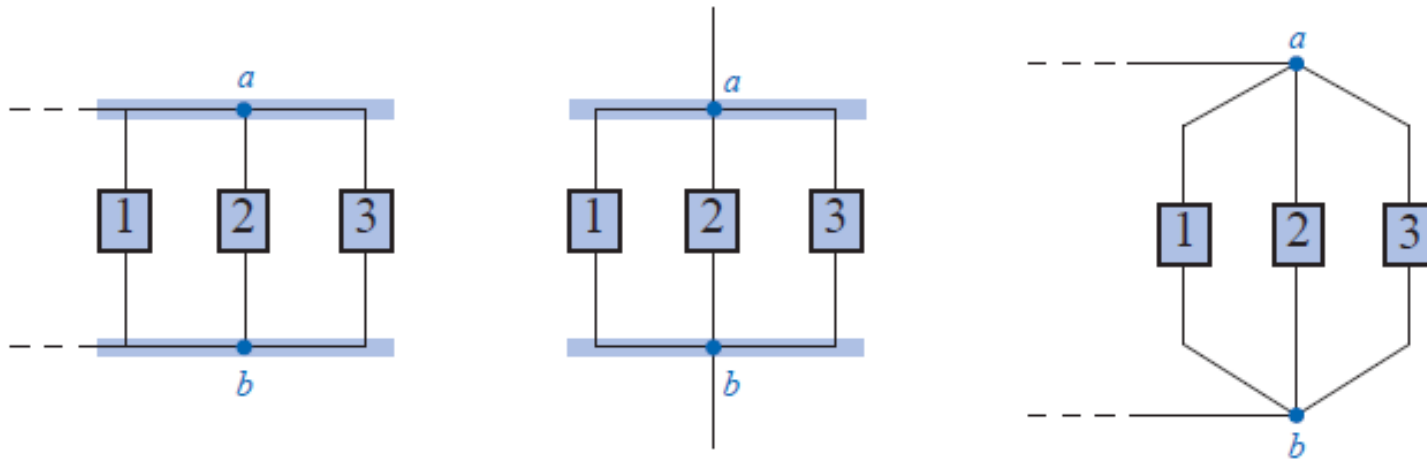
(voltage divider rule)

Reminder: Elements in Parallel

- ***Two elements, branches, or networks are in parallel if they have two points in common***



Reminder: Elements in Parallel



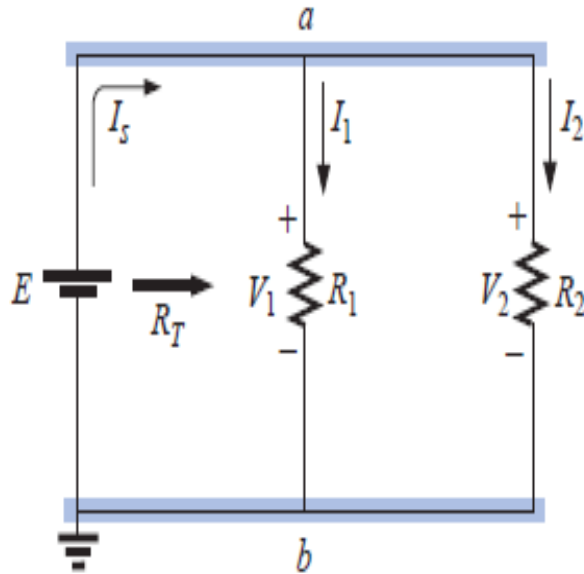
Reminder: Resistors in parallel

- ***For parallel resistors, the total conductance is the sum of the individual conductances.***
- $G = \frac{1}{R}$

$$G_T = G_1 + G_2 + G_3 + \cdots + G_N$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_N}$$

Reminder: Resistors in parallel



The voltage across parallel elements is the same.

$$V_1 = V_2 = E$$

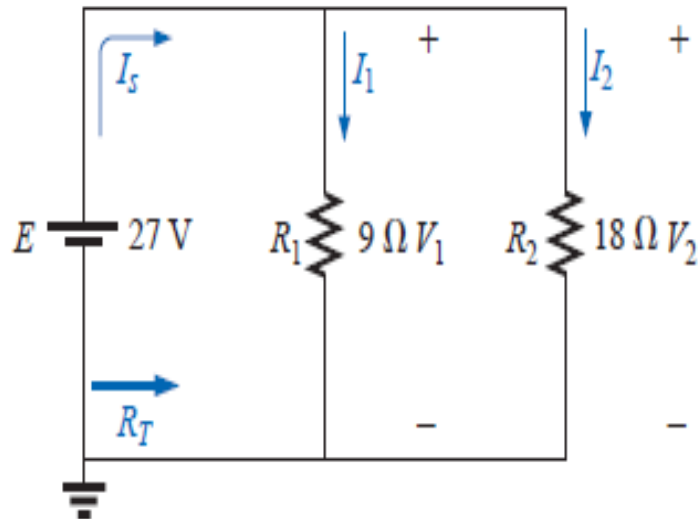
$$I_1 = \frac{V_1}{R_1} = \frac{E}{R_1}$$

$$I_2 = \frac{V_2}{R_2} = \frac{E}{R_2}$$

For single-source parallel networks, the source current (I_s) is equal to the sum of the individual branch currents.

$$I_s = I_1 + I_2$$

Question: Resistors in parallel



- Calculate R_T .
- Determine I_s .
- Calculate I_1 and I_2 , and demonstrate that $I_s = I_1 + I_2$.
- Determine the power to each resistive load.
- Determine the power delivered by the source, and compare it to the total power dissipated by the resistive elements.

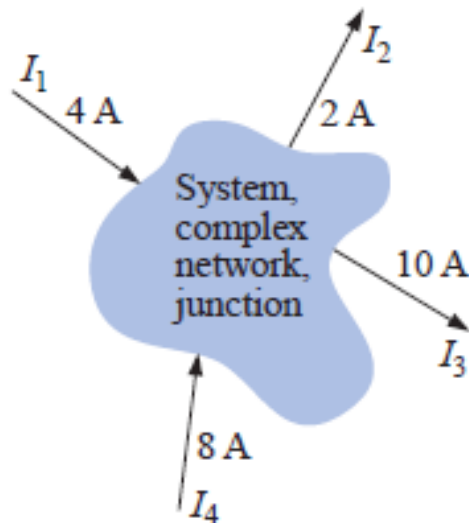
KIRCHHOFF'S CURRENT LAW

- Kirchhoff's current law (KCL) states that the algebraic sum of the currents entering and leaving an area, system, or junction is zero.
- In other words, the sum of the currents entering an area, system, or junction must equal the sum of the currents leaving the area, system, or junction.

$$\sum I_{\text{entering}} = \sum I_{\text{leaving}}$$

KIRCHHOFF'S CURRENT LAW

$$\sum I_{\text{entering}} = \sum I_{\text{leaving}}$$



$$\begin{aligned} I_1 + I_4 &= I_2 + I_3 \\ 4\text{ A} + 8\text{ A} &= 2\text{ A} + 10\text{ A} \\ 12\text{ A} &= 12\text{ A} \end{aligned}$$

Current Divider Principle

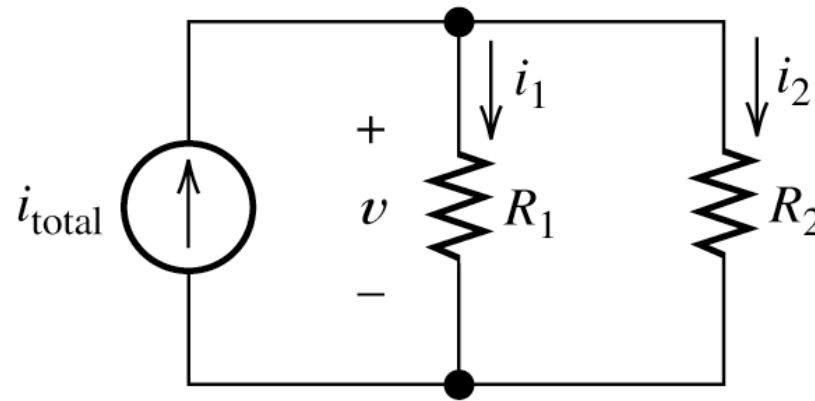
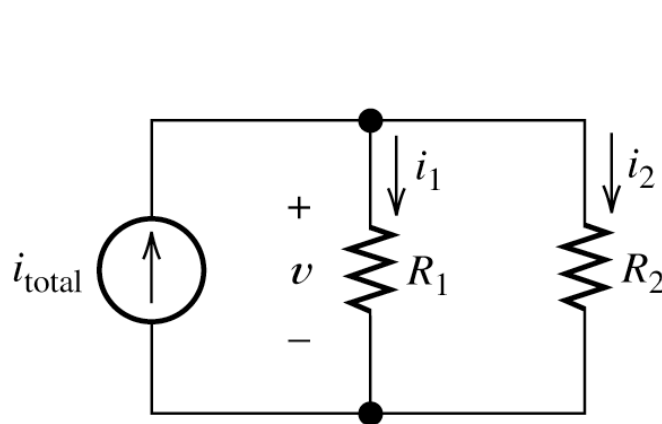


Figure 2.10 Circuit used to derive the current-division principle.

Current Division Principle



$$i_1 = \frac{v}{R_1} = \frac{R_2}{R_1 + R_2} i_{\text{total}}$$

$$i_2 = \frac{v}{R_2} = \frac{R_1}{R_1 + R_2} i_{\text{total}}$$

It can be also simplified as
(Especially considering
multiple resistors in parallel)

$$I_x = \frac{R_T}{R_x} I$$

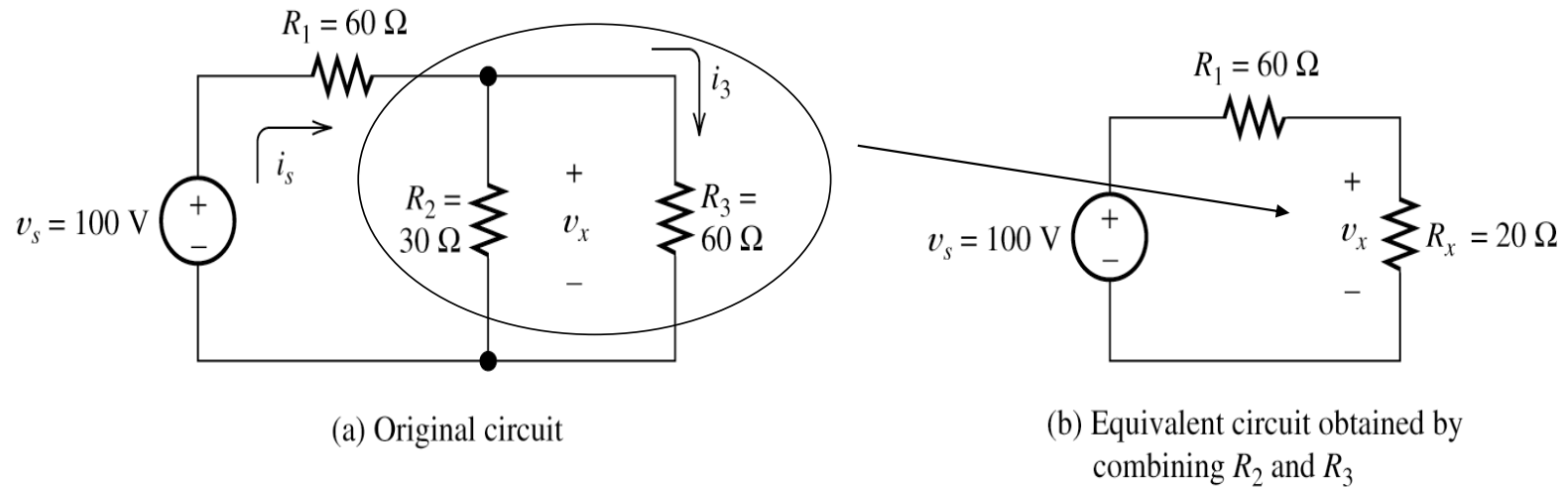
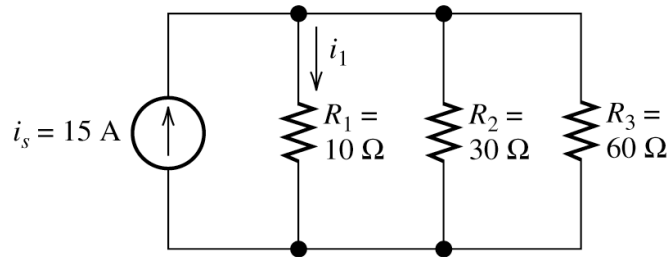


Figure 2.11 Circuit for Example 2.4.

Application of the Current-Division Principle



(a) Original circuit

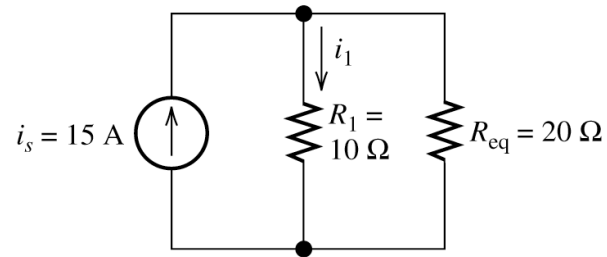
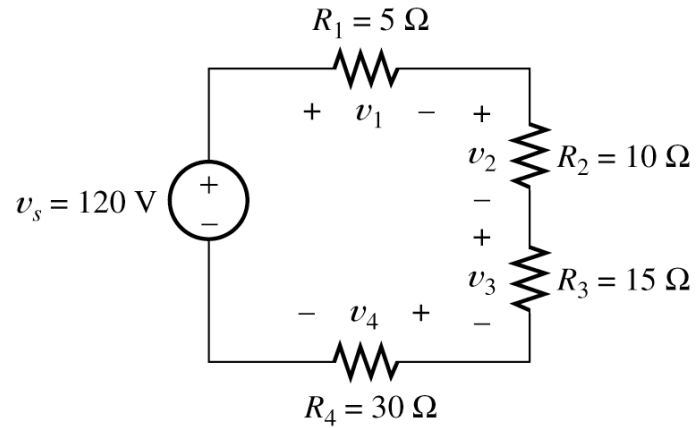
(b) Circuit after combining R_2 and R_3

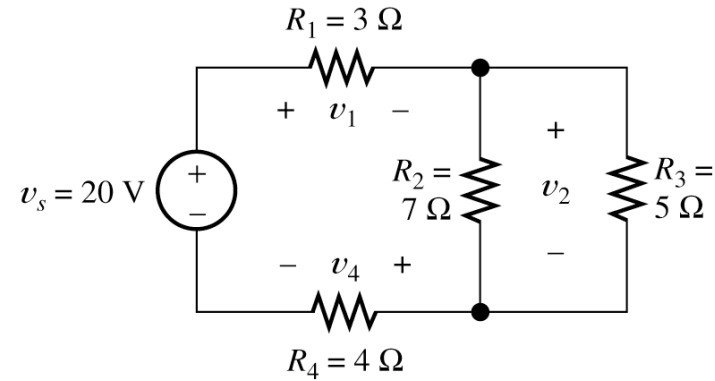
Figure 2.12 Circuit for Example 2.5.

$$R_{eq} = \frac{R_2 R_3}{R_2 + R_3} = \frac{30 \times 60}{30 + 60} = 20 \Omega$$

$$i_1 = \frac{R_{eq}}{R_1 + R_{eq}} i_s = \frac{20}{10 + 20} 15 = 10 \text{ A}$$



(a)



(b)

Figure 2.14 Circuits for Exercise 2.3.

- Voltage division

- Voltage division and
- current division

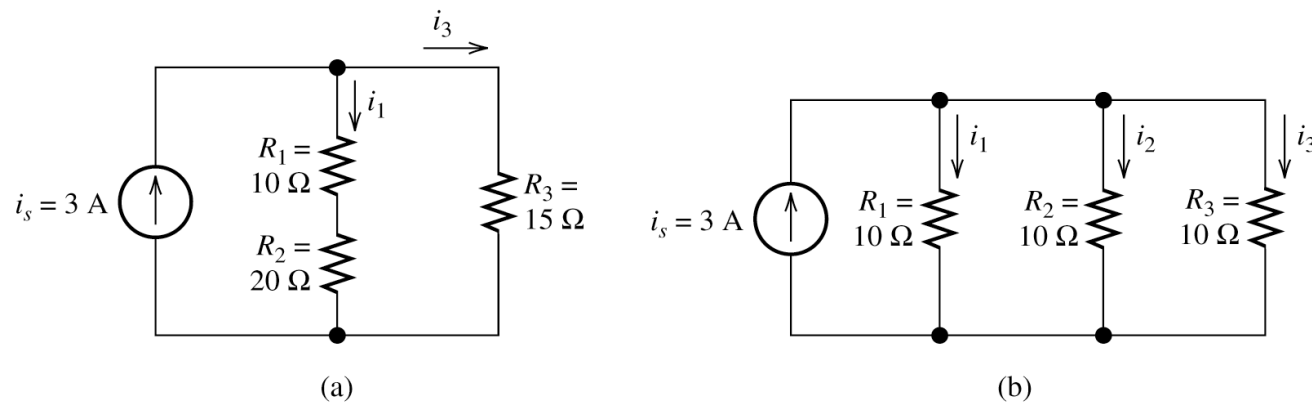


Figure 2.15 Circuits for Exercise 2.4.

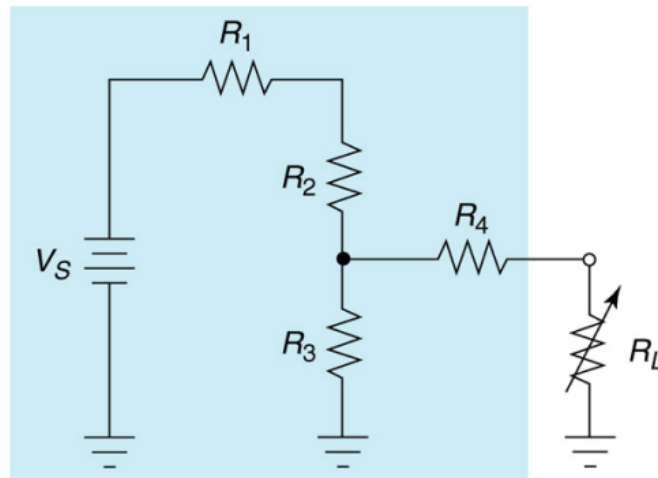
•Current division



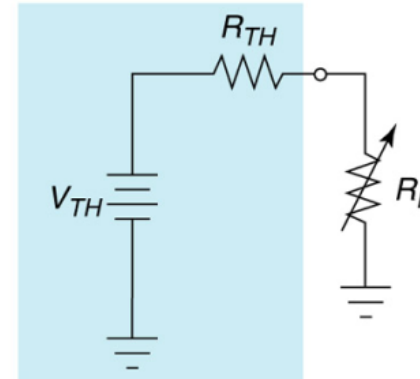
Although they are very important concepts, series/parallel equivalents and the current/voltage division principles are not sufficient to solve all circuits.

Thevenin's Theorem

- Thevenin's Theorem – any resistive circuit or network, no matter how complex, can be represented as a voltage source in series with a source resistance



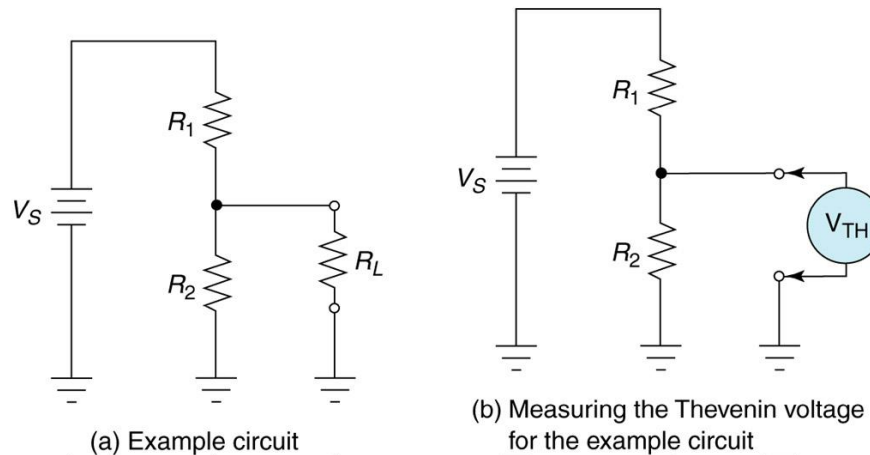
(a) Source circuit



(b) Thevenin equivalent

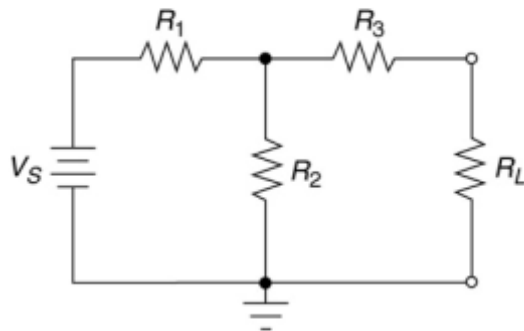
Thevenin's Theorem

- Thevenin Voltage (V_{TH}) – the voltage present at the output terminals of the circuit when the load is removed

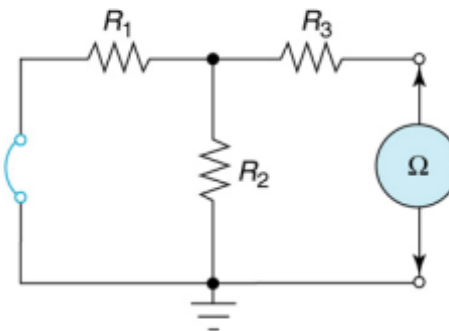


Thevenin's Theorem

- Thevenin Resistance (R_{TH}) – the resistance measured across the output terminals with the load removed



(a) Example circuit



(b) Measuring the Thevenin resistance for the example circuit. (Note: that the source has been replaced by a jumper wire.)

Thévenin Equivalent Circuits

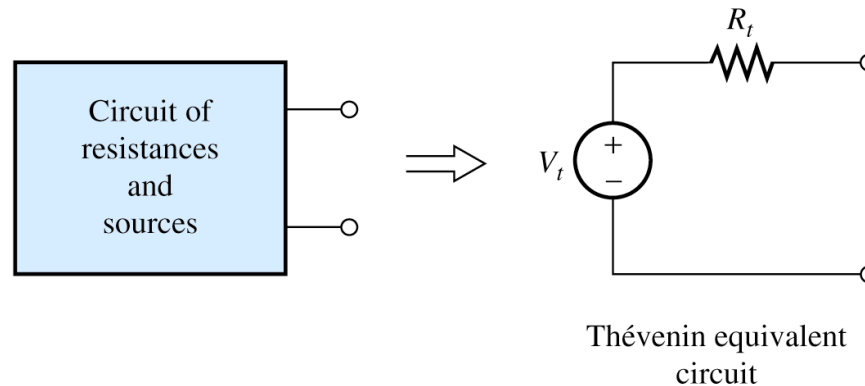


Figure 2.40 A two-terminal circuit consisting of resistances and sources can be replaced by a Thévenin equivalent circuit.

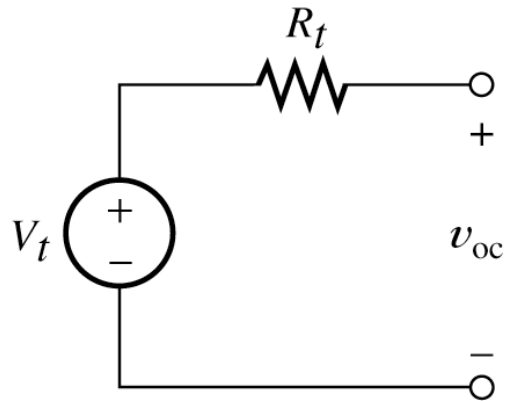


Figure 2.41 Thévenin equivalent circuit with open-circuited terminals. The open-circuit voltage v_{oc} is equal to the Thévenin voltage V_t .

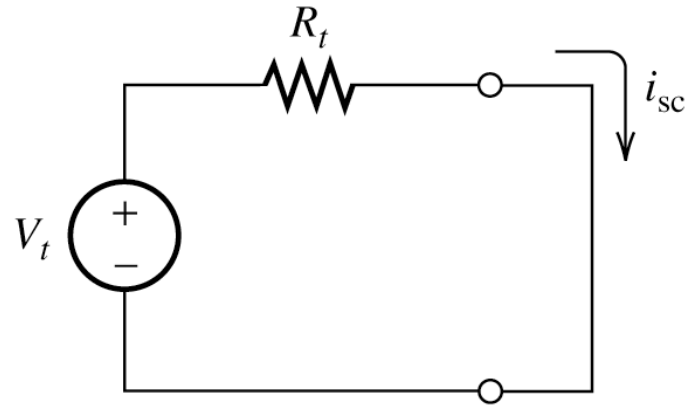
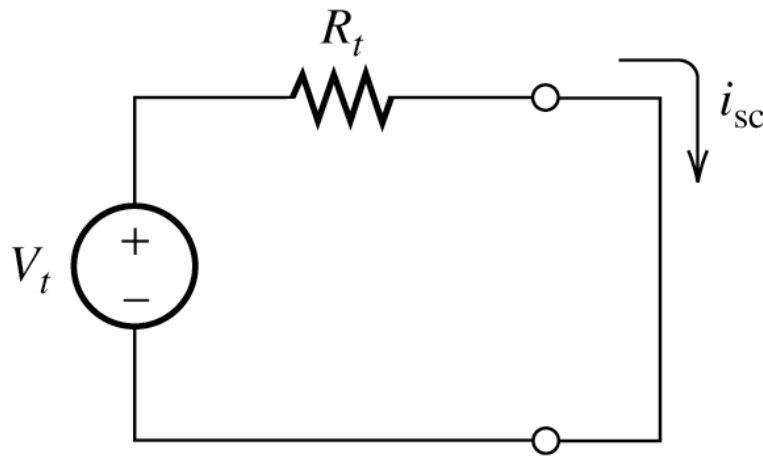


Figure 2.42 Thévenin equivalent circuit with short-circuited terminals. The short-circuit current is $i_{sc} = V_t/R_t$.

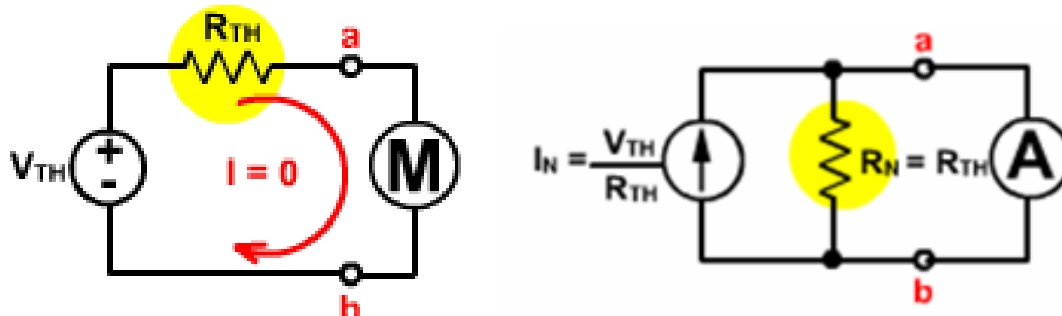
Thévenin Equivalent Circuits



$$V_t = v_{oc}$$

$$R_t = \frac{v_{oc}}{i_{sc}}$$

Thévenin Equivalent Circuits



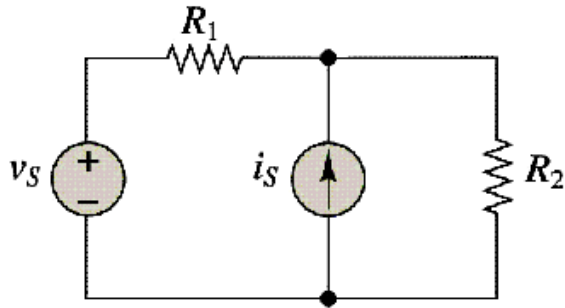
1. **Connect a voltmeter to the box**
Meter measured V_{TH} , since $I = 0$
2. **Connect an ammeter to the box**
Meter measured $I_N = \left(\frac{V_{TH}}{R_{TH}} \right)$
3. $R_{TH} = \frac{V_{MEASURED}}{I_{MEASURED}}$

Finding the Thévenin Resistance Directly

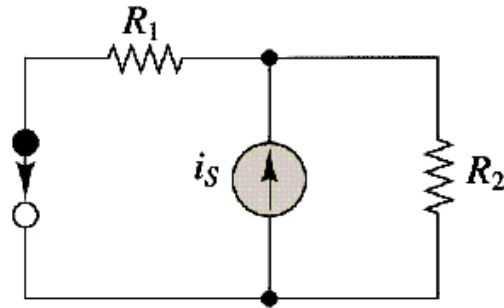
When **zeroing a voltage source**, it becomes a short circuit. When **zeroing a current source**, it becomes an open circuit.

*We can find the Thévenin resistance by **zeroing the sources in the original network** and then computing the resistance between the terminals.*

1. In order to set a voltage source equal to zero, we replace it with a short circuit.

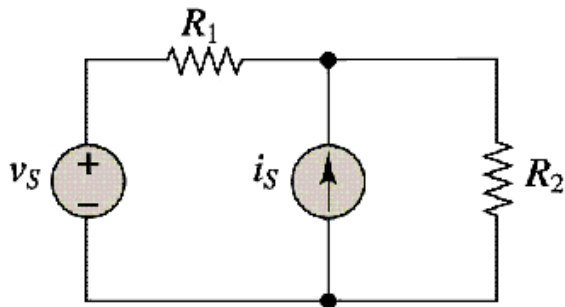


A circuit

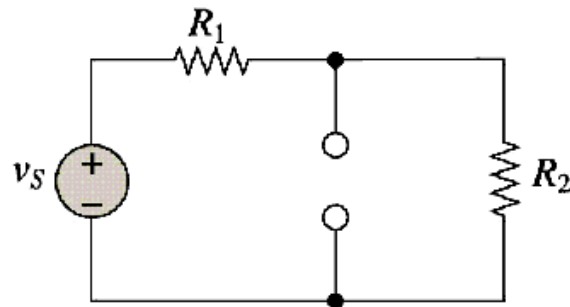


The same circuit with $v_S = 0$

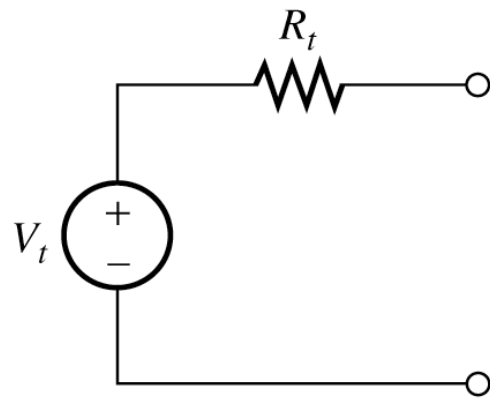
2. In order to set a current source equal to zero, we replace it with an open circuit.



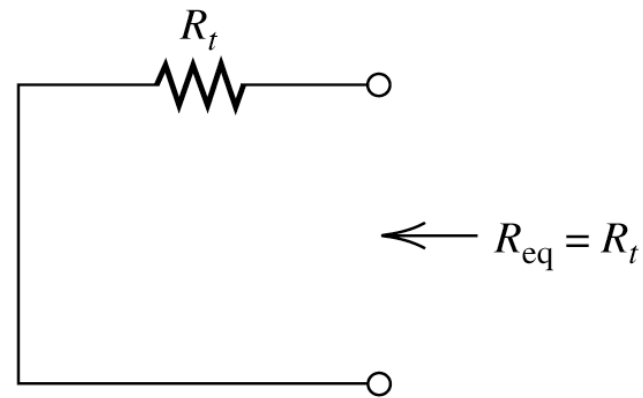
A circuit



The same circuit with $i_S = 0$



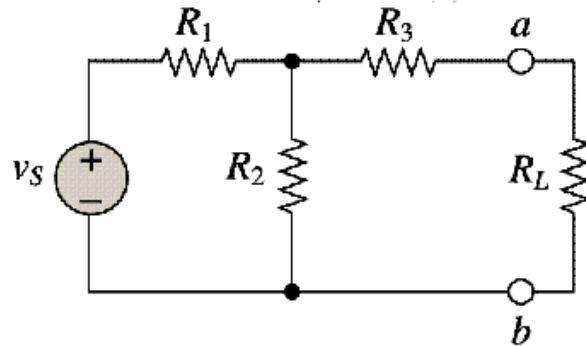
(a) Thévenin equivalent



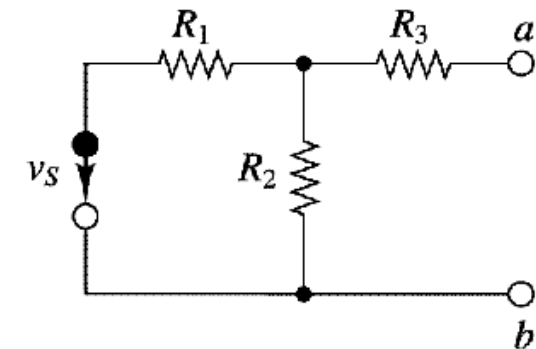
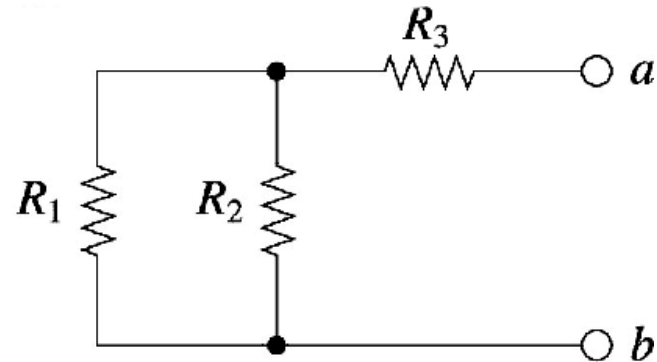
(b) Thévenin equivalent with its source zeroed

Figure 2.45 When the source is zeroed, the resistance seen from the circuit terminals is equal to the Thévenin resistance.

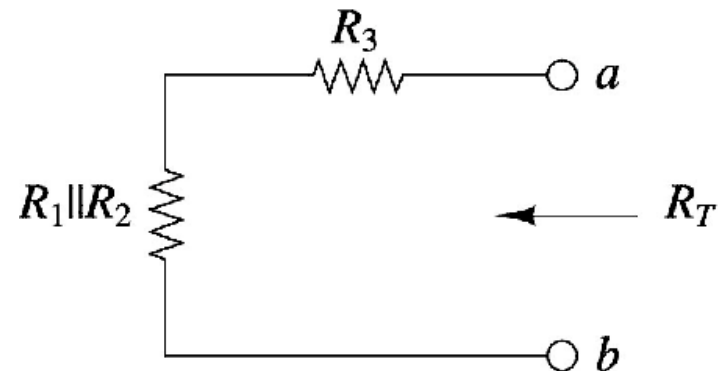
Computation of Thevenin resistance

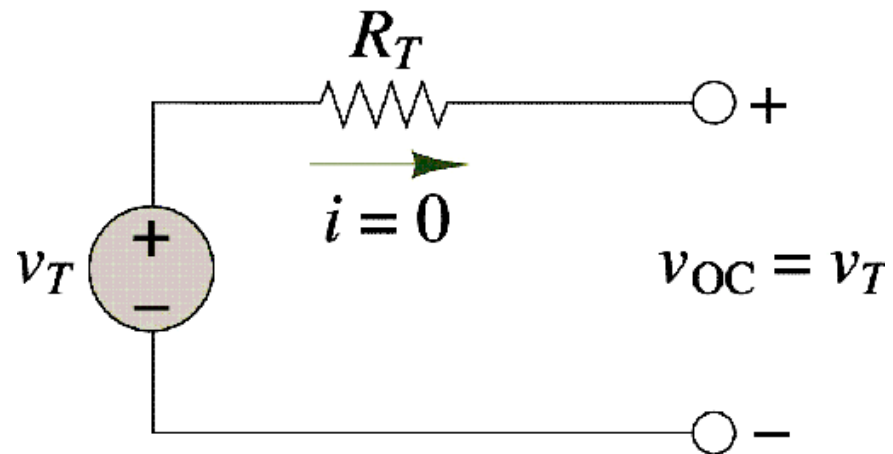
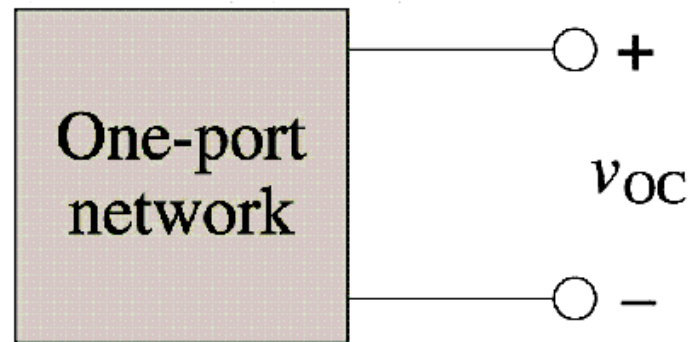


Complete circuit

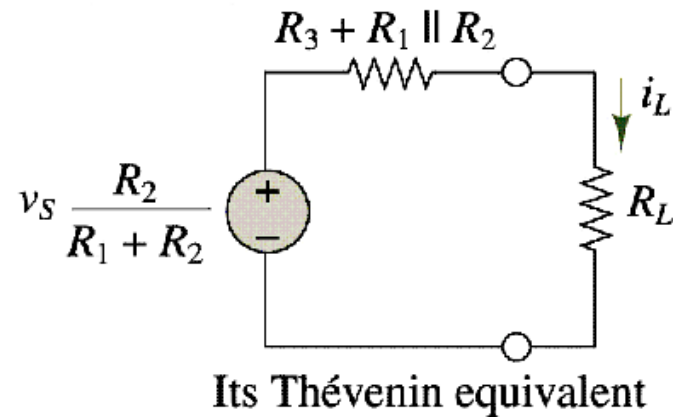
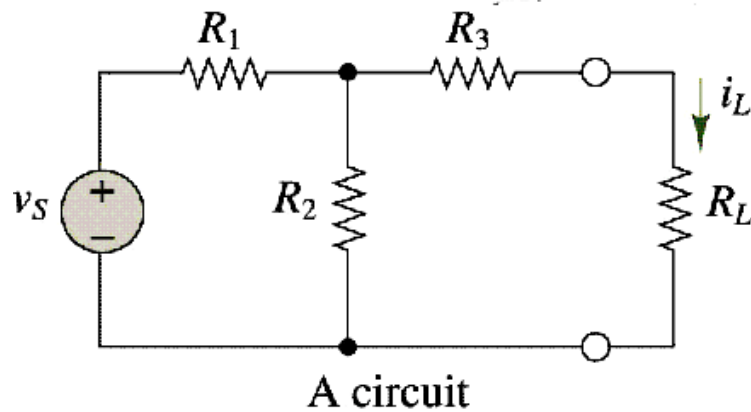


Circuit with load removed for computation of R_T . The voltage source is replaced by a short circuit.





A circuit and its Thévenin equivalent



Applications of Thevenin's Theorem

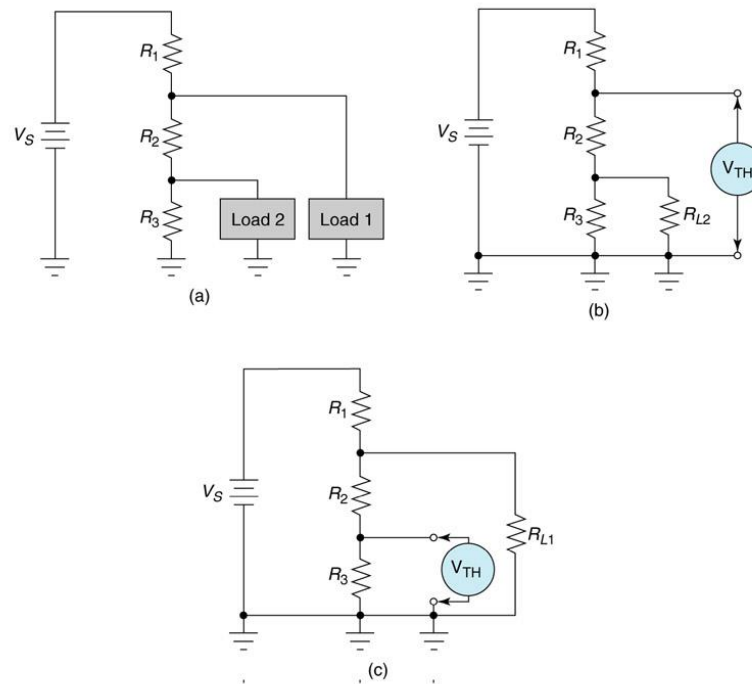
- Load Voltage Ranges – Thevenin's theorem is most commonly used to predict the change in load voltage that will result from a change in load resistance

Applications of Thevenin's Theorem

- Maximum Power Transfer
 - Maximum power transfer from a circuit to a variable load occurs when the load resistance equals the source resistance
 - For a series-parallel circuit, maximum power occurs when $R_L = R_{TH}$

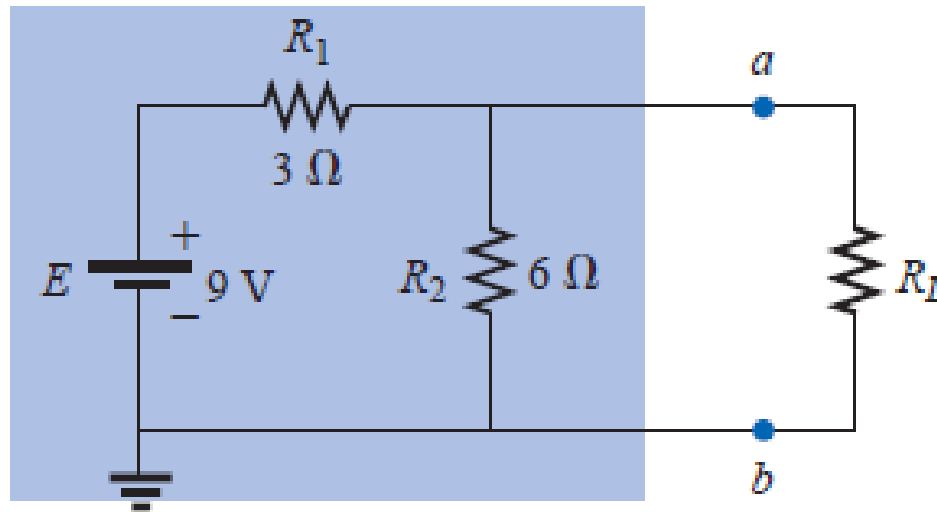
Applications of Thevenin's Theorem

- Multiload Circuits



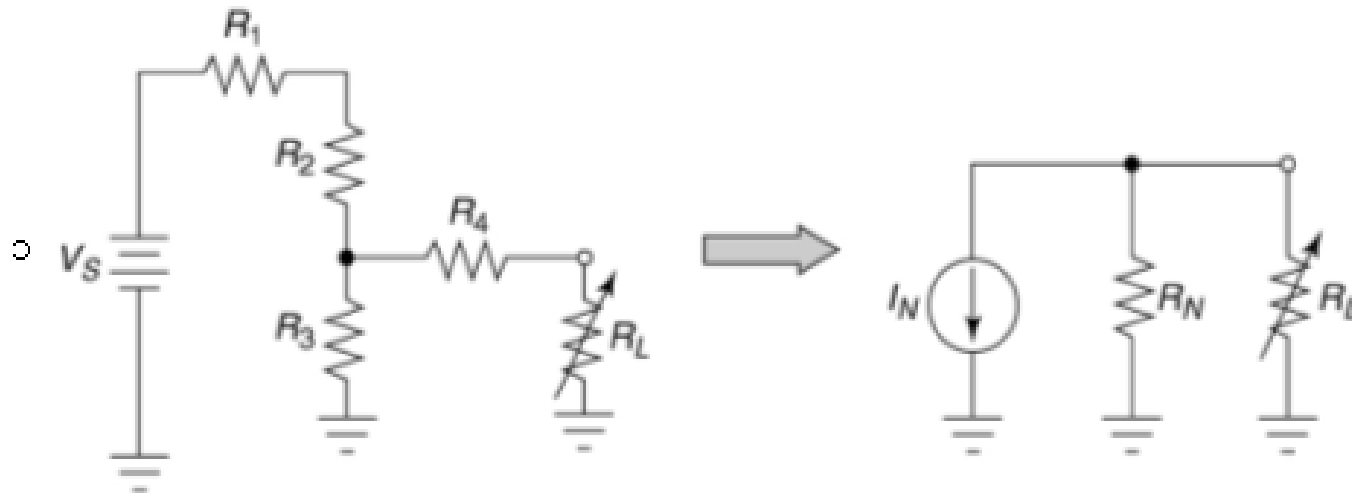
Applications of Thevenin's Theorem

- Example: Find the Thevenin equivalent circuit for the network in the shaded area of the circuit below



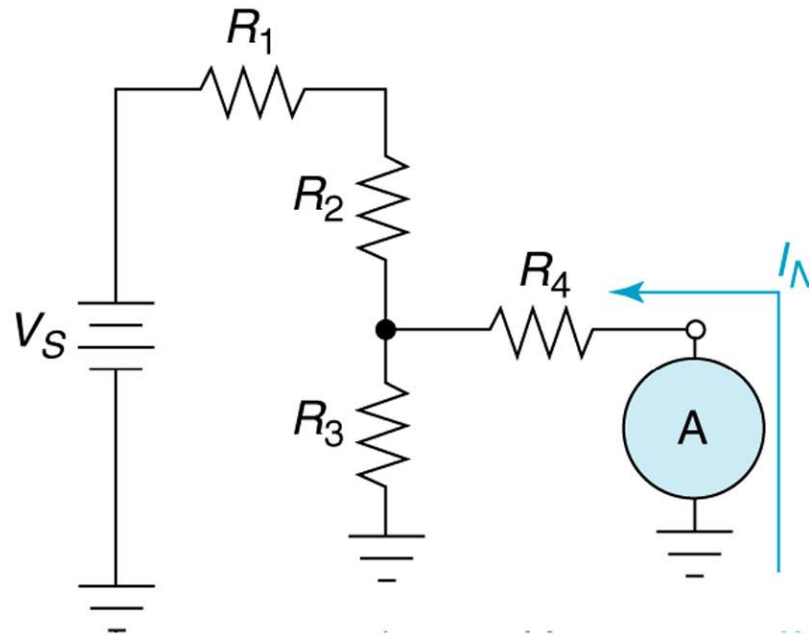
Norton's Theorem

- Norton's Theorem – any resistive circuit or network, no matter how complex, can be represented as a current source in parallel with a source resistance



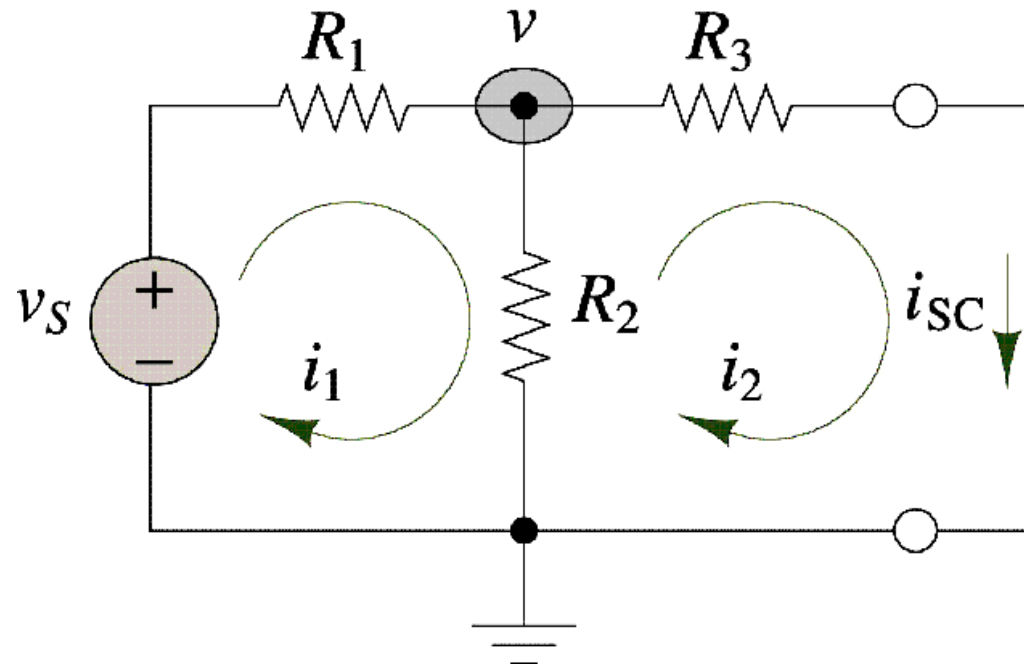
Norton's Theorem

- Norton Current (I_N) – the current through the shorted load terminals



Computation of Norton current

Short circuit
replacing the load

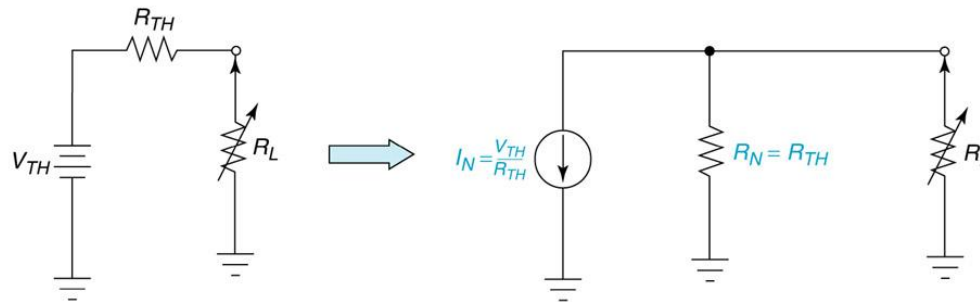


Norton's Theorem

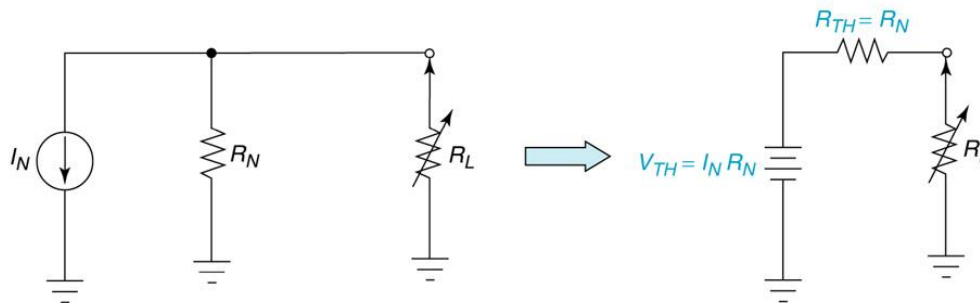
- Norton Resistance (R_N) – the resistance measured across the open load terminals (measured and calculated exactly like R_{TH})

Norton's Theorem

- Norton-to-Thevenin and Thevenin-to-Norton Conversions



(a) Thevenin to Norton

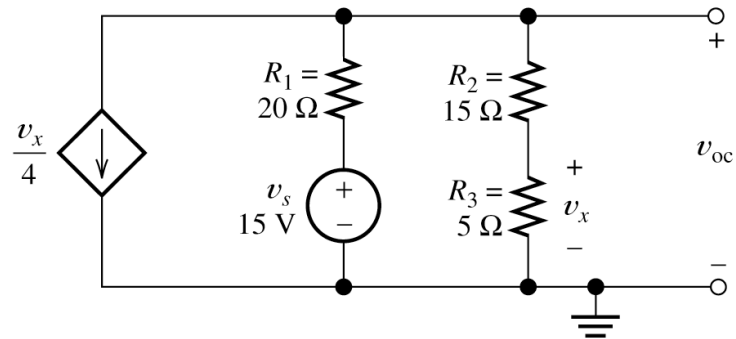


(b) Norton to Thevenin

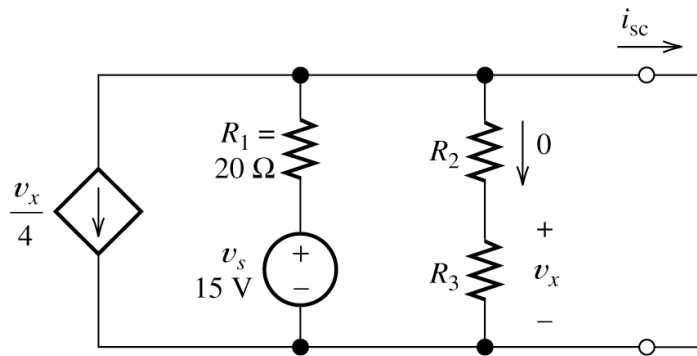
Step-by-step Thévenin/Norton-Equivalent-Circuit Analysis

1. Perform two of these:
 - a. Determine the open-circuit voltage $V_t = v_{oc}$.
 - b. Determine the short-circuit current $I_n = i_{sc}$.
 - c. Zero the sources and find the Thévenin resistance R_t looking back into the terminals.

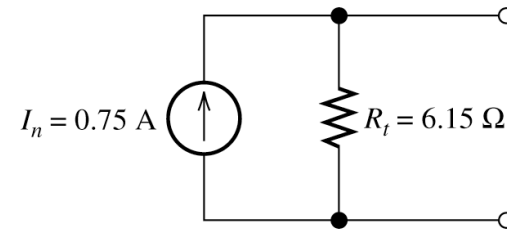
2. Use the equation $V_t = R_t I_n$ to compute the remaining value.
3. The Thévenin equivalent consists of a voltage source V_t in series with R_t .
4. The Norton equivalent consists of a current source I_n in parallel with R_t .



(a) Original circuit under open-circuit conditions



(b) Circuit with a short circuit



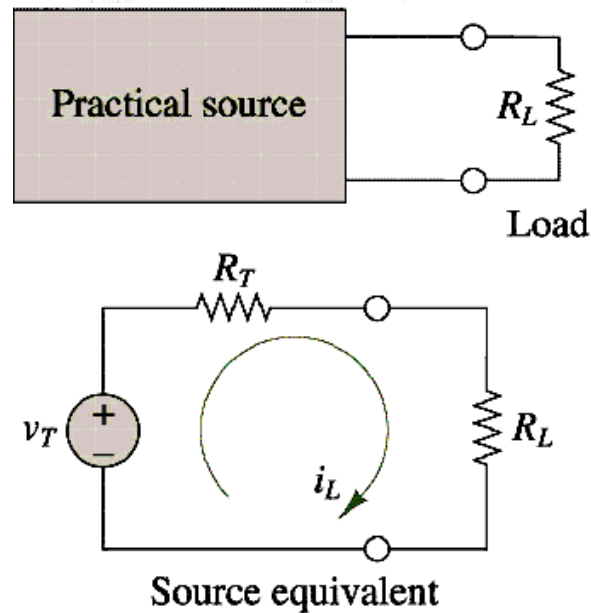
(c) Norton equivalent circuit

Figure 2.51 Circuit of Example 2.17.

Maximum Power Transfer

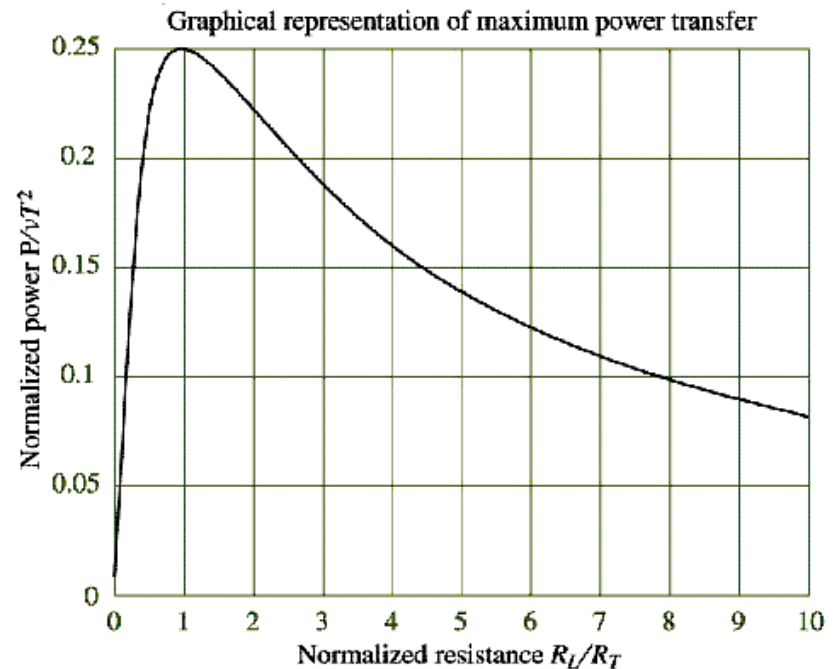
The load resistance that absorbs the maximum power from a two-terminal circuit is equal to the Thévenin resistance.

Power transfer between source and load



Given v_T and R_T , what value of R_L will allow for maximum power transfer?

Graphical representation of maximum power transfer



...that's all folks...
...thanks for your time...



EE 152: Basic Electronics (Semiconductor Basics)



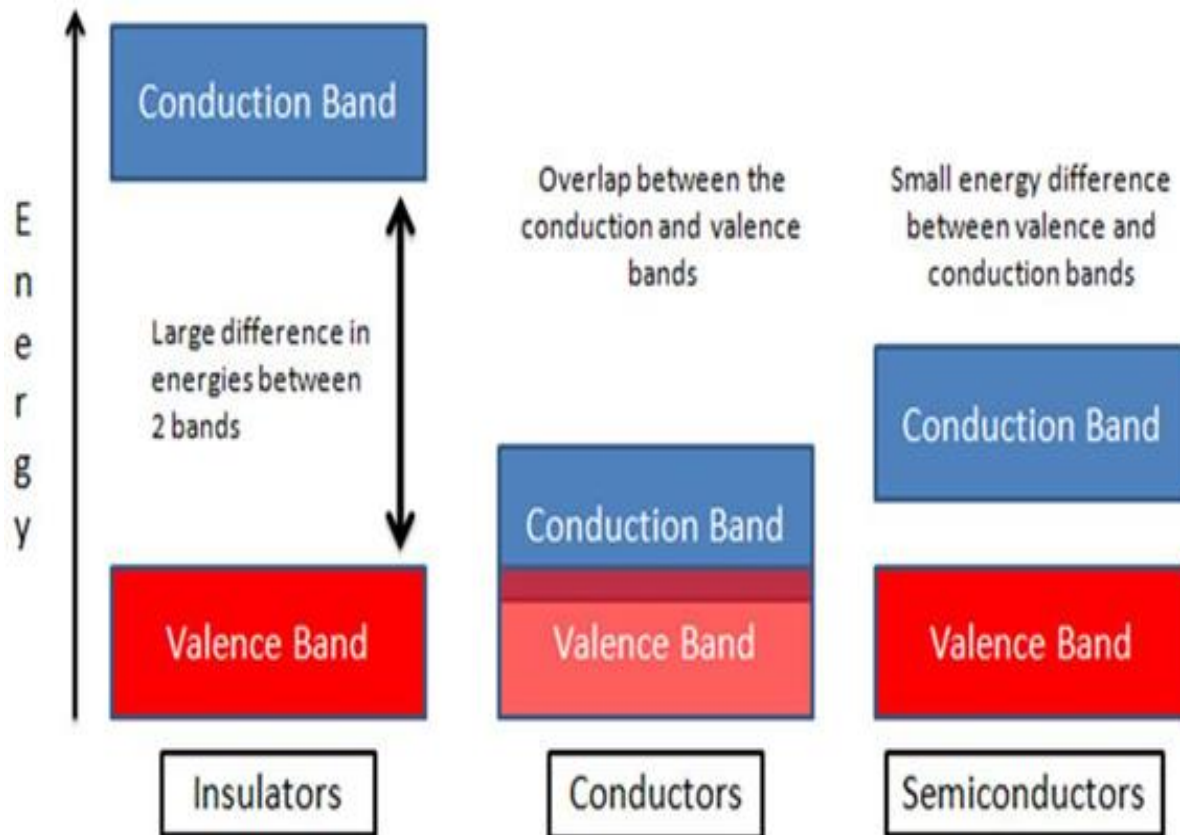
Outline

- Introduction
- Basic Semiconductor Concepts
 - Intrinsic
 - Doping
 - Extrinsic
 - N-type
 - P-type
 - Carrier movement

Electronic Materials

- The goal of electronic materials is to generate and control the flow of an electrical current.
- Electronic materials include:
 1. Conductors: have low resistance which allows electrical current flow
 2. Insulators: have high resistance which suppresses electrical current flow
 3. Semiconductors: can allow or suppress electrical current flow

Conductors, semiconductors and insulators

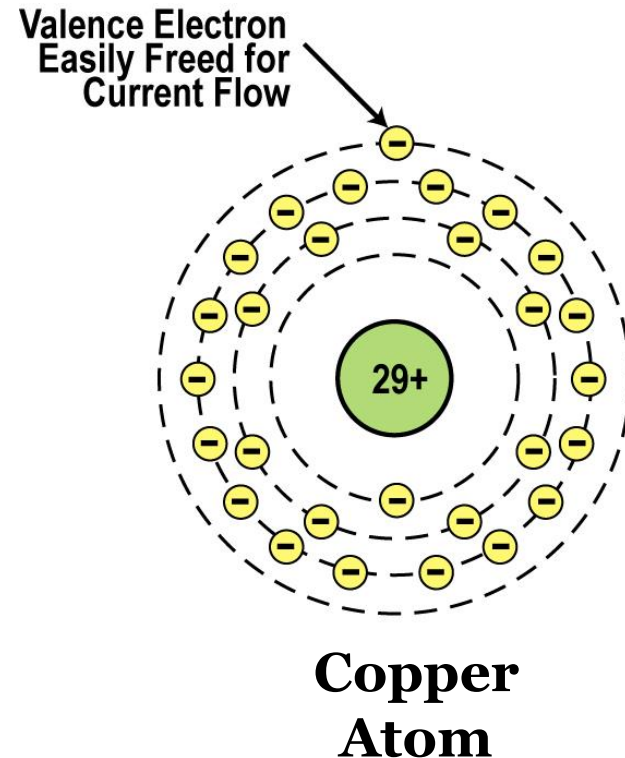


Conductors

- Good conductors have low resistance so electrons flow through them with ease.
- **Best** element conductors include:
 - Copper, silver, gold, aluminum, & nickel
- Alloys are also good conductors:
 - Brass & steel
- Good conductors can also be liquid:
 - Salt water

Conductor Atomic Structure

- The atomic structure of good conductors usually includes only one electron in their outer shell.
 - It is called a valence electron.
 - It is easily stripped from the atom, producing current flow.



Insulators

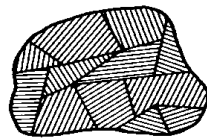
- Insulators have a high resistance so current does not flow in them.
- Good insulators include:
 - Glass, ceramic, plastics, & wood
- Most insulators are compounds of several elements.
- The atoms are tightly bound to one another so electrons are difficult to strip away for current flow.

Semiconductors

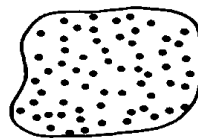
- Semiconductors are materials that essentially can be conditioned to act as good conductors, or good insulators, or any thing in between.
- Common elements such as **carbon**, **silicon**, and **germanium** are semiconductors.
- Silicon is the best and most widely used semiconductor.

What is a Semiconductor?

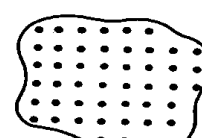
- A semiconductor is a material with conducting properties between those of a good insulator (e.g. glass) and a good conductor (e.g. copper).
- The most commonly used semiconductor is silicon.
- Low resistivity => “conductor”
- High resistivity => “insulator”
- Intermediate resistivity => “semiconductor”
 - conductivity lies between that of conductors and insulators
 - generally crystalline in structure for IC devices
 - In recent years, however, non-crystalline semiconductors have become commercially very important



polycrystalline

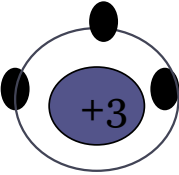
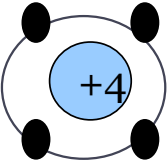
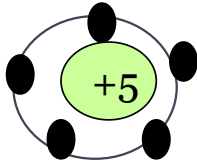


amorphous

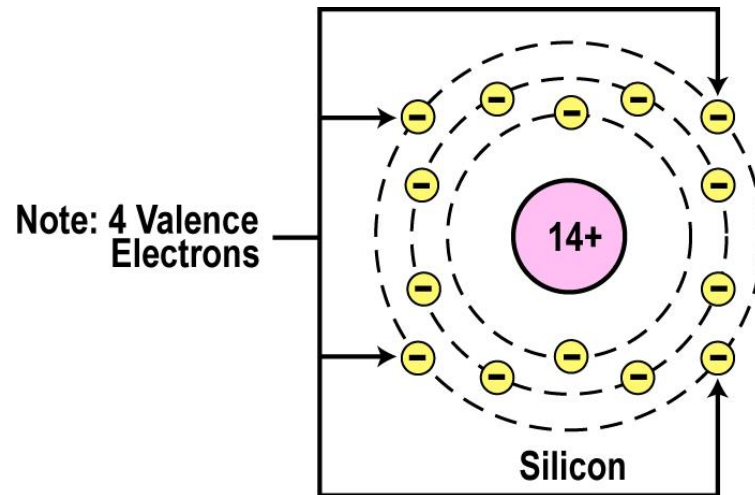


crystalline

Semiconductor Elements in the Periodic Table

Group III	Group IV	Group V
		
Boron (B)	Carbon (C)	Nitrogen (N)
Aluminium (Al)	Silicon (Si)	Phosphorus (P)
Gallium (Ga)	Germanium (Ge)	Arsenic (As)
Indium (In)	Tin (Sn)	Antimony (Sb)

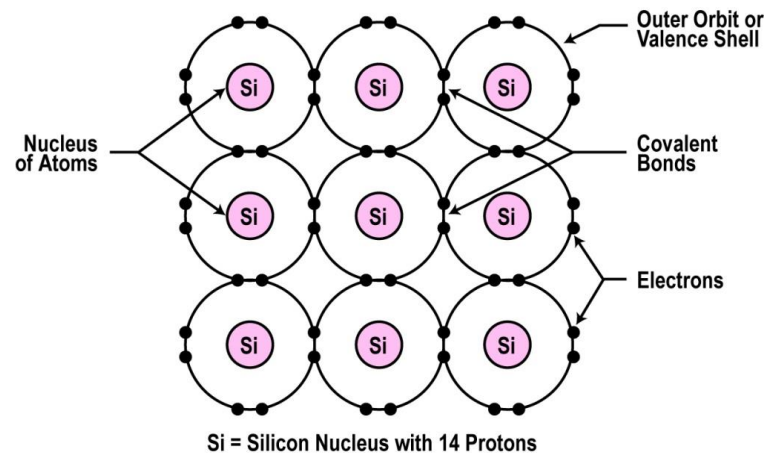
Semiconductor Valence Orbit



- The main characteristic of a semiconductor element is that it has four electrons in its outer or valence orbit.

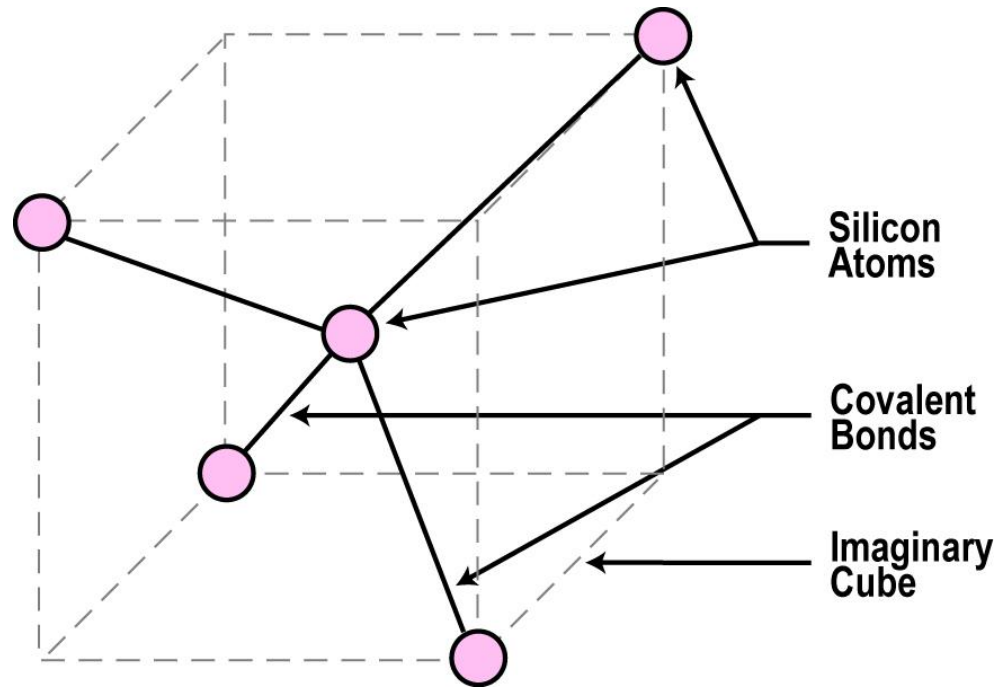
Crystal Lattice Structure

- The unique capability of semiconductor atoms is their ability to link together to form a physical structure called a crystal lattice.
- The atoms link together with one another sharing their outer electrons.
- These links are called covalent bonds.



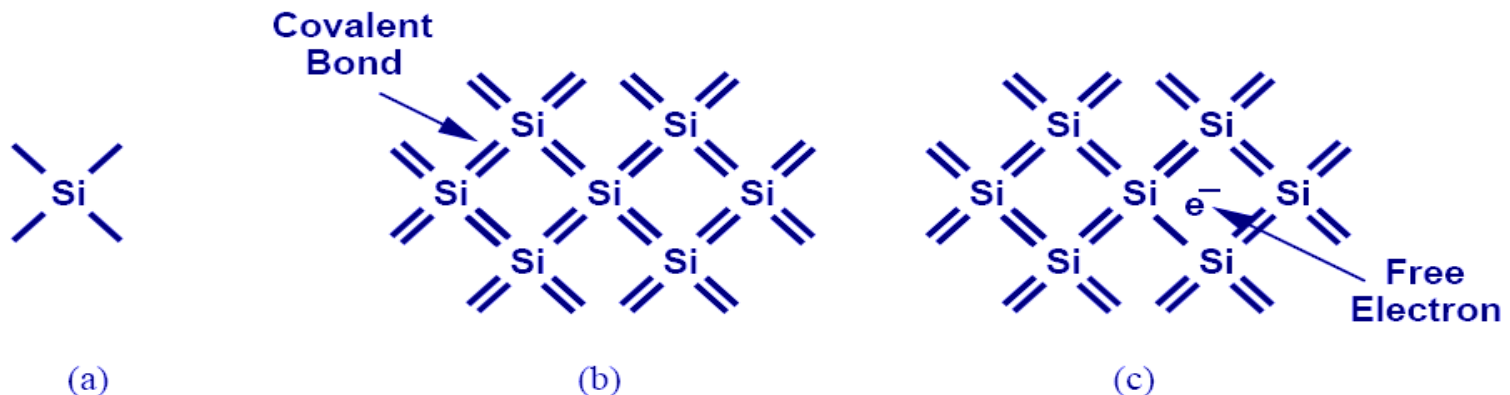
2D Crystal Lattice Structure

3D Crystal Lattice Structure



Silicon

- Atomic density: 5×10^{22} atoms/cm³
- Each silicon atom has an outer shell with four valence electrons and four vacancies (It is a *tetravalent* element).
- In *intrinsic* (pure) silicon, atoms join together by forming *covalent bonds*. Each atom shares its valence electrons with each of four adjacent neighbours effectively filling its outer shell.
- When temperature goes up, electrons can become free to move about the Si lattice.



Electronic Properties of Si

- **Silicon is a semiconductor material.**
 - Pure Si has a relatively high electrical resistivity at room temperature.
- **There are 2 types of mobile charge-carriers in Si:**
 - *Conduction electrons* are negatively charged;
 - *Holes* are positively charged.
- **The concentration ($\#/cm^3$) of conduction electrons & holes in a semiconductor can be modulated in several ways:**
 1. by adding special impurity atoms (*dopants*)
 2. by applying an electric field
 3. by changing the temperature
 4. by irradiation

- Thermal ionization
 - *Valence electron*---each silicon atom has four valence electrons
 - *Covalent bond*---two valence electrons from different two silicon atoms form the covalent bond
 - Be intact at sufficiently low temperature
 - Be broken at room temperature
 - *Free electron*---produced by thermal ionization, move freely in the lattice structure.
 - *Hole*---empty position in broken covalent bond, can be filled by free electron, positive charge
- Carriers

A free electron is negatively charge and a hole is positively charge. Both of them can move in the crystal structure. They can conduct electric circuit.

- Recombination

Some free electrons filling the holes results in the disappearance of free electrons and holes.

- Thermal equilibrium

At a certain temperature, the recombination rate is equal to the ionization rate. So the concentration of the carriers is able to be calculated.

- Carrier concentration in thermal equilibrium

$$n = p = n_i$$

$$n_i^2 = BT^3 e^{-E_G/kT}$$

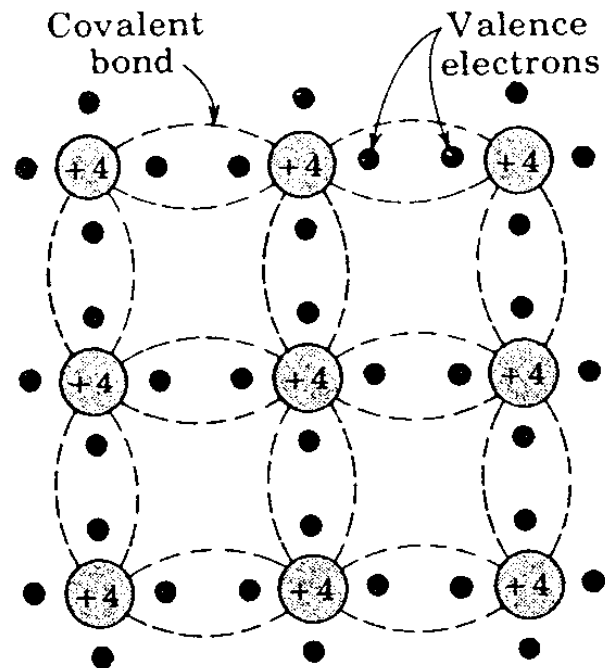
- At room temperature($T=300\text{K}$)

$$n_i \cong 1.5 \times 10^{10} \quad \text{carriers/cm}^3$$

Semiconductors can be Insulators

- If the material is pure semiconductor material like silicon, the crystal lattice structure forms an excellent insulator since all the atoms are bound to one another and are not free for current flow.
- Good insulating semiconductor material is referred to as intrinsic.
- Since the outer valence electrons of each atom are tightly bound together with one another, the electrons are difficult to dislodge for current flow.
- Silicon in this form is a great insulator.
- Semiconductor material is often used as an insulator.

Intrinsic Semiconductors



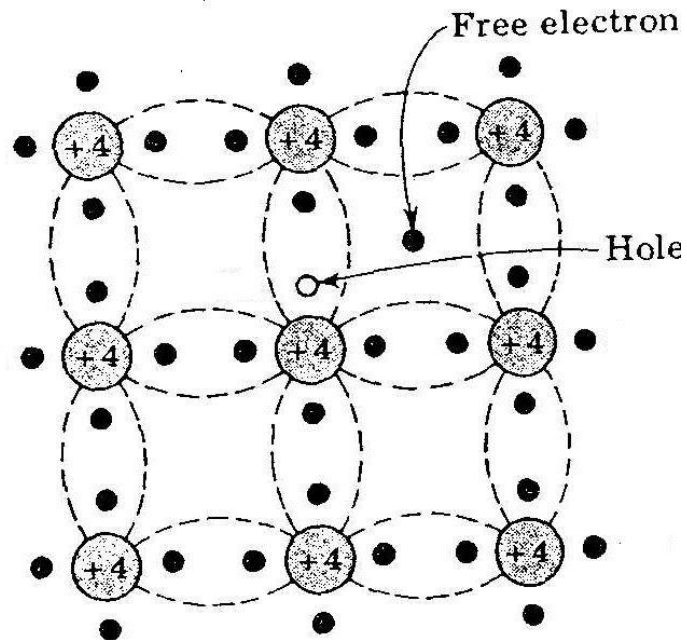
Intrinsic Semiconductors

- The structure has zero overall charge
- The complete nature of the structure means that at absolute zero temperature (0 K) none of the electrons is available for conduction...thus far the material is an insulator.

Intrinsic Semiconductors

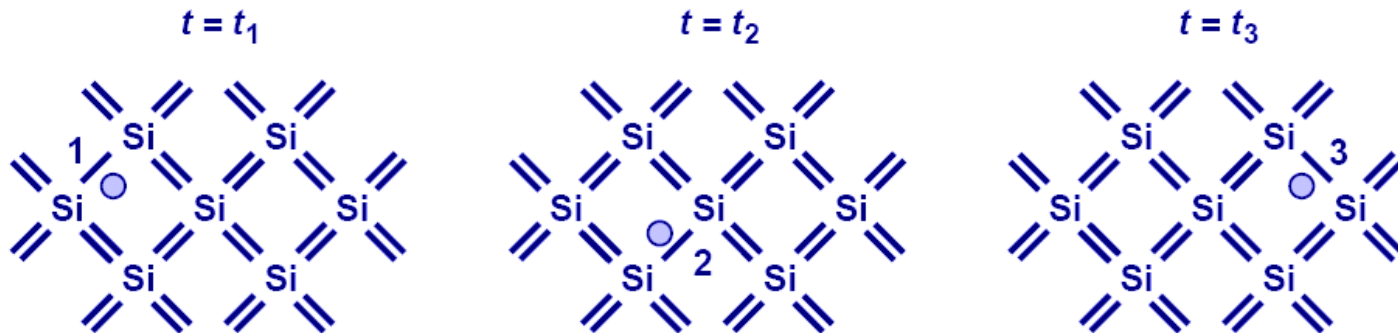
- At room temperature some of the electrons are able to acquire sufficient thermal energy to break free from their bond.
- Whenever an electron leaves its position in the lattice it leaves a vacancy known as a *hole*.
- The process is known as *electron-hole pair generation*

Intrinsic Semiconductors



Electron-Hole Pair Generation

- When a conduction electron is thermally generated, a “hole” is also generated.
- A hole is associated with a positive charge, and is free to move about the Si lattice as well.



Carrier Concentrations in Intrinsic Si

- The “band-gap energy” E_g is the amount of energy needed to remove an electron from a covalent bond.
- The concentration of conduction electrons in intrinsic silicon, n_i , depends exponentially on E_g and the absolute temperature (T):

$$n_i = 5.2 \times 10^{15} T^{3/2} \exp \frac{-E_g}{2kT} \text{ electrons / cm}^3$$

$$n_i \cong 1 \times 10^{10} \text{ electrons / cm}^3 \text{ at } 300\text{K}$$

$$n_i \cong 1 \times 10^{15} \text{ electrons / cm}^3 \text{ at } 600\text{K}$$

Intrinsic Semiconductors

- A freed electron can move through the body of the material until it encounters another broken bond where it is drawn in to complete the bond or *recombines*.

Intrinsic Semiconductors

- At a given temperature there is a dynamic equilibrium between thermal electron-hole *generation* and the *recombination* of electrons and holes
- As a result the concentration of electrons and holes in an intrinsic semiconductor is constant at any given temperature.
- The higher the temperature the more electron-hole pairs that are present.

Intrinsic Semiconductors

- **Two** mechanisms for conduction become possible when a bond breaks:
 - 1. Due to the movement of the freed electron.
 - 2. Due to neighbouring electrons moving into the hole leaving a space behind it. (This can be most simply thought of as movement of the hole, a single moving positive charge carrier even though it is actually a series of electrons that move.

Intrinsic Semiconductors

- When an electric field (voltage) is applied, the holes move in one direction and the electrons in the other.
- However both current components are in the direction of the field.
- The conduction is ohmic, i.e. current is proportional to the applied voltage (field)

Intrinsic Semiconductors

- The proportion of freed electrons is very small indeed:
- In silicon the energy E_G required to free an electron is 1.2eV
- The mean thermal energy (kT) is only 25meV at room temperature (1/40 eV)
- The proportion of freed electrons varies exponentially ($-E_G/kT$).

Intrinsic Semiconductors

- For an intrinsic semiconductor the number of electron and hole carriers, and thus the conductivity, increases rapidly with temperature.
- This is not very useful.
- Hence we **dope** the material to produce an **extrinsic semiconductor**.

Doping

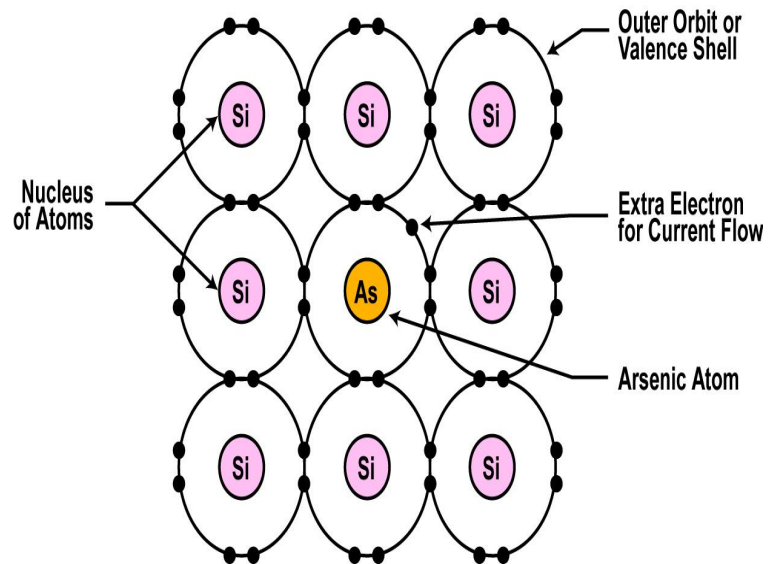
- To make the semiconductor conduct electricity, other atoms called impurities must be added.
- “Impurities” are different elements.
- This process is called doping.

Extrinsic Semiconductors

- Intrinsic conduction is very small.
- Conductivity levels can be raised and controlled by *doping* with minute levels of impurity atoms to give *extrinsic* or *doped* semiconductors.
- Extrinsic semiconductors may be further divided into either n-type or p-type

Semiconductors can be Conductors

- An impurity, or element like arsenic, has 5 valence electrons.
- Adding arsenic (doping) will allow four of the arsenic valence electrons to bond with the neighboring silicon atoms.
- The one electron left over for each arsenic atom becomes available to conduct current flow.

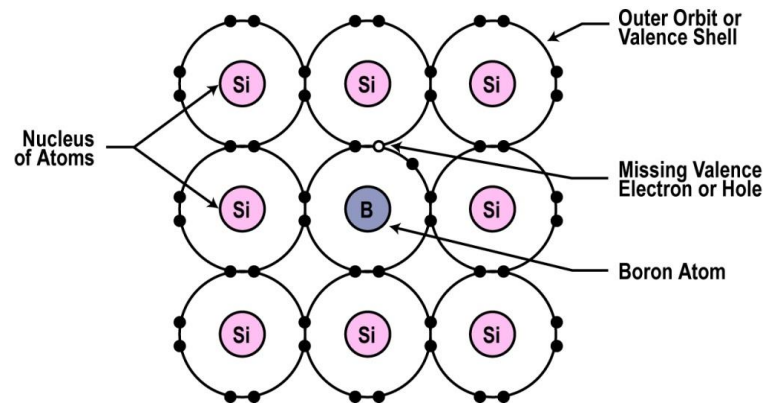


Resistance Effects of Doping

- If you use lots of arsenic atoms for doping, there will be lots of extra electrons so the resistance of the material will be low and current will flow freely.
- If you use only a few boron atoms, there will be fewer free electrons so the resistance will be high and less current will flow.
- By controlling the doping amount, virtually any resistance can be achieved.

Another Way to Dope

- You can also dope a semiconductor material with an atom such as boron that has only 3 valence electrons.
- The 3 electrons in the outer orbit do form covalent bonds with its neighboring semiconductor atoms as before. But one electron is missing from the bond.
- This place where a fourth electron should be is referred to as a hole.
- The hole assumes a positive charge so it can attract electrons from some other source.
- Holes become a type of current carrier like the electron to support current flow.



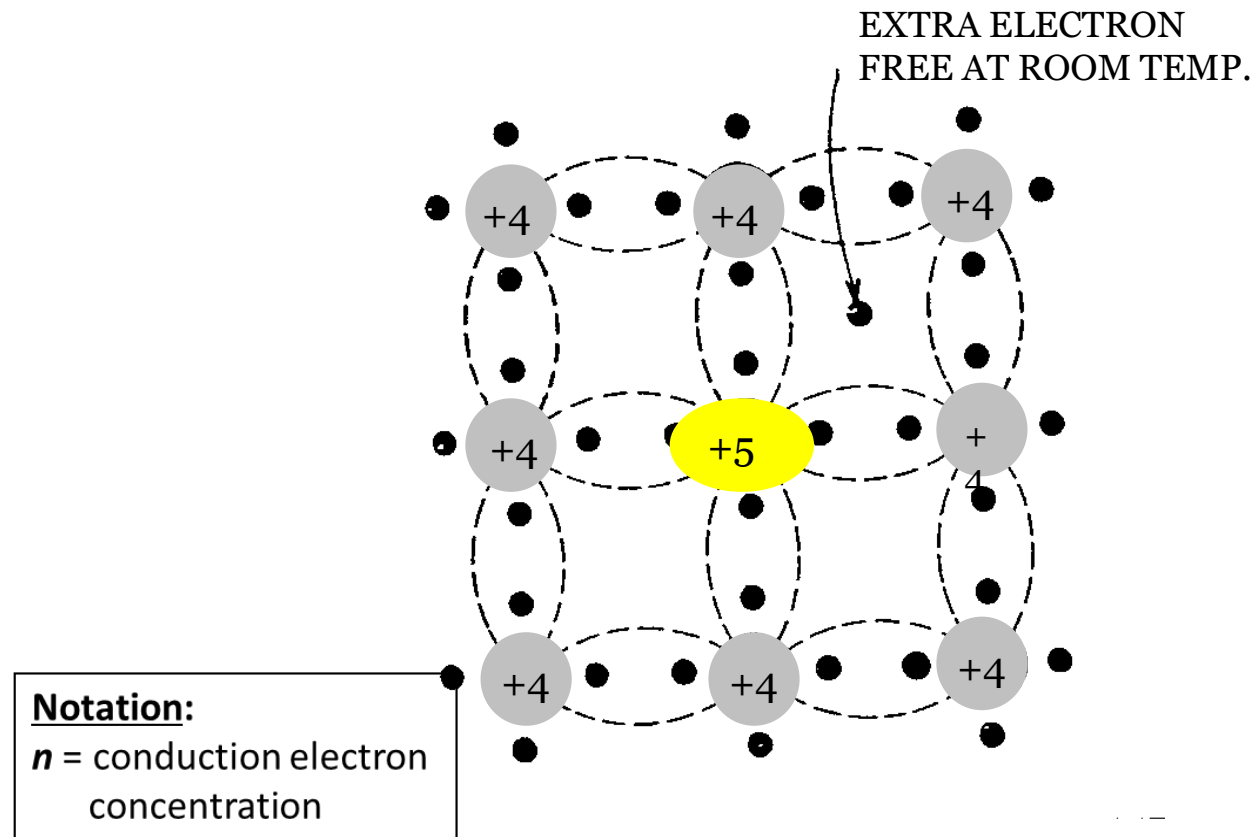
Types of Semiconductor Materials

- The silicon doped with extra electrons is called an “N type” semiconductor.
 - “N” is for negative, which is the charge of an electron.
- Silicon doped with material missing electrons that produce locations called holes is called “P type” semiconductor.
 - “P” is for positive, which is the charge of a hole.

N-type Semiconductors

- An n-type impurity atom has five outer (valence) electrons, rather than the four of silicon.
- Only four of the outer electrons are required for covalent bonding. The fifth is much more easily detached from the parent atom.
- As the energy needed to free the fifth electron is smaller than the thermal energy at room temperature virtually all are freed.

N-type Semiconductors



Carrier concentration for n type

a) Thermal equilibrium equation

$$n_{n0} \cdot p_{n0} = n_i^2$$

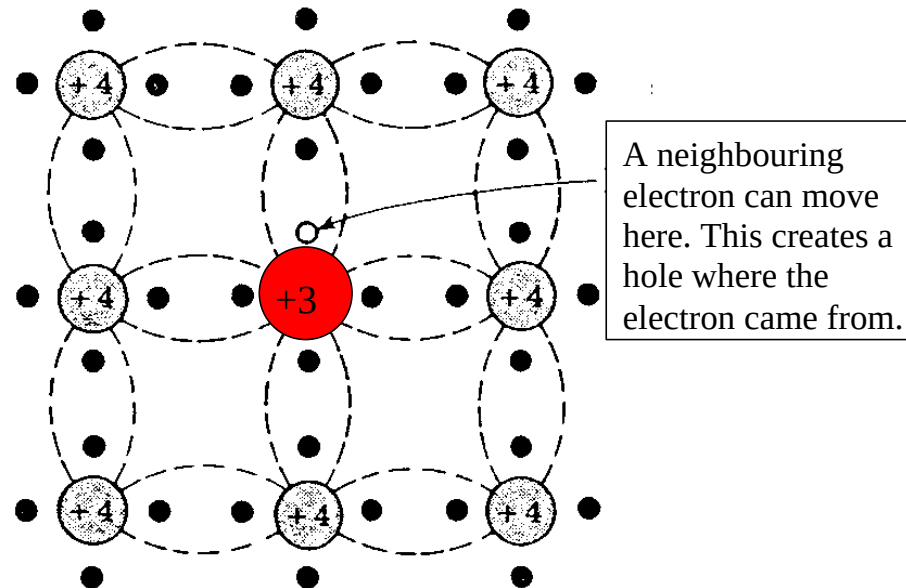
b) Electric neutral equation

$$n_{n0} = p_{n0} + N_D$$

P-type Semiconductors

- Here the doping atom has only three electrons in its outer shell.
- It is relatively easy for an electron from a neighbouring atom to move in, so releasing a hole at its parent atom. The freed hole is available for conduction.
- The energy needed to free the electron from its parent is usually small compared to the thermal energy so each impurity atom contributes one hole for conduction (fully ionised).

P-type Semiconductors



Notation:

p = hole concentration

Carrier concentration for p type

a) Thermal equilibrium equation

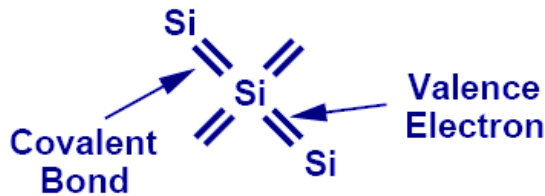
$$p_{p0} \cdot n_{p0} = n_i^2$$

b) Electric neutral equation

$$p_{p0} = n_{p0} + N_A$$

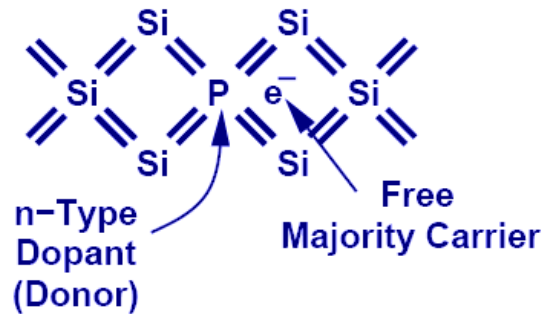
Summary of Charge Carriers

Intrinsic Semiconductor

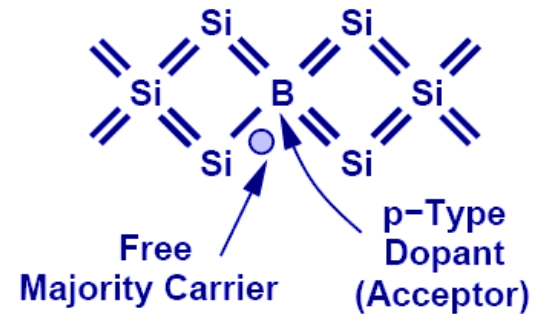


Extrinsic Semiconductor

Silicon Crystal
 N_D Donors/cm³



Silicon Crystal
 N_A Acceptors/cm³



Electron and Hole Concentrations

- Under thermal equilibrium conditions, the product of the conduction-electron density and the hole density is ALWAYS equal to the square of n_i :

$$np = n_i^2$$

N-type material

$$n \approx N_D$$

$$p \approx \frac{n_i^2}{N_D}$$

P-type material

$$p \approx N_A$$

$$n \approx \frac{n_i^2}{N_A}$$

Electron and Hole Densities

$$np = n_i^2$$

Majority Carriers : $p \approx N_A$

Minority Carriers : $n \approx \frac{n_i^2}{N_A}$

Majority Carriers : $n \approx N_D$

Minority Carriers : $p \approx \frac{n_i^2}{N_D}$

- The product of electron and hole densities is ALWAYS equal to the square of intrinsic electron density regardless of doping levels.

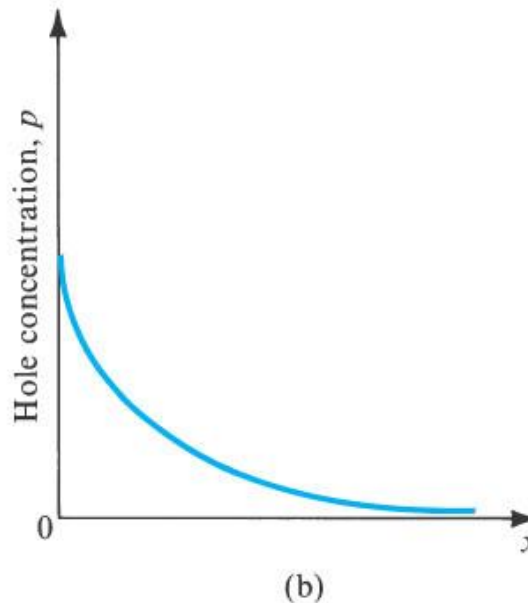
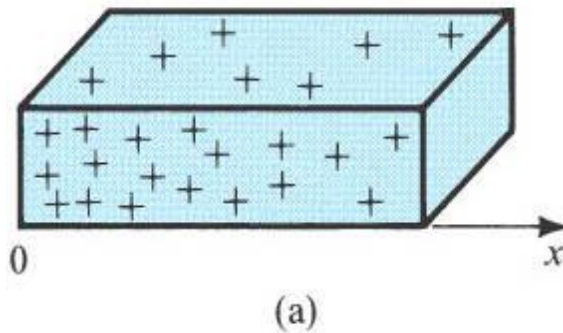
Carriers Movement

There are two mechanisms by which holes and free electrons move through a silicon crystal.

- Drift--- The carrier motion is generated by the electrical field across a piece of silicon. This motion will produce drift current.
- Diffusion--- The carrier motion is generated by the different concentration of carrier in a piece of silicon. The diffused motion, usually carriers diffuse from high concentration to low concentration, will give rise to diffusion current.

Diffusion and Diffusion Current

- diffusion

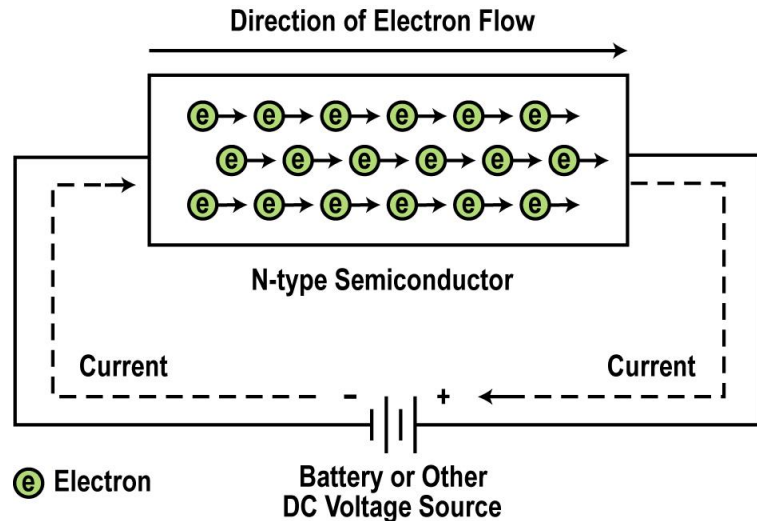


A bar of intrinsic silicon (a) in which the hole concentration profile shown in (b) has been created along the x -axis by some unspecified mechanism.

The diffusion current density is proportional to the slope of the concentration curve, or the concentration gradient.

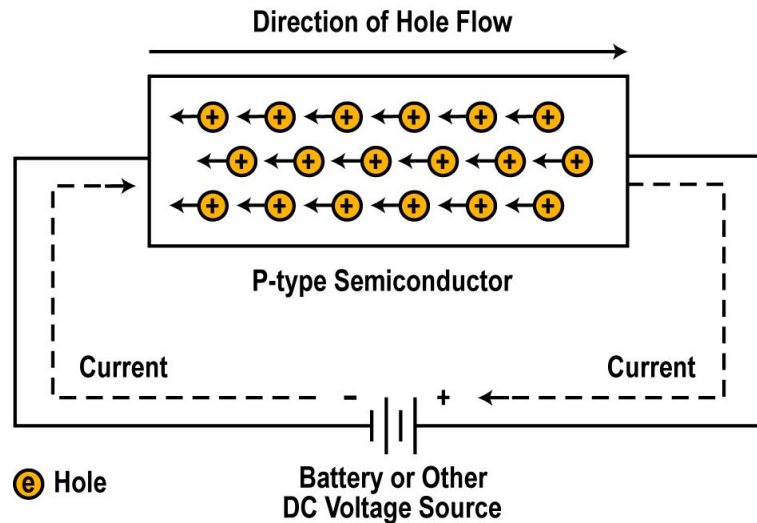
Current Flow in N-type Semiconductors

- The DC voltage source has a positive terminal that attracts the free electrons in the semiconductor and pulls them away from their atoms leaving the atoms charged positively.
- Electrons from the negative terminal of the supply enter the semiconductor material and are attracted by the positive charge of the atoms missing one of their electrons.
- Current (electrons) flows from the positive terminal to the negative terminal.



Current Flow in P-type Semiconductors

- Electrons from the negative supply terminal are attracted to the positive holes and fill them.
- The positive terminal of the supply pulls the electrons from the holes leaving the holes to attract more electrons.
- Current (electrons) flows from the negative terminal to the positive terminal.
- Inside the semiconductor current flow is actually by the movement of the holes from positive to negative.



Temperature sensitivity

- In both types of extrinsic semiconductor virtually all available charge carriers are freed from their parent atoms at room temperature. Temperature variations thus make little difference to the conductivity σ .
- For intrinsic conductivity the number of carriers, and thus σ , increases rapidly with temperature.
- For both extrinsic and intrinsic mechanisms the conductivity is zero at $T=0$ K

Terminology

donor: impurity atom that increases n

acceptor: impurity atom that increases p

N-type material: contains more electrons than holes

P-type material: contains more holes than electrons

majority carrier: the most abundant carrier

minority carrier: the least abundant carrier

intrinsic semiconductor: $n = p = n_i$ 

extrinsic semiconductor: doped semiconductor

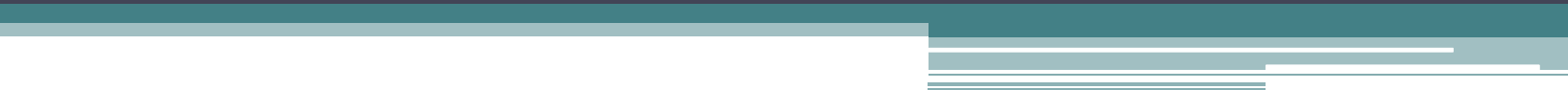
In Summary

- In its pure state, semiconductor material is an excellent insulator.
- The commonly used semiconductor material is silicon.
- Semiconductor materials can be doped with other atoms to add or subtract electrons.
- An N-type semiconductor material has extra electrons.
- In an N-type semiconductor, conduction is mainly due to electrons (negative charges). Positive charges (holes) are the minority carriers.
- A P-type semiconductor material has a shortage of electrons with vacancies called holes.
- In a P-type semiconductor, conduction is mainly due to holes (positive charges). Negative charges (electrons) are the minority carriers.
- The heavier the doping, the greater the conductivity or the lower the resistance.
- By controlling the doping of silicon the semiconductor material can be made as conductive as desired.

...that's all folks...
...thanks for your time...



PN-Junction Diode



Outline

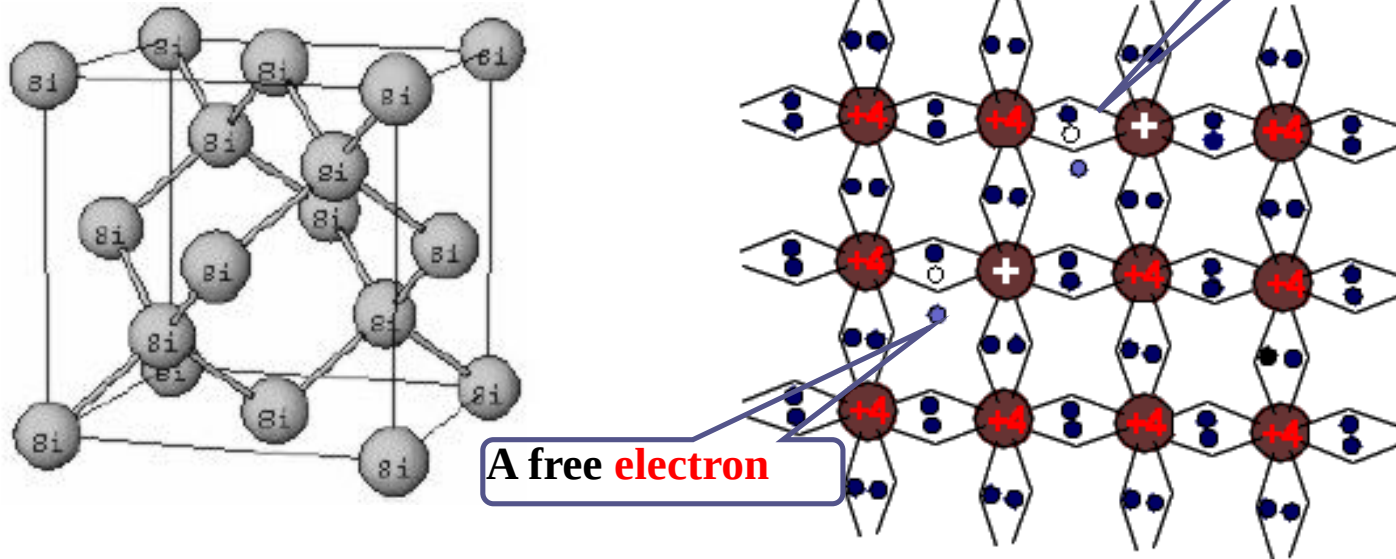
- The PN-Junction Diode
- Analysis of diode circuits
- I-V characteristic of *pn* junction
 - Terminal characteristic of junction diode.
 - Physical operation of diode.
- Applications of diode circuits
 - Rectification
 - Half wave Rectifiers
 - Full wave Rectifiers
 - Centre-tap
 - Bridge

Previous Lecture

- Semiconductor
 - Intrinsic
 - Doping
 - Extrinsic
 - N-type
 - P-type

Intrinsic (pure) Semiconductors

- Intrinsic(pure) silicon



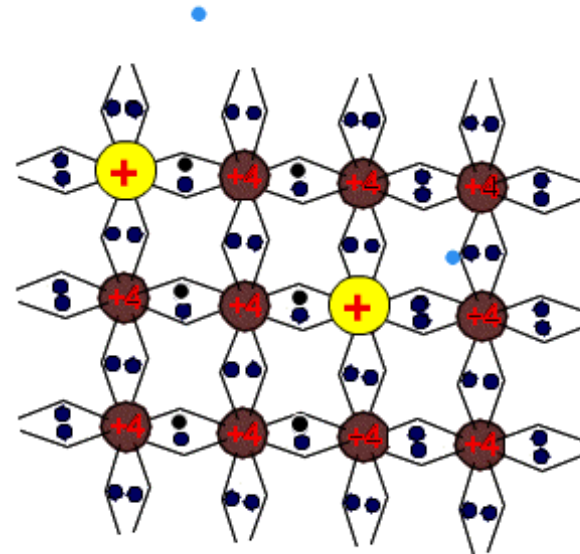
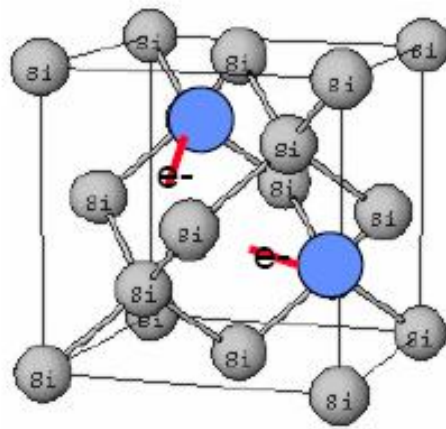
- An electron-hole pair is created when an electron get excited by thermal or light energy;
- Recombination occurs when an electron loses energy and falls back into a hole.

Intrinsic (pure) Semiconductors

- Holes also conduct current. In reality, it's the movement of all the other electrons. The hole allows this motion.
 - Holes have positive charge.
 - Current flows in the same direction as the holes move.
-
- Both electrons and holes carry current-- **carriers**.
 - In intrinsic semiconductors the electron and hole concentrations are equal because carriers are created in pairs
 - The intrinsic concentration depends exponentially on temperature.
 - At room temp (300K), the intrinsic carrier concentration of silicon is:

$$n_i = 1.5 \times 10^{10} / \text{cm}^3$$

Phosphorus Doping (N-type)



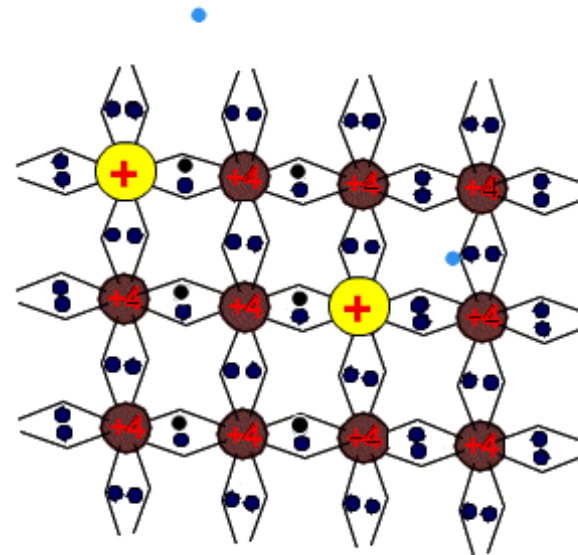
- Phosphorus has 5 valence electrons.
- P atoms will sit in the location of a Si atom in the lattice, to avoid breaking symmetry, but each will have an extra electron that does not bond in the same way. And these extra electrons are easier to excite (and can move around more easily)
- These electrons depends on the amounts of the two materials.

Phosphorus Doping (N-type)

Electrons---**Majority carrier.**

Holes---**Minority carrier**

Phosphorus---**Donor materials.**



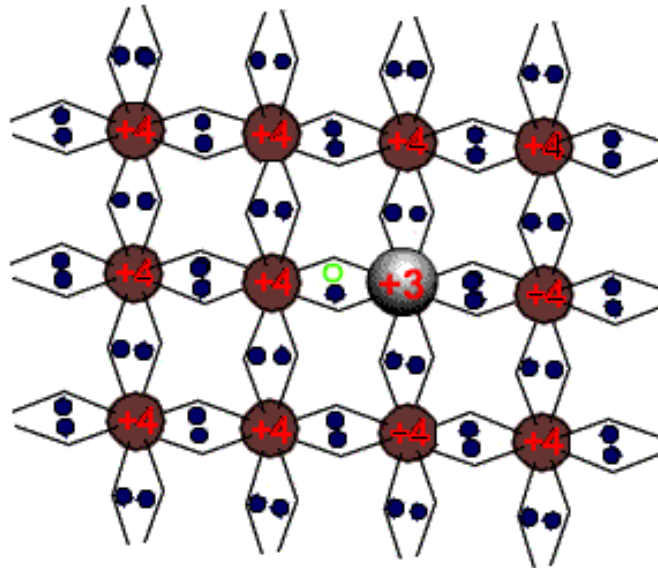
- In equilibrium, $pn = p_i n_i = p_i^2 = n_i^2$
- At room temp (300K), if $1/10^{10}$ donors are added to the intrinsic silicon, then the electron carrier concentration is about 10^{13}cm^{-3} ; the hole carrier concentration is about 10^6cm^{-3} .

Phosphorus $\rho \approx 89.3\Omega \cdot \text{cm}$; Intrinsic silicon $\rho \approx 2.14 \times 10^5 \Omega \cdot \text{cm}$

Boron Doping (P-type)

Holes---**Majority carrier;**
Electrons---**Minority carrier**

Boron---acceptor materials.

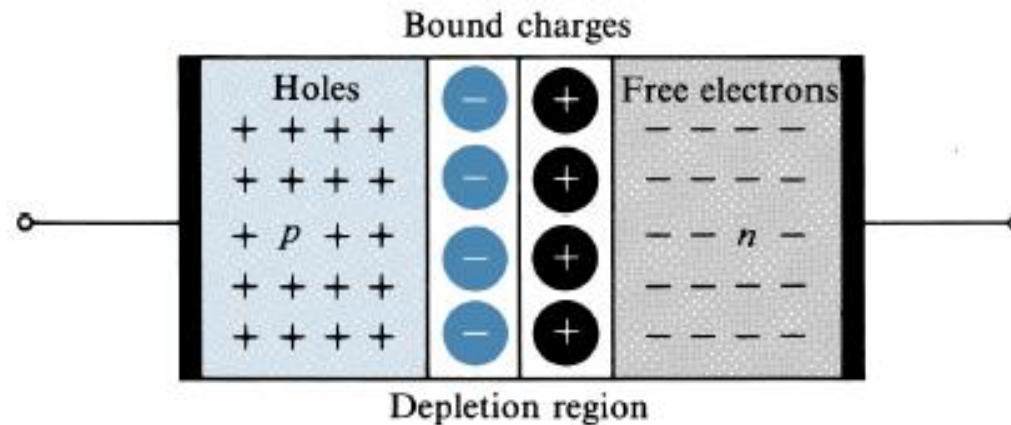


- Boron has 3 valence electrons.
- B will sit at a lattice site, but the adjacent Si atoms lack an electron to fill its shell. This creates a hole.

PN Junction

- **N-type** materials: Doping Si with a Group V element, providing extra electrons (n for negative) .
- **P-type** materials: Doping Si with a Group III element, providing extra holes (p for positive).

What happens when P-type meets N-type?



PN Junction

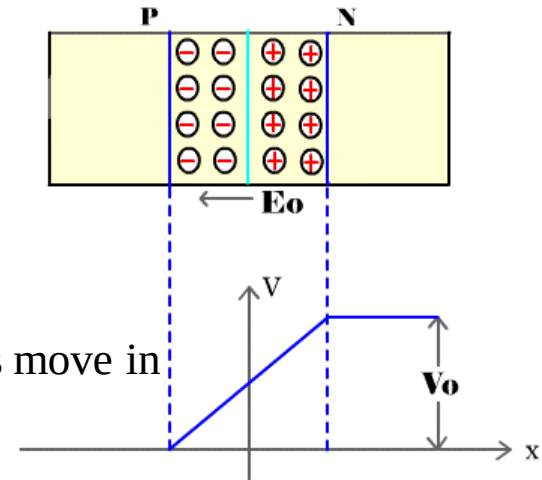
What happens when P-type meets N-type?

- Holes diffuse from the P-type into the N-type, electrons diffuse from the N-type into the P-type, creating a diffusion current.
- Once the holes [electrons] cross into the N-type [P-type] region, they recombine with the electrons [holes].
- This recombination “strips” the n-type [P-type] of its electrons near the boundary, creating an electric field due to the positive and negative bound charges.
- The region “stripped” of carriers is called the space-charge region, or depletion region.
- V_0 is the contact potential that exists due to the electric field. Typically, at room temp, V_0 is 0.5~0.8V.
- Some carriers are generated (thermally) and make their way into the depletion region where they are whisked away by the electric field, creating a drift current.

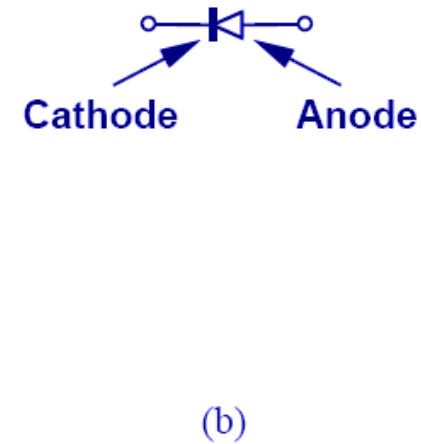
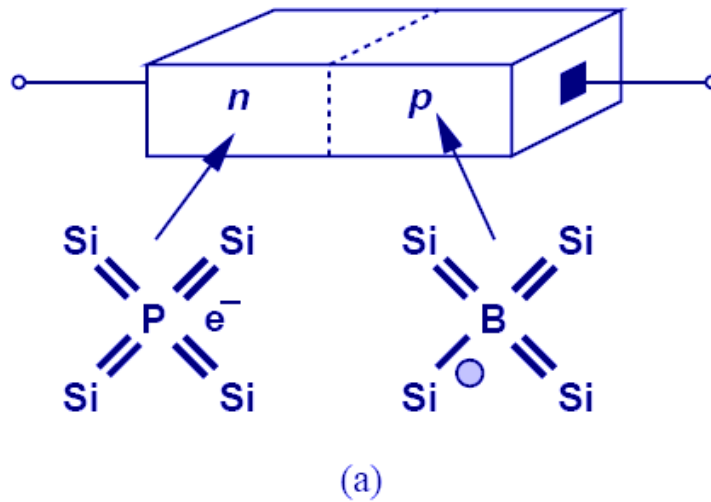
PN Junction

What happens when P-type meets N-type?

- There are two mechanisms by which mobile carriers move in semiconductors – resulting in current flow
 - Diffusion
 - Majority carriers move (diffuse) from a place of higher concentration to a place of lower concentration
 - Drift
 - Minority carrier movement is induced by the electric field.
- In equilibrium, diffusion current (I_D) is balanced by drift current (I_S).
So, there is no net current flow.

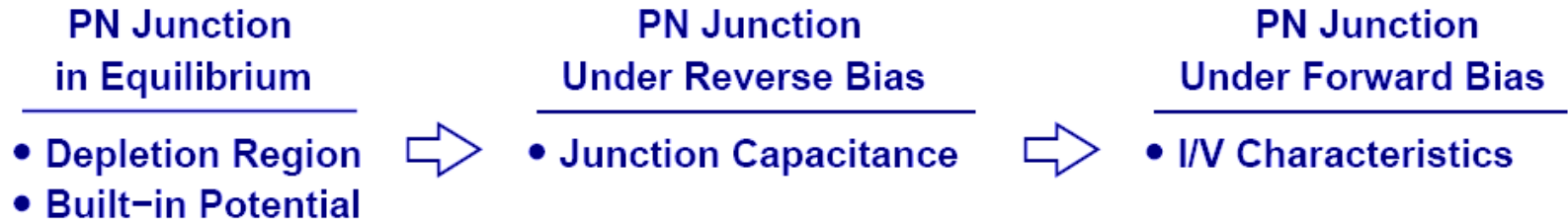


PN Junction (Diode)



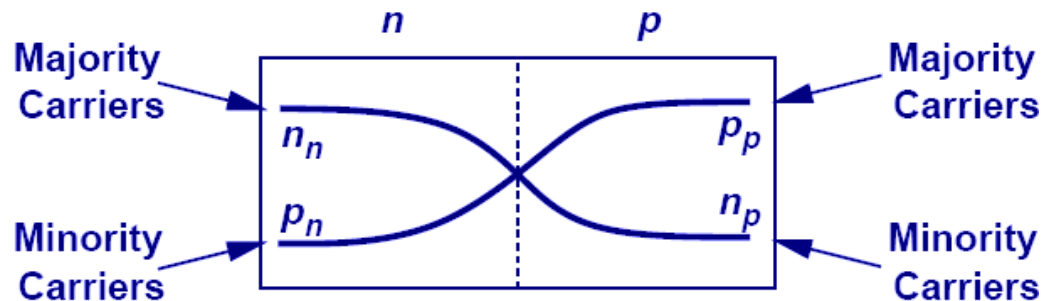
- When N-type and P-type dopants are introduced side-by-side in a semiconductor, a PN junction or a diode is formed.

Diode's Three Operation Regions



- In order to understand the operation of a diode, it is necessary to study its three operation regions: equilibrium, reverse bias, and forward bias.

Current Flow Across Junction: Diffusion



n_n : Concentration of electrons
on n side

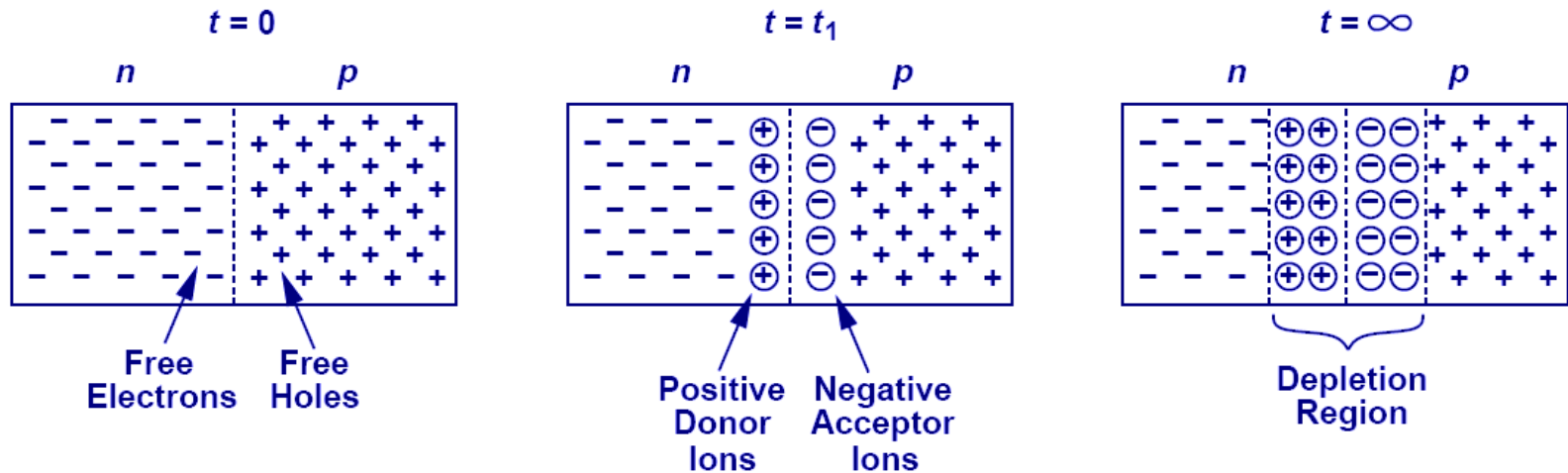
p_n : Concentration of holes
on n side

p_p : Concentration of holes
on p side

n_p : Concentration of electrons
on p side

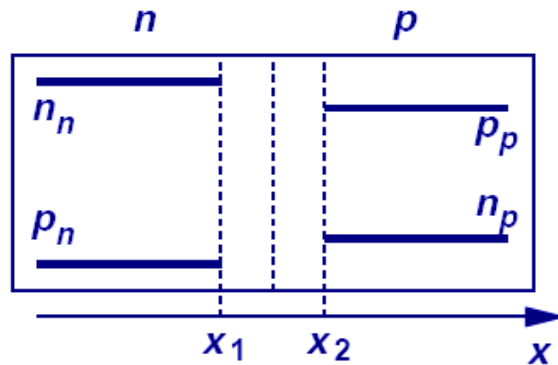
- Because each side of the junction contains an excess of holes or electrons compared to the other side, there exists a large concentration gradient. Therefore, a diffusion current flows across the junction from each side.

Depletion Region



- As free electrons and holes diffuse across the junction, a region of fixed ions is left behind. This region is known as the “depletion region.”

Current Flow Across Junction: Equilibrium

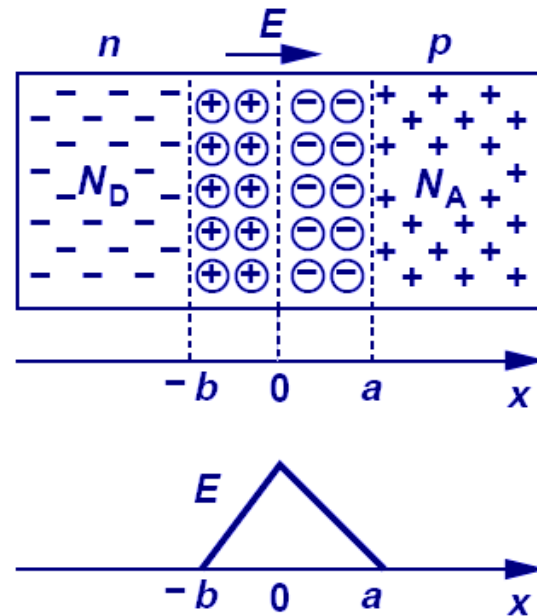


$$I_{drift,p} = I_{diff,p}$$

$$I_{drift,n} = I_{diff,n}$$

- At equilibrium, the drift current flowing in one direction cancels out the diffusion current flowing in the opposite direction, creating a net current of zero.
- The figure shows the charge profile of the PN junction.

Current Flow Across Junction: Drift

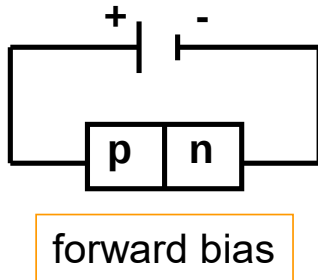


- The fixed ions in depletion region create an electric field that results in a drift current.

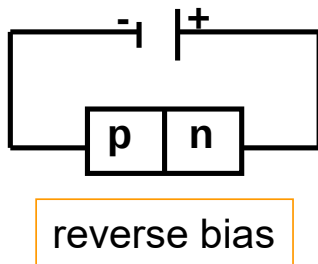
Biassing a PN Junction Diode



Applying bias to p-n junction

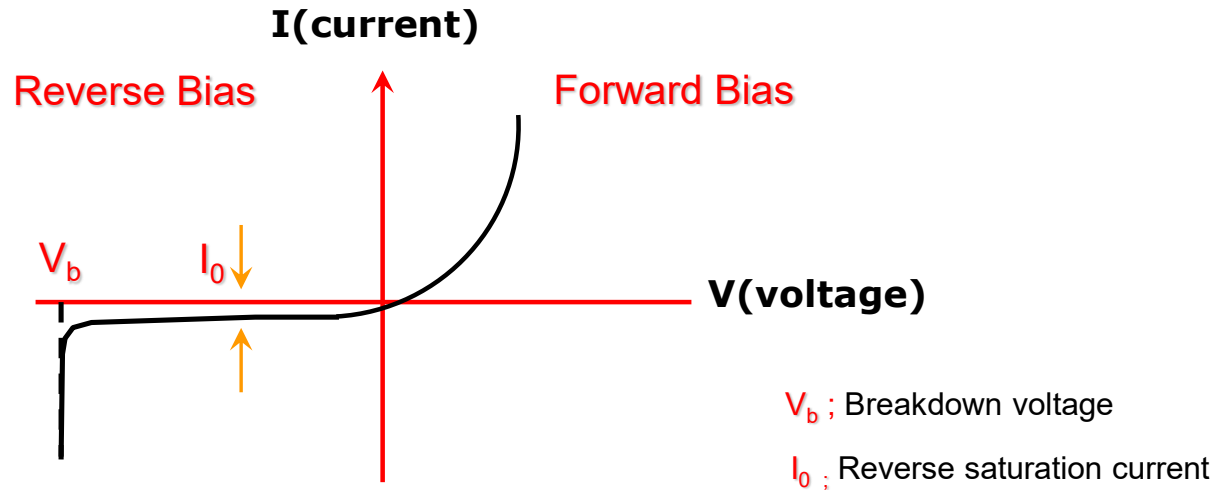


❖ How current flows through the p-n junction when a bias (voltage) is applied.



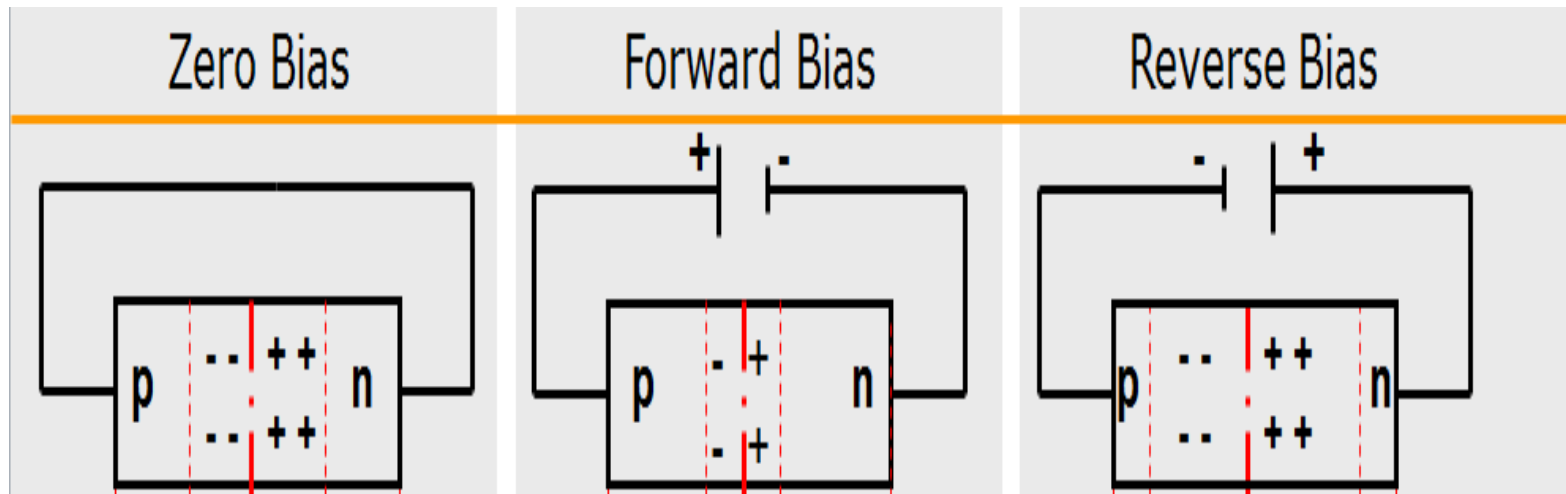
❖ The current flows all the time whenever a voltage source is connected to the diode. But the current flows rapidly in forward bias, however a very small constant current flows in reverse bias case.

Applying bias to p-n junction



- ❖ There is no turn-on voltage because current flows in any case. However, the turn-on voltage can be defined as the forward bias required to produce a given amount of forward current.
- ❖ If 1 m A is required for the circuit to work, 0.7 volt can be called as turn-on voltage.

Biasing PN Junction Diode



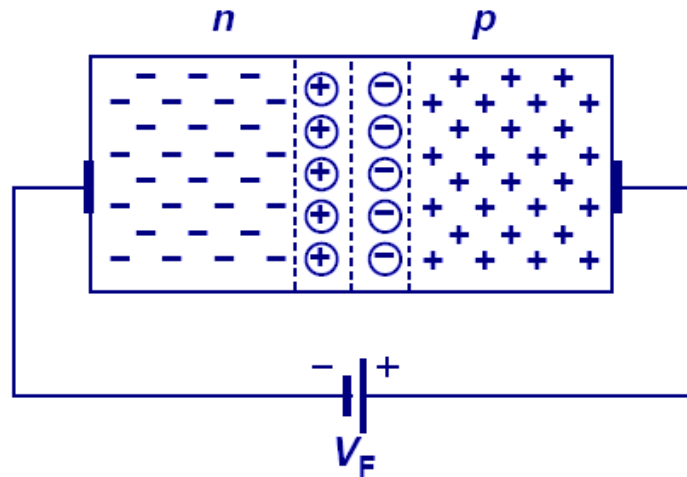
Applying bias to p-n junction

$V_F \rightarrow$ forward voltage

$V_R \rightarrow$ reverse voltage

- ❖ When a voltage is applied to a diode , bands move and the behaviour of the bands with applied forward and reverse fields are shown in previous diagram.

Diode in Forward Bias

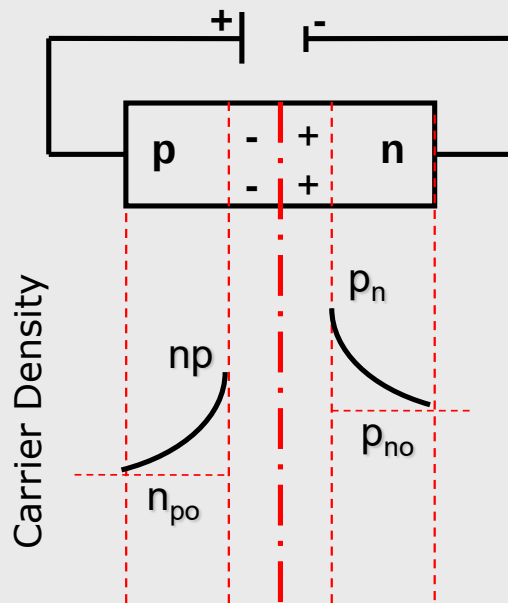


Add more majority carriers to both sides
 → shrink the depletion region → lower V_0
 → diffusion current increases.

- Decrease the built-in potential, lower the barrier height.
 - Increase the number of carriers able to diffuse across the barrier
 - Diffusion current increases
 - Drift current remains the same. The drift current is essentially constant, as it is dependent on temperature.
 - Current flows from p to n
- When the N-type region of a diode is at a lower potential than the P-type region, the diode is in forward bias.
 - The depletion width is shortened and the built-in electric field decreased.

- ❖ Junction potential reduced
 - ❖ Enhanced hole diffusion from p-side to n-side compared with the equilibrium case.
 - ❖ Enhanced electron diffusion from n-side to p-side compared with the equilibrium case.
 - ❖ Drift current flow is similar to the equilibrium case.
 - ❖ Overall, a large diffusion current is able to flow.
 - ❖ Mnemonic. Connect positive terminal to p-side for forward bias.
-
- ❖ Drift current is very similar to that of the equilibrium case. This current is due to the minority carriers on each side of the junction and the movement minority carriers is due to the built in field accross the depletion region.

Qualitative explanation of forward bias

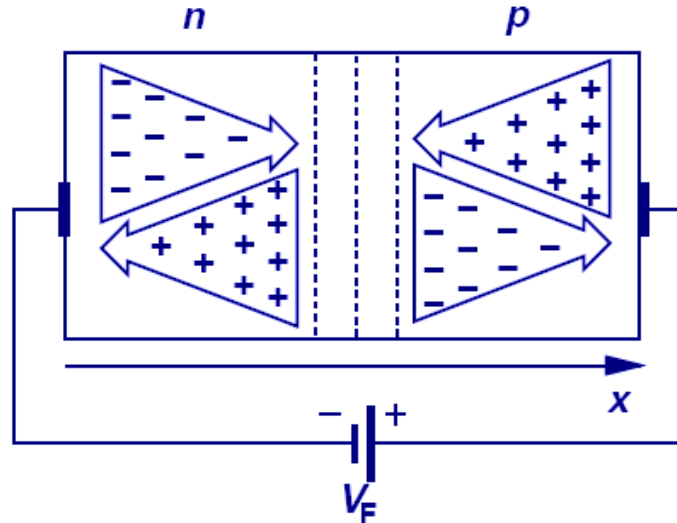


p-n junction in forward bias

- ❖ Junction potential is reduced
- ❖ By forward biasing a large number of electrons are injected from **n-side to p-side** across the depletion region and these electrons become minority carriers on **p-side**, and the minority recombine with majority holes so that the number of injected minority electrons decreases (decays) exponentially with distance into the **p-side**.

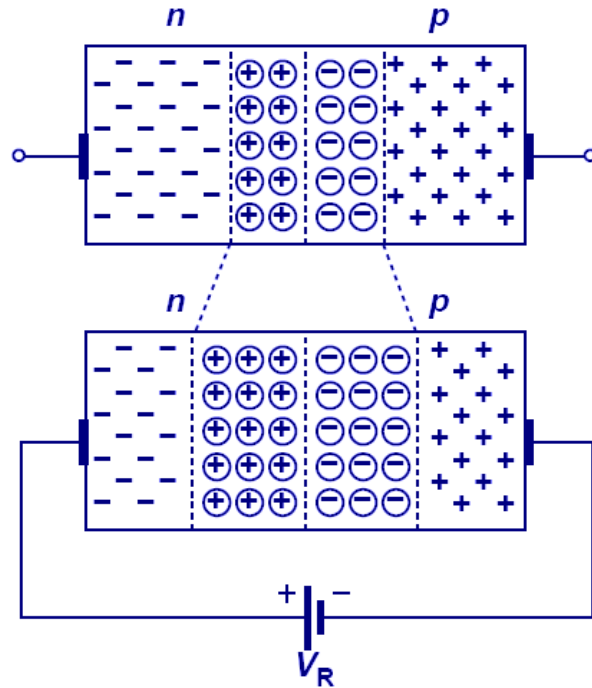
- ❖ Similarly, by forward biasing a large number of holes are injected from p-side to n-side across the DR. These holes become minority carriers at the depletion region edge at the n-side so that their number (number of injected excess holes) decreases with distance into the neutral n-side.
- ❖ In summary, by forward biasing in fact one injects minority carriers to the opposite sides. These injected minorities recombine with majorities.

Forward Bias Condition: Summary



- In forward bias, there are large diffusion currents of minority carriers through the junction. However, as we go deep into the P and N regions, recombination currents from the majority carriers dominate. These two currents add up to a constant value.

Diode in Reverse Bias



- Increase the built-in potential, increase the barrier height.
- Decrease the number of carriers able to diffuse across the barrier.
- Diffusion current decreases.
- Drift current remains the same
- Almost no current flows. Reverse leakage current, I_S , is the drift current, flowing from N to P.

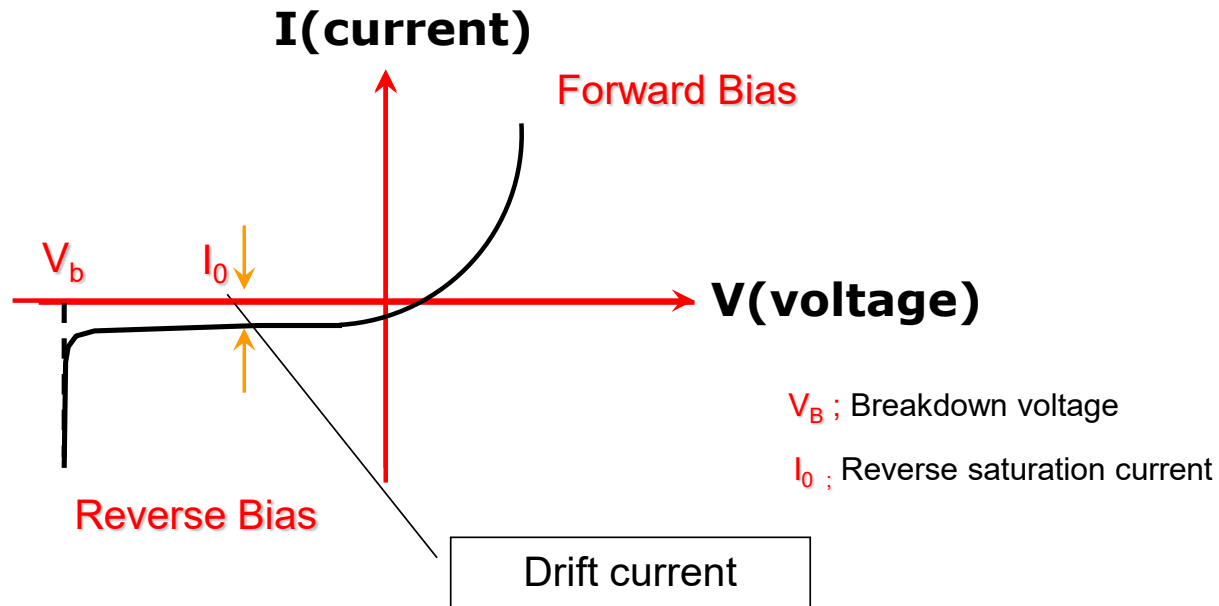
- When the N-type region of a diode is connected to a higher potential than the P-type region, the diode is under reverse bias, which results in wider depletion region and larger built-in electric field across the junction.

Reverse Bias

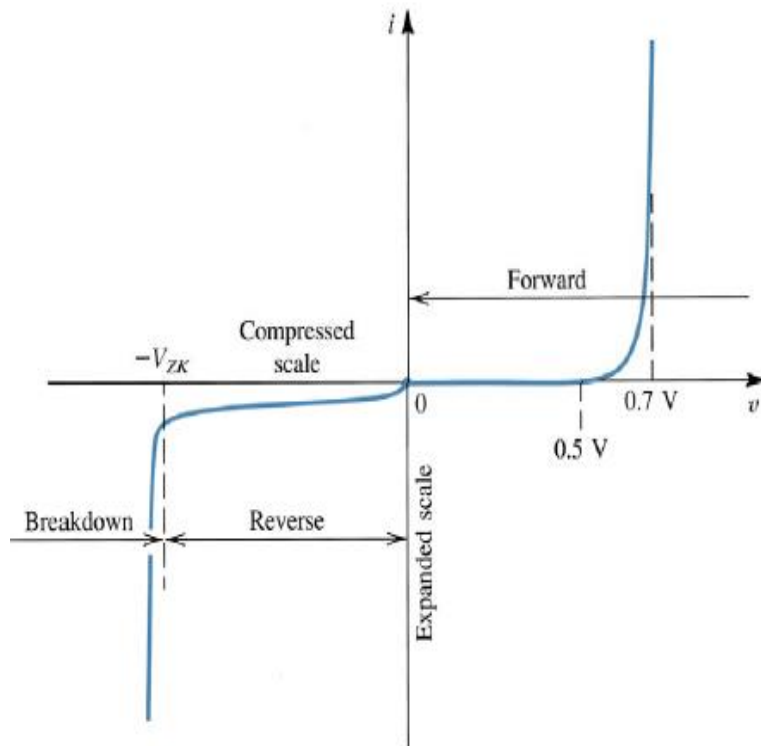
- ❖ Junction potential increased
- ❖ Reduced hole diffusion from p-side to n-side compared with the equilibrium case.
- ❖ Reduced electron diffusion from n-side to p-side compared with the equilibrium case
- ❖ Drift current flow is similar to the equilibrium case.
- ❖ Overall a very small reverse saturation current flows.
- ❖ Mnemonic. Connect positive terminal to n-side for reverse bias.

p-n junction in reverse bias

- The flow of these minorities produces the reverse saturation current and this current increases exponentially with temperature but it is independent of applied reverse voltage.



PN Junction Diode I-V Characteristic



Typical PN junction diode volt-ampere characteristic is shown on the left.

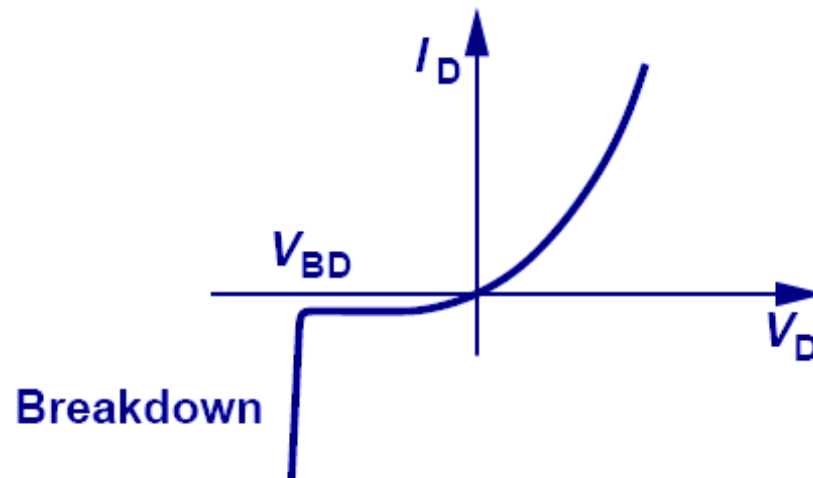
- In forward bias, the PN junction has a “turn on” voltage based on the “built-in” potential of the PN junction. turn on voltage is typically in the range of 0.5V to 0.8V
- In reverse bias, the PN junction conducts essentially no current until a critical breakdown voltage is reached. The breakdown voltage can range from 1V to 100V. Breakdown mechanisms include avalanche and zener tunneling.

The current and voltage relationship of a PN junction is exponential in forward bias region, and relatively constant in reverse bias region.

Junction breakdown or reverse breakdown

- An applied reverse bias (voltage) will result in a small current to flow through the device.
 - At a particular high voltage value, which is called as **breakdown voltage V_B** , large currents start to flow. If there is no current limiting resistor which is connected in series to the diode, the diode will be destroyed. There are **two physical effects** which cause this breakdown.
- 1) Zener breakdown is observed in highly doped p-n junctions and occurs for voltages of about 5 V or less.
 - 2) Avalanche breakdown is observed in less highly doped p-n junctions.

Reverse Breakdown



- When a large reverse bias voltage is applied, breakdown occurs and an enormous current flows through the diode.

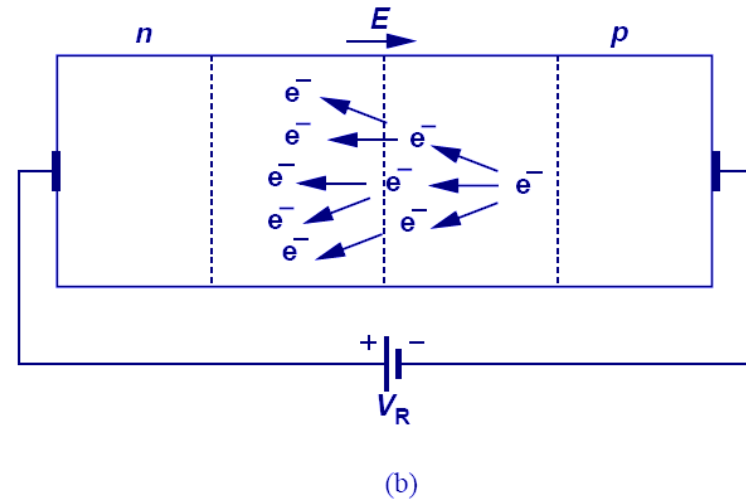
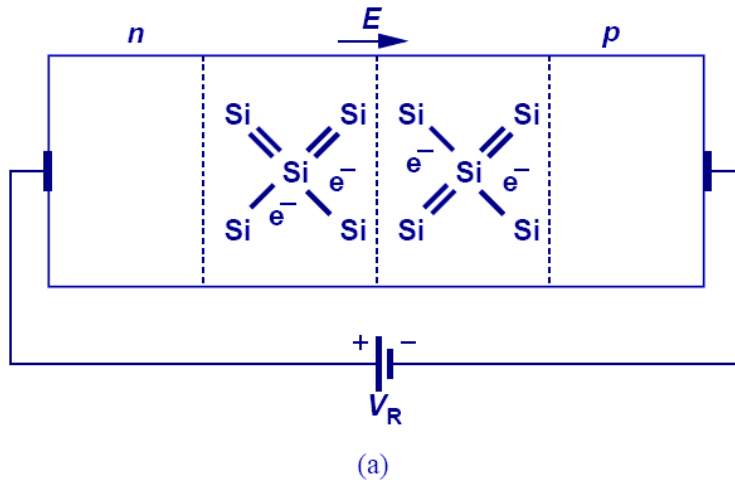
Zener breakdown

- **Zener breakdown** occurs at highly doped p-n junctions with a tunneling mechanism.
- In a **highly doped p-n junction** the conduction and valance bands on opposite side of the junction become so close during the reverse-bias that the electrons on the p-side can tunnel into the n-side.

Avalanche Breakdown

- **Avalanche breakdown** mechanism occurs when electrons and holes moving through the DR and acquire sufficient energy from the electric field to **break a bond i.e. create electron-hole** pairs by colliding with atomic electrons within the depletion region.
- The newly created electrons and holes move in opposite directions due to the electric field and thereby **add to the existing reverse bias current**. This is the most important breakdown mechanism in p-n junction.

Zener vs. Avalanche Breakdown



- Zener breakdown is a result of the large electric field inside the depletion region that breaks electrons or holes off their covalent bonds.
- Avalanche breakdown is a result of electrons or holes colliding with the fixed ions inside the depletion region.

Lab Experiment

I-V Characteristics of PN Junction Diode

Components

- Diode
- Variable Resistor
- Ammeter
- Voltmeter
- Voltage Supply (Signal Generator)

Diode

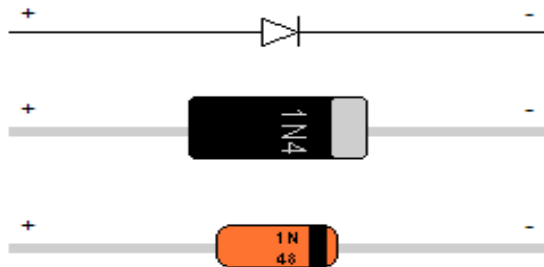
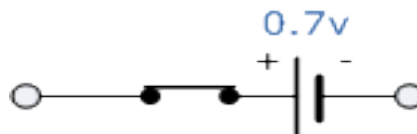


Figure 2: Orientation of diodes

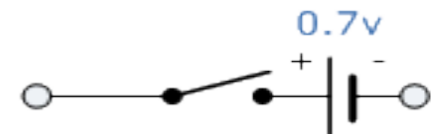


Forward Biased



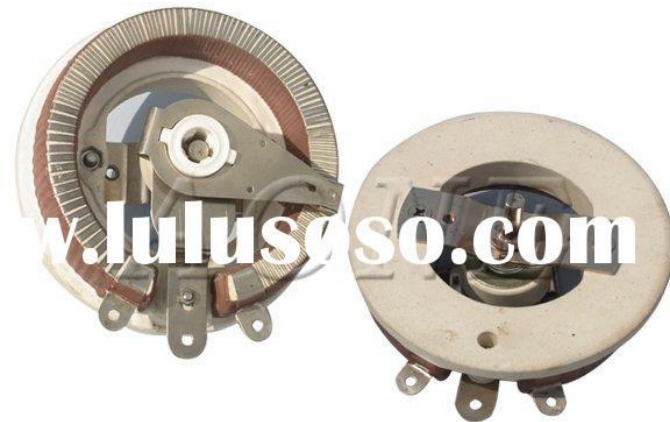
Forward Bias
(switch closed)

Reversed Biased



Reverse Bias
(switch open)

Variable Resistor



www.topresistor.com

Ammeter



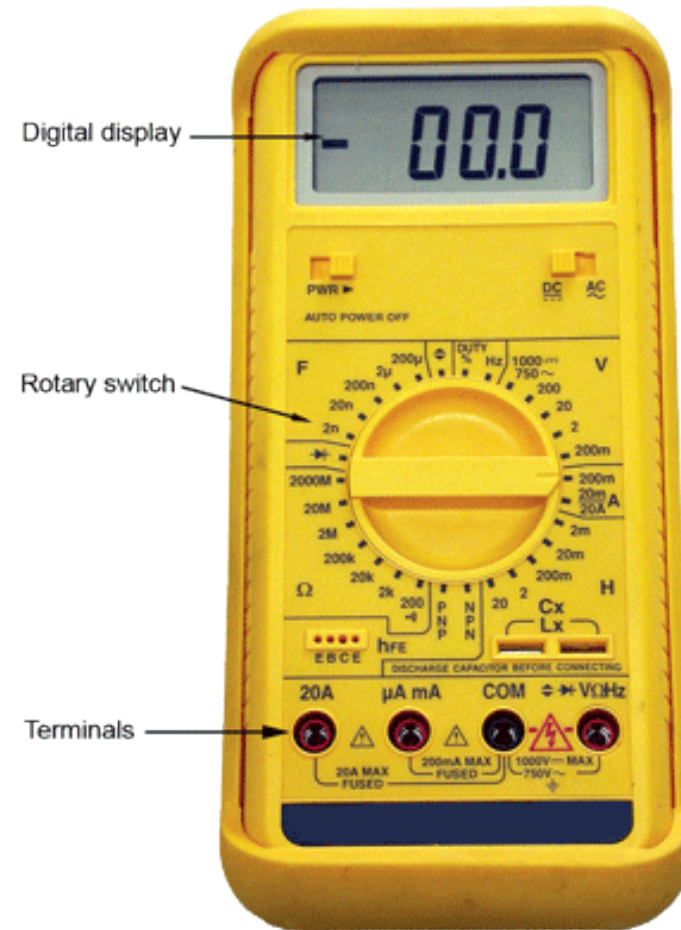
Digital display

Rotary switch

Terminals



Voltmeter

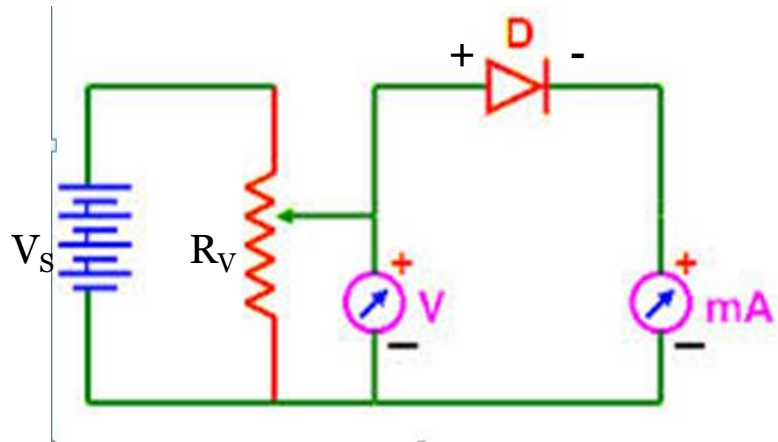


Voltage Supply (Signal Generator)

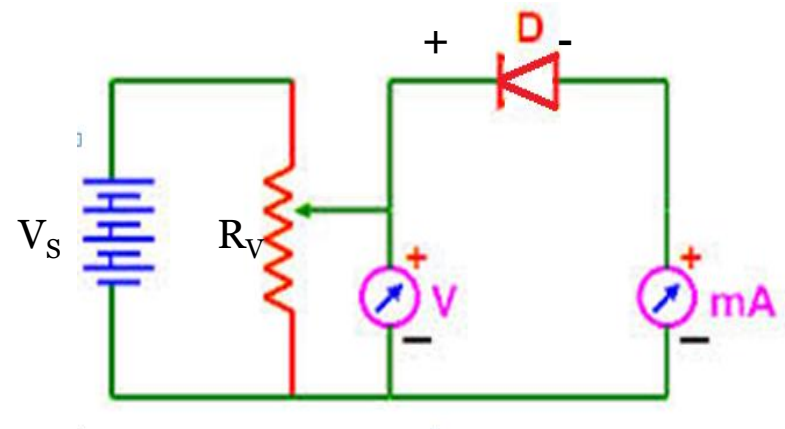


Experimental Setup

Forward Bias

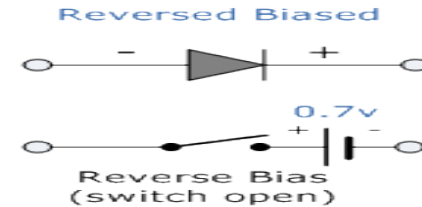
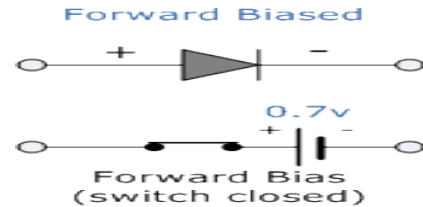


Reverse Bias



V_s = Supply Voltage
 R_v = Variable Resistor
 V = Voltmeter
 mA/uA = Ammeter

I-V characteristics of PN junction Diode



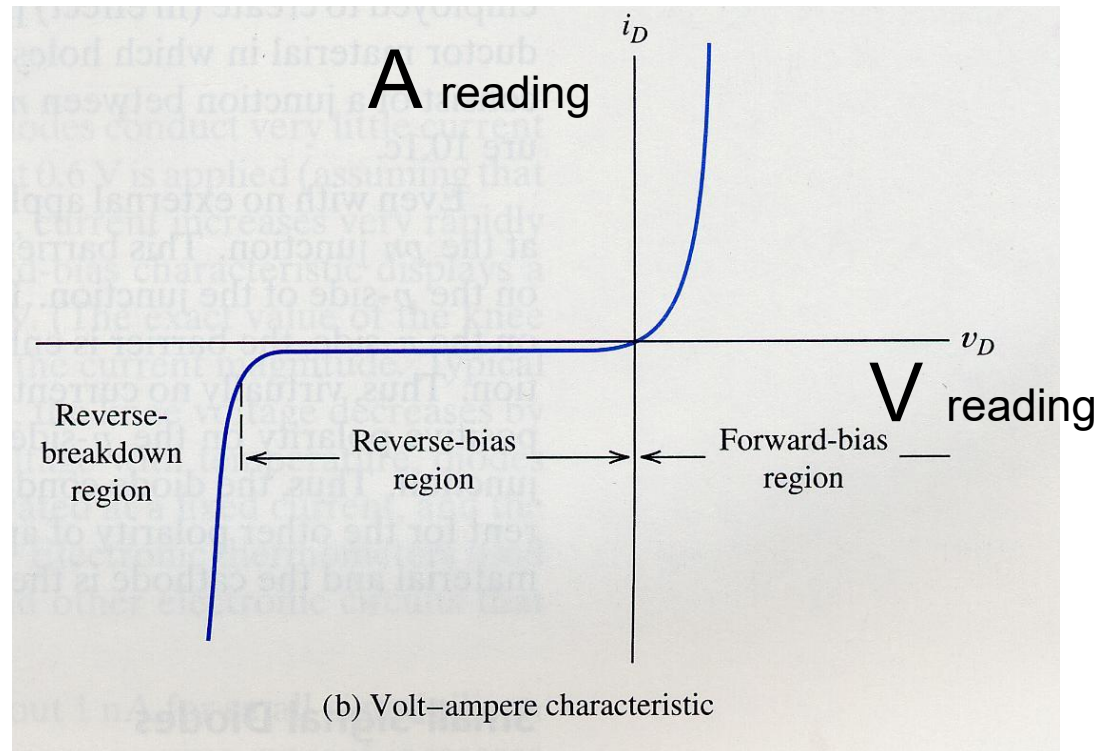
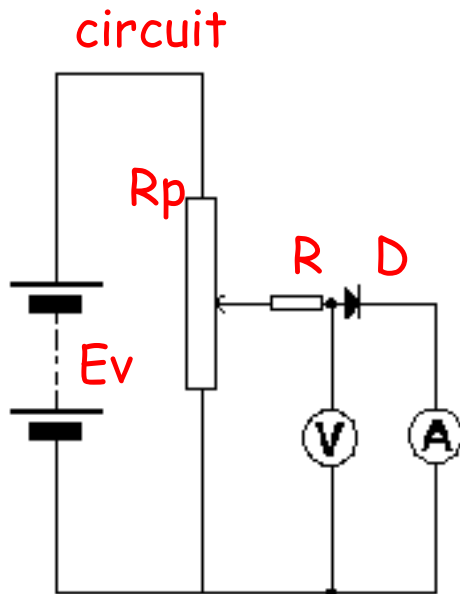
Forward Bias

Test	Voltmeter Reading	Ammeter Reading

Reverse Bias

Test	Voltmeter Reading	Ammeter Reading

Diode Characteristic

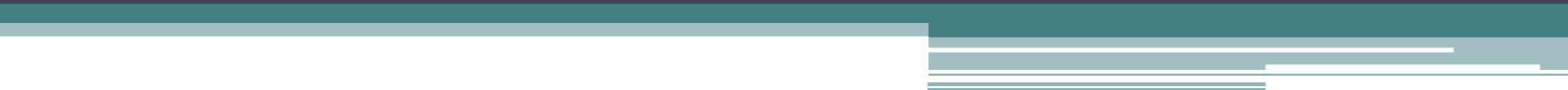


A diode is a nonlinear device and typical linear circuit analysis methods do not apply!

...that's all folks...
...thanks for your time...



PN-Junction Diode Applications



Outline

- Analysis of Diode Circuit
 - Models
 - Circuit model
- Applications of diode
 - Rectification
 - Half wave Rectifiers
 - Full wave Rectifiers
 - Centre-tap
 - Bridge
 - Filtration
 - Voltage Regulators
 - Zener Diode

Diode Circuits



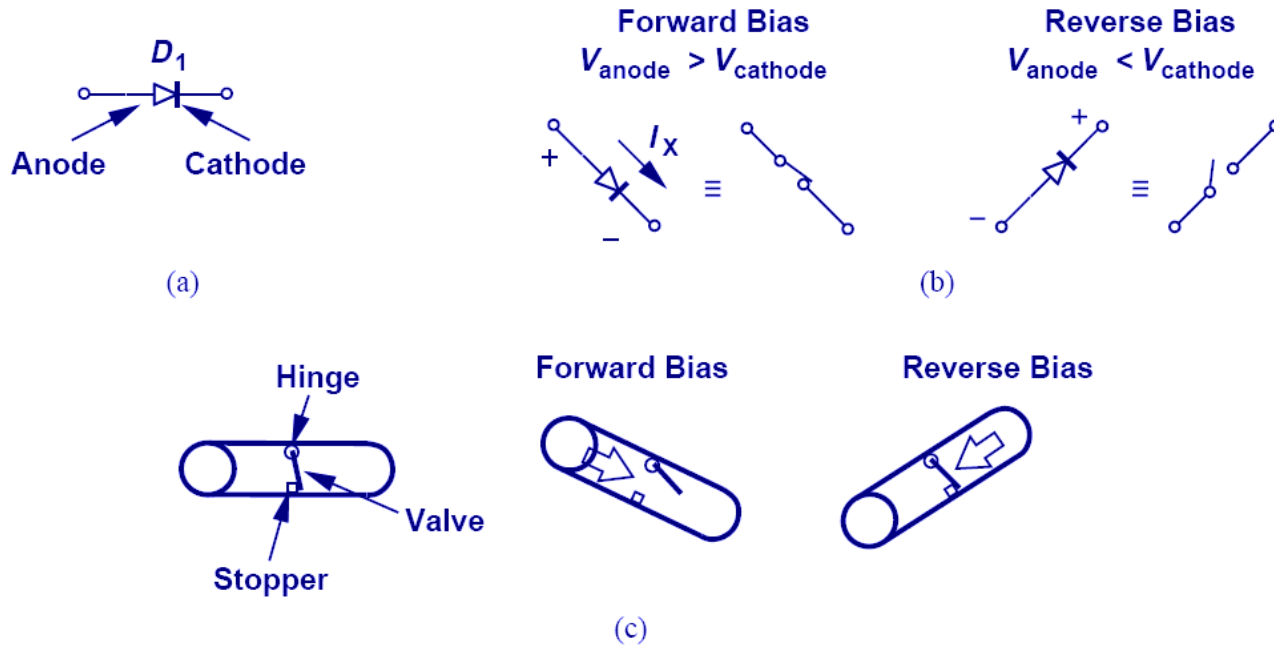
- After we have studied in detail the physics of a diode, it is time to study its behavior as a circuit element and its many applications.

The Diode Models

Circuit Model

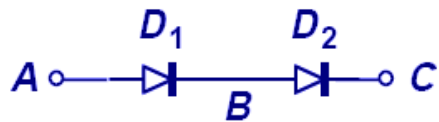
- a) Simplified diode model
- b) The constant-voltage-drop model
- c) Zener Diode Model

Ideal Diode

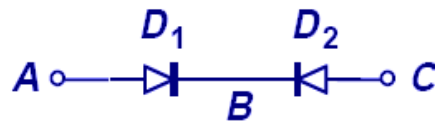


- In an ideal diode, if the voltage across it tends to exceed zero, current flows.
- It is analogous to a water pipe that allows water to flow in only one direction.

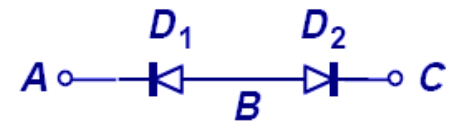
Diodes in Series



(a)



(b)

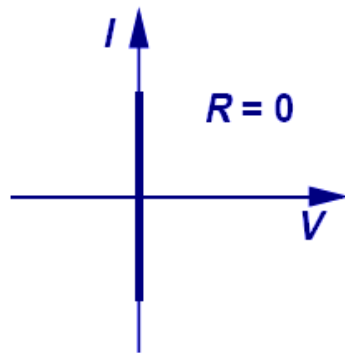


(c)

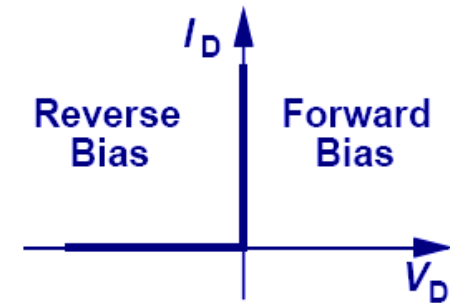
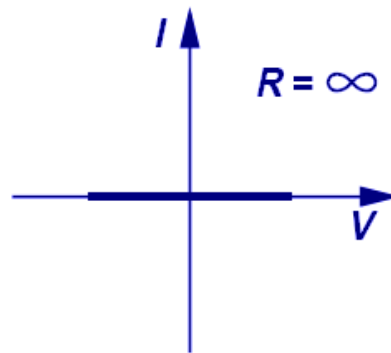
- Diodes cannot be connected in series randomly. For the circuits above, only a) can conduct current from A to C.

IV Characteristics of an Ideal Diode

$$R = 0 \Rightarrow I = \frac{V}{R} = \infty \quad R = \infty \Rightarrow I = \frac{V}{R} = 0$$



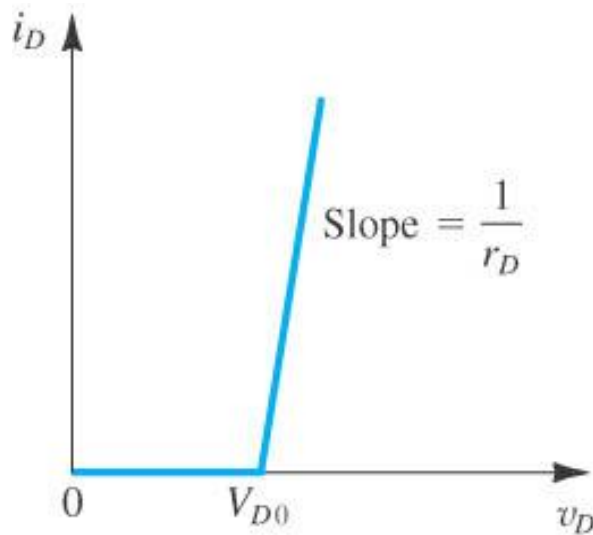
(a)



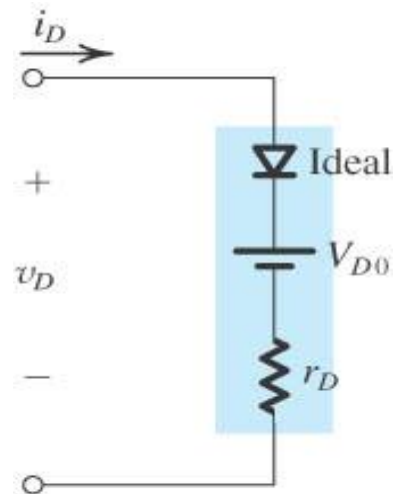
(b)

- If the voltage across anode and cathode is greater than zero, the resistance of an ideal diode is zero and current becomes infinite. However, if the voltage is less than zero, the resistance becomes infinite and current is zero.

Simplified Diode Model



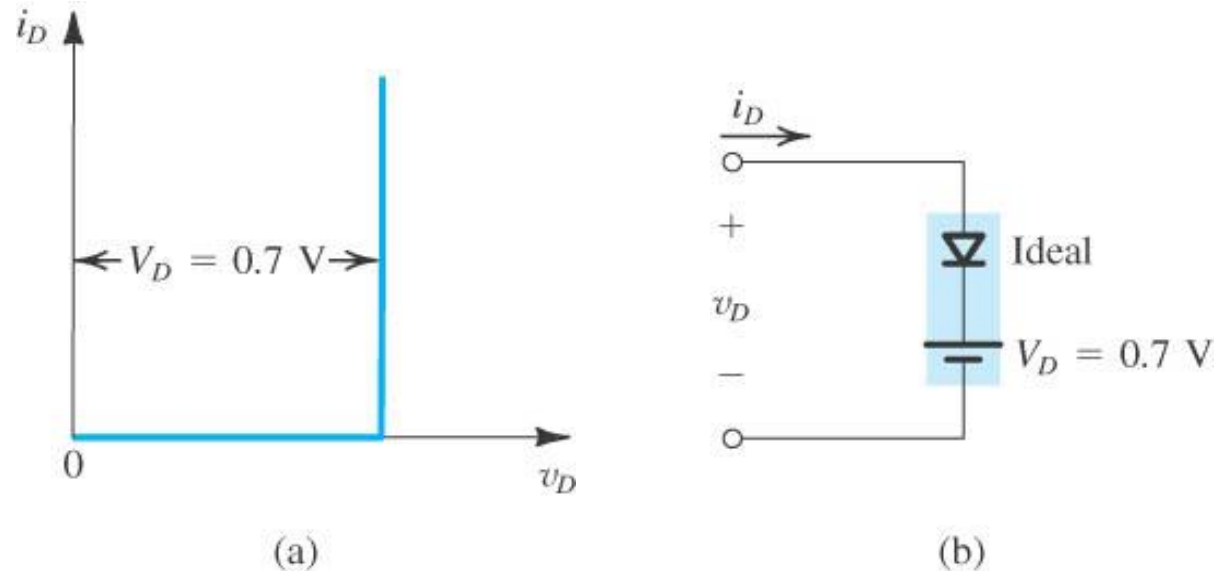
(a)



(b)

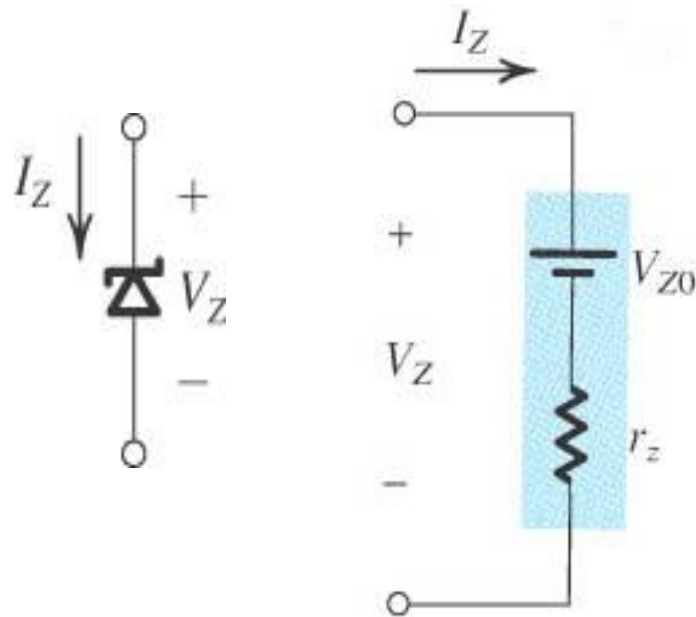
Piecewise-linear model of the diode forward characteristic and its equivalent circuit representation.

The Constant-Voltage-Drop Model



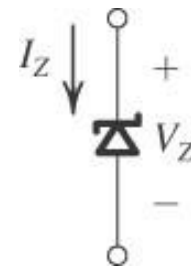
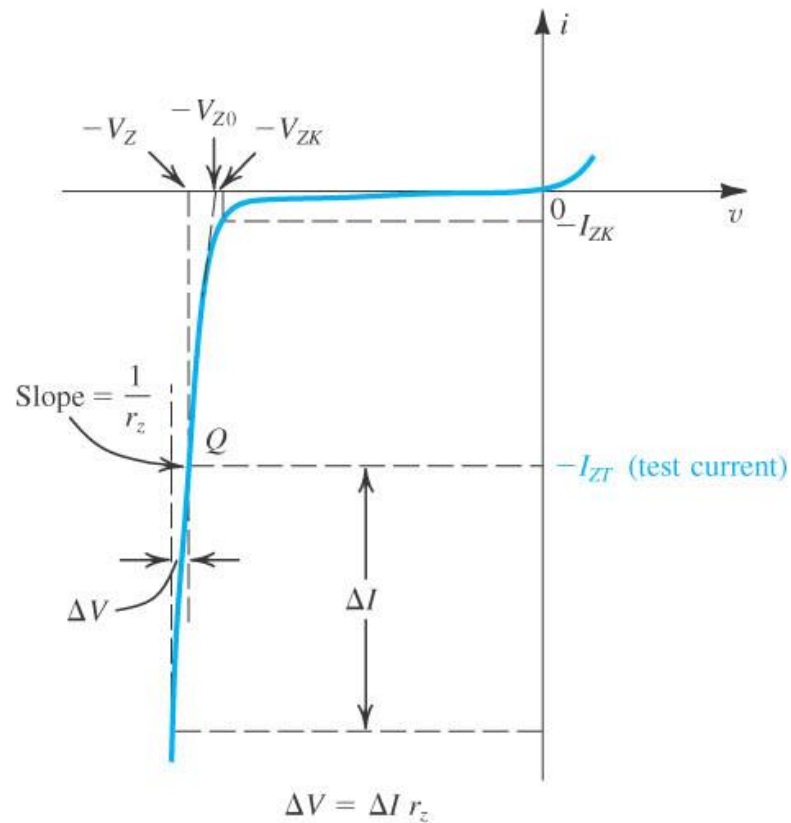
The constant-voltage-drop model of the diode forward characteristics and its equivalent-circuit representation.

Zener Diode Model



$$V_Z = V_{Z0} + I_Z r_z$$

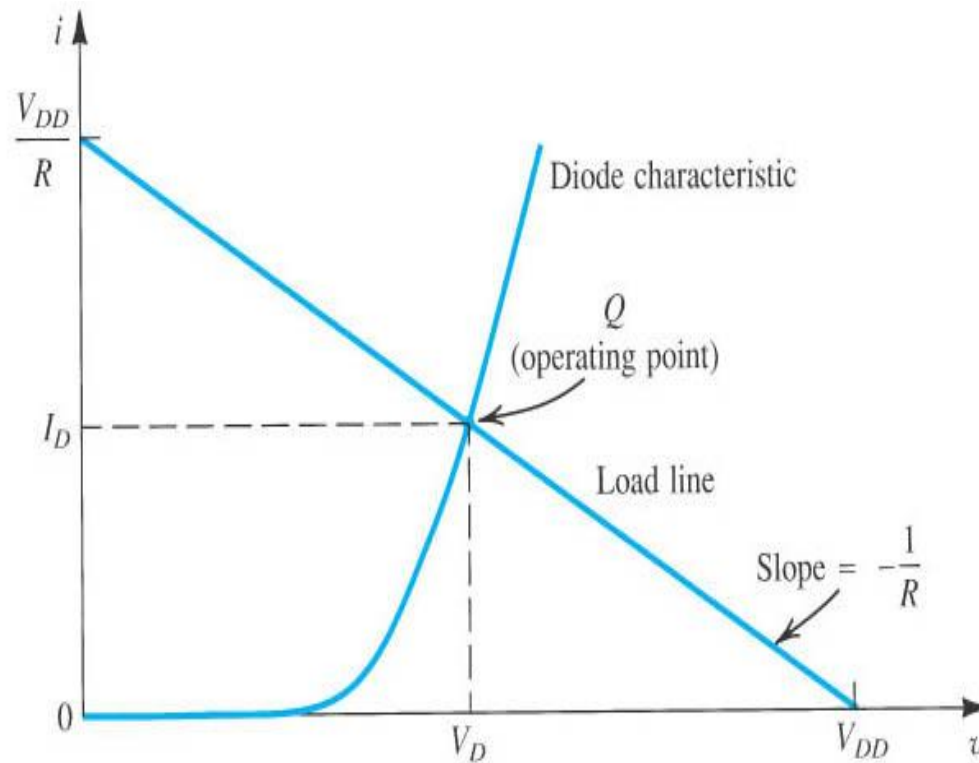
Zener Diode



Circuit symbol

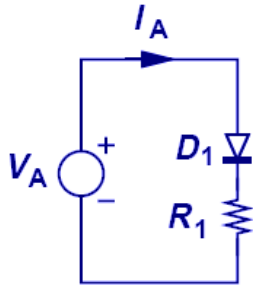
The diode $i-v$ characteristic with the breakdown region shown in some detail.

Method of Analysis

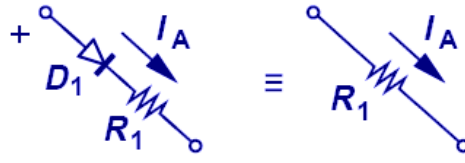


- Load line
- Diode characteristic
- Q is the intersect point
- Visualization

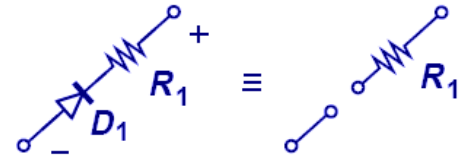
Diode-Resistor Combination



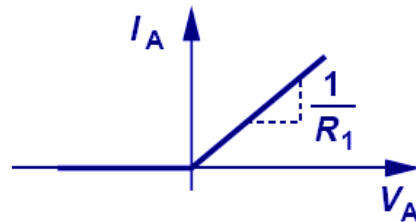
(a)



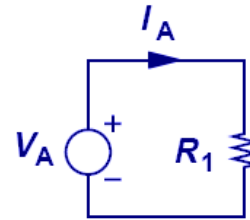
(b)



(c)



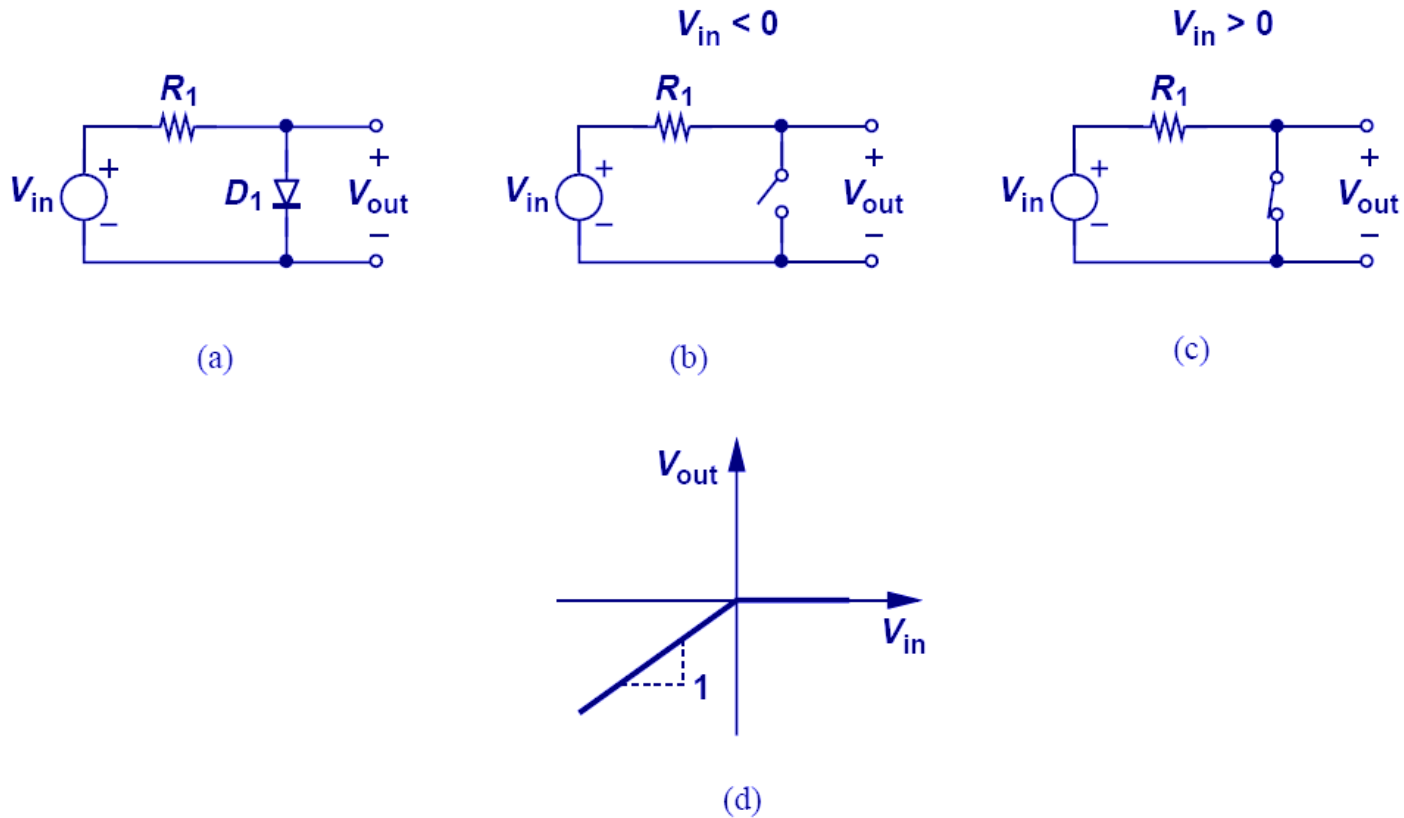
(d)



(e)

- The IV characteristic of this diode-resistor combination is zero for negative voltages and Ohm's law for positive voltages.

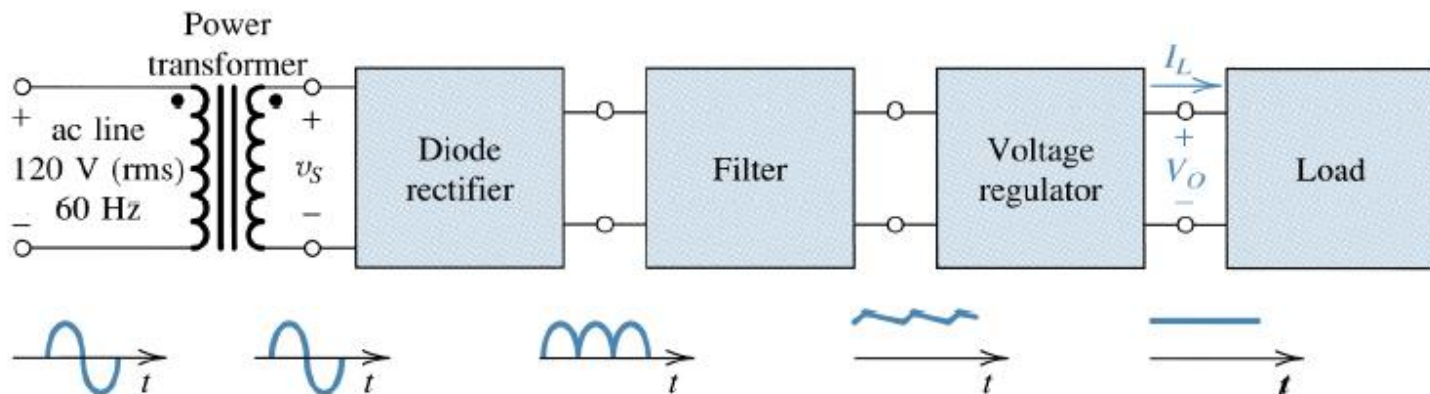
Input/Output Characteristics



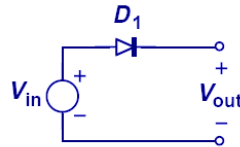
- When V_{in} is less than zero, the diode opens, so $V_{out} = V_{in}$.
- When V_{in} is greater than zero, the diode shorts, so $V_{out} = 0$.

Rectifier Circuits

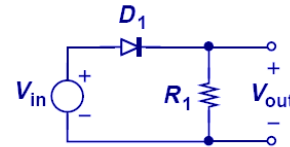
One of the most important applications of diodes is in the design of rectifier circuits. Used to convert an AC signal into a DC voltage used by most electronics.



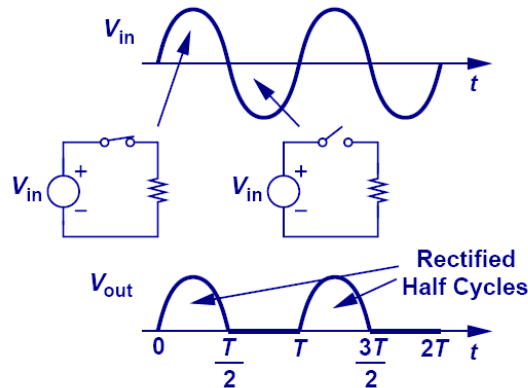
Diode's Application: Rectifier



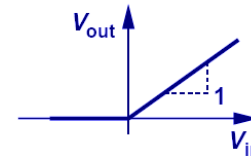
(a)



(b)



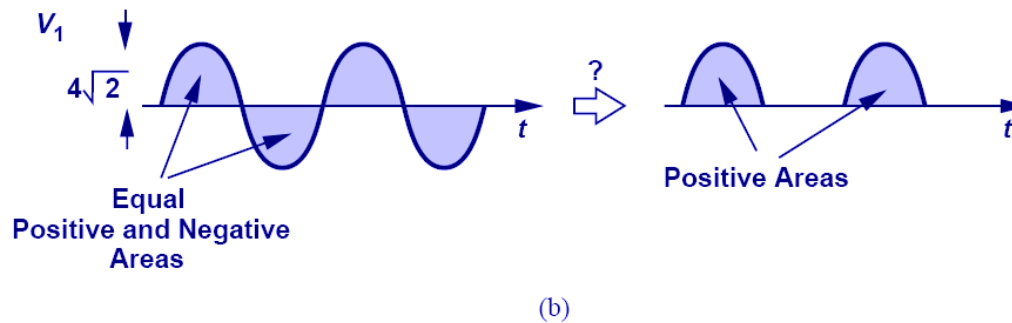
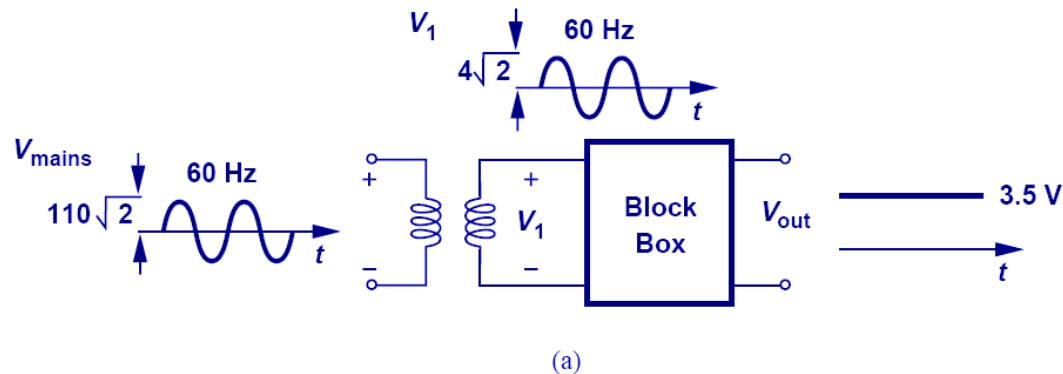
(c)



(d)

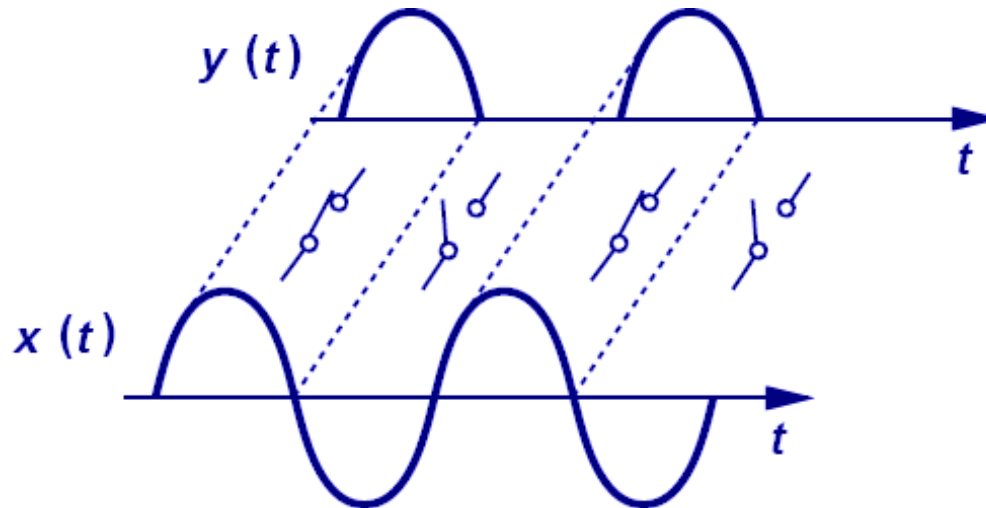
- A rectifier is a device that passes positive-half cycle of a sinusoid and blocks the negative half-cycle or vice versa.
- When V_{in} is greater than 0, diode shorts, so $V_{out} = V_{in}$; however, when V_{in} is less than 0, diode opens, no current flows thru R_1 , $V_{out} = I_{R1} R_1 = 0$.

Diode's Application: Cell Phone Charger



- An important application of diode is chargers.
- Diode acts as the black box (after transformer) that passes only the positive half of the stepped-down sinusoid.

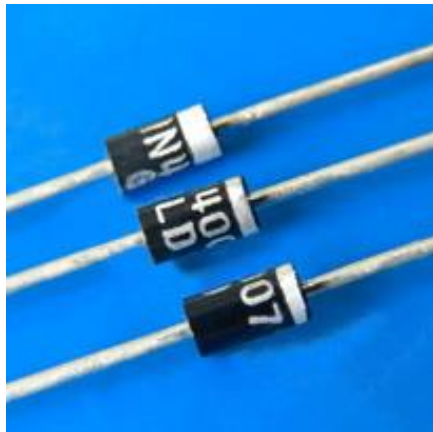
Diode's Action in The Black Box (Ideal Diode)



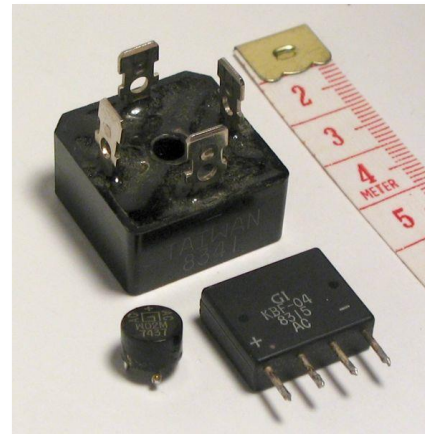
- The diode behaves as a short circuit during the positive half cycle (voltage across it tends to exceed zero), and an open circuit during the negative half cycle (voltage across it is less than zero).

Rectification Technologies

- Rectification Based on Diode
 - Rectification is **most popular application** of diode
 - Diodes provide **compact and inexpensive** means of rectification
 - Can **create rectifiers** from multiple diodes or purchase integrated module

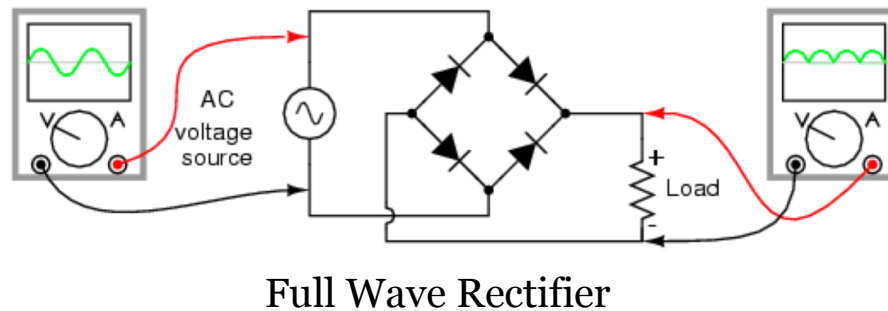
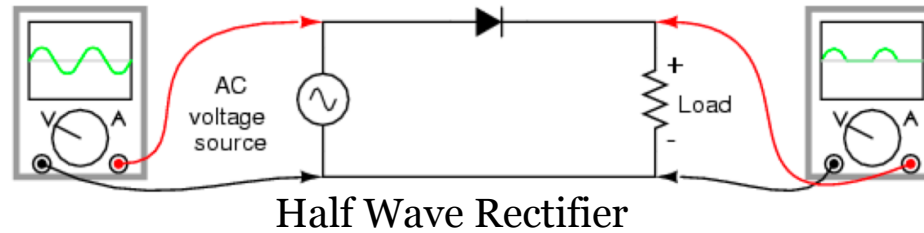


Diodes



Diode Rectifier Modules

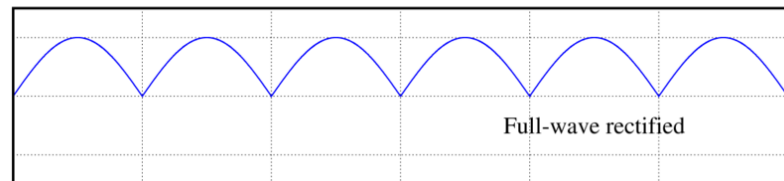
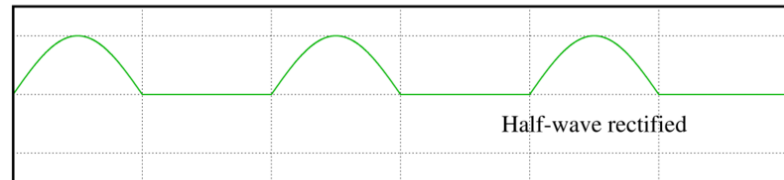
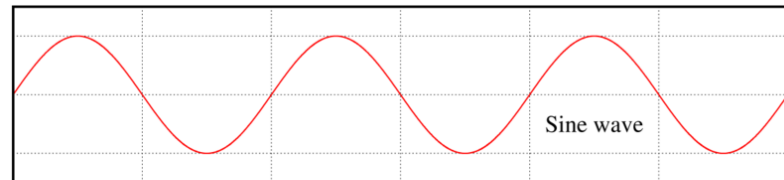
Types of Rectification



- While output of the rectifiers is now DC (**current only flows in one direction**), output oscillates

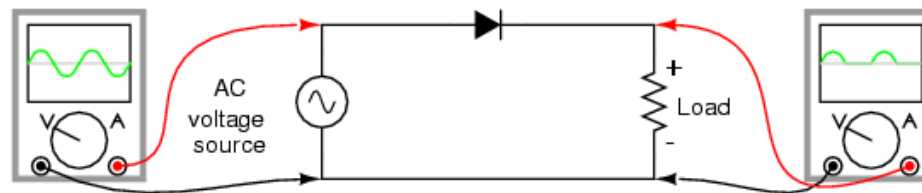
Types of Rectification

- **Half Wave:**
 - Negative components of sine wave are discarded
- **Full Wave:**
 - Negative components are inverted



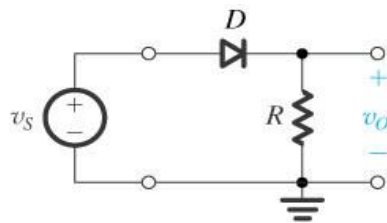
Rectification Circuit: Half-Wave

- Simplest kind of **rectifier circuit** is *half-wave* rectifier.
- **Allows one half of AC waveform** to pass through to load.
- **Converts** alternating current (AC) to direct current (DC).
- Involves device that only **allows one-way flow of electrons**, and this is exactly what semiconductor diode does.

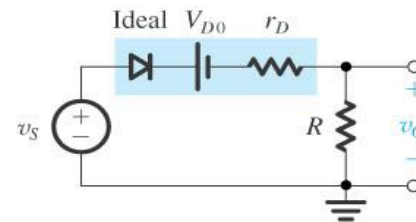


Half-wave rectifier circuit

Half-Wave Rectifier



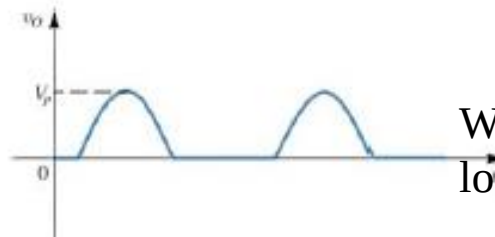
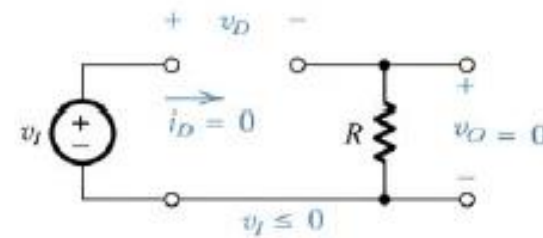
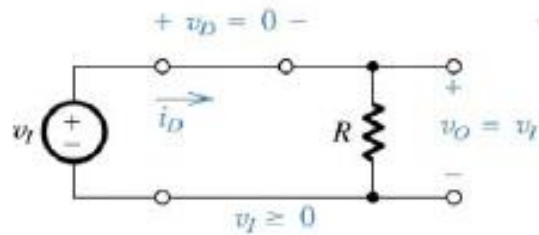
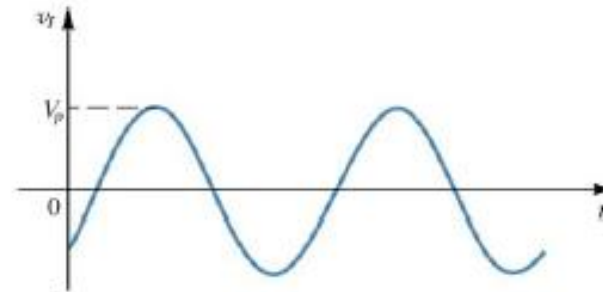
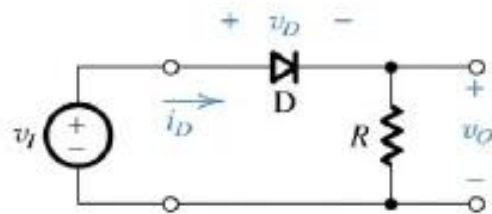
(a)



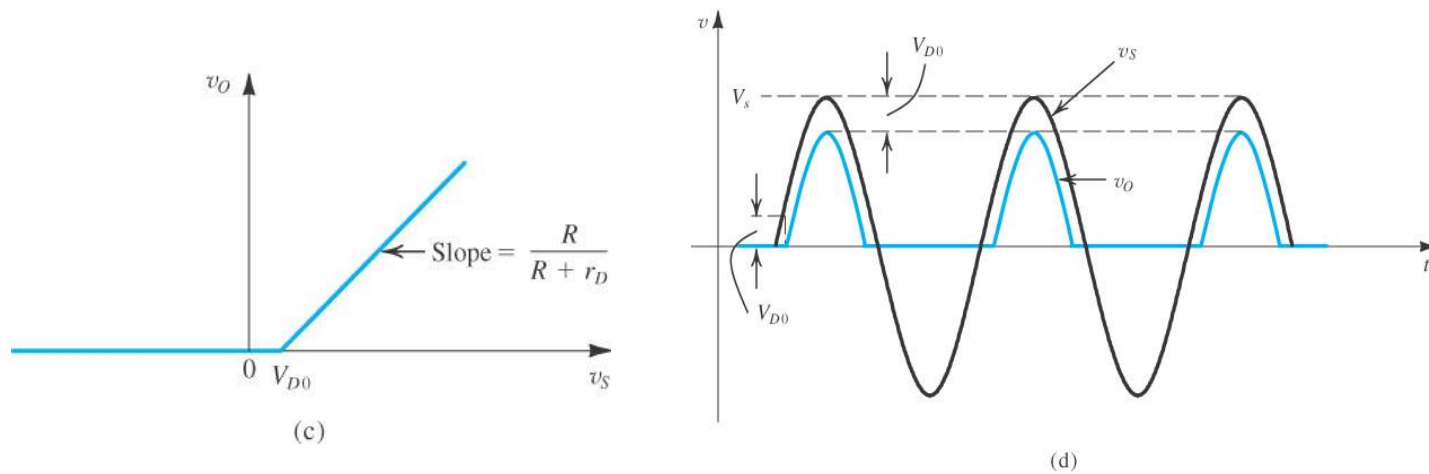
(b)

- (a) Half-wave rectifier.
- (b) Equivalent circuit of the half-wave rectifier with the diode replaced with its battery-plus-resistance model.

Simple Half-Wave Rectifier



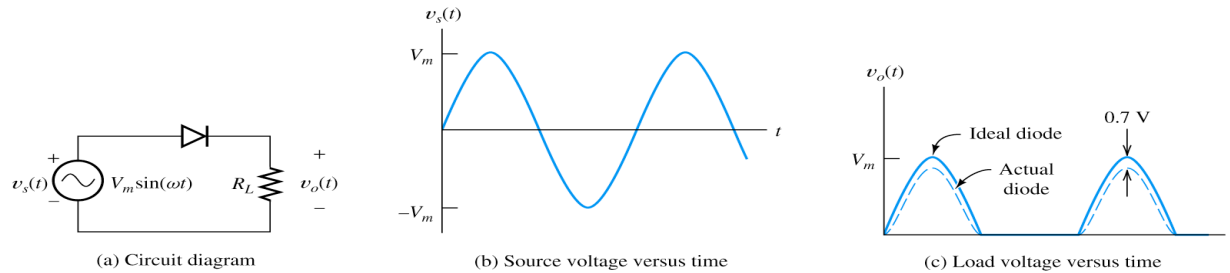
What would the waveform look like if not an ideal diode?



(c) Transfer characteristic of the rectifier circuit.

(d) Input and output waveforms, assuming that $r_D \ll R$

* *Half-Wave Rectifier with Smoothing Capacitor*



* To place a large capacitance across the output terminals:

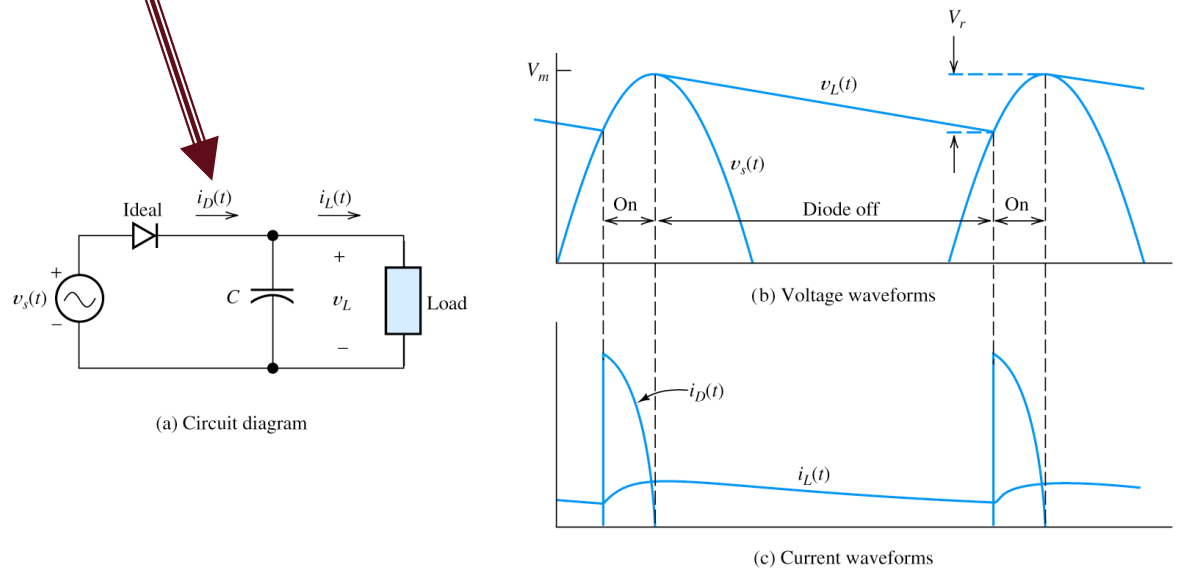


Figure 10.26 Half-wave rectifier with smoothing capacitor.

* *Battery-Charging Circuit*

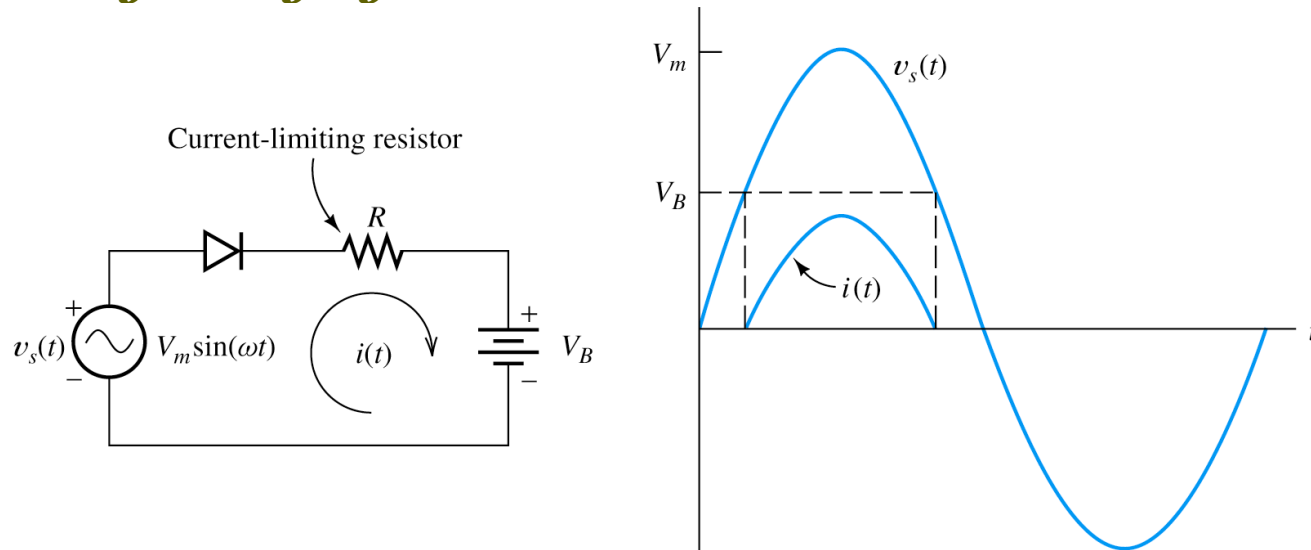


Figure 10.25 Half-wave rectifier used to charge a battery.

* The current flows only in the direction that charges the battery.

Average or DC Value

$$V_{out} = V_p \sin \omega t = 0 \quad \text{for} \quad 0 \leq t \leq \frac{T}{2}$$

$$\begin{aligned} V_{out,avg} &= \frac{1}{T} \int_0^T V_{out}(t) dt = \frac{1}{T} \int_0^{T/2} V_p \sin \omega t dt \\ &= \frac{1}{T} \frac{V_p}{\omega} [-\cos \omega t]_0^{T/2} = \frac{V_p}{\pi} \quad \text{for} \quad \frac{T}{2} \leq t \leq T \end{aligned}$$

- The averaged value of a rectifier output can be used as a signal strength indicator for the input, since $V_{out,avg}$ is proportional to V_p , the input signal's amplitude.

$$V_p = V_m \quad I_p = I_m$$

$$I_{avg} = I_{dc} = \frac{V_{dc}}{R_L}$$

$$V_{dc} = \frac{V_m}{\pi}$$

$$I_{dc} = \frac{V_m}{\pi R_L} = \frac{I_m}{\pi}$$

$$V_{dc} = \frac{V_m}{\pi} - I_{dc} \cdot R_f$$

$$I_{dc} = \frac{V_m}{\pi(R_L + R_f)}$$

R_f = forward resistance of Diode

R_L = Load Resistance

V_p/V_m = Peak Voltage

I_p/I_m = Peak Current

I_{dc}/I_{avg} = Load Current

V_{dc} = output dc Voltage

The rms value of the current, I_{rms} can be derived as:

$$I_{rms} = \sqrt{\frac{1}{2\pi} \int_0^\pi I_m^2 \sin^2 \omega t d(\omega t)}$$

$$I_{rms} = \sqrt{\frac{I_m^2}{2\pi} \int_0^\pi \frac{1}{2} (1 - \cos 2\omega t) d(\omega t)}$$

$$I_{rms} = \sqrt{\frac{I_m^2}{2\pi} \times \frac{1}{2} \left[\omega t + \frac{1}{2} \sin 2\omega t \right]_0^\pi}$$

$$I_{rms} = \sqrt{\frac{I_m^2}{4\pi} \times [\pi]}$$

$$I_{rms} = \frac{I_m}{2} = 0.5 I_m$$

The voltage across the load is given by:

$$V_R = Ri = RI_m \sin \omega t = V_m \sin \omega t$$

Similarly, the mean value of the load voltage and the rms value of the effective input voltage are respectively given by:

$$V_{dc} = \frac{V_m}{\pi} = 0.318V_m$$

$$V_{rms} = \frac{V_m}{2} = 0.5V_m$$

Efficiency of a half-wave rectifier

Rectifier efficiency, $\eta = \frac{\text{output dc power}}{\text{input ac power}}$

$$\eta = \frac{I_{dc}^2 R}{I_{rms}^2 (r_f + R)}$$

$$\eta = \frac{\left[\frac{I_m}{\pi} \right]^2 R}{\left[\frac{I_m}{2} \right]^2 (r_f + R)}$$

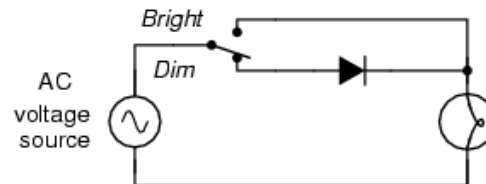
$$\eta = \frac{4}{\pi^2} \left[\frac{R}{(r_f + R)} \right]$$

$$\eta = \frac{0.405}{1 + \frac{r_f}{R}}$$

For maximum efficiency, the value of r_f should be negligible compared to R . Therefore, the maximum efficiency of the half-wave rectifier is 40.5%.

Rectification Circuit: Half-Wave

- For most power applications, **half-wave rectification is insufficient** for task.
 - Harmonic content of **rectifier's output waveform is very large and** consequently difficult to filter.
 - AC power source only **supplies power to load once every half-cycle**, meaning that much of its capacity is unused.
 - Half-wave rectification is, however, **very simple way to reduce power to resistive load**.
- Two-position **lamp dimmer switches** apply full AC power to lamp filament for “full” brightness and **then half-wave rectify it for a lesser light output**.



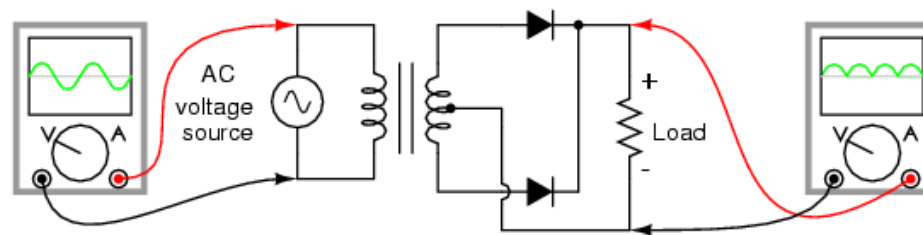
Half-wave rectifier application: Two level lamp dimmer.

Rectification Circuit: Half-Wave

- In “Dim” switch position, **incandescent lamp receives approximately one-half power** it would normally receive operating on full-wave AC.
 - Because half-wave rectified power pulses far more rapidly than filament has time to heat up and cool down, lamp does not blink.
 - Instead, its filament merely operates at lesser temperature than normal, providing less light output.
- This principle of “pulsing” power rapidly to slow-responding load device to control electrical power sent to it is common in world of industrial electronics.
- Since controlling device (diode, in this case) is either fully conducting or fully non-conducting at any given time, it dissipates little heat energy while controlling load power, making this **method of power control very energy-efficient**.
- This circuit is perhaps crudest possible method of pulsing power to a load, but it suffices as a proof-of-concept application.

Rectifier Circuit: Full-Wave

- If we need to rectify AC power:
 - In order obtain full use of *both* half-cycles of sine wave
 - *Full-wave* rectifier must be used
- Types of full-wave rectifier:
 - Center-tap design
 - Full-wave bridge.
- *Center-tap* design
 - Uses transformer with **center-tapped secondary winding** and **two diodes**



Full-wave rectifier, center-tapped design.

Full-Wave Rectifier Circuits

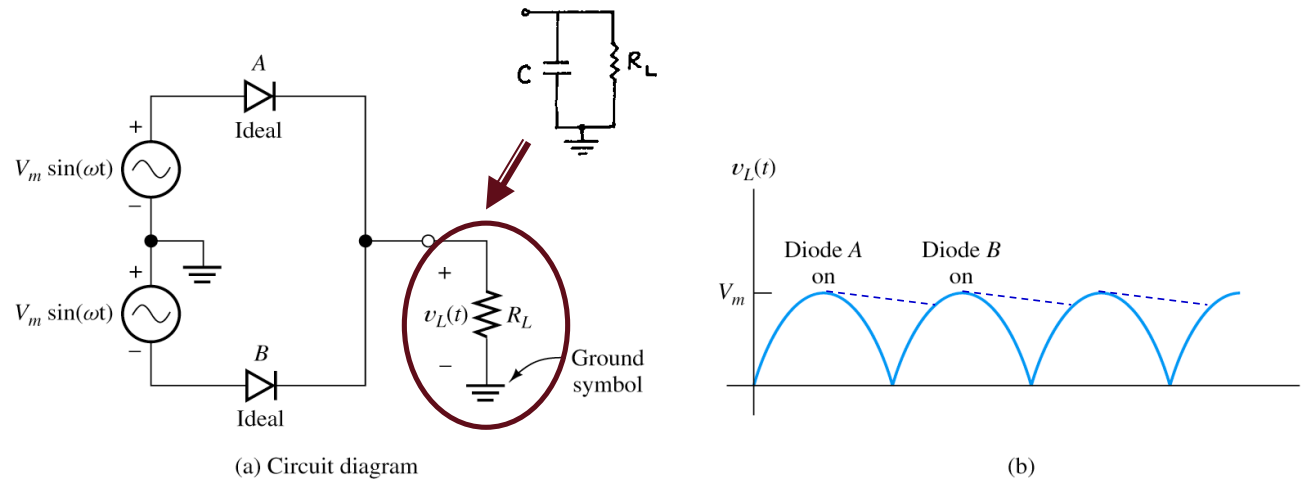
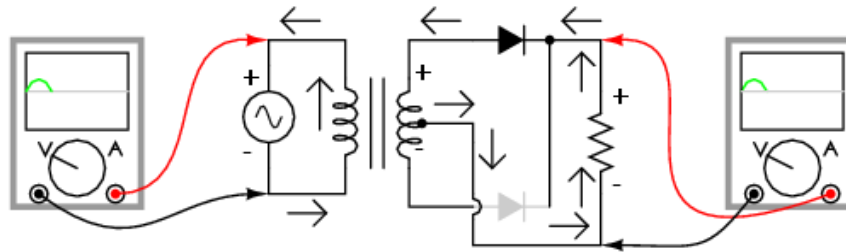


Figure 10.27 Full-wave rectifier.

- * **Centre-Tapped Full-Wave Rectifier** – two half-wave rectifier with out-of-phase source voltages and a common ground.
- * When upper source supplies “+” voltage to diode A, the lower source supplies “-” voltage to diode B; and vice versa.
- * We can also smooth the output by using a large capacitance.

Rectifier circuit

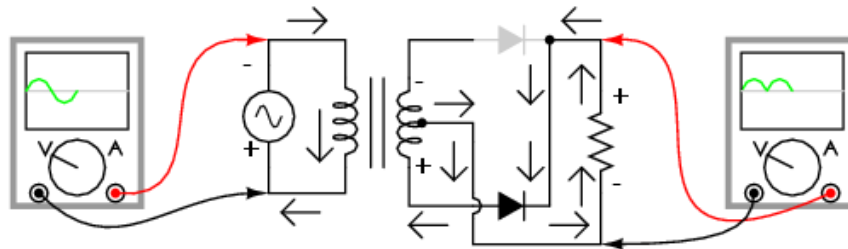
- Look at its circuit operation **one half-cycle at a time.**
- **Consider first half-cycle:**
 - Source voltage polarity is positive (+) on top and negative (-) on bottom.
 - Only **top diode is conducting**; bottom diode is blocking current, and load “sees” first half of **sine wave.**
 - Only **top half of transformer's secondary winding carries current** during this half-cycle.



Full-wave center-tap rectifier: Top half of secondary winding conducts during positive half-cycle of input, delivering positive half-cycle to load.

Rectifier circuit

- Consider next (second) half-cycle:
 - AC polarity reverses
 - Other diode and other half of transformer's secondary winding now carry current
 - Portions of circuit formerly carrying current during first half-cycle sit idle
 - Load still “sees” half of sine wave, of same polarity as before.



Full-wave center-tap rectifier: During negative input half-cycle, bottom half of secondary winding conducts, delivering a positive half-cycle to the load.

$$V_{dc} = \frac{2V_m}{\pi}$$

$$V_m = \frac{V_{p(sec)}}{2}$$

$$\Rightarrow V_{p(sec)} = 2V_m$$

$$PIV = V_{p(sec)} = 2V_m$$

Rectifier Circuit: Full-Wave

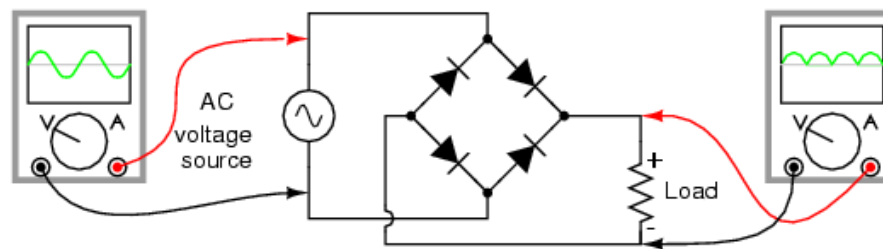
- One disadvantage of this full-wave rectifier design is:
 - Necessity of transformer with center-tapped secondary winding.
- If circuit in question is one of high power:
 - Size and expense of suitable transformer will be significant.
- Consequently:
 - Center-tap rectifier design is only seen in low-power applications.

Disadvantages

1. Since, each diode uses only one-half of the transformers secondary voltage, the d.c. output is comparatively small.
2. It is difficult to locate the centre-tap on secondary winding of the transformer.
3. The diodes used must have high peak-inverse voltage.

- **Full-wave Bridge Rectifier.**

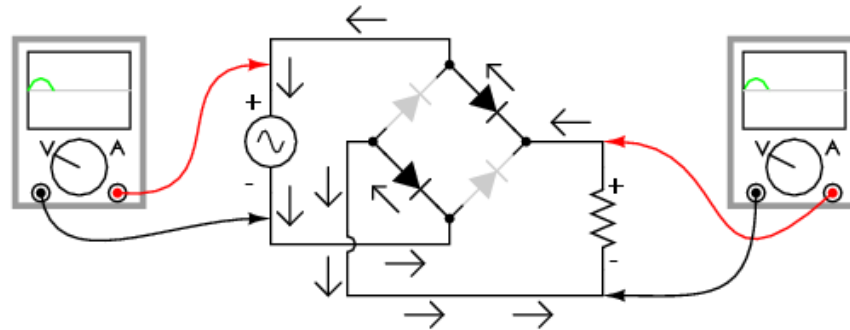
- More popular full-wave rectifier design
- Built around four-diode bridge configuration.
- For obvious reasons, this design is called *full-wave bridge*.



Full-wave bridge rectifier.

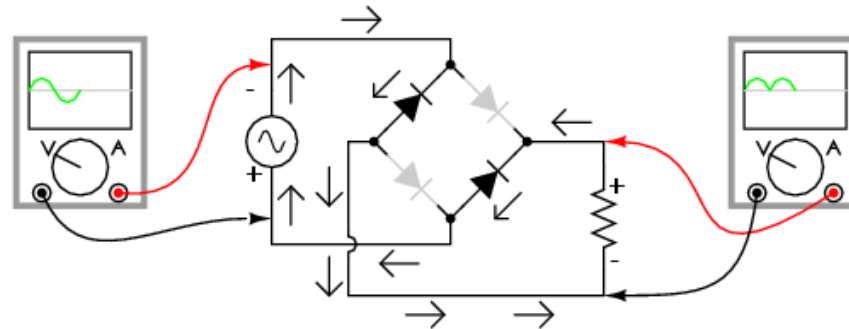
- **Full-wave Bridge**

- Current directions for positive and negative half-cycles of:
 - AC source waveform are shown below and next page respectively.
- **Note:** regardless of polarity of input, current flows in same direction through load.
- That is, negative half-cycle of source is positive half-cycle at load.



*Full-wave bridge rectifier: **Electron flow for positive half-cycles***

Rectifier circuit: Full-Wave Bridge

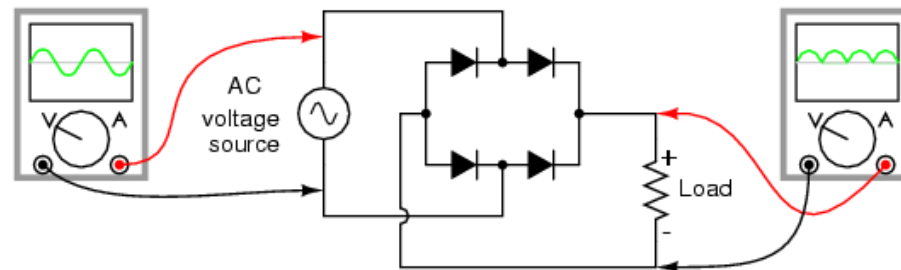


Full-wave bridge rectifier: Electron flow for negative half-cycles.

- Current flow is through two diodes in series for both polarities.
- Thus, sum of voltage drops for two diodes is $2(0.7)$ volts
- This is **disadvantage when compared to full-wave center-tap design.**
 - Will only be problem in very low voltage power supplies

- **Full-wave Bridge: with horizontal altitude**

- Remembering proper layout of diodes in full-wave bridge rectifier circuit can often be frustrating some times.
- Alternative representation of this circuit is easier both to remember and to comprehend.
- It is exact same circuit, except all diodes are drawn in horizontal altitude, all “pointing” same direction.



Alternative layout style for Full-wave bridge rectifier.

- One advantage of remembering this layout for bridge rectifier circuit is that it expands easily into poly-phase version (beyond basic electronics).

Disadvantages

- It requires four diodes.
- The use of two extra diodes cause an additional voltage drop thereby reducing the output voltage.

Efficiency of a full-wave rectifier

Rectifier efficiency, $\eta = \frac{\text{output dc power}}{\text{input ac power}}$

$$\eta = \frac{I_{dc}^2 R}{I_{rms}^2 (2r_f + R)}$$

$$\eta = \frac{\left[\frac{2I_m}{\pi} \right]^2 R}{\left[\frac{I_m}{\sqrt{2}} \right]^2 (2r_f + R)}$$

$$\eta = \frac{8}{\pi^2} \left[\frac{R}{(r_f + R)} \right]$$

$$\eta = \frac{0.81}{1 + \frac{2r_f}{R}}$$

For a bridge rectification circuit, the supply passes through two diodes ($2r_f$) at any particular time

In the case of a centre-tap rectification circuit, only one diode conducts at any particular time.

Thus, its efficiency will be

$$\eta = \frac{0.81}{1 + \frac{r_f}{R}}$$

For maximum efficiency, the value of r_f should be negligible compared to R . Therefore, the maximum efficiency of the full-wave rectifier is 81% and is twice as efficient as the half-wave rectifier.

The **peak inverse voltage (PIV)** of the diode is the peak value of the voltage that a diode can withstand when it is reversed biased

Type of Rectifier	PIV
Half Wave	Peak value of the input secondary voltage, $V_{m(\text{peak})}$
Full Wave : Center-Tapped	$2V_{m(\text{peak})}$
Full Wave: Bridge	$V_{m(\text{peak})}$

Rectifier circuit

- In any case of rectification: single-phase or poly-phase
 - Amount of AC voltage mixed with rectifier's DC output is called *ripple voltage*.
- In most cases, since “pure” DC is desired goal
 - *Ripple voltage is undesirable.*
- If power levels are not too great
 - *Filtering networks may be employed* to reduce amount of ripple in output voltage.

Output Ripple

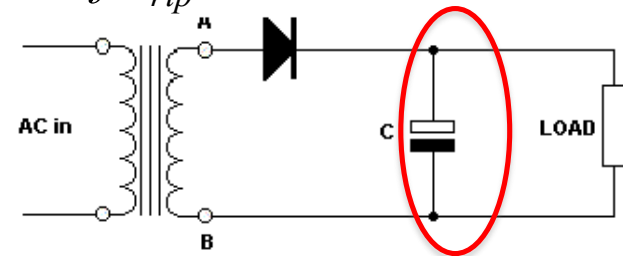
- Output ripple **will always be present** in circuits shown above
- Amplitude of ripple **can be reduced** by adding smoothing capacitor
- Capacitor and load (shown here as resistor) form **low pass filter with time constant** : $T = RC$
- Time constant **should be much longer** than one ripple
- For given **ripple amplitude**: capacitor size (in microfarads) is given by

$$C = \frac{I_{load}}{fV_{rip}} \times 10^6 \text{ (Half wave)} \quad \text{or} \quad C = \frac{I_{load}}{2fV_{rip}} \times 10^6 \text{ (Full wave)}$$

f : line frequency

I_{load} : Load Current

V_{rip} : Amplitude of ripple voltage



NOTE: Voltage rating of the capacitor must be $> 1.4 \times V_{out}$

Large capacitors should have bleeder resistors for safety!

Rectifier Circuit: Output Voltage

- Full wave rectification will produce voltage roughly equal to

$$V_o \approx \sqrt{2}V_{i,RMS}$$

- In practice, **there will be small voltage drop across diodes** that will reduce this voltage
- For accurate supplies, regulation is necessary

Rectifier circuit

REVIEW:

- *Rectification* is conversion of alternating current (AC) to direct current (DC).
- A *half-wave* rectifier is circuit that allows only one half-cycle of AC voltage waveform to be applied to load, resulting in one non-alternating polarity across it.
 - The resulting DC delivered to load “pulsates” significantly.
- A *full-wave* rectifier is circuit that converts both half-cycles of AC voltage waveform to unbroken series of voltage pulses of same polarity.
 - The resulting DC delivered to load doesn't “pulsate” as much.

- The *ripple factor* indicates how close the rectified output is to the pure ideal dc voltage waveform
- The rms value of a full-wave signal is $V_{\text{rms}} = 0.707V_p$, which is the same as V_{rms} for a full-wave sine wave
- *Efficiency* of a rectifier is defined as the ratio of dc power delivered to the load to the ac power from the secondary winding of the transformer

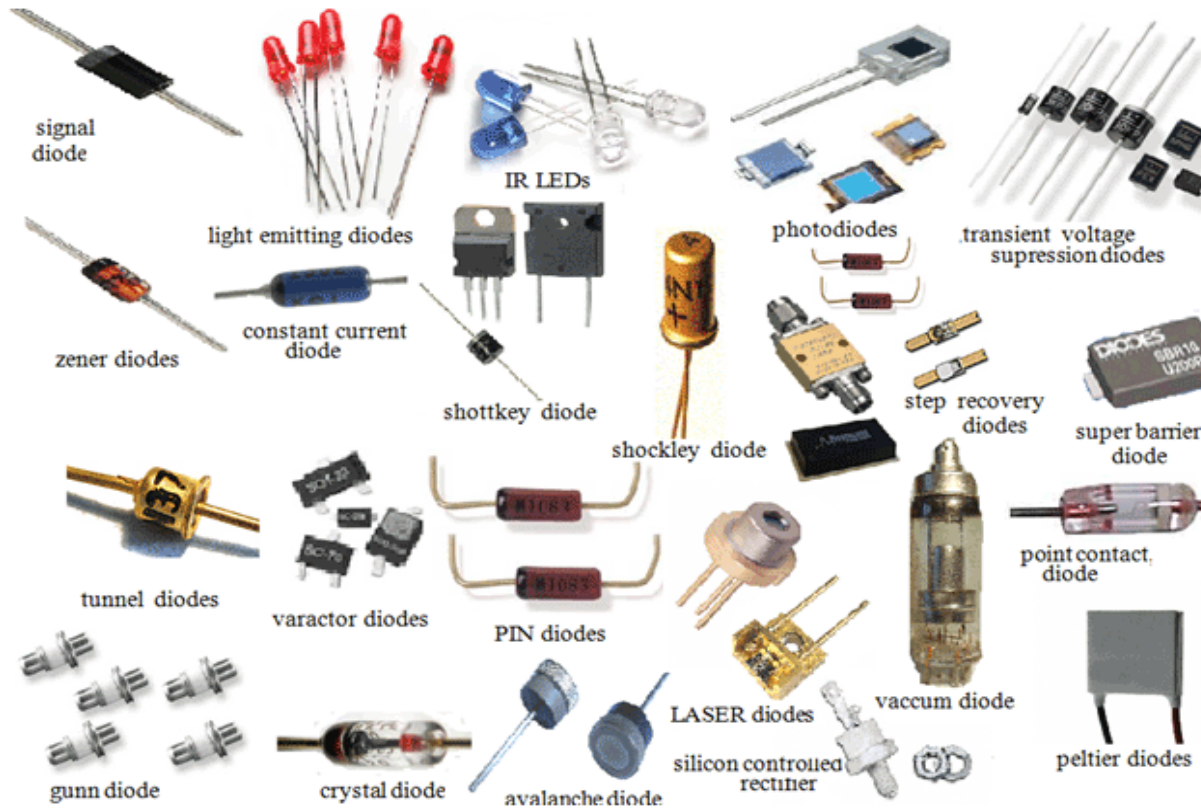
...that's all folks...
...thanks for your time...



Zener Diode Voltage Regulators

KLOGO, Griffith Selorm
(PhD Candidate)

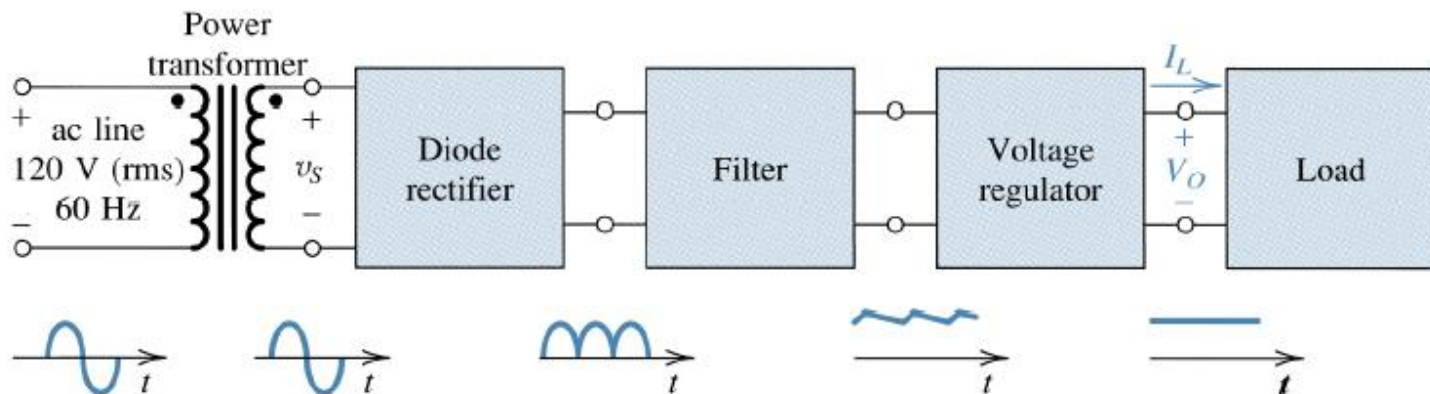
Types of Diodes



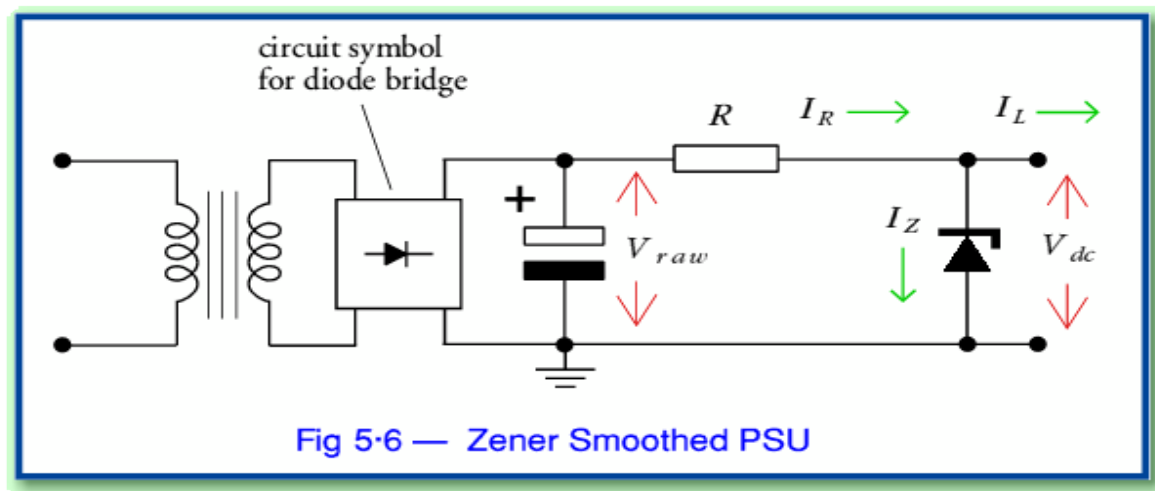
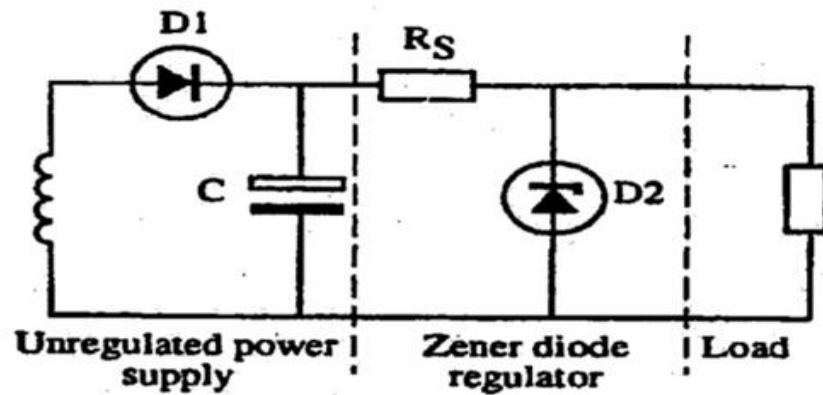
Types of Diode

Rectifier Circuits

One of the most important applications of diodes is in the design of rectifier circuits. Used to convert an AC signal into a DC voltage used by most electronics.



Regulator Circuits



The Zener Diode

- * *Zener diode* is designed for operation in the reverse-breakdown region.
- * The *breakdown voltage* is controlled by the doping level (-1.8 V to -200 V).
- * The major application of Zener diode is to provide an output reference that is stable despite changes in input voltage – power supplies, voltmeter,...

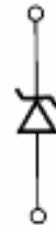
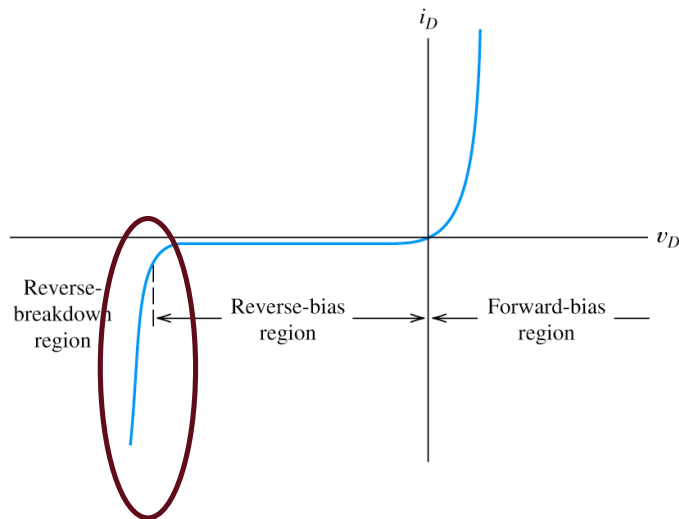


Figure 7.3 Zener diode symbol.

Zener-Diode Voltage-Regulator Circuits

- * Sometimes, a circuit that produces constant output voltage while operating from a variable supply voltage is needed. Such circuits are called *voltage regulator*.
- * The Zener diode has a breakdown voltage equal to the desired output voltage.
- The resistor limits the diode current to a safe value so that Zener diode does not overheat.
- Zener Diode is a reversed-biased heavily-doped PN-Junction diode which operates in the breakdown region.
- The breakdown voltage of a zener diode can be set by controlling the doping level.
- For zener diodes, silicon is preferred to Ge because of its higher temperature and current capability

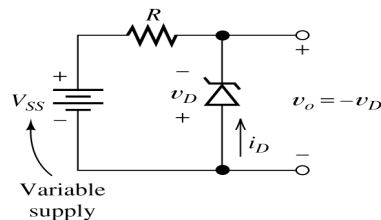
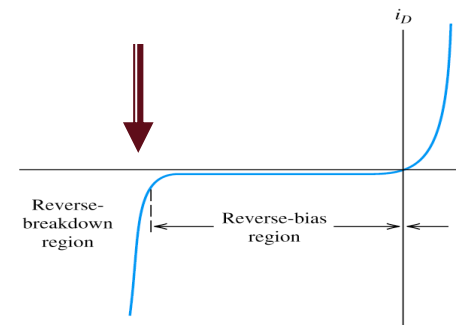
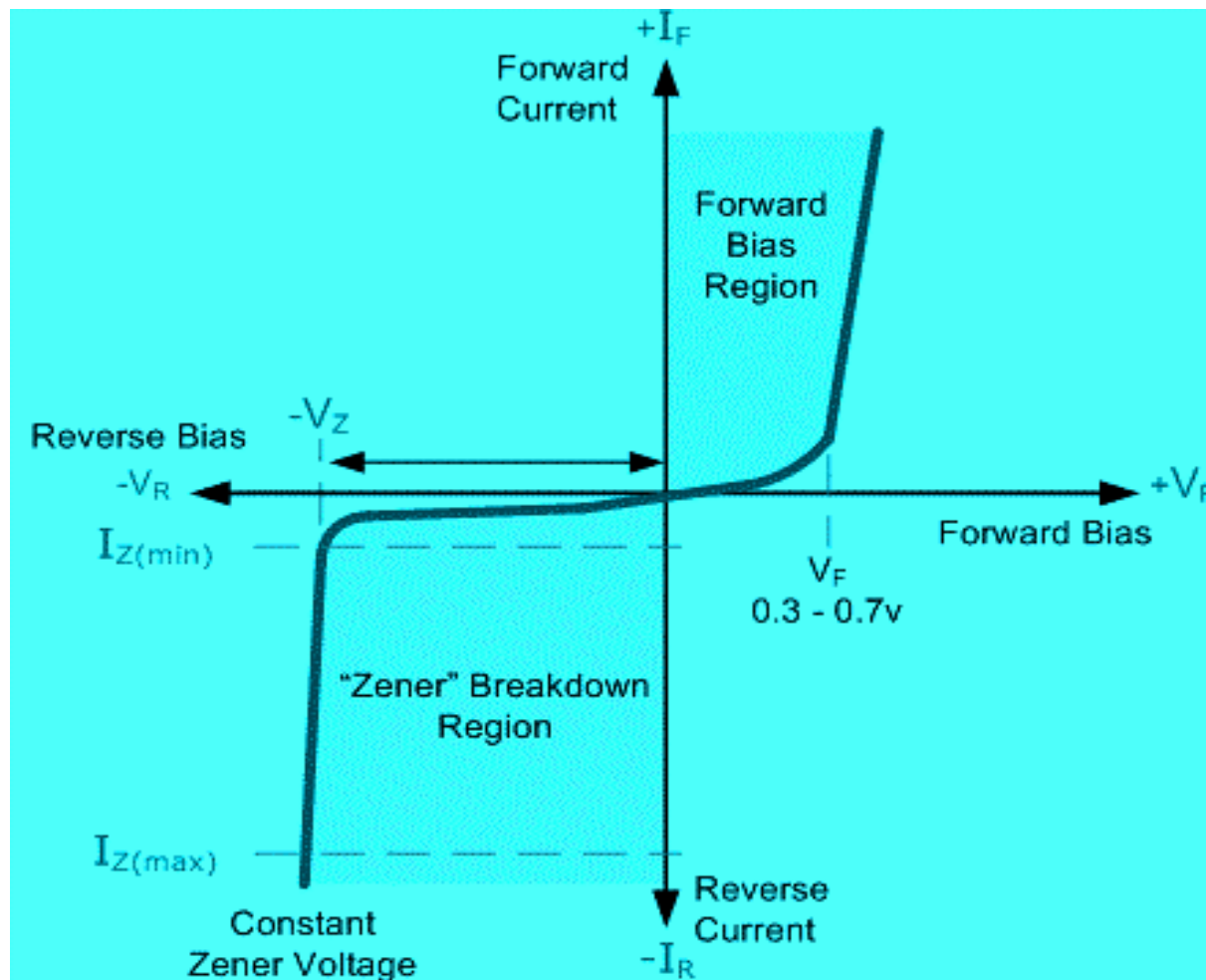


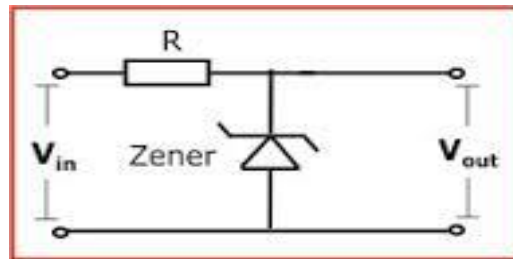
Figure 10.9 A simple regulator circuit that provides a nearly constant output voltage v_o from a variable supply voltage.



I-V Characteristics



Zener Diode - Voltage Regulator (reverse biased)

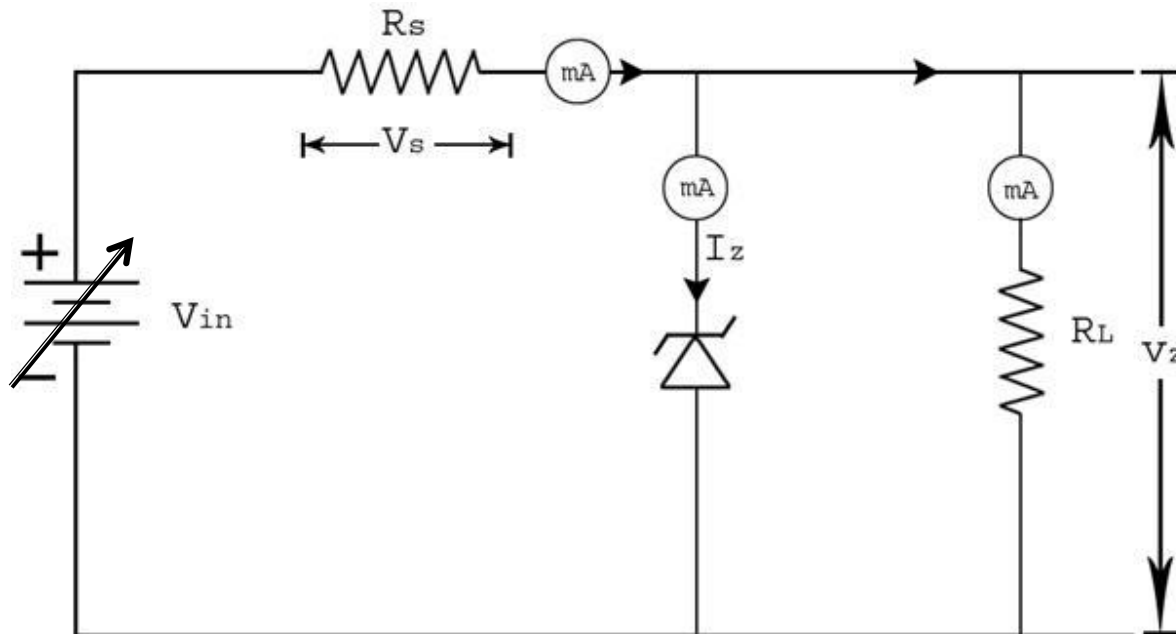


- A **Zener diode** is a type of [diode](#) that permits [current](#) not only in the forward direction like a normal diode, but also in the reverse direction if the voltage is larger than the [breakdown voltage](#) known as "Zener knee voltage" or "Zener voltage".
- The zener breakdown voltage or zener voltage can be precisely controlled by controlling the doping levels of P and N regions at the time of manufacturing a zener diode
- After breakdown has occurred, the voltage across zener diode remains constant equal to V_z .
- Any increase in the source voltage will result in the increase in the reverse zener current.
- The zener current after reverse breakdown must be controlled by connecting a resistor R as shown. This is essential to avoid any damage to the device due to excessive heating

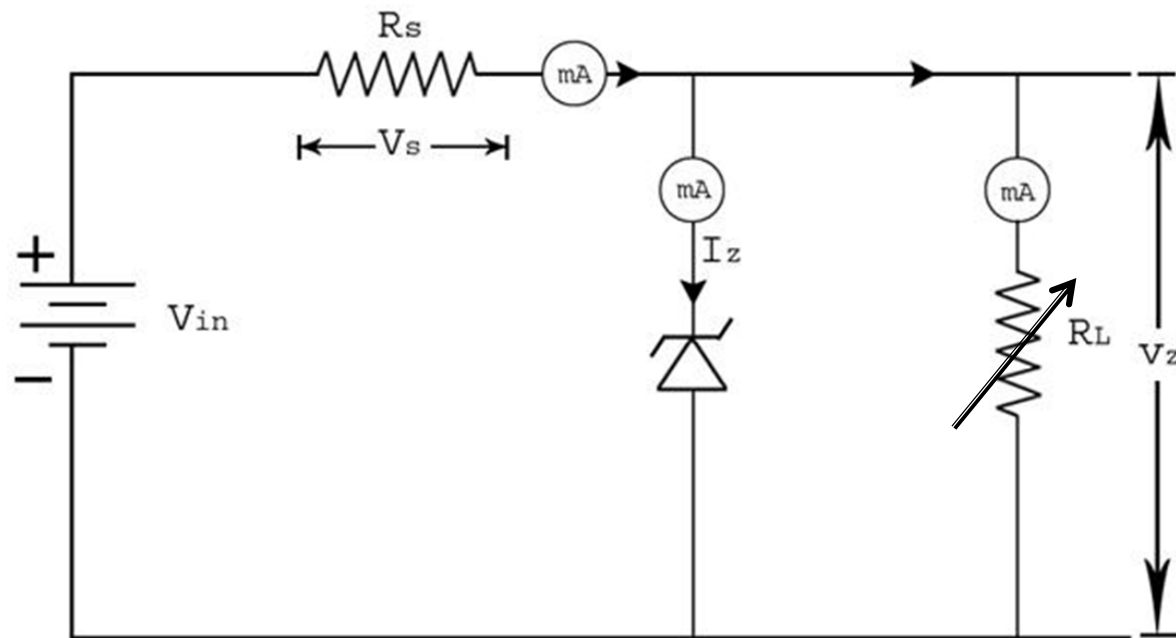
Working of Zener Diode

- Case 1: Regulation when input voltage is varied
- Case 2: Regulation when load resistance is varied

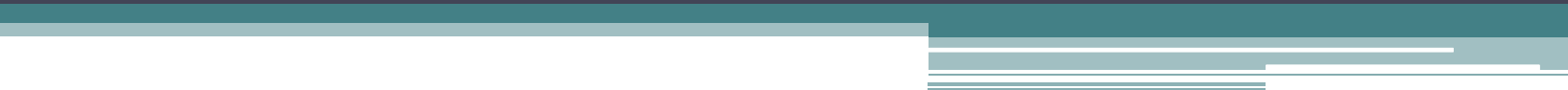
Case 1: Regulation when input voltage is varied



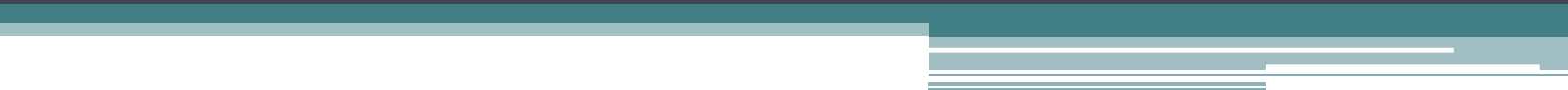
Case 2: Regulation when load resistance is varied



Tutorials



Bipolar Junction Transistors (BJTs)



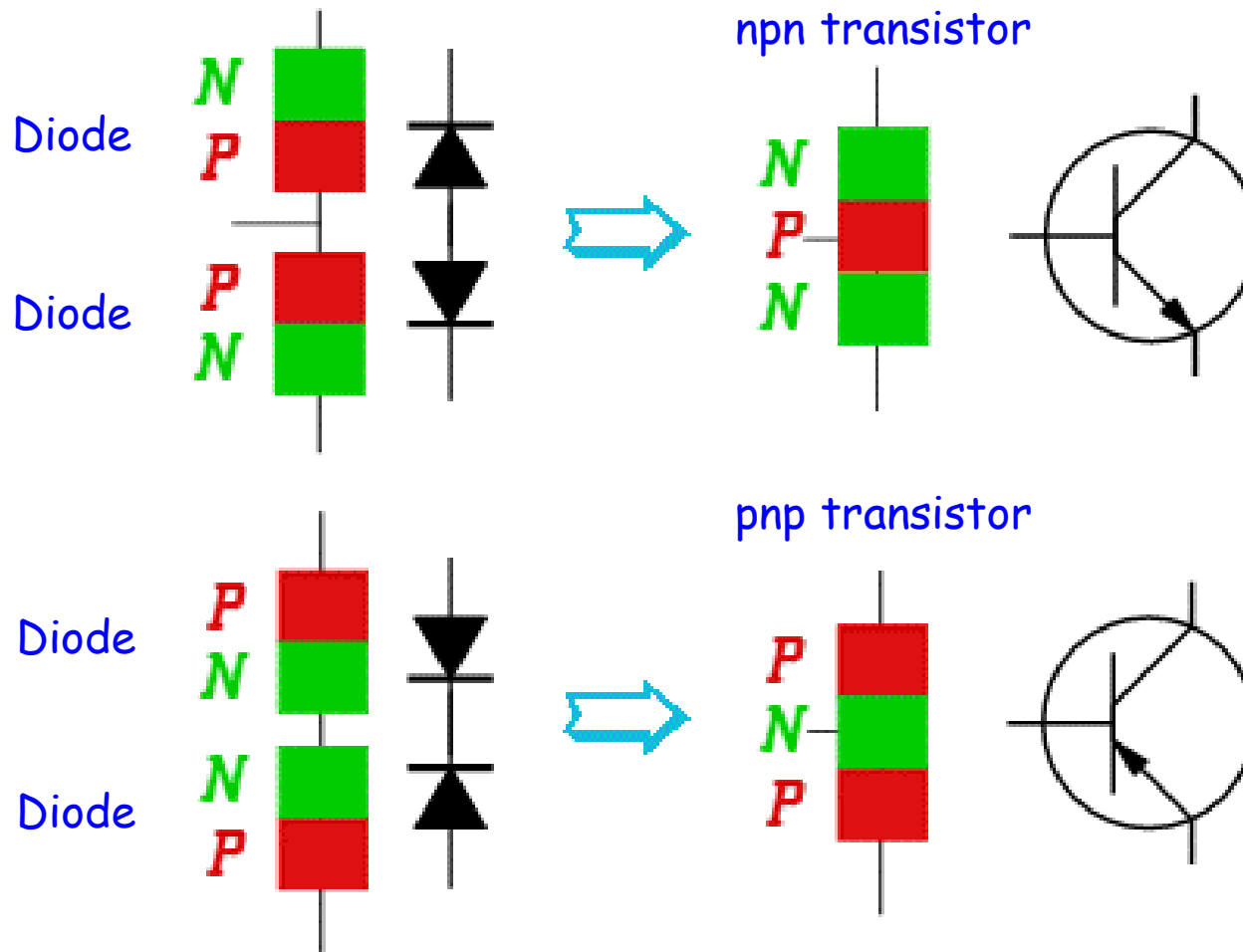
Introduction

- The basic of electronic system nowadays is semiconductor device.
- The famous and commonly use of this device is BJTs (Bipolar Junction Transistors).
- It can be use as amplifier and logic switches.
- BJT consists of three terminal:
 - collector : C
 - base : B
 - emitter : E
- Two types of BJT : pnp and npn

Transistor Construction

- 3 layer semiconductor device consisting:
 - 2 n- and 1 p-type layers of material → npn transistor
 - 2 p- and 1 n-type layers of material → pnp transistor
- The term bipolar reflects the fact that holes and electrons participate in the injection process into the oppositely polarized material
- A single pn junction has two different types of bias:
 - forward bias
 - reverse bias
- Thus, a two-pn-junction device has four types of bias.

Basic models of BJT



Qualitative basic operation of BJTs

A BJT consists of two back-to-back p-n junctions.

The middle region, the base, is very thin.

The three regions are the emitter, base, and collector.

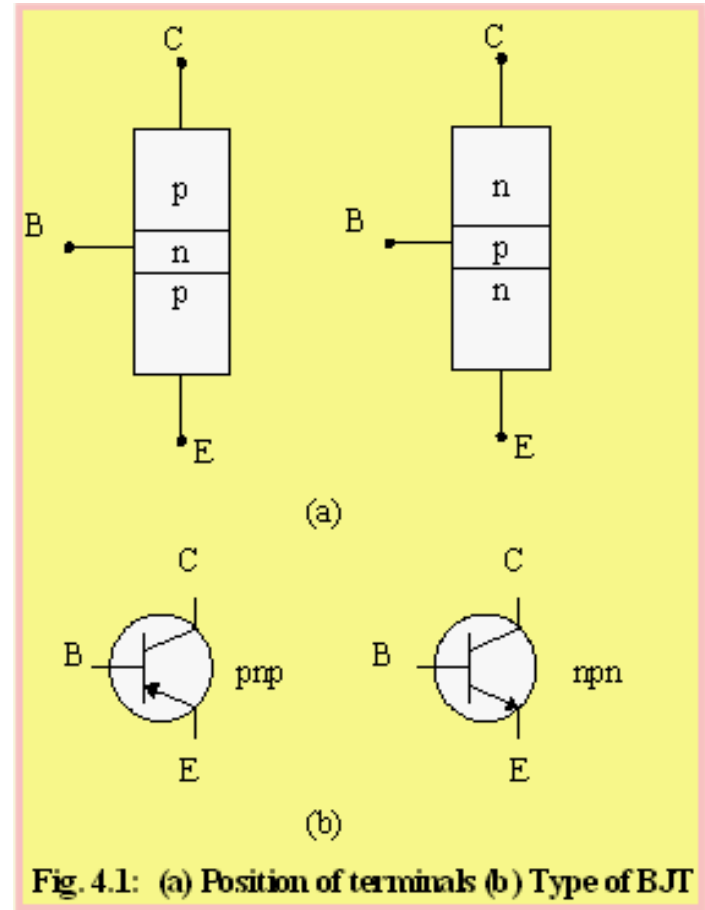
Carriers are injected (“emitted”) into the base from the emitter.

Since the base is thin, most carriers injected into base diffuse into collector.

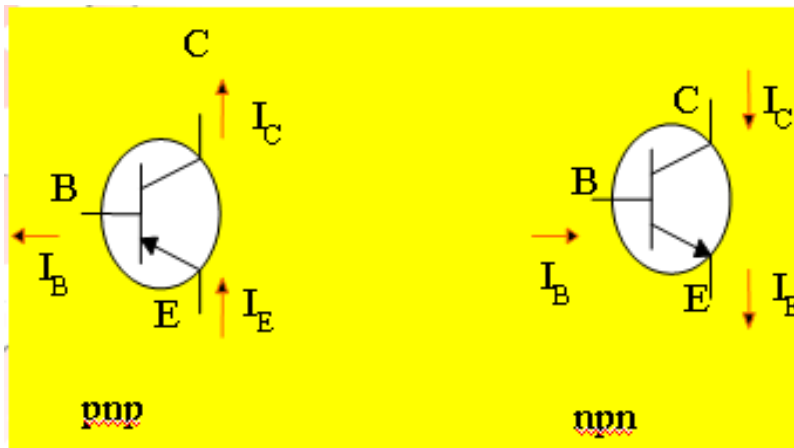
What does a “thin base thickness” mean? → Base thickness is much thinner than the diffusion length of carriers injected from the emitter.

Position of the terminals and symbol of BJT.

- Base is located at the middle and more thin from the level of collector and emitter
- The emitter and collector terminals are made of the same type of semiconductor material, while the base of the other type of material



Transistor currents



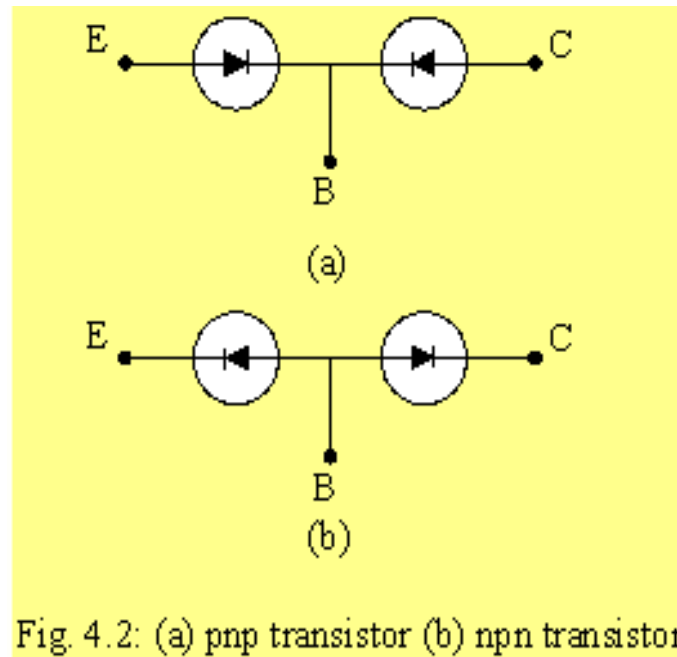
I_C = the collector current
 I_B = the base current
 I_E = the emitter current

- The arrow is always drawn on the emitter
- The arrow always point toward the n-type
- The arrow indicates the direction of the emitter current:

pnp: $E \rightarrow B$

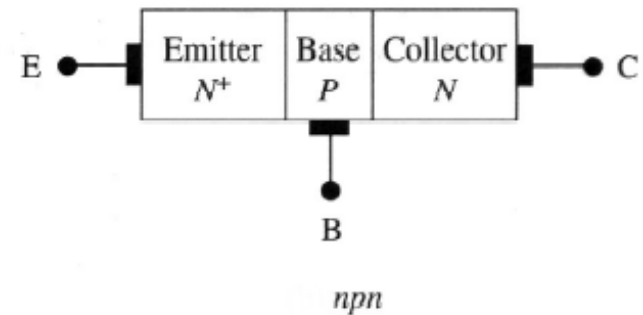
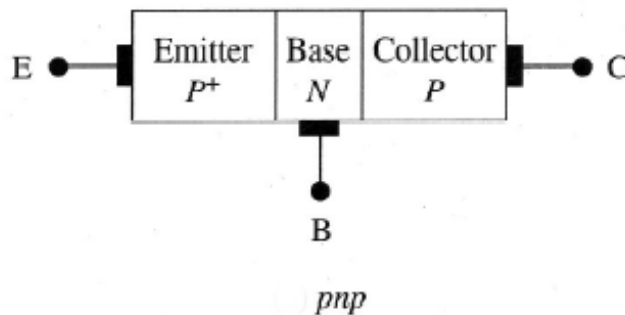
nnp: $B \rightarrow E$

- By imaging the analogy of diode, transistor can be construct like two diodes that connetecd together.
- It can be conclude that the work of transistor is base on work of diode.

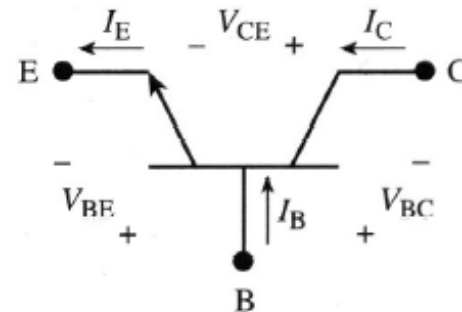
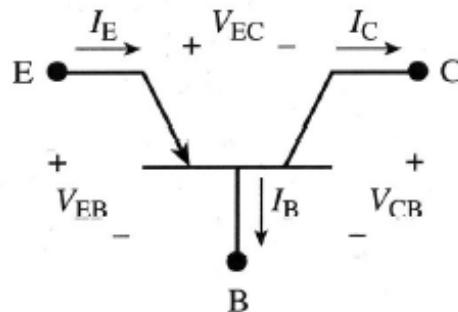


Basic models of BJT

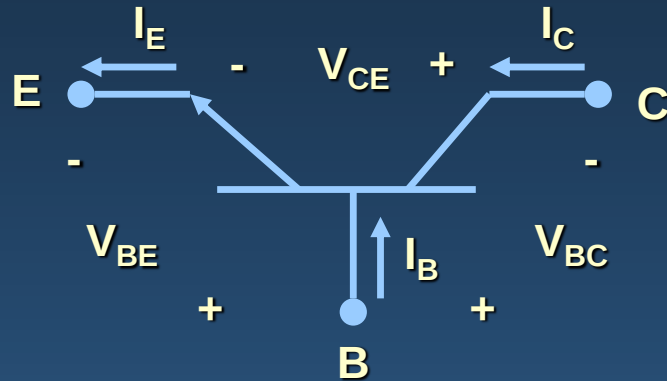
Bipolar Junction Transistor Fundamentals



Looks sort of
like two diodes
back to back



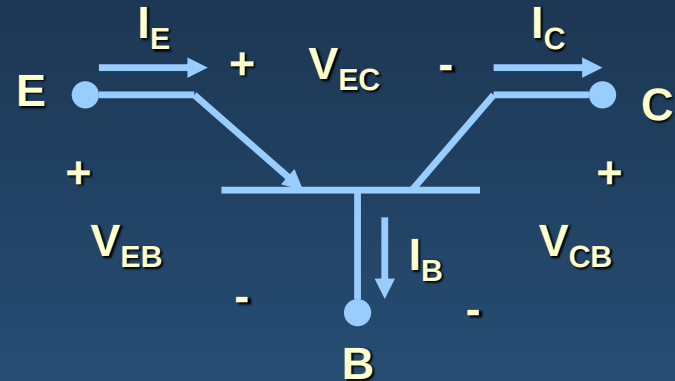
BJT Relationships - Equations



npn

$$I_E = I_B + I_C$$

$$V_{CE} = -V_{BC} + V_{BE}$$



pnp

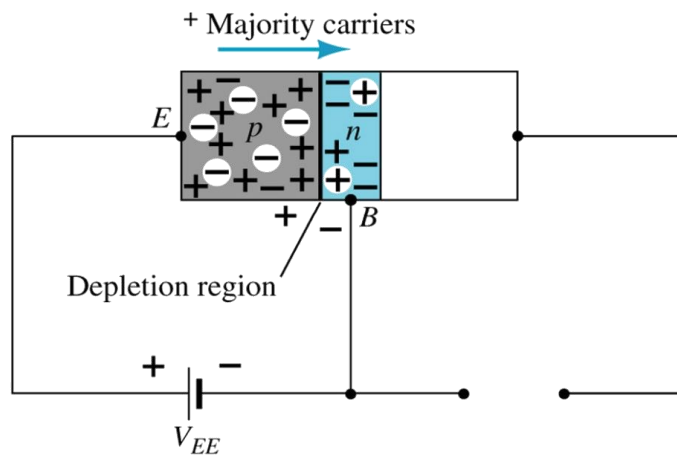
$$I_E = I_B + I_C$$

$$V_{EC} = V_{EB} - V_{CB}$$

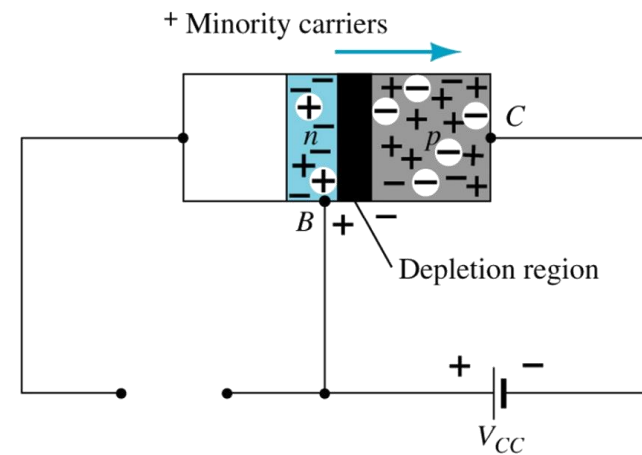
Note: The equations seen above are for the transistor, not the circuit.

Transistor Operation

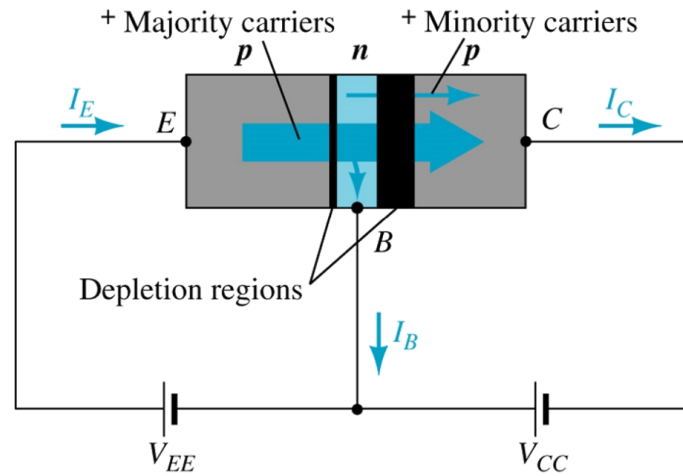
- The basic operation will be described using the pnp transistor. The operation of the pnp transistor is exactly the same if the roles played by the electron and hole are interchanged.
- One p-n junction of a transistor is reverse-biased, whereas the other is forward-biased.



Forward-biased junction
of a pnp transistor



Reverse-biased junction
of a pnp transistor



- Both biasing potentials have been applied to a pnp transistor and resulting majority and minority carrier flows indicated.
- Majority carriers (+) will diffuse across the forward-biased p-n junction into the n-type material.
- A very small number of carriers (+) will through n-type material to the base terminal. Resulting I_B is typically in order of microamperes.
- The large number of majority carriers will diffuse across the reverse-biased junction into the p-type material connected to the collector terminal.

- Majority carriers can cross the reverse-biased junction because the injected majority carriers will appear as minority carriers in the n-type material.
- Applying KCL to the transistor :

$$I_E = I_C + I_B$$

- The I_C comprises of two components – the majority and minority carriers

$$I_C = I_{Cmajority} + I_{Cminority}$$

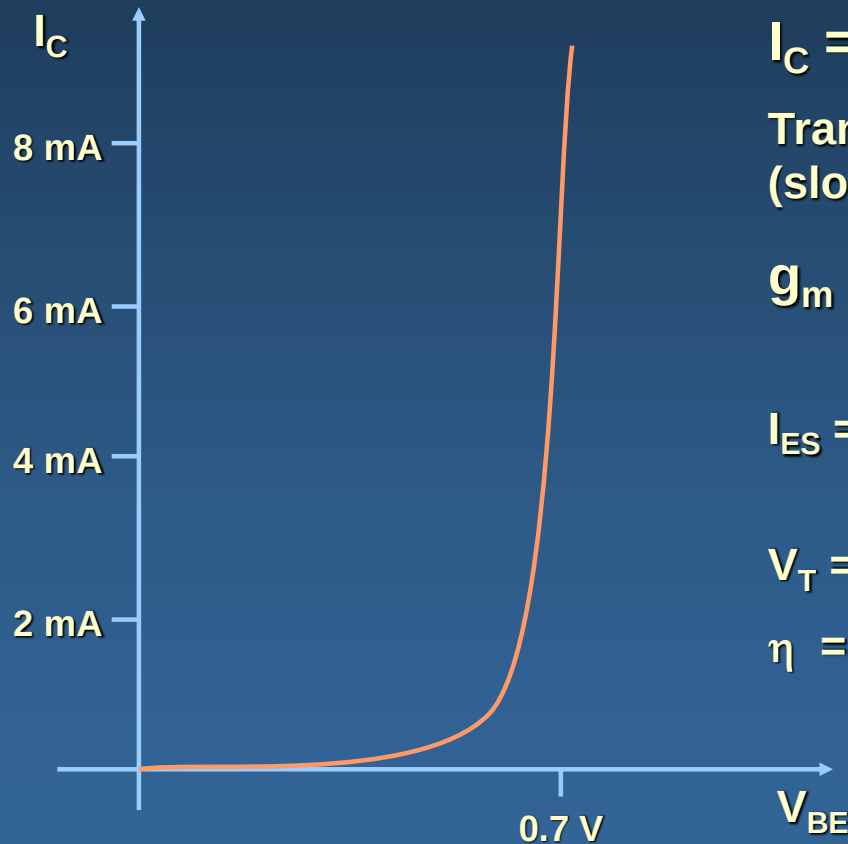
- I_{CO} – I_C current with emitter terminal open and is called leakage current.

Transistor Characteristics

- The complete behavior of a transistor may be observed with the help of some types of curves known as characteristics curves
- Two types of characteristics curves
 - **Input characteristics curves**
 - Relates input current and input voltage for a given value of output voltage
 - **Output characteristics curves**
 - Relates output current and output voltage for a given value of input voltage

BJT Transconductance Curve

Typical NPN Transistor ¹



Collector Current:

$$I_C = \alpha I_{ES} e^{V_{BE}/\eta V_T}$$

Transconductance:
(slope of the curve)

$$g_m = \frac{dI_C}{dV_{BE}}$$

I_{ES} = The reverse saturation current of the B-E Junction.

$$V_T = kT/q = 26 \text{ mV (@ } T=300\text{K)}$$

η = the emission coefficient and is usually ~ 1

Modes of Operation

Active:

- Most important mode of operation
- Central to amplifier operation
- The region where current curves are practically flat

Saturation:

- Barrier potential of the junctions cancel each other out causing a virtual short

Cutoff:

- Current reduced to zero
- Ideal transistor behaves like an open switch

*** Note:** There is also a mode of operation called inverse active, but it is rarely used.

Three Types of BJT Biasing

Biasing the transistor refers to applying voltage to get the transistor to achieve certain operating conditions.

Common-Base Biasing (CB) :

input = V_{EB} & I_E

output = V_{CB} & I_C

Common-Emitter Biasing (CE):

input = V_{BE} & I_B

output = V_{CE} & I_C

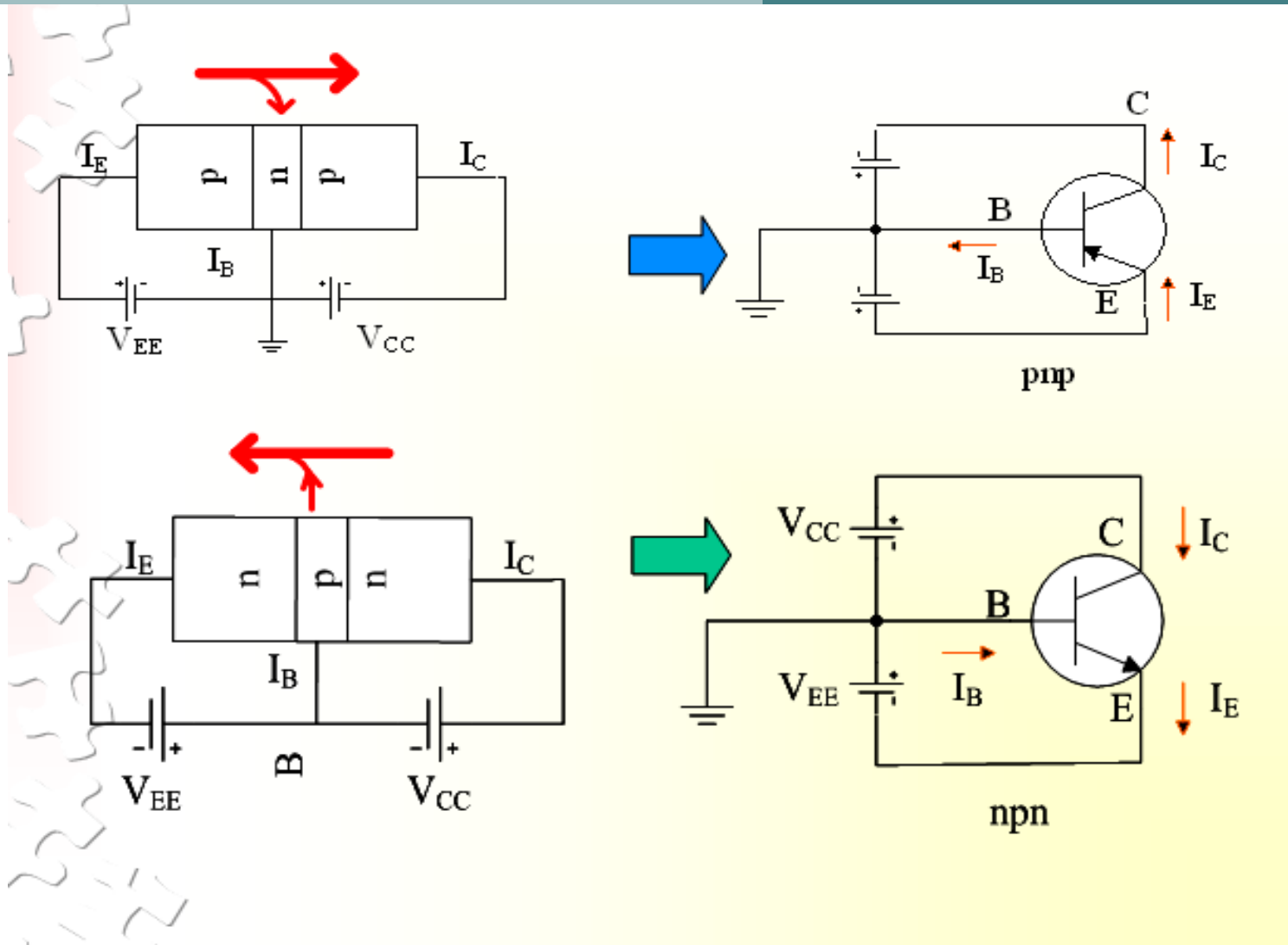
Common-Collector Biasing (CC):

input = V_{BC} & I_B

output = V_{EC} & I_E

Common-Base Configuration

- Common-base terminology is derived from the fact that the :
 - base is common to both input and output of the configuration.
 - base is usually the terminal closest to or at ground potential.
- All current directions will refer to **conventional** (hole) flow and the arrows in all electronic symbols have a direction defined by this convention.
- Note that the applied **biasing** (voltage sources) are such as to establish current in the direction indicated for each branch.



Transistor Characteristics in CB Configuration

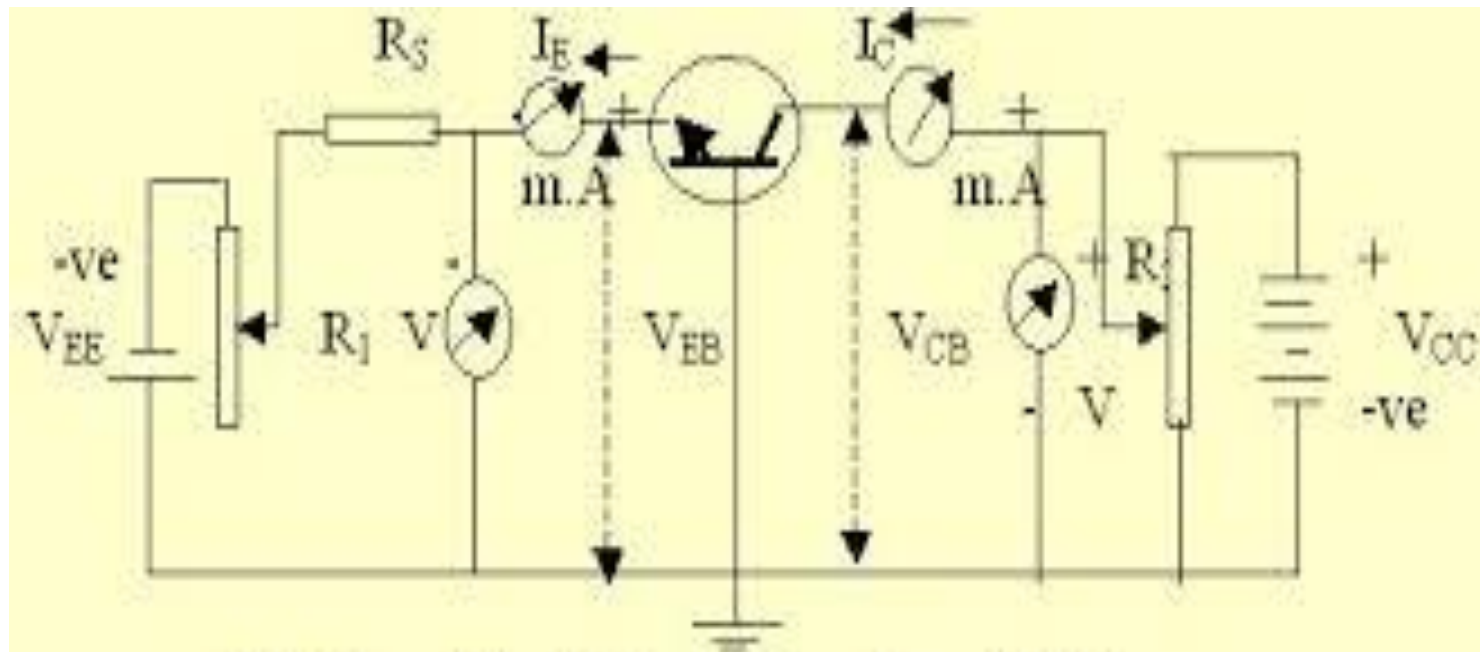
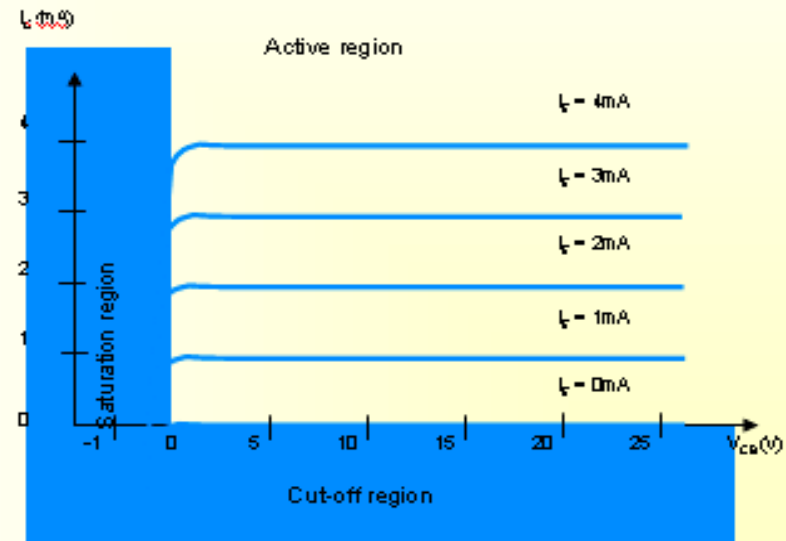
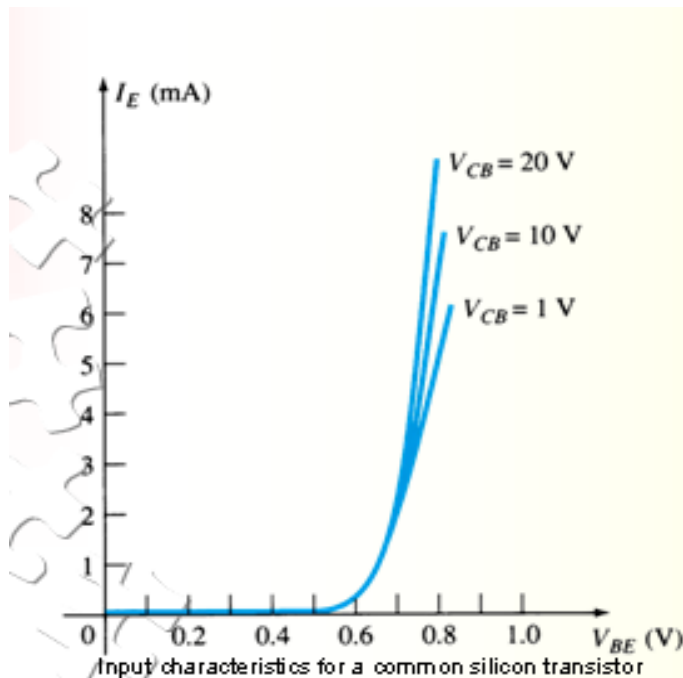


Fig: Circuit for determining the characteristics of common base n-p-n transistor.

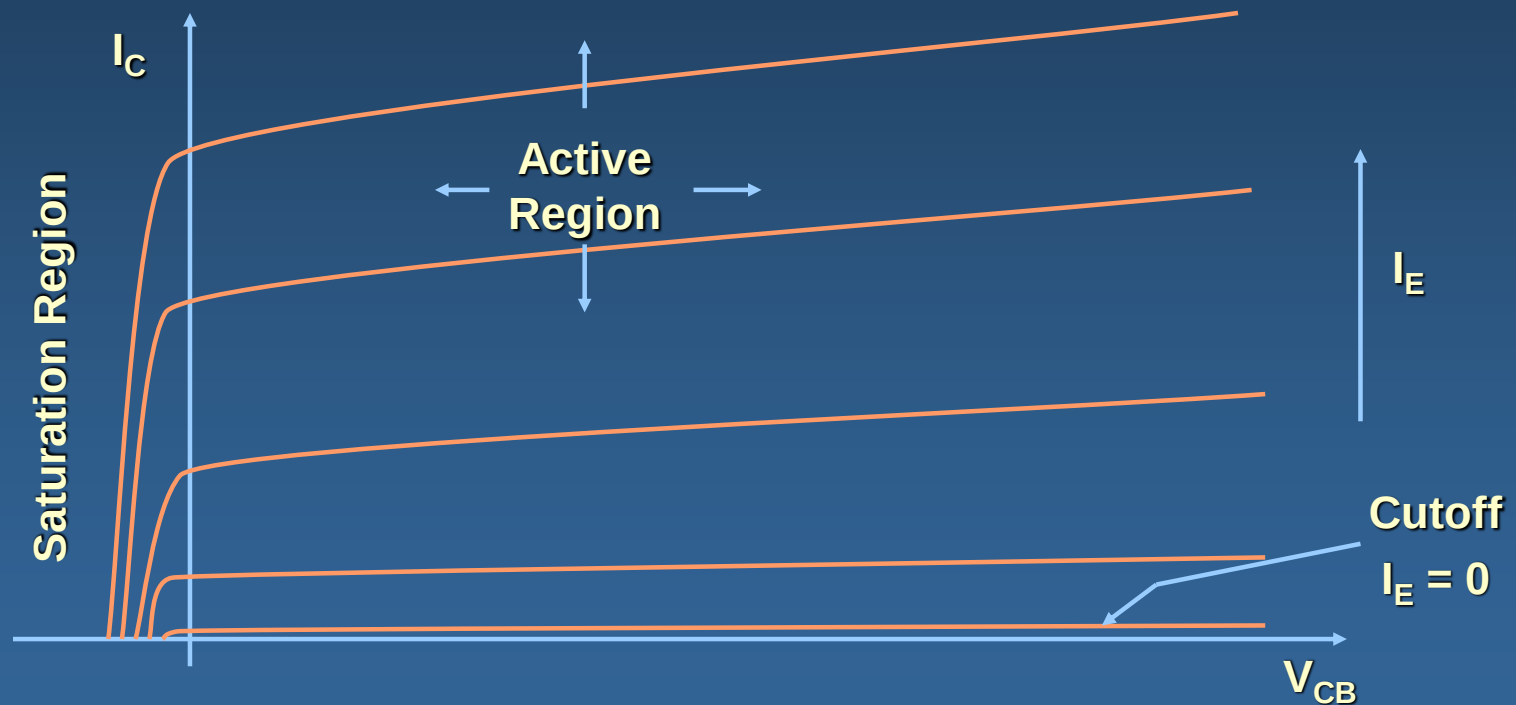
- To describe the behavior of common-base configuration requires two set of characteristics:
 - Input or driving point characteristics.
 - Output or collector characteristics
- The output characteristics has 3 basic regions:
 - Active region –defined by the biasing arrangements
 - Cutoff region – region where the collector current is 0A
 - Saturation region- region of the characteristics to the left of $V_{CB} = 0V$



Common-Base

Although the Common-Base configuration is not the most common biasing type, it is often helpful in the understanding of how the BJT works.

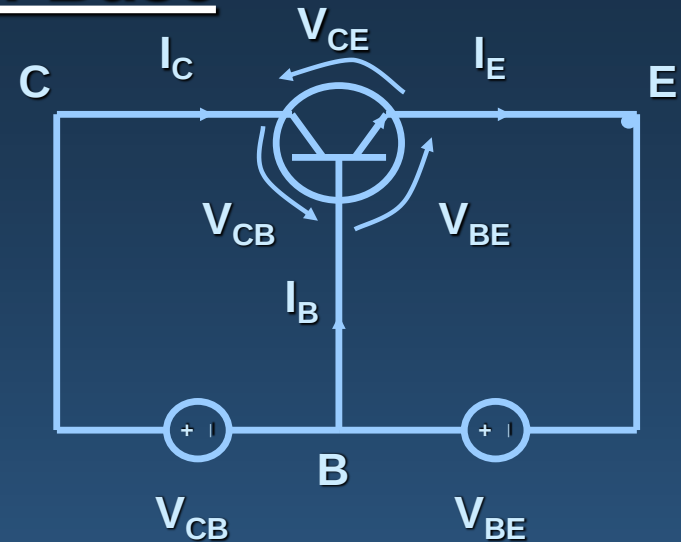
Emitter-Current Curves



Common-Base

Circuit Diagram: NPN Transistor

The Table Below lists assumptions that can be made for the attributes of the common-base biased circuit in the different regions of operation. Given for a Silicon NPN transistor.



Region of Operation	I_C	V_{CE}	V_{BE}	V_{CB}	C-B Bias	E-B Bias
Active	βI_B	$=V_{BE}+V_{CE}$	$\sim 0.7V$	$\bigcirc 0V$	Rev.	Fwd.
Saturation	Max	$\sim 0V$	$\sim 0.7V$	$-0.7V < V_{CE} < 0$	Fwd.	Fwd.
Cutoff	~ 0	$=V_{BE}+V_{CE}$	$\bigcirc 0V$	$\bigcirc 0V$	Rev.	None /Rev.

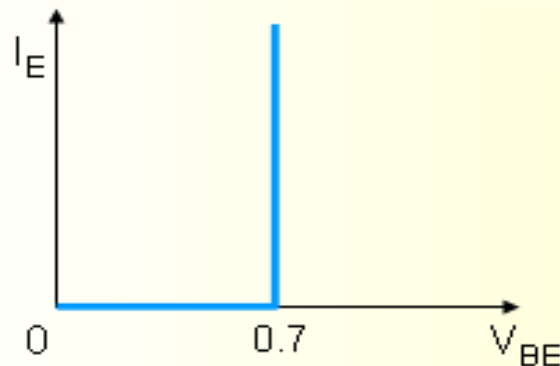
Active region	Saturation region	Cut-off region
• I_E increased, I_C increased • BE junction forward bias and CB junction reverse bias • Refer to the <u>graf</u>, $I_C \approx I_E$ • I_C not depends on V_{CB} • Suitable region for the transistor working as amplifier	• BE and CB junction is forward bias • Small changes in V_{CB} will cause big different to I_C • The allocation for this region is to the left of $V_{CB} = 0$ V.	• Region below the line of $I_E = 0$ A • BE and CB is reverse bias • no current flow at collector, only leakage current

- The curves (output characteristics) clearly indicate that a first approximation to the relationship between I_E and I_C in the active region is given by

$$I_C \approx I_E$$

- Once a transistor is in the 'on' state, the base-emitter voltage will be assumed to be

$$V_{BE} = 0.7V$$



- In the dc mode the level of I_C and I_E due to the majority carriers are related by a quantity called alpha

$$\alpha = \frac{I_C}{I_E}$$

$$I_C = \alpha I_E + I_{CBO}$$

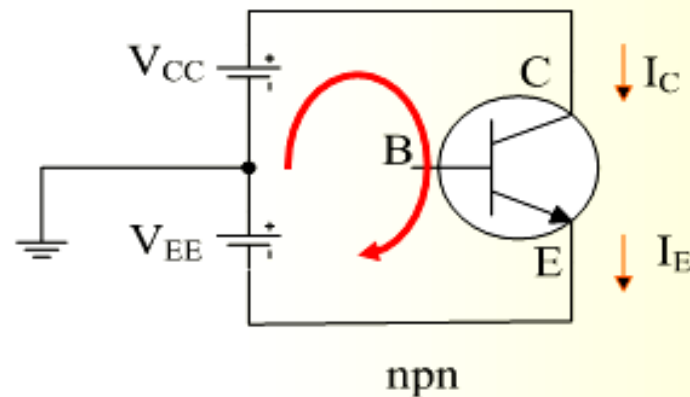
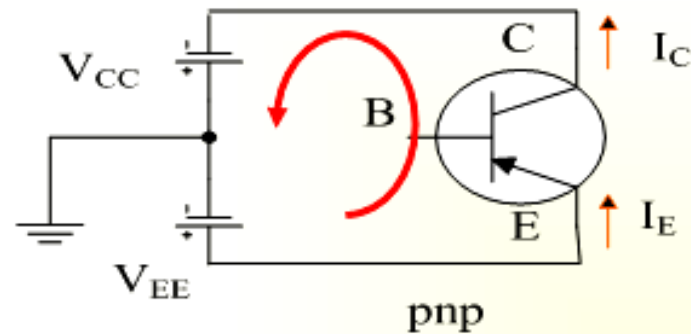
- It can then be summarize to $I_C = \alpha I_E$ (ignore I_{CBO} due to small value)
- For ac situations where the point of operation moves on the characteristics curve, an ac alpha is defined by

$$\alpha = \frac{\Delta I_C}{\Delta I_E}$$

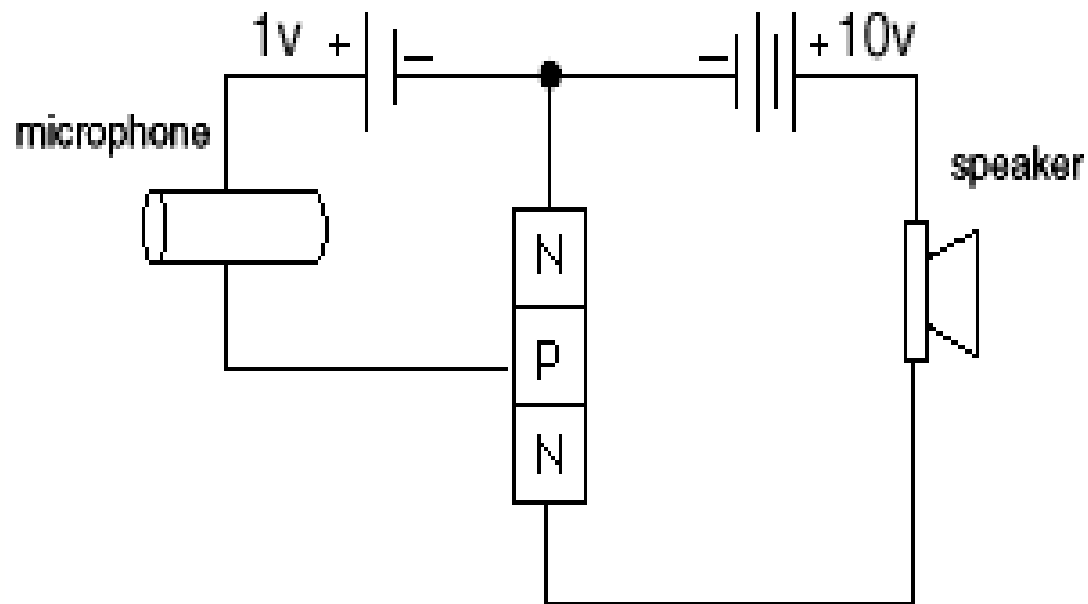
- Alpha is a common base current gain factor that shows the efficiency by calculating the current percent from current flow from emitter to collector. The value of α is typical from $0.9 \sim 0.998$.

Biasing

- Proper biasing CB configuration in active region by approximation $I_C \approx I_E$ ($I_B \approx 0 \mu A$)



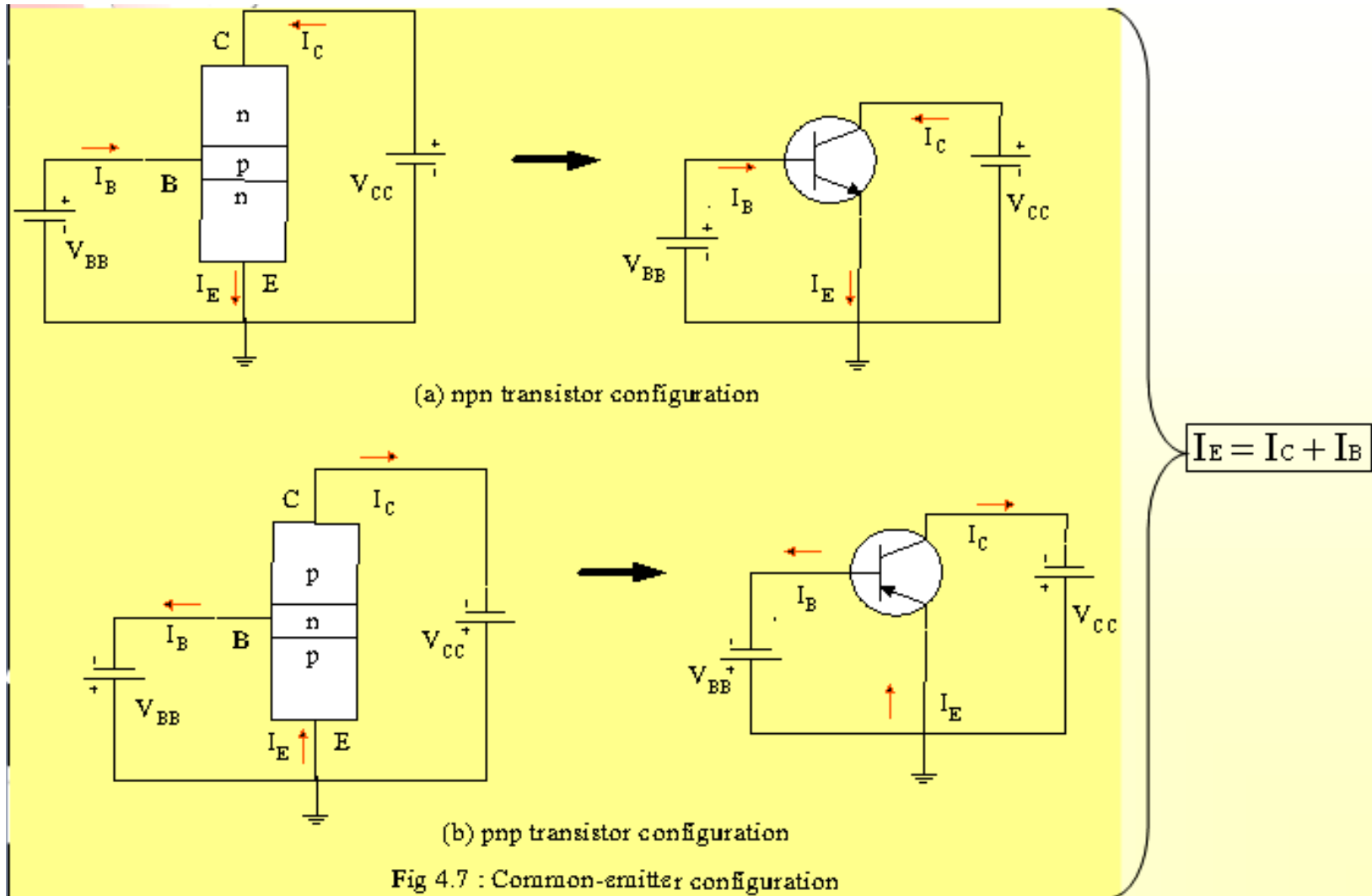
Simulation of transistor as an amplifier

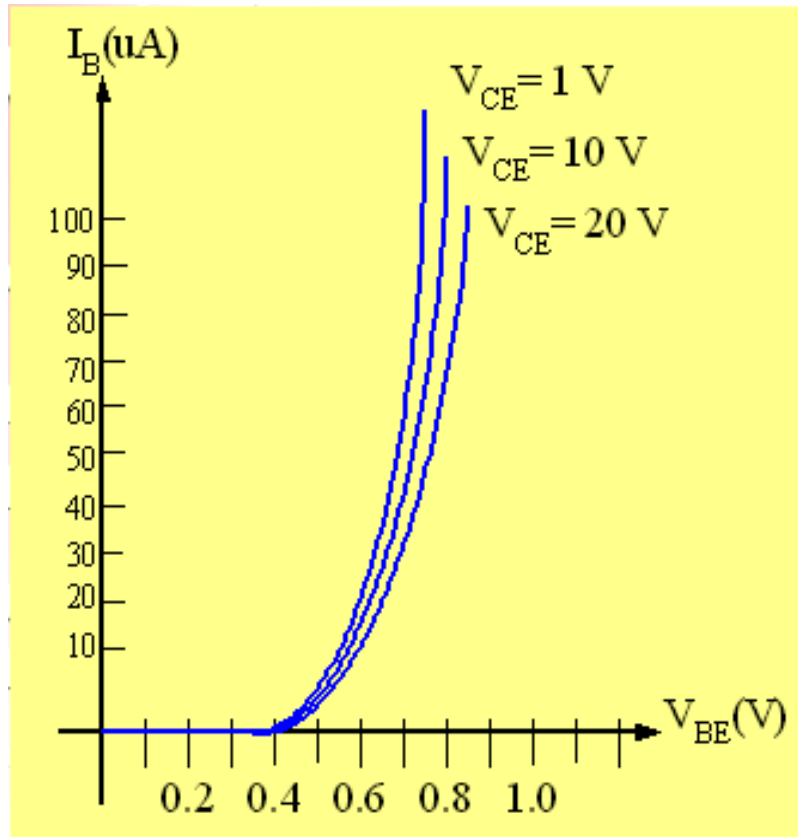


Common-Emitter Configuration

- It is called common-emitter configuration since :
 - emitter is common or reference to both input and output terminals.
 - emitter is usually the terminal closest to or at ground potential.
- Almost amplifier design is using connection of CE due to the high gain for current and voltage.
- Two set of characteristics are necessary to describe the behavior for CE ;input (base terminal) and output (collector terminal) parameters.

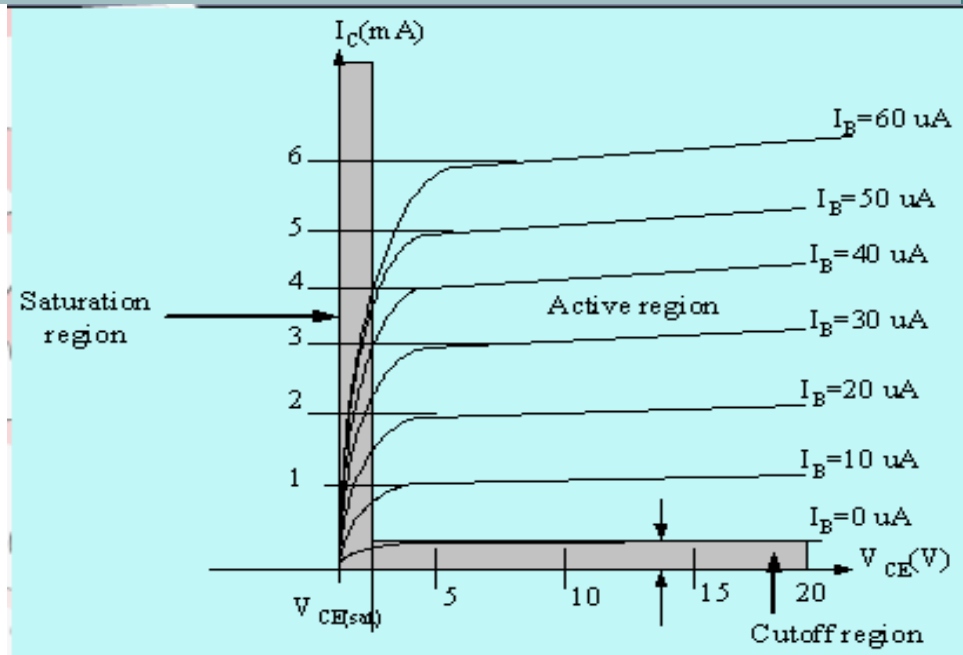
Proper Biasing common-emitter configuration in active region





Input characteristics for a common-emitter NPN transistor

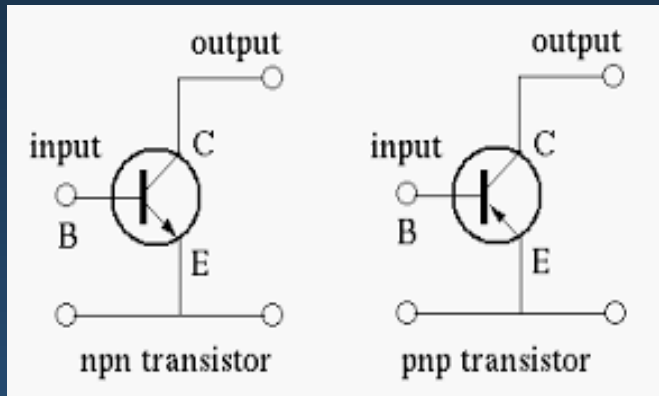
- I_B is microamperes compared to milliamperes of I_C .
- I_B will flow when $V_{BE} > 0.7\text{V}$ for silicon and 0.3V for germanium
- Before this value I_B is very small and no I_B .
- Base-emitter junction is forward bias
- Increasing V_{CE} will reduce I_B for different values.



Output characteristics for a common-emitter npn transistor

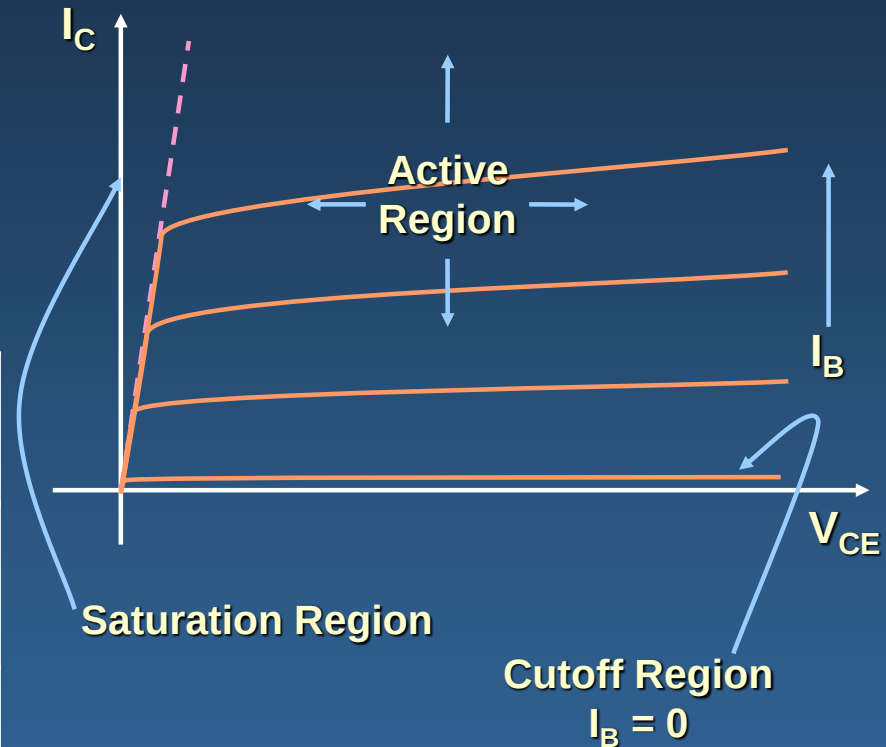
- For small V_{CE} ($V_{CE} < V_{CE(sat)}$, I_C increase linearly with increasing of V_{CE}
- $V_{CE} > V_{CE(sat)}$ I_C not totally depends on $V_{CE} \rightarrow$ constant I_C
- I_B (μA) is very small compare to I_C (mA). Small increase in I_B cause big increase in I_C
- $I_B = 0 A \rightarrow I_{CEO}$ occur.
- Noticing the value when $I_C = 0 A$. There is still some value of current flows.

Common-Emitter

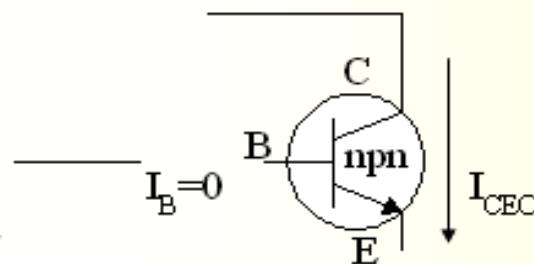


Region of Operation	Description
Active	Small base current controls a large collector current
Saturation	$V_{CE(sat)} \sim 0.2V$, V_{CE} increases with I_C
Cutoff	Achieved by reducing I_B to 0, Ideally, I_C will also equal 0.

Collector-Current Curves



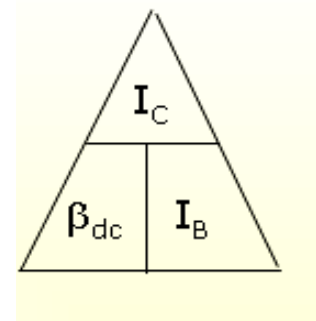
Active region	Saturation region	Cut-off region
• B-E junction is forward bias • C-B junction is reverse bias • can be employed for voltage, current and power amplification	• B-E and C-B junction is forward bias, thus the values of I_B and I_C is too big. • The value of V_{CE} is so small. • Suitable region when the transistor as a logic switch. • NOT and avoid this region when the transistor as an amplifier.	• region below $I_B = 0 \mu A$ is to be avoided if an undistorted o/p signal is required • B-E junction and C-B junction is reverse bias • $I_B = 0$, I_C not zero, during this condition $I_C = I_{CEO}$ where is this current flow when B-E is reverse bias.



Beta (β) or amplification factor

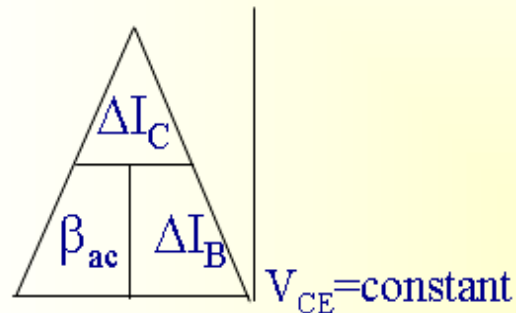
- The ratio of dc collector current (I_C) to the dc base current (I_B) is dc beta (β_{dc}) which is dc current gain where I_C and I_B are determined at a particular operating point, Q-point (quiescent point).
- It's define by the following equation:

$$30 < \beta_{dc} < 300 \rightarrow 2N3904$$



- On data sheet, $\beta_{dc} = h_{FE}$ with h is derived from ac hybrid equivalent cct. FE are derived from forward-current amplification and common-emitter configuration respectively.

- For ac conditions an ac beta has been defined as the changes of collector current (I_C) compared to the changes of base current (I_B) where I_C and I_B are determined at operating point.
- On data sheet, $\beta_{ac} = h_{fe}$
- It can be defined by the following equation:



Relationship analysis between α and β

CASE 1

$$I_E = I_C + I_B \quad (1)$$

substitute equ. $I_C = \beta I_B$ into (1) we get

$$\underline{\underline{I_E = (\beta + 1)I_B}}$$

CASE 2

$$\text{known : } \alpha = \frac{I_C}{I_E} \Rightarrow I_E = \frac{I_C}{\alpha} \quad (2)$$

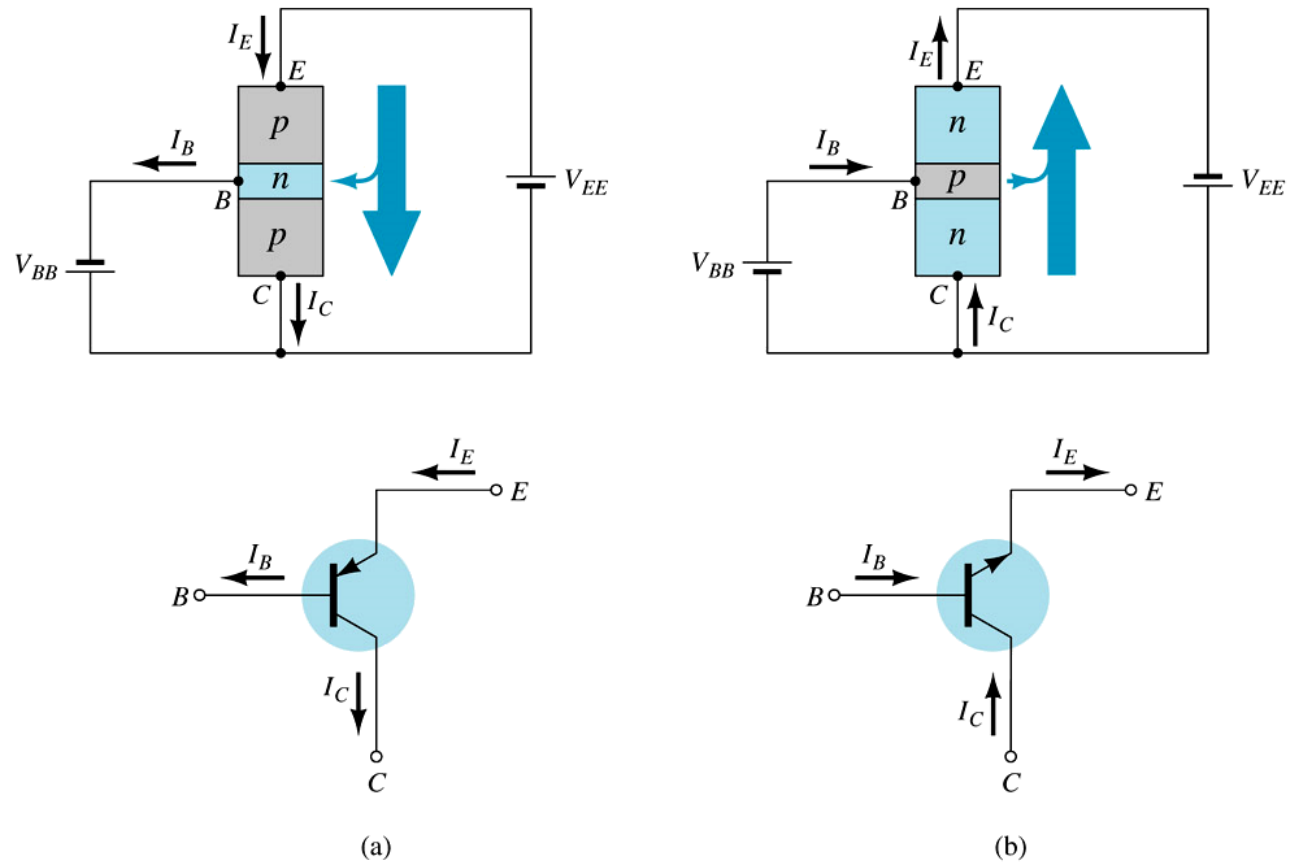
$$\text{known : } \beta = \frac{I_C}{I_B} \Rightarrow I_B = \frac{I_C}{\beta} \quad (3)$$

substitute (2) and (3) into (1) we get,

$$\underline{\underline{\alpha = \frac{\beta}{\beta + 1}}} \quad \text{and} \quad \underline{\underline{\beta = \frac{\alpha}{1 - \alpha}}}$$

Common – Collector Configuration

- Also called emitter-follower (EF).
- It is called common-emitter configuration since both the signal source and the load share the collector terminal as a common connection point.
- The output voltage is obtained at emitter terminal.
- The input characteristic of common-collector configuration is similar with common-emitter. configuration.
- Common-collector circuit configuration is provided with the load resistor connected from emitter to ground.
- It is used primarily for impedance-matching purpose since it has high input impedance and low output impedance.



Notation and symbols used with the common-collector configuration:
 (a) pnp transistor ; (b) npn transistor.

- For the common-collector configuration, the output characteristics are a plot of I_E vs V_{CE} for a range of values of I_B .

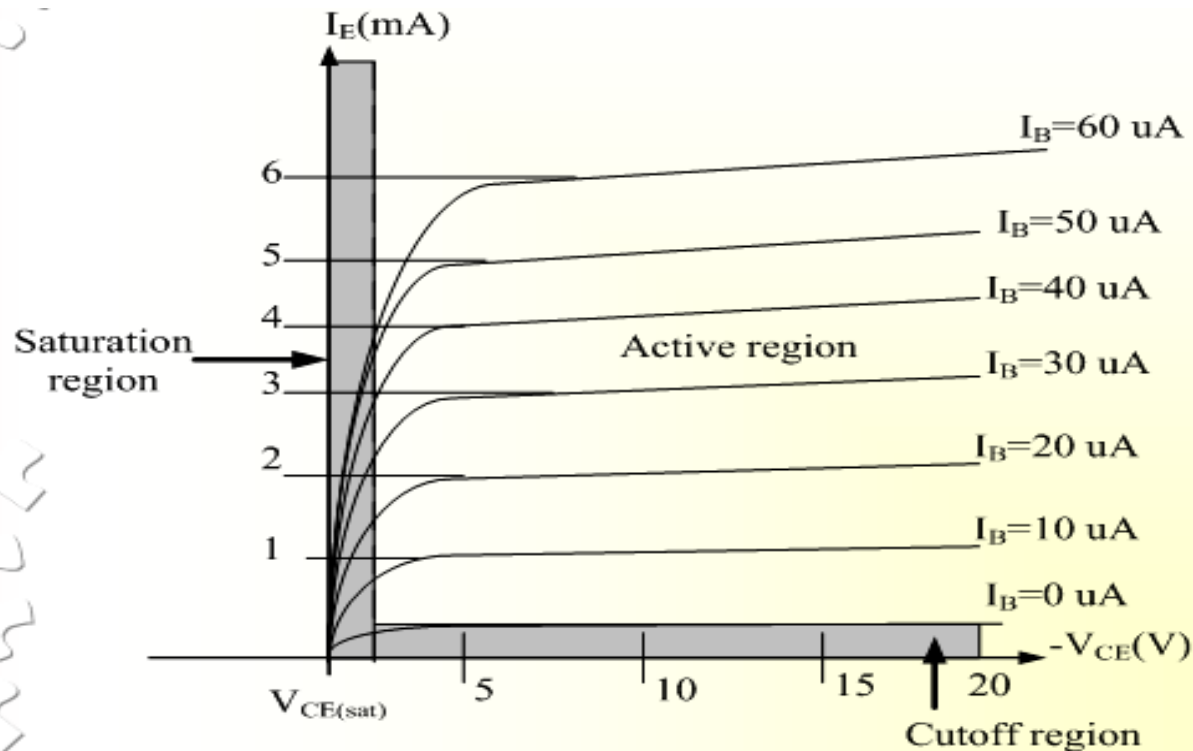
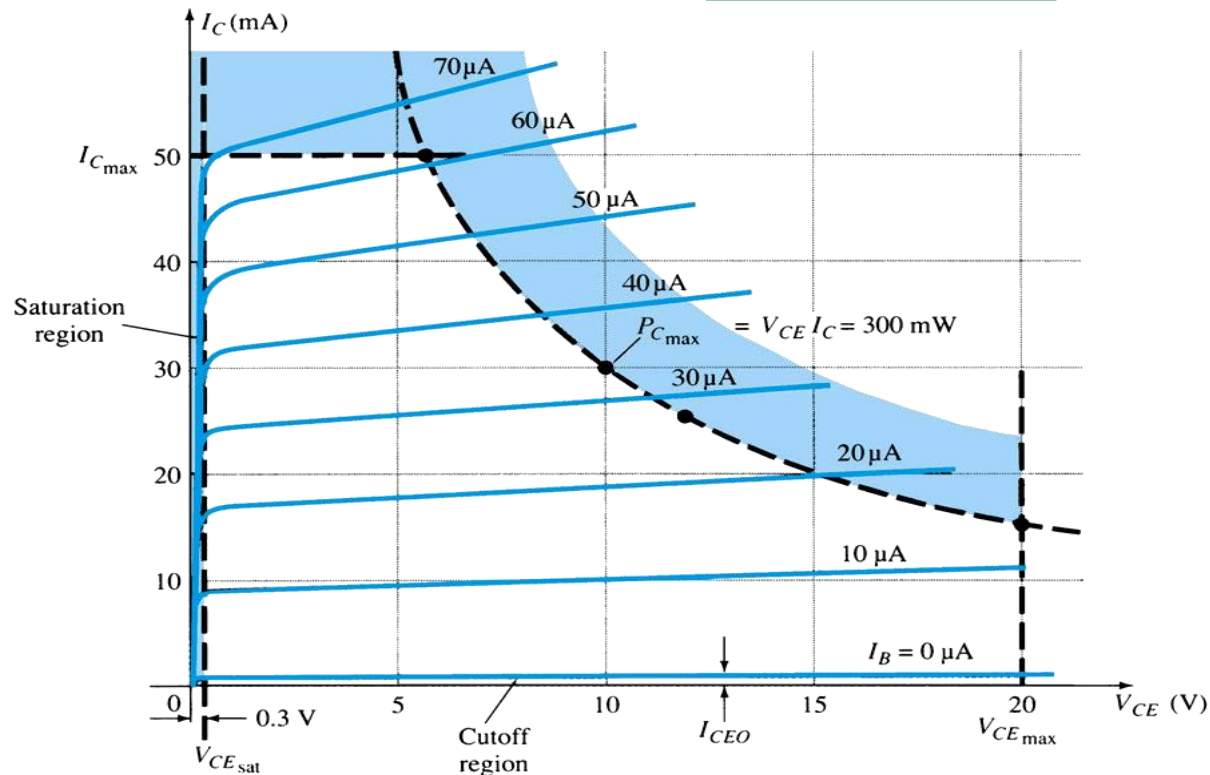


Fig 4.9 : Output characteristic in CC configuration for npn transistor

Limits of Operation

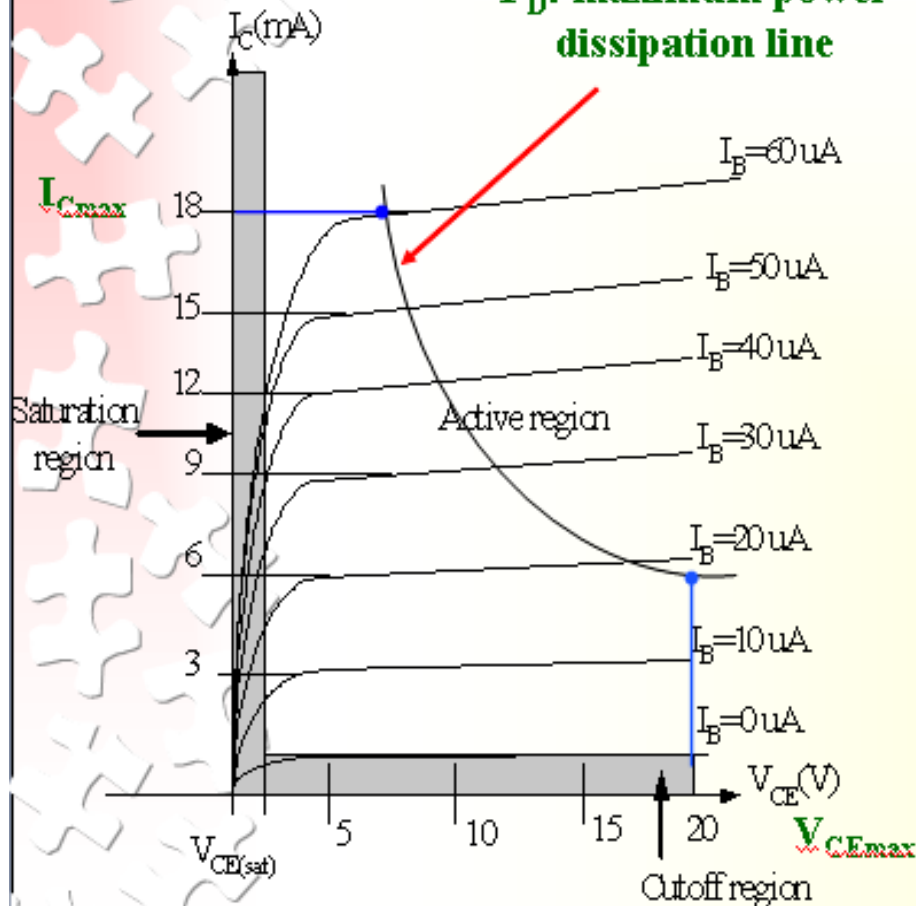
- Many BJT transistor are used as an amplifier. Thus it is important to notice the limits of operations.
- At least 3 maximum values is mentioned in data sheet.
- These are:
 - a) Maximum power dissipation at collector: P_{Cmax} or P_D
 - b) Maximum collector-emitter voltage: V_{CEmax} sometimes named as $V_{BR(CEO)}$ or V_{CEO} .
 - c) Maximum collector current: I_{Cmax}
- There are few rules that need to be followed for BJT transistor used as an amplifier. The rules are:
 - i) transistor need to be operate in active region!
 - ii) $I_C < I_{Cmax}$
 - ii) $P_C < P_{Cmax}$



Note: V_{CE} is at maximum and I_C is at minimum ($I_{Cmax}=I_{CEO}$) in the cutoff region. I_C is at maximum and V_{CE} is at minimum ($V_{CE\ max} = V_{CEsat} = V_{CEO}$) in the saturation region. The transistor operates in the active region between saturation and cutoff.

Example 1:

P_D : maximum power dissipation line



Refer to the fig.

Step 1:

The maximum collector power dissipation,

$$P_D = I_{Cmax} \times V_{CEmax} \quad (1)$$

$$= 18\text{m} \times 20 = 360 \text{ mW}$$

Step 2:

At any point on the characteristics the product of and must be equal to 360 mW.

Ex. 1. If choose $I_{Cmax} = 5 \text{ mA}$, substitute into the (1), we get

$$V_{CEmax} I_{Cmax} = 360 \text{ mW}$$

$$V_{CEmax} (5 \text{ m}) = 360 / 5 = \underline{7.2 \text{ V}}$$

Ex.2. If choose $V_{CEmax} = 18 \text{ V}$, substitute into (1), we get

$$V_{CEmax} I_{Cmax} = 360 \text{ mW}$$

$$(10) I_{Cmax} = 360\text{m} / 18 = \underline{20 \text{ mA}}$$

Derating P_{Dmax}

- P_{Dmax} is usually specified at 25°C.
- The higher temperature goes, the less is P_{Dmax}
- Example;
 - A derating factor of 2mW/°C indicates the power dissipation is reduced 2mW each degree centigrade increase of temperature.

Transistor Specification Sheet

MAXIMUM RATINGS

Rating	Symbol	2N4123	Unit
Collector-Emitter Voltage	V_{CE0}	30	Vdc
Collector-Base Voltage	V_{CB0}	40	Vdc
Emitter-Base Voltage	V_{EB0}	5.0	Vdc
Collector Current – Continuous	I_C	200	mAdc
Total Device Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	625 5.0	mW mW/°C
Operating and Storage Junction Temperature Range	T_j, T_{stg}	-55 to +150	°C

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction to Case	$R_{\theta JC}$	83.3	°C/W
Thermal Resistance, Junction to Ambient	$R_{\theta JA}$	200	°C/W

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

Characteristic	Symbol	Min	Max	Unit
----------------	--------	-----	-----	------

OFF CHARACTERISTICS

Collector-Emitter Breakdown Voltage (1) ($I_C = 1.0\text{ mAdc}$, $I_E = 0$)	$V_{(BR)CEO}$	30		Vdc
Collector-Base Breakdown Voltage ($I_C = 10\text{ }\mu\text{Adc}$, $I_E = 0$)	$V_{(BR)CBO}$	40		Vdc
Emitter-Base Breakdown Voltage ($I_E = 10\text{ }\mu\text{Adc}$, $I_C = 0$)	$V_{(BR)EBO}$	5.0	–	Vdc
Collector Cutoff Current ($V_{CB} = 20\text{ Vdc}$, $I_E = 0$)	I_{CBO}	–	50	nAdc
Emitter Cutoff Current ($V_{BE} = 3.0\text{ Vdc}$, $I_C = 0$)	I_{EBO}	–	50	nAdc

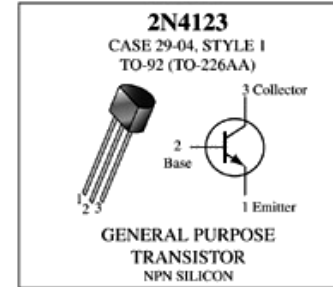
ON CHARACTERISTICS

DC Current Gain (1) ($I_C = 2.0\text{ mAdc}$, $V_{CE} = 1.0\text{ Vdc}$) ($I_C = 50\text{ mAdc}$, $V_{CE} = 1.0\text{ Vdc}$)	h_{FE}	50 25	150 –	–
Collector-Emitter Saturation Voltage(1) ($I_C = 50\text{ mAdc}$, $I_B = 5.0\text{ mAdc}$)	$V_{CE(sat)}$	–	0.3	Vdc
Base-Emitter Saturation Voltage(1) ($I_C = 50\text{ mAdc}$, $I_B = 5.0\text{ mAdc}$)	$V_{BE(sat)}$	–	0.95	Vdc

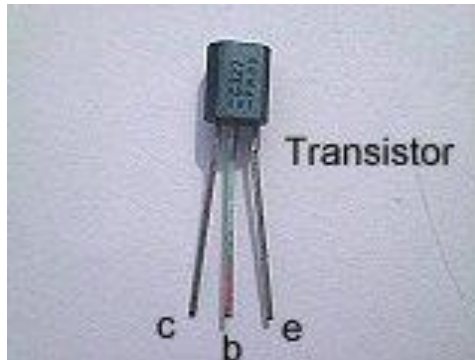
SMALL-SIGNAL CHARACTERISTICS

Current-Gain – Bandwidth Product ($I_C = 10\text{ mAdc}$, $V_{CE} = 20\text{ Vdc}$, $f = 100\text{ MHz}$)	f_T	250		MHz
Output Capacitance ($V_{CB} = 5.0\text{ Vdc}$, $I_E = 0$, $f = 100\text{ MHz}$)	C_{obo}	–	4.0	pF
Input Capacitance ($V_{BE} = 0.5\text{ Vdc}$, $I_C = 0$, $f = 100\text{ kHz}$)	C_{ibo}	–	8.0	pF
Collector-Base Capacitance ($I_E = 0$, $V_{CB} = 5.0\text{ V}$, $f = 100\text{ kHz}$)	C_{cb}	–	4.0	pF
Small-Signal Current Gain ($I_C = 2.0\text{ mAdc}$, $V_{CE} = 10\text{ Vdc}$, $f = 1.0\text{ kHz}$)	h_{ie}	50	200	–
Current Gain – High Frequency ($I_C = 10\text{ mAdc}$, $V_{CE} = 20\text{ Vdc}$, $f = 100\text{ MHz}$) ($I_C = 2.0\text{ mAdc}$, $V_{CE} = 10\text{ V}$, $f = 1.0\text{ kHz}$)	h_{fo}	2.5 50	– 200	–
Noise Figure ($I_C = 100\text{ }\mu\text{Adc}$, $V_{CE} = 5.0\text{ Vdc}$, $R_S = 1.0\text{ k ohm}$, $f = 1.0\text{ kHz}$)	NF	–	6.0	dB

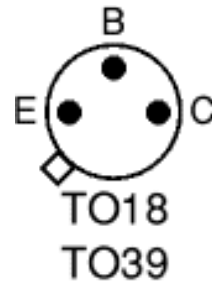
(1) Pulse Test: Pulse Width = 300 μs . Duty Cycle = 2.0%



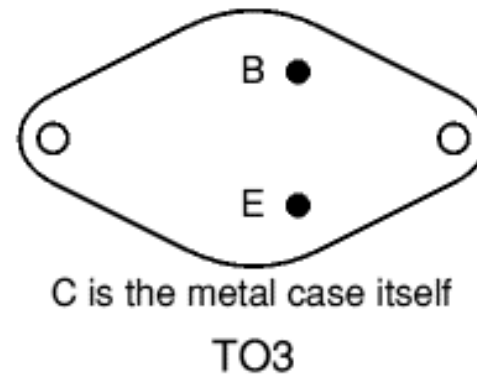
BJTs - Practical Aspects



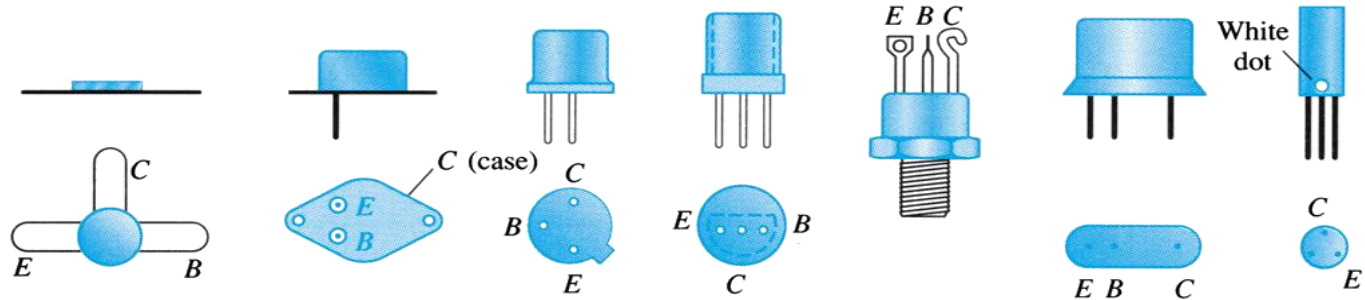
Heat sink



Views are from below with the leads towards you.



Transistor Terminal Identification



Transistor Testing

1. Curve Tracer

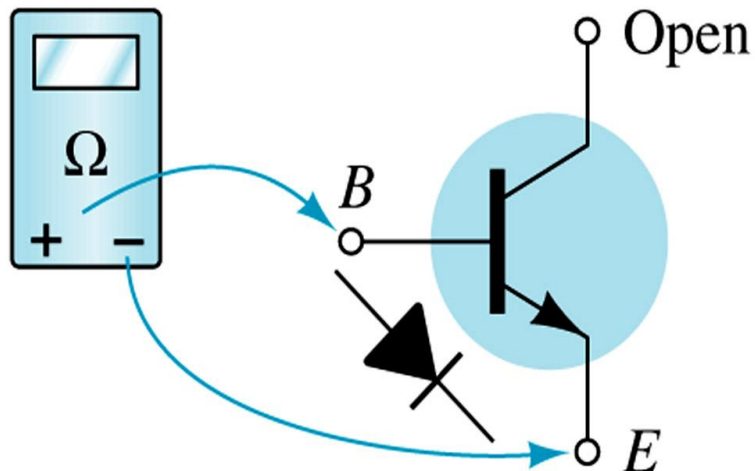
Provides a graph of the characteristic curves.

2. DMM

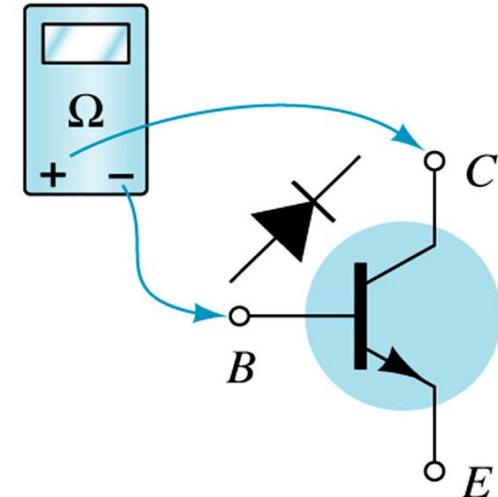
Some DMM's will measure β_{DC} or HFE.

3. Ohmmeter

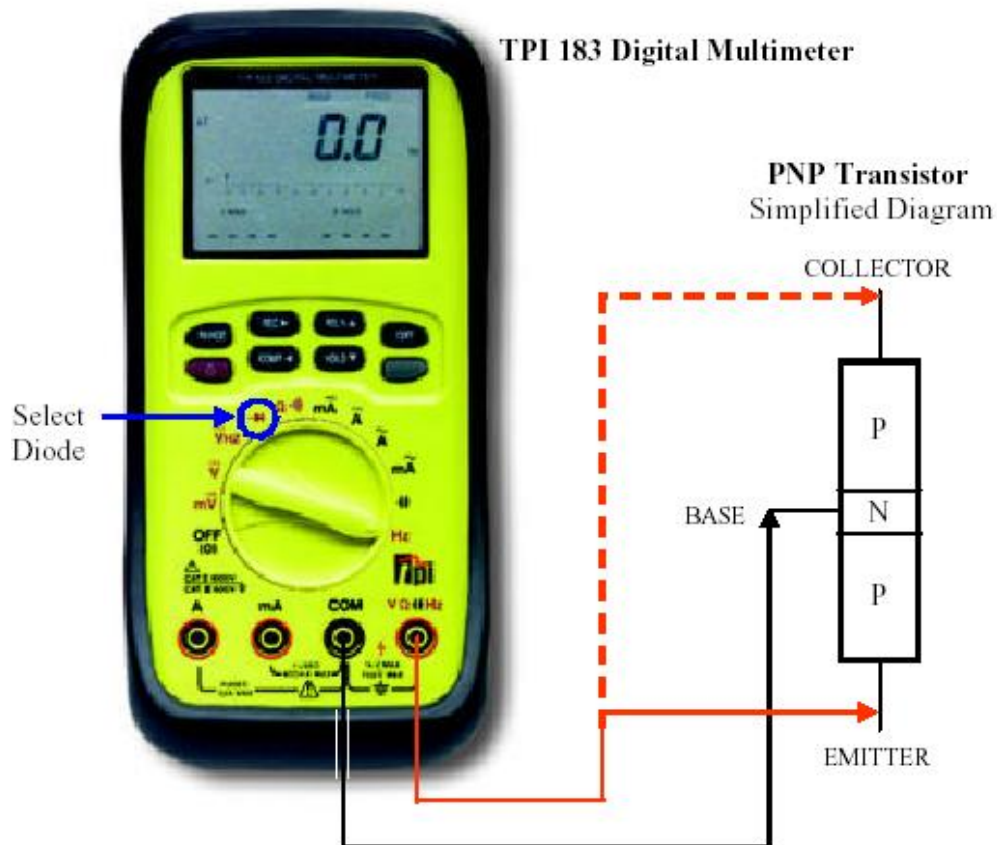
Low R



High R



BJTs - Testing



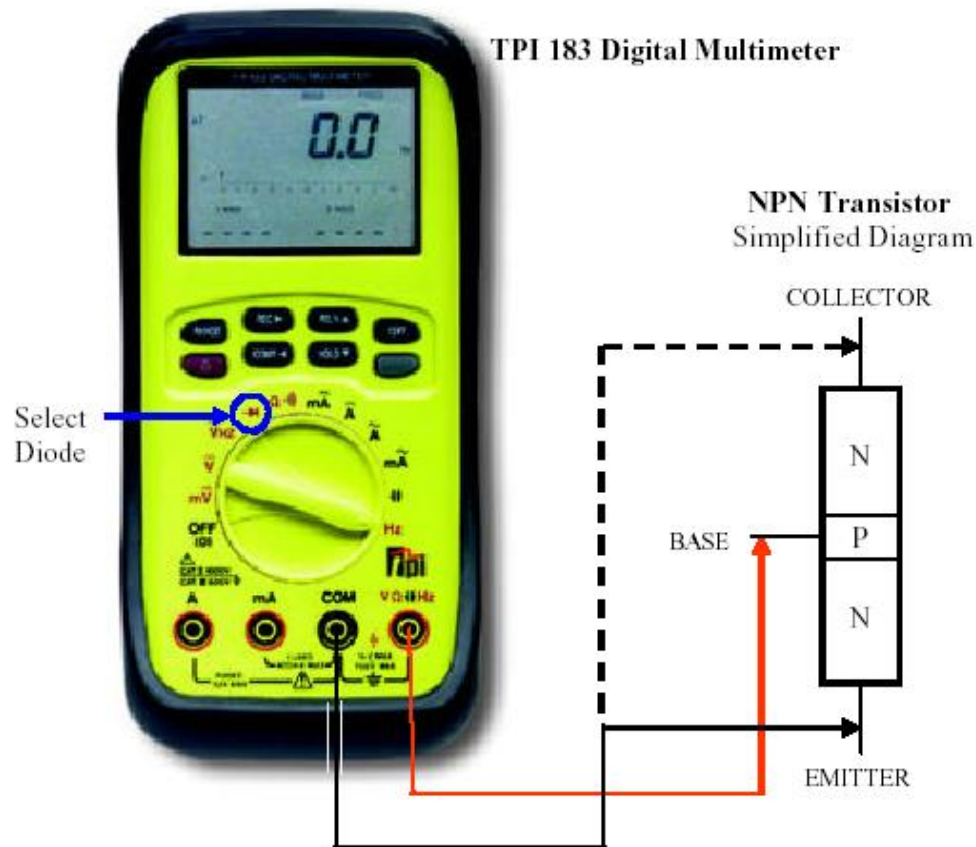
PNP Test Procedure

Connect the meter leads with the polarity as shown and verify that the base-to-emitter and base-to-collector junctions read as a forward biased diode: 0.5 to 0.8 VDC.

Reverse the meter connections to the transistor and verify that both PN junctions do not conduct. Meter should indicate an open circuit. (Display = OUCH or OL.)

Finally read the resistance from emitter to collector and verify an open circuit reading in both directions. (Note: A short can exist from emitter to collector even if the individual PN junctions test properly.)

BJTs - Testing



PNP Test Procedure

Connect the meter leads with the polarity as shown and verify that the base-to-emitter and base-to-collector junctions read as a forward biased diode: 0.5 to 0.8 VDC.

Reverse the meter connections to the transistor and verify that both PN junctions do not conduct. Meter should indicate an open circuit. (Display = OUCH or OL.)

Finally read the resistance from emitter to collector and verify an open circuit reading in both directions. (Note: A short can exist from emitter to collector even if the individual PN junctions test properly.)

...that's all folks...
...thanks for your time...



Transistor DC Biasing Circuits

Pictures are redrawn (with some modifications) from
Introductory Electronic Devices and Circuits

Objectives

- State the purpose of dc biasing circuits.
- Plot the dc load line given the value of V_{CC} and the total collector-emitter circuit resistance.
- Describe the Q-point of an amplifier.
- Describe and analyze the operations of various bias circuits:
 - base-bias circuits
 - voltage-divider bias circuits
 - emitter-bias circuits
 - collector-feedback bias circuits
 - emitter-feedback bias circuits

Transistor Biasing

The basic function of transistor is amplification. The process of raising the strength of weak signal without any change in its general shape is referred as faithful amplification. For faithful amplification it is essential that:-

1. Emitter-Base junction is forward biased
2. Collector- Base junction is reversed biased
3. Proper zero signal collector current

The proper flow of zero signal collector current and the maintenance of proper collector emitter voltage during the passage of signal is called transistor biasing.

WHY BIASING?

If the transistor is not biased properly, it would work inefficiently and produce distortion in output signal.

HOW A TRANSISTOR CAN BE BIASSED?

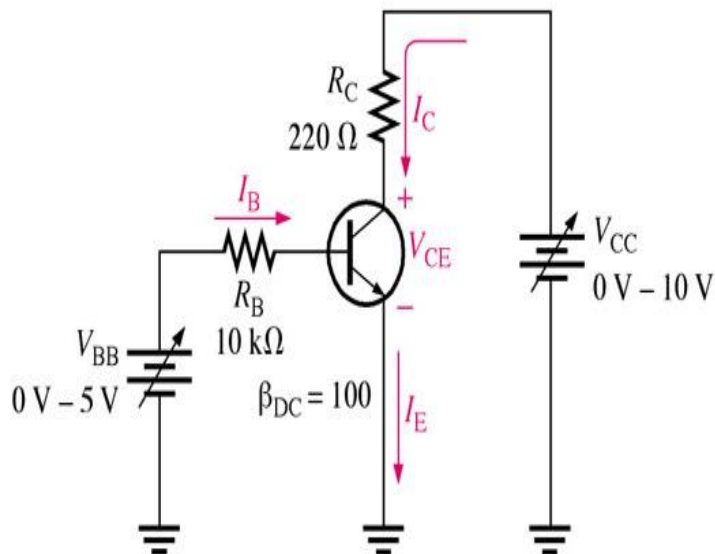
A transistor is biased either with the help of battery or associating a circuit with the transistor. The later method is more efficient and is frequently used. The circuit used for transistor biasing is called the biasing circuit.

BIAS STABILITY

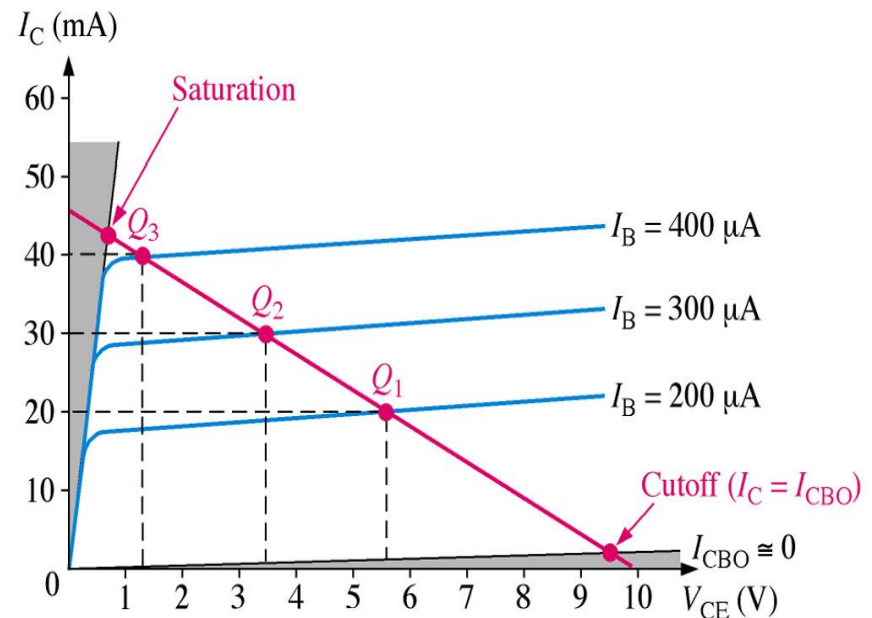
- ❖ Through proper biasing, a desired quiescent operating point of the transistor amplifier in the active region (linear region) of the characteristics is obtained. It is desired that once selected the operating point should remain stable. The maintenance of operating point stable is called Stabilisation.
- ❖ The selection of a proper quiescent point generally depends on the following factors:
 - (a) The amplitude of the signal to be handled by the amplifier and distortion level in signal
 - (b) The load to which the amplifier is to work for a corresponding supply voltage
- ❖ The operating point of a transistor amplifier shifts mainly with changes in temperature, since the transistor parameters — β , I_{CO} and V_{BE} (where the symbols carry their usual meaning)—are functions of temperature.

The DC Operating Point

For a transistor circuit to amplify it must be properly biased with dc voltages. The dc operating point between saturation and cutoff is called the **Q-point**. The goal is to set the Q-point such that that it does not go into saturation or cutoff when an ac signal is applied.



(a) DC biased circuit



Requirements of biasing network

- Ensuring proper zero signal collector current.
- Ensuring V_{CE} not falling below 0.5V for Ge transistor and 1V for Silicon transistor at any instant.
- Ensuring Stabilization of operating point. (zero signal I_C and V_{CE})

The Thermal Stability of Operating Point ($S_{I_{CO}}$)

❖ **Stability Factor S** :- The stability factor S , as the change of collector current with respect to the reverse saturation current, keeping β and V_{BE} constant. This can be written as:

The Thermal Stability Factor : $S_{I_{CO}}$

$$S_{I_{CO}} = \left. \frac{\partial I_C}{\partial I_{CO}} \right|_{V_{BE}, \beta}$$

This equation signifies that I_C Changes $S_{I_{CO}}$ times as fast as I_{CO}

Differentiating the equation of Collector Current $I_C = (1+\beta)I_{CO} + \beta I_B$ & rearranging the terms we can write

$$S_{I_{CO}} = \frac{1+\beta}{1 - \beta (\partial I_B / \partial I_C)}$$

It may be noted that Lower is the value of $S_{I_{CO}}$ better is the stability

Various Biasing Circuits

- **Fixed Bias Circuit**
- **Fixed Bias with Emitter Resistor**
- **Collector to Base Bias Circuit**
- **Potential Divider Bias Circuit**

Fig 7.1 Typical amplifier operation.

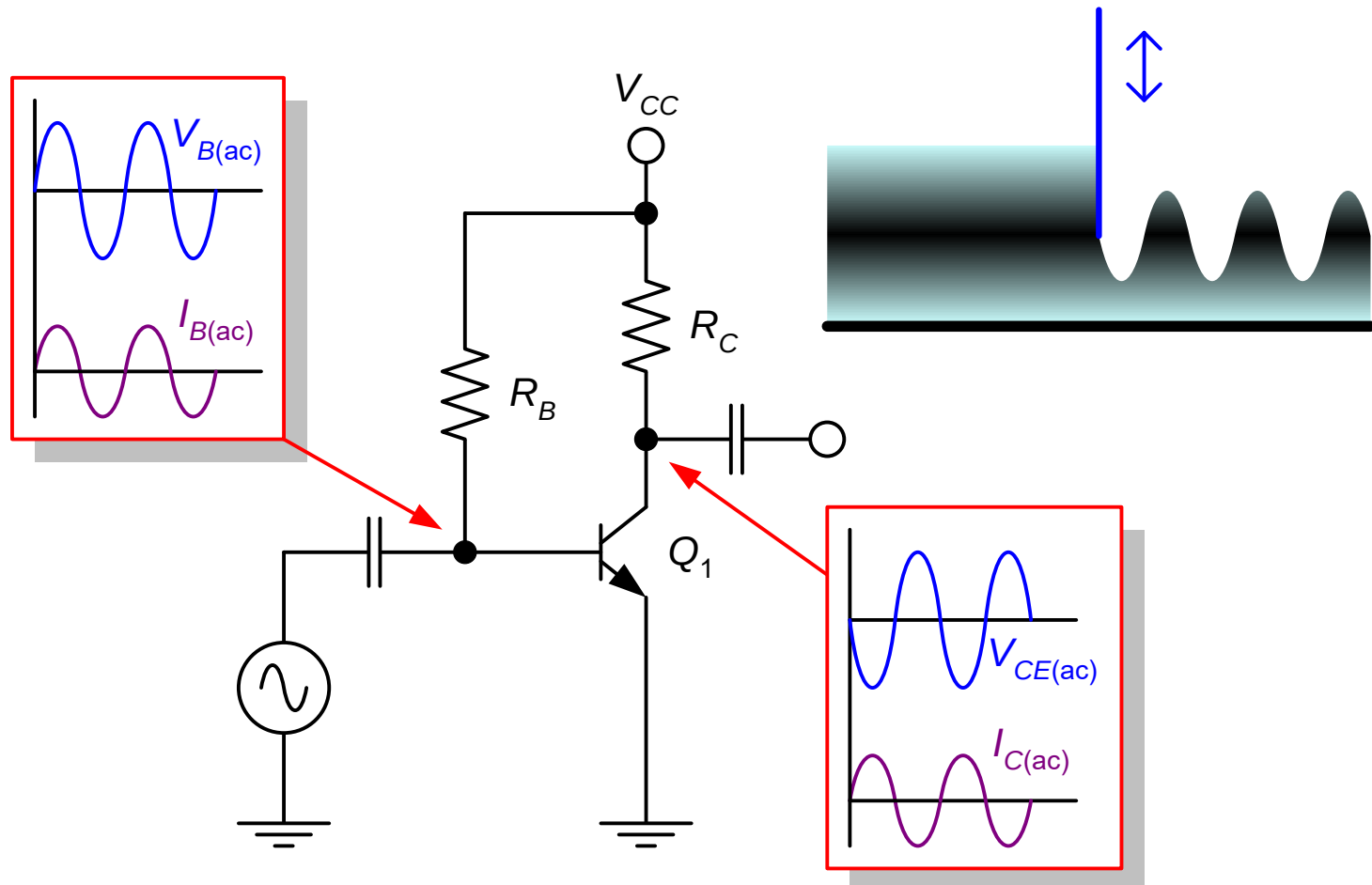


Fig 7.2 A generic dc load line.

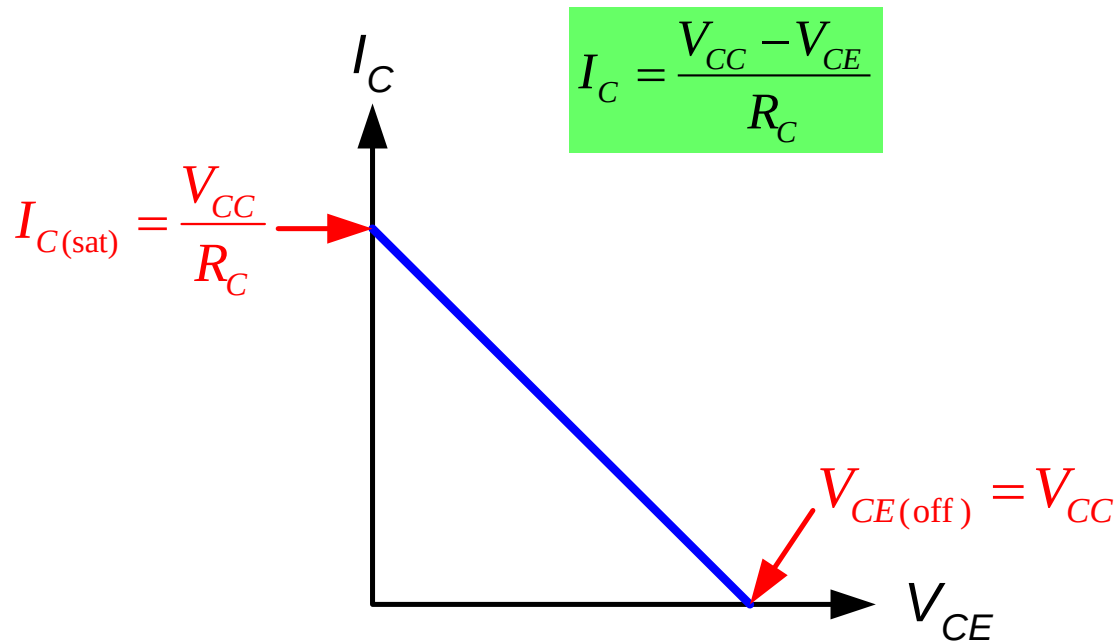


Fig 7.3 Example 7.1.

Plot the dc load line for the circuit shown in Fig. 7.3a.

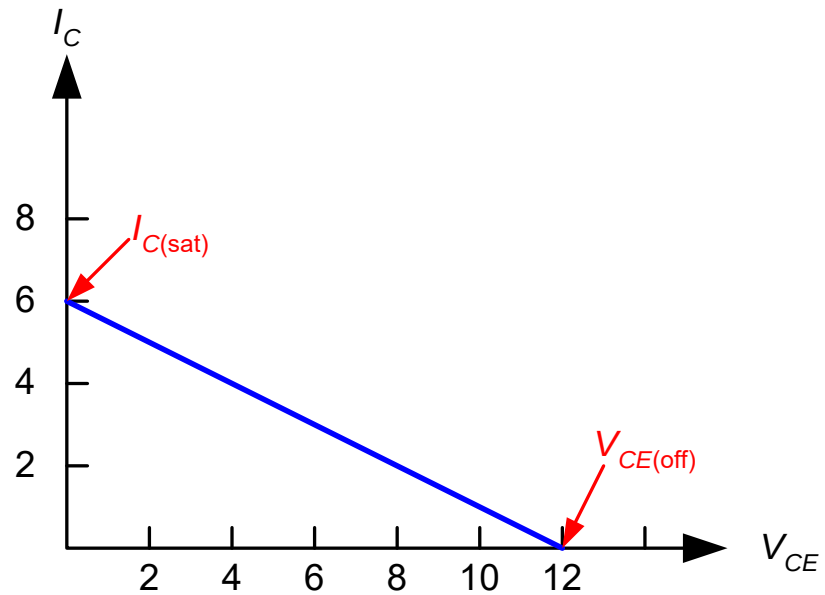
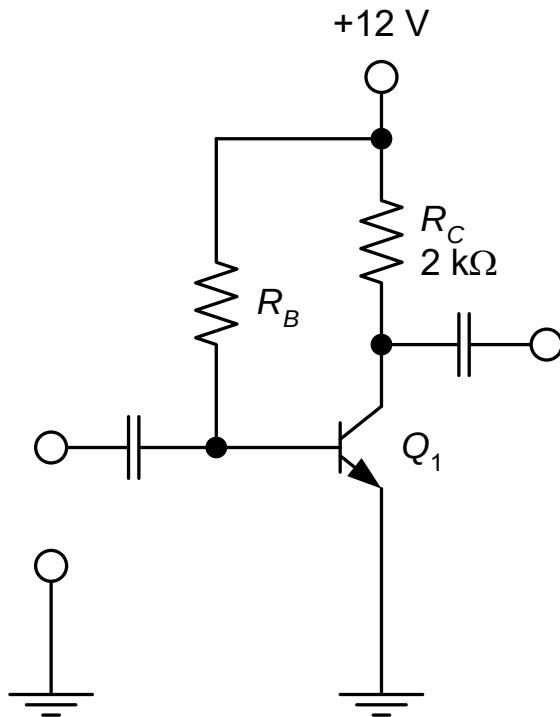
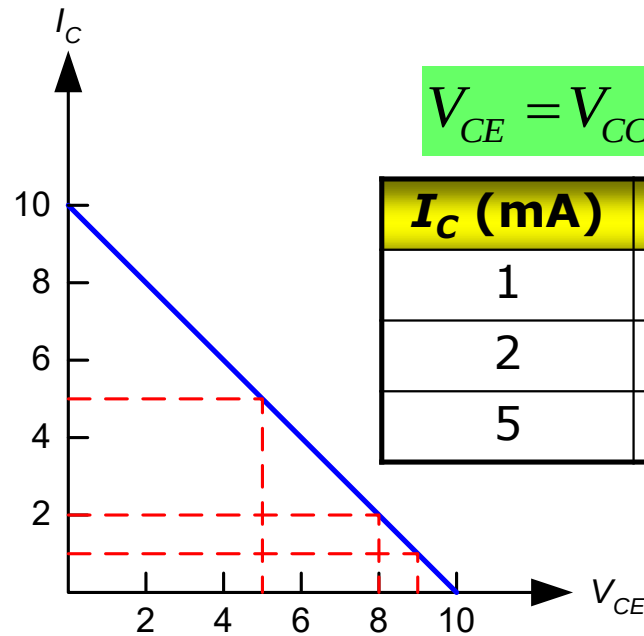
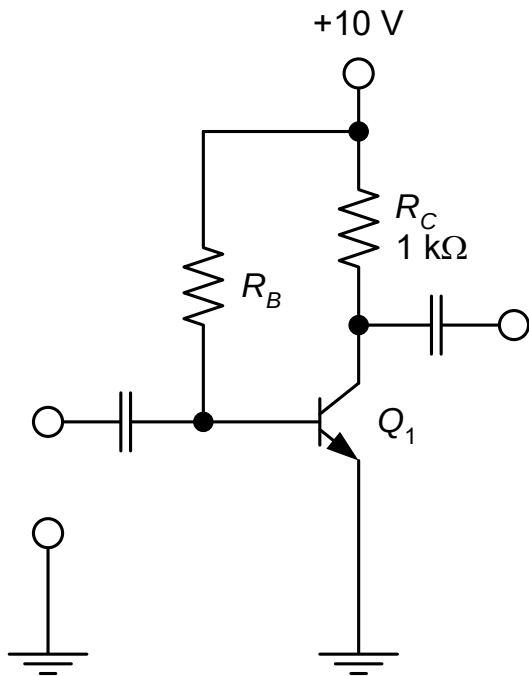


Fig 7.4 Example 7.2.

Plot the dc load line for the circuit shown in Fig. 7.4. Then, find the values of V_{CE} for $I_C = 1, 2, 5$ mA respectively.



$$V_{CE} = V_{CC} - I_C R_C$$

I_C (mA)	V_{CE} (V)
1	9
2	8
5	5

Fig 7.6-8 Optimum Q-point with amplifier operation.

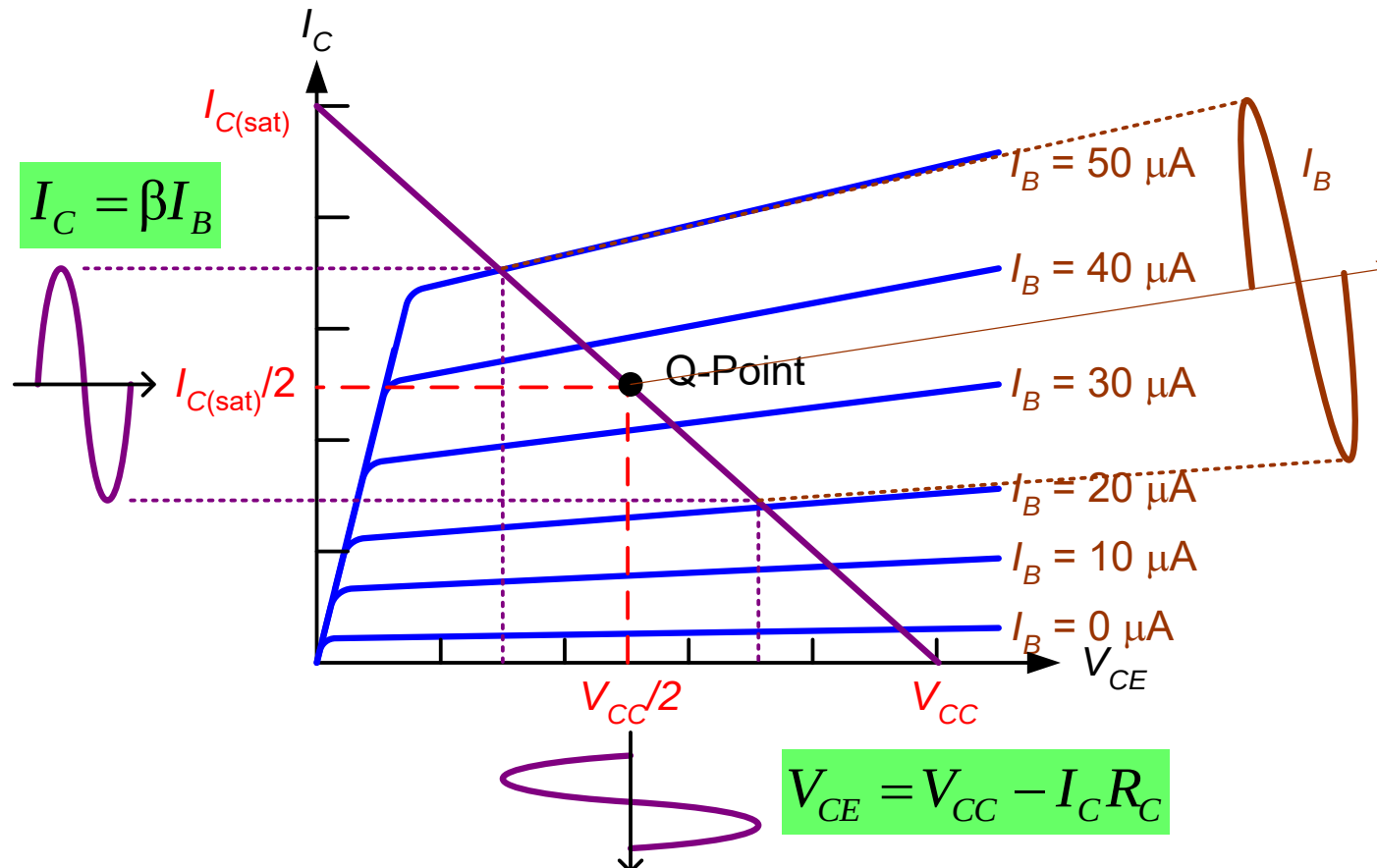
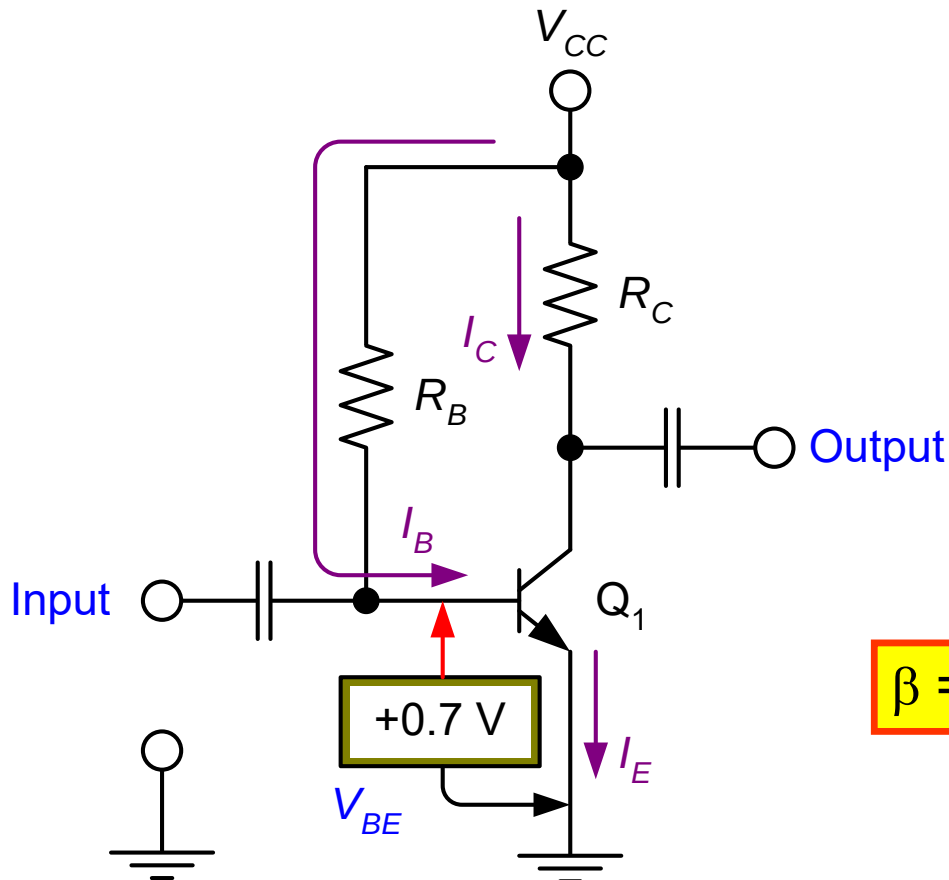


Fig 7.9 Base bias (fixed bias).



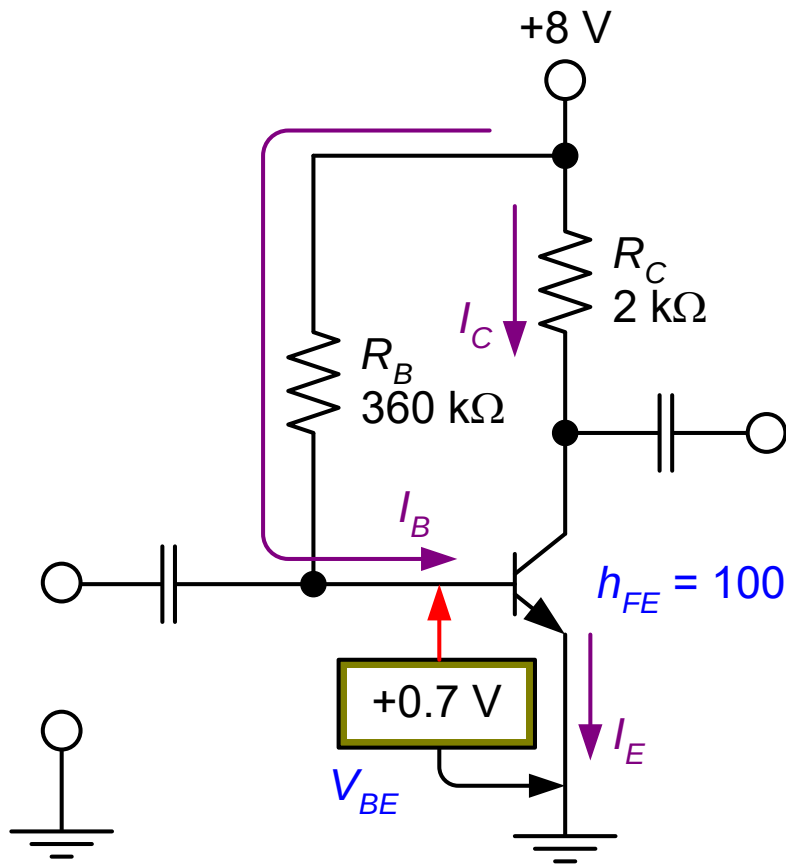
$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$I_C = \beta I_B$$

$$V_{CE} = V_{CC} - I_C R_C$$

$$\beta = \text{dc current gain} = h_{FE}$$

Fig 7.10 Example 7.3.



$$I_B = \frac{V_{CC} - 0.7\text{V}}{R_B} = \frac{8\text{V} - 0.7\text{V}}{360\text{k}\Omega} = 20.28\mu\text{A}$$

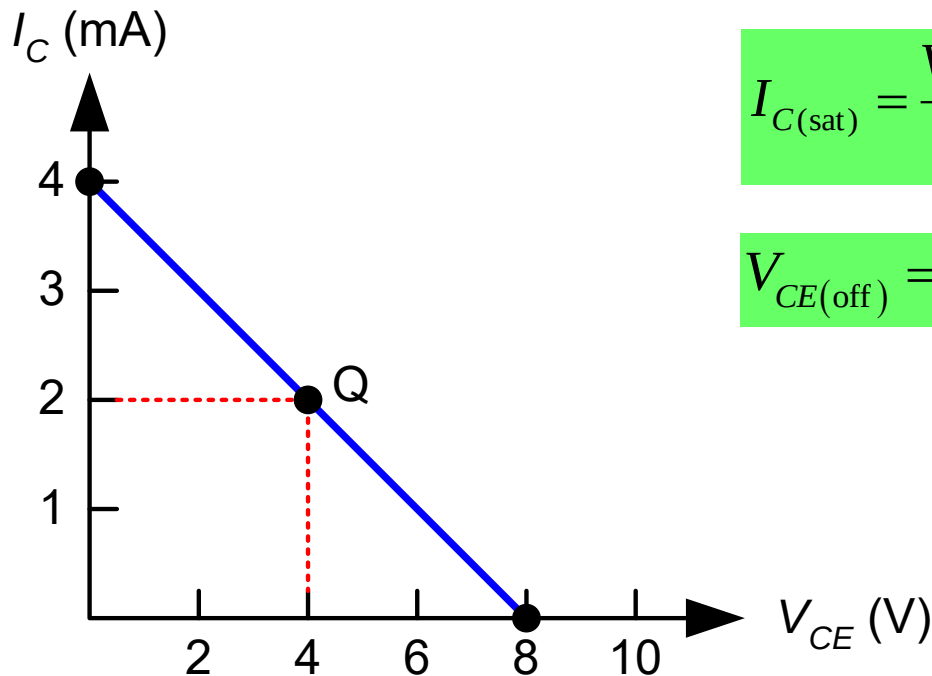
$$I_C = h_{FE} I_B = (100)(20.28\mu\text{A}) = 2.028\text{mA}$$

$$V_{CE} = V_{CC} - I_C R_C = 8\text{V} - (2.028\text{mA})(2\text{k}\Omega) = 3.94\text{V}$$

The circuit is midpoint biased.

Fig 7.11 Example 7.4.

Construct the dc load line for the circuit shown in Fig. 7.10, and plot the Q-point from the values obtained in Example 7.3. Determine whether the circuit is midpoint biased.

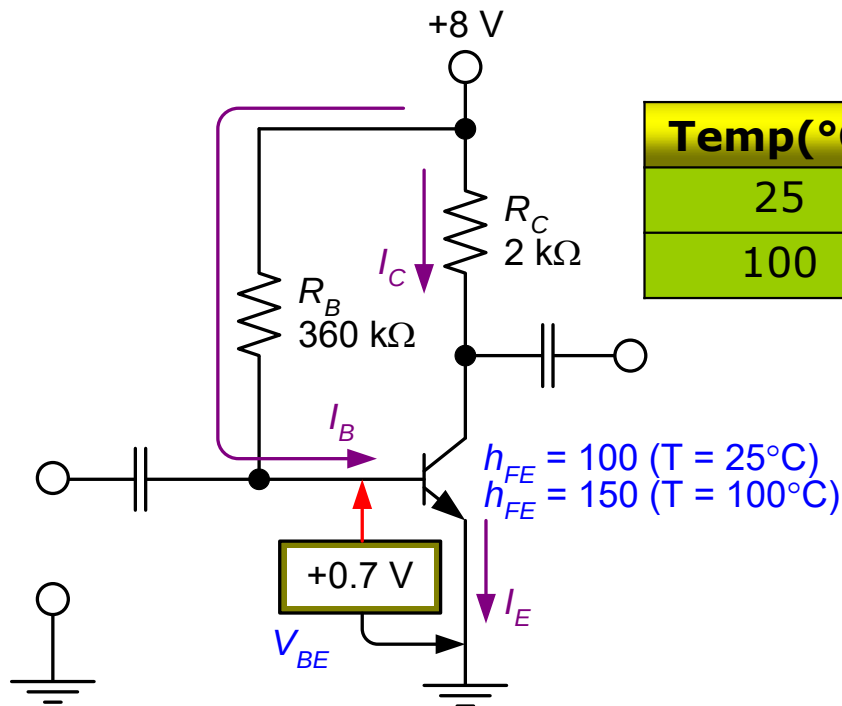


$$I_{C(\text{sat})} = \frac{V_{CC}}{R_C} = \frac{8\text{V}}{2\text{k}\Omega} = 4\text{mA}$$

$$V_{CE(\text{off})} = V_{CC} = 8\text{V}$$

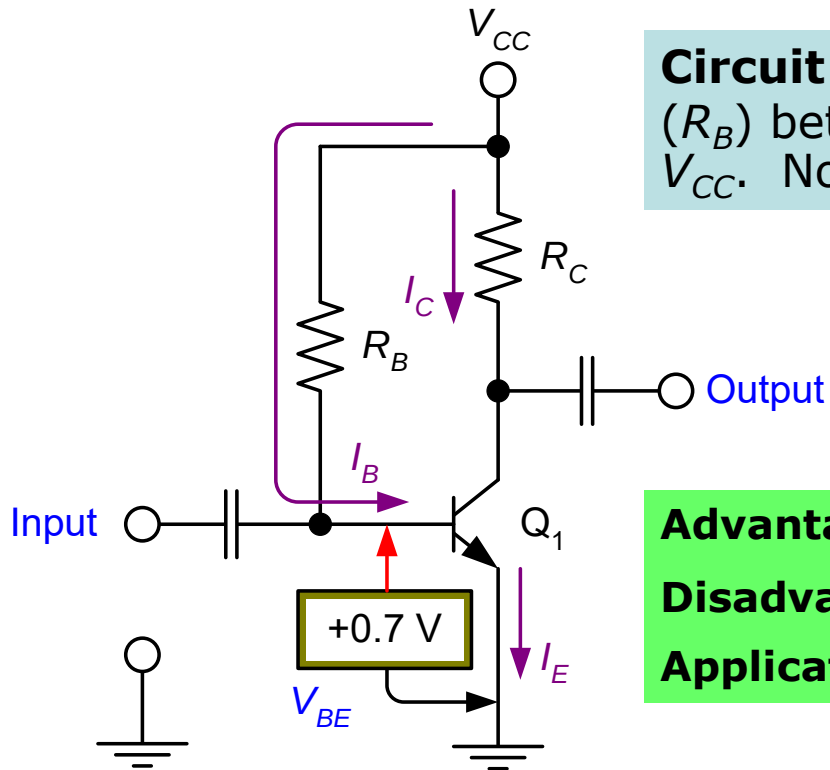
Fig 7.12 Example 7.6. (Q-point shift.)

The transistor in Fig. 7.12 has values of $h_{FE} = 100$ when $T = 25^\circ\text{C}$ and $h_{FE} = 150$ when $T = 100^\circ\text{C}$. Determine the Q-point values of I_C and V_{CE} at both of these temperatures.



Temp($^\circ\text{C}$)	I_B (μA)	I_C (mA)	V_{CE} (V)
25	20.28	2.028	3.94
100	20.28	3.04	1.92

Fig 7.13 Base bias characteristics. (1)



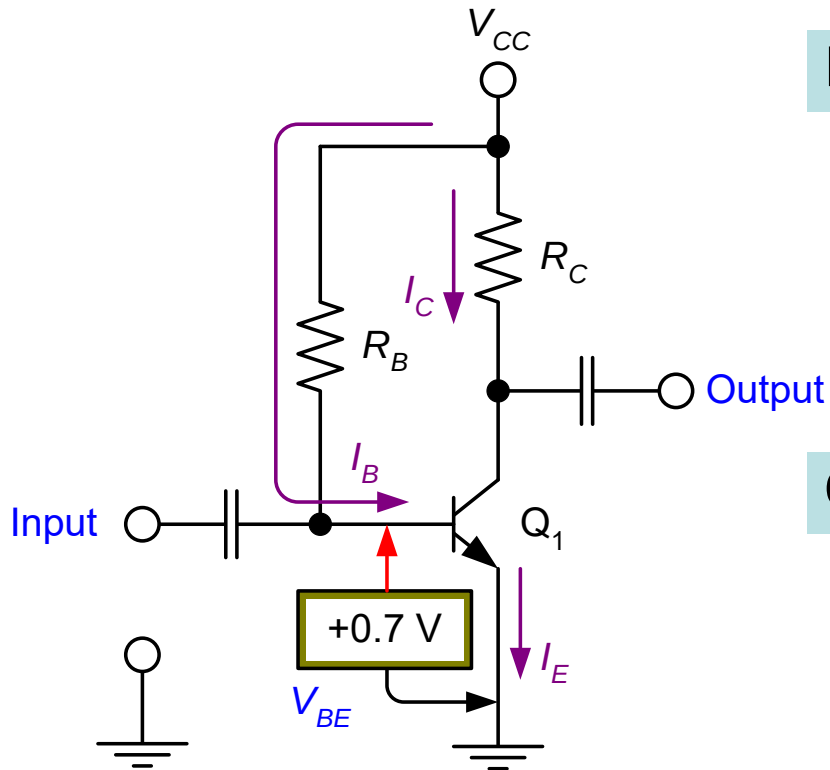
Circuit recognition: A single resistor (R_B) between the base terminal and V_{CC} . No emitter resistor.

Advantage: Circuit simplicity.

Disadvantage: Q-point shift with temp.

Applications: Switching circuits only.

Fig 7.13 Base bias characteristics. (2)



Load line equations:

$$I_{C(\text{sat})} \cong \frac{V_{CC}}{R_C}$$

$$V_{CE(\text{off})} = V_{CC}$$

Q-point equations:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$I_C = h_{FE} I_B$$

$$V_{CE} = V_{CC} - I_C R_C$$

Merits:

- It is simple to shift the operating point anywhere in the active region by merely changing the base resistor (R_B).
- A very small number of components are required.

Demerits:

- The collector current does not remain constant with variation in temperature or power supply voltage. Therefore the operating point is unstable.
- When the transistor is replaced with another one, considerable change in the value of β can be expected. Due to this change the operating point will shift.
- For small-signal transistors (e.g., not power transistors) with relatively high values of β (i.e., between 100 and 200), this configuration will be prone to thermal runaway. In particular, the stability factor, which is a measure of the change in collector current with changes in reverse saturation current, is approximately $\beta+1$. To ensure absolute stability of the amplifier, a stability factor of less than 25 is preferred, and so small-signal transistors have large stability factors.

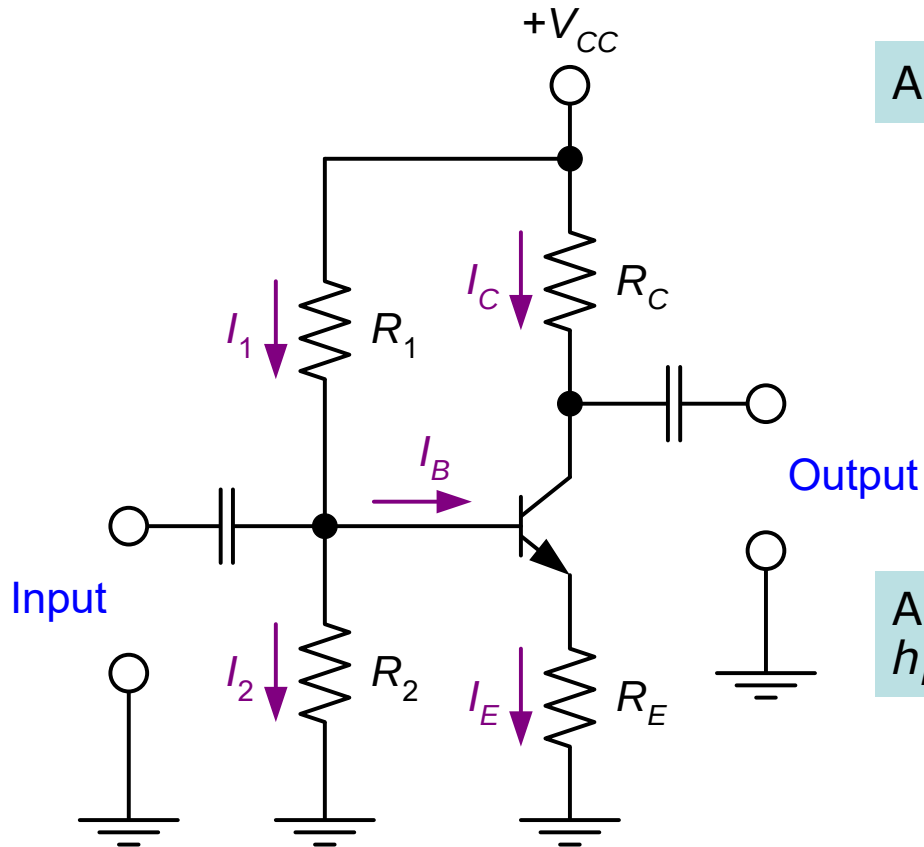
Usage:

- Due to the above inherent drawbacks, fixed bias is rarely used in linear circuits (i.e., those circuits which use the transistor as a current source). Instead, it is often used in circuits where transistor is used as a switch. However, one application of fixed bias is to achieve crude automatic gain control in the transistor by feeding the base resistor from a DC signal derived from the AC output of a later stage.

The Potential Divider Bias Circuit

This is the most commonly used arrangement for biasing as it provides good bias stability. In this arrangement the emitter resistance ' R_E ' provides stabilization. The resistance ' R_E ' causes a voltage drop in a direction so as to reverse bias the emitter junction. Since the emitter-base junction is to be forward biased, the base voltage is obtained from R_1 - R_2 network. The net forward bias across the emitter base junction is equal to V_B - dc voltage drop across ' R_E '. The base voltage is set by V_{CC} and R_1 and R_2 . The dc bias circuit is independent of transistor current gain. In case of amplifier, to avoid the loss of ac signal, a capacitor of large capacitance is connected across R_E . The capacitor offers a very small reactance to ac signal and so it passes through the condenser.

Fig 7.14 Voltage divider bias. (1)



Assume that $I_2 > 10I_B$.

$$V_B = \frac{R_2}{R_1 + R_2} V_{CC}$$

$$V_E = V_B - 0.7V$$

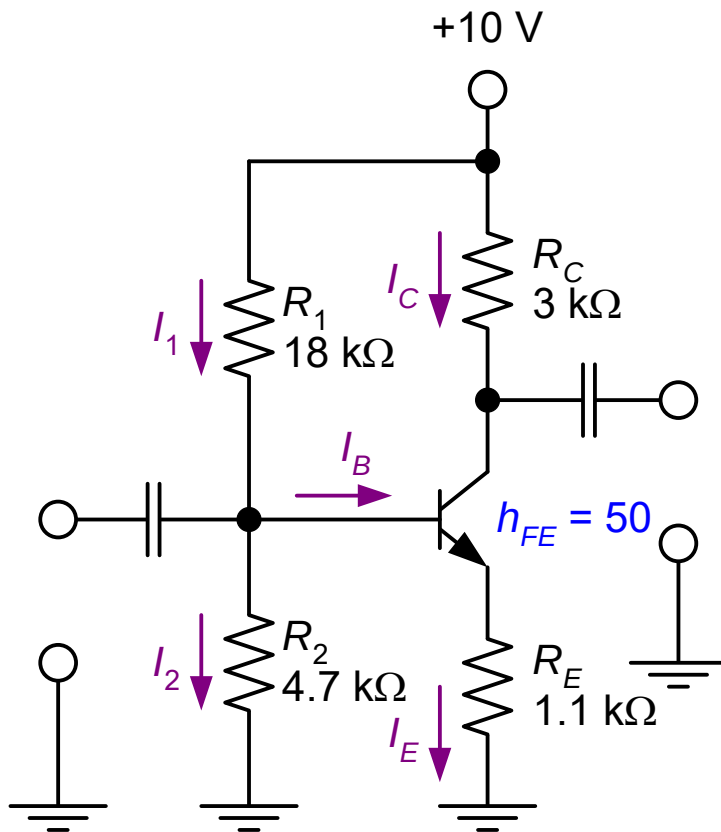
$$I_E = \frac{V_E}{R_E}$$

Assume that $I_{CQ} \cong I_E$ (or $h_{FE} \gg 1$). Then

$$V_{CEQ} = V_{CC} - I_{CQ} (R_C + R_E)$$

Fig 7.15 Example 7.7. (1)

Determine the values of I_{CQ} and V_{CEQ} for the circuit shown in Fig. 7.15.



$$V_B = V_{CC} \frac{R_2}{R_1 + R_2}$$

$$= (10\text{V}) \frac{4.7\text{k}\Omega}{22.7\text{k}\Omega} = 2.07\text{V}$$

$$V_E = V_B - 0.7\text{V}$$

$$= 2.07\text{V} - 0.7\text{V} = 1.37\text{V}$$

Because $I_{CQ} \cong I_E$ (or $h_{FE} \gg 1$),

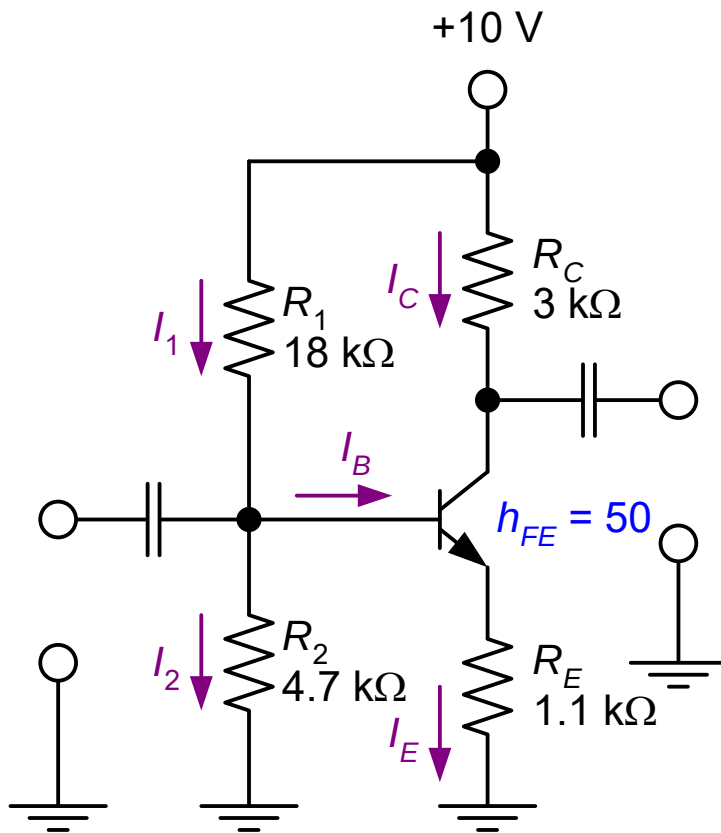
$$I_{CQ} \cong \frac{V_E}{R_E} = \frac{1.37\text{V}}{1.1\text{k}\Omega} = 1.25\text{mA}$$

$$V_{CEQ} = V_{CC} - I_{CQ} (R_C + R_E)$$

$$= 10\text{V} - (1.25\text{mA})(4.1\text{k}\Omega) = 4.87\text{V}$$

Fig 7.15 Example 7.7. (2)

Verify that $I_2 > 10 I_B$.



$$I_2 = \frac{V_B}{R_2} = \frac{2.07\text{V}}{4.7\text{k}\Omega} = 440.4\mu\text{A}$$

$$I_B = \frac{I_E}{h_{FE} + 1} = \frac{1.25\text{mA}}{50 + 1} = 24.51\mu\text{A}$$

$$\therefore I_2 > 10 I_B$$

Which value of h_{FE} do I use?

Transistor specification sheet may list any combination of the following h_{FE} : max. h_{FE} , min. h_{FE} , or **typ. h_{FE}** . Use **typical** value if there is one. Otherwise, use

$$h_{FE(ave)} = \sqrt{h_{FE(min)} \times h_{FE(max)}}$$

Example 7.9

A voltage-divider bias circuit has the following values: $R_1 = 1.5 \text{ k}\Omega$, $R_2 = 680 \Omega$, $R_C = 260 \Omega$, $R_E = 240 \Omega$ and $V_{CC} = 10 \text{ V}$. Assuming the transistor is a 2N3904, determine the value of I_B for the circuit.

$$V_B = V_{CC} \frac{R_2}{R_1 + R_2} = (10\text{V}) \frac{680\Omega}{2180\Omega} = 3.12\text{V}$$

$$V_E = V_B - 0.7\text{V} = 3.12\text{V} - 0.7\text{V} = 2.42\text{V}$$

$$I_{CQ} \cong I_E = \frac{V_E}{R_E} = \frac{2.42\text{V}}{240\Omega} = 10\text{mA}$$

$$h_{FE(\text{ave})} = \sqrt{h_{FE(\text{min})} \times h_{FE(\text{max})}} = \sqrt{100 \times 300} = 173$$

$$I_B = \frac{I_E}{h_{FE(\text{ave})} + 1} = \frac{10\text{mA}}{174} = 57.5\mu\text{A}$$

Stability of Voltage Divider Bias Circuit

The Q-point of voltage divider bias circuit is less dependent on h_{FE} than that of the base bias (fixed bias).

For example, if I_E is exactly 10 mA, the range of h_{FE} is 100 to 300. Then

$$\text{At } h_{FE} = 100, I_B = \frac{I_E}{h_{FE} + 1} = \frac{10\text{mA}}{101} \cong 100\mu\text{A} \text{ and } I_{CQ} = I_E - I_B \cong 9.90\text{mA}$$

$$\text{At } h_{FE} = 300, I_B = \frac{I_E}{h_{FE} + 1} = \frac{10\text{mA}}{301} \cong 33\mu\text{A} \text{ and } I_{CQ} = I_E - I_B \cong 9.97\text{mA}$$

I_{CQ} hardly changes over the entire range of h_{FE} .

Fig 7.18 Load line for voltage divider bias circuit.

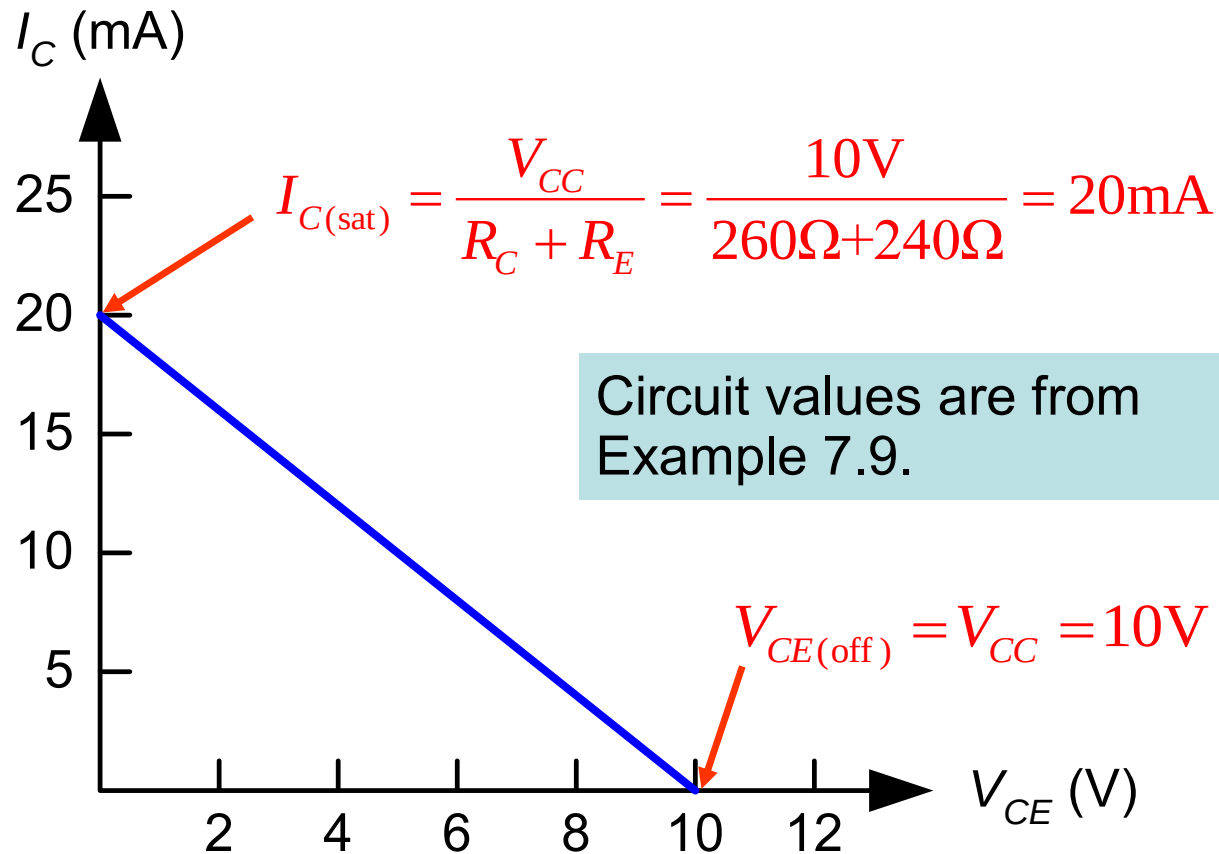


Fig 7.19-20 Base input resistance. (1)

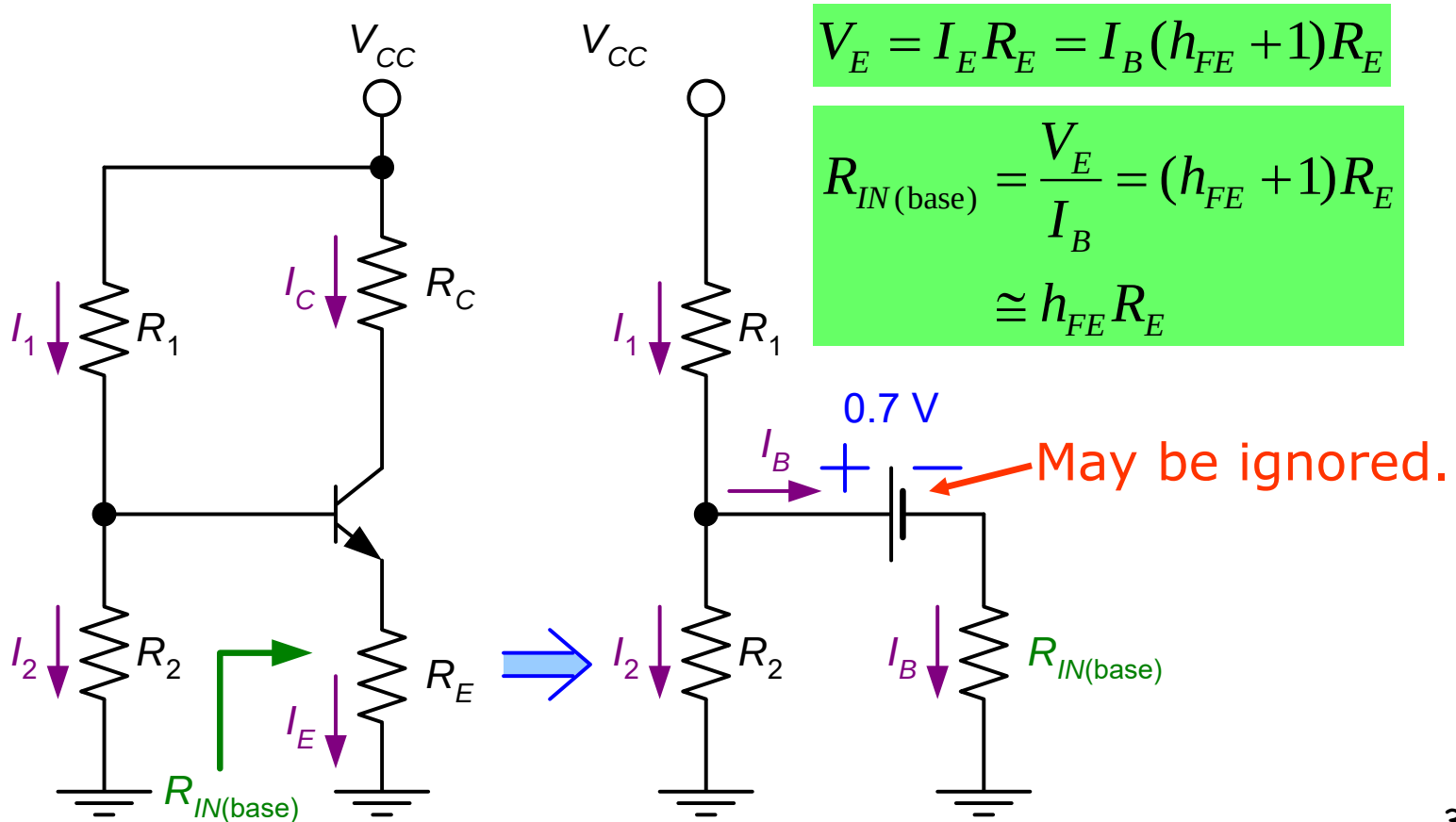
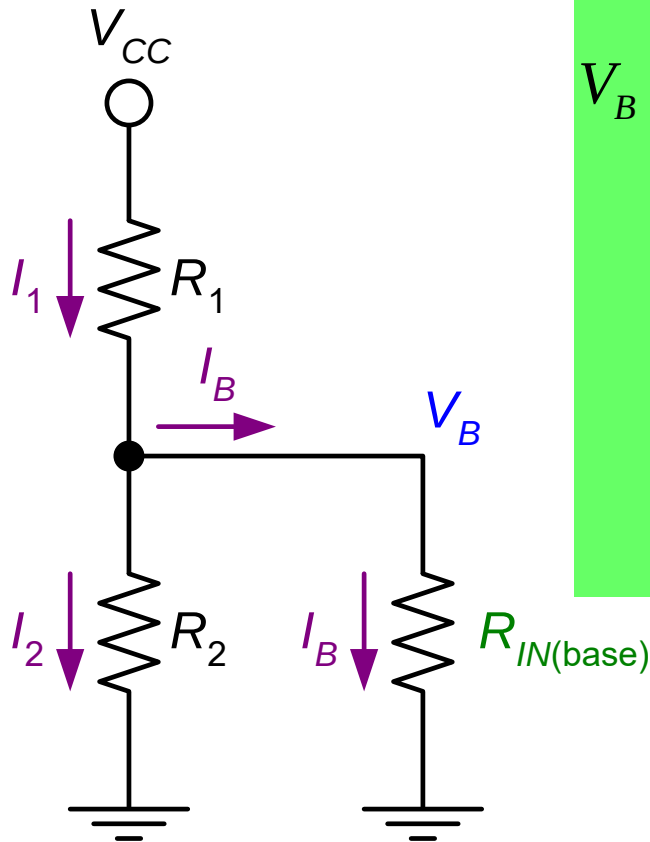
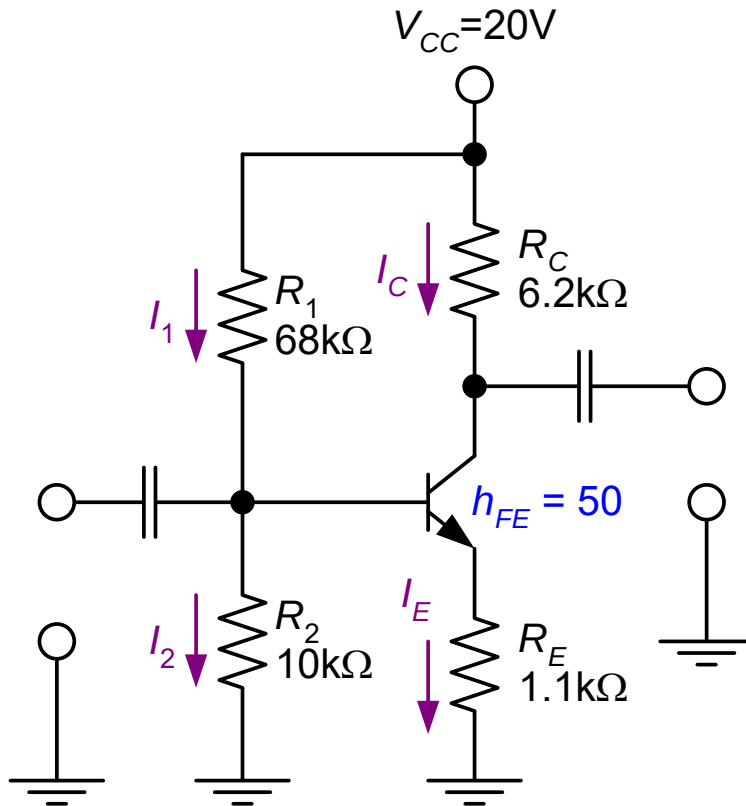


Fig 7.19-20 Base input resistance. (2)



$$\begin{aligned}
 V_B &= \frac{R_2 \parallel R_{IN(base)}}{R_1 + R_2 \parallel R_{IN(base)}} V_{CC} \\
 &= \frac{R_2 \parallel (h_{FE} R_E)}{R_1 + R_2 \parallel (h_{FE} R_E)} V_{CC} \\
 &= \frac{R_{EQ}}{R_1 + R_{EQ}} V_{CC} \quad \left| \quad R_{EQ} = R_2 \parallel (h_{FE} R_E) \right.
 \end{aligned}$$

Fig 7.21 Example 7.11.



$$R_{EQ} = R_2 // (h_{FE} R_E)$$

$$= 10k\Omega // (50 \times 1.1k\Omega) = 8.46k\Omega$$

$$V_B \cong V_{CC} \frac{R_{EQ}}{R_1 + R_{EQ}}$$

$$= (20V) \frac{8.46k\Omega}{68k\Omega + 8.46k\Omega} = 2.21V$$

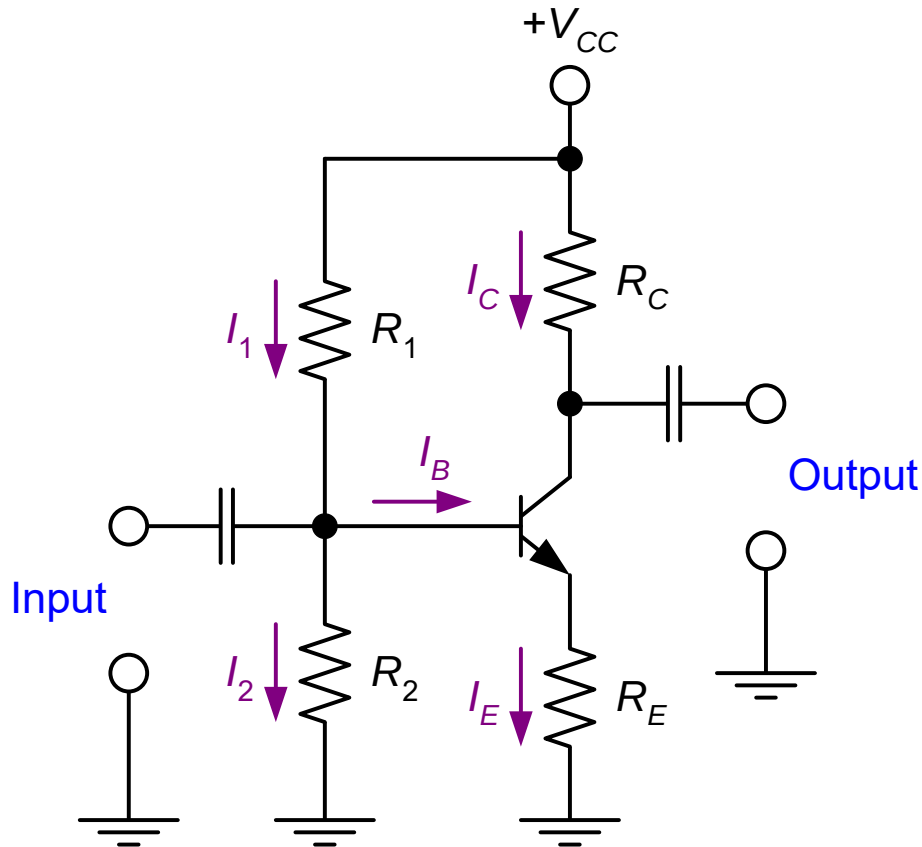
$$I_{CQ} \cong I_E = \frac{V_E}{R_E} = \frac{V_B - 0.7V}{R_E}$$

$$= \frac{2.21V - 0.7V}{1.1k\Omega} = 1.37mA$$

$$V_{CEQ} = V_{CC} - I_{CQ} (R_C + R_E)$$

$$= 20V - (1.37mA)(7.3k\Omega) = 9.99V$$

Fig 7.24 Voltage-divider bias characteristics. (1)



Circuit recognition: The voltage divider in the base circuit.

Advantages: The circuit Q-point values are stable against changes in h_{FE} .

Disadvantages: Requires more components than most other biasing circuits.

Applications: Used primarily to bias linear amplifier.

Merits:

- Operating point is almost independent of β variation.
- Operating point stabilized against shift in temperature.

Demerits:

- As β -value is fixed for a given transistor, this relation can be satisfied either by keeping R_E fairly large, or making $R_1 || R_2$ very low.

➤ If R_E is of large value, high V_{CC} is necessary. This increases cost as well

as precautions necessary while handling.

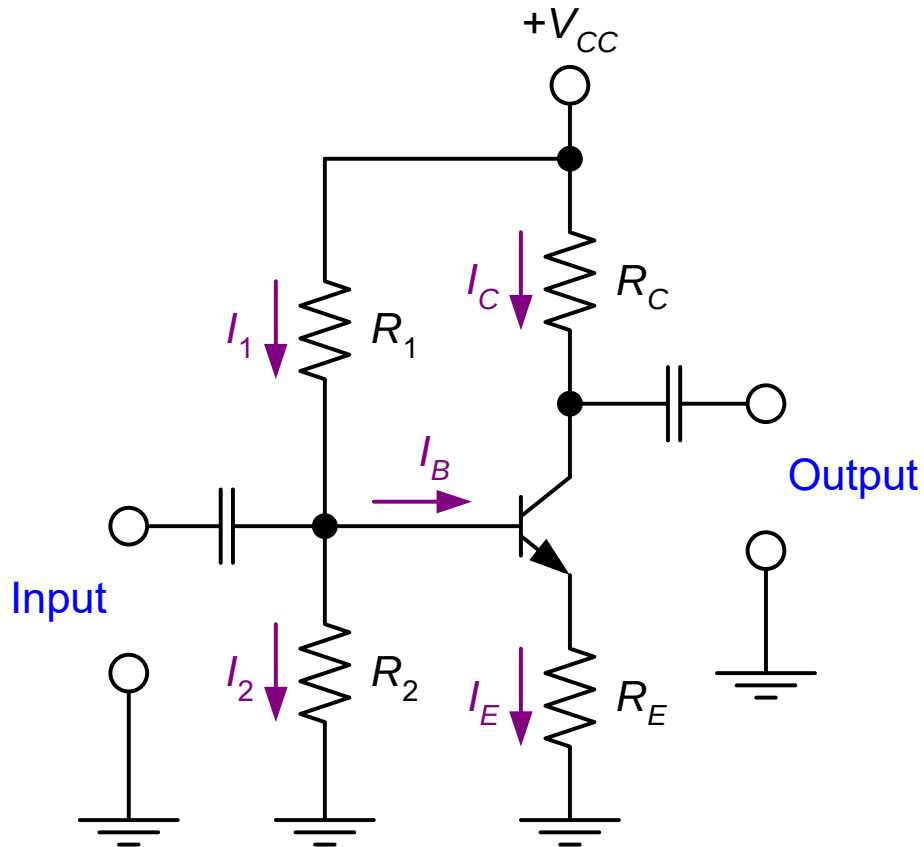
➤ If $R_1 || R_2$ is low, either R_1 is low, or R_2 is low, or both are low. A low R_1 raises V_B closer to V_C , reducing the available swing in collector voltage, and limiting how large R_C can be made without driving the transistor out of active mode. A low R_2 lowers V_{be} , reducing the allowed collector current. Lowering both resistor values draws more current from the power supply and lowers the input resistance of the amplifier as seen from the base.

➤ AC as well as DC feedback is caused by R_E , which reduces the AC voltage gain of the amplifier. A method to avoid AC feedback while retaining DC feedback is discussed below.

Usage:

The circuit's stability and merits as above make it widely used for linear circuits.

Fig 7.24 Voltage-divider bias characteristics. (2)



Load line equations:

$$I_{C(\text{sat})} = \frac{V_{CC}}{R_C + R_E}$$

$$V_{CE(\text{off})} = V_{CC}$$

Q-point equations (assume that $h_{FE}R_E > 10R_2$):

$$V_B = V_{CC} \frac{R_2}{R_1 + R_2}$$

$$V_E = V_B - 0.7V$$

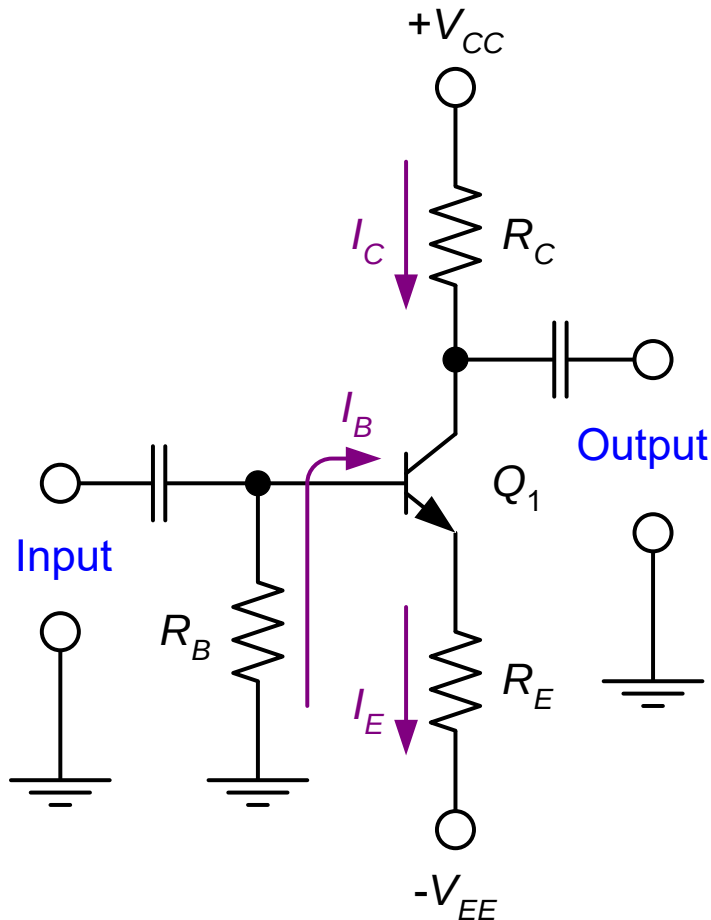
$$I_{CQ} \cong I_E = \frac{V_E}{R_E}$$

$$V_{CEQ} = V_{CC} - I_{CQ} (R_C + R_E)$$

Other Transistor Biasing Circuits

- Emitter-bias circuits
- Feedback-bias circuits
 - Collector-feedback bias
 - Emitter-feedback bias

Fig 7.25-6 Emitter bias.



Assume that the transistor operation is in active region.

$$I_B = \frac{V_{EE} - 0.7V}{R_B + (h_{FE} + 1)R_E}$$

$$I_C = h_{FE} I_B$$

$$I_E = (h_{FE} + 1) I_B$$

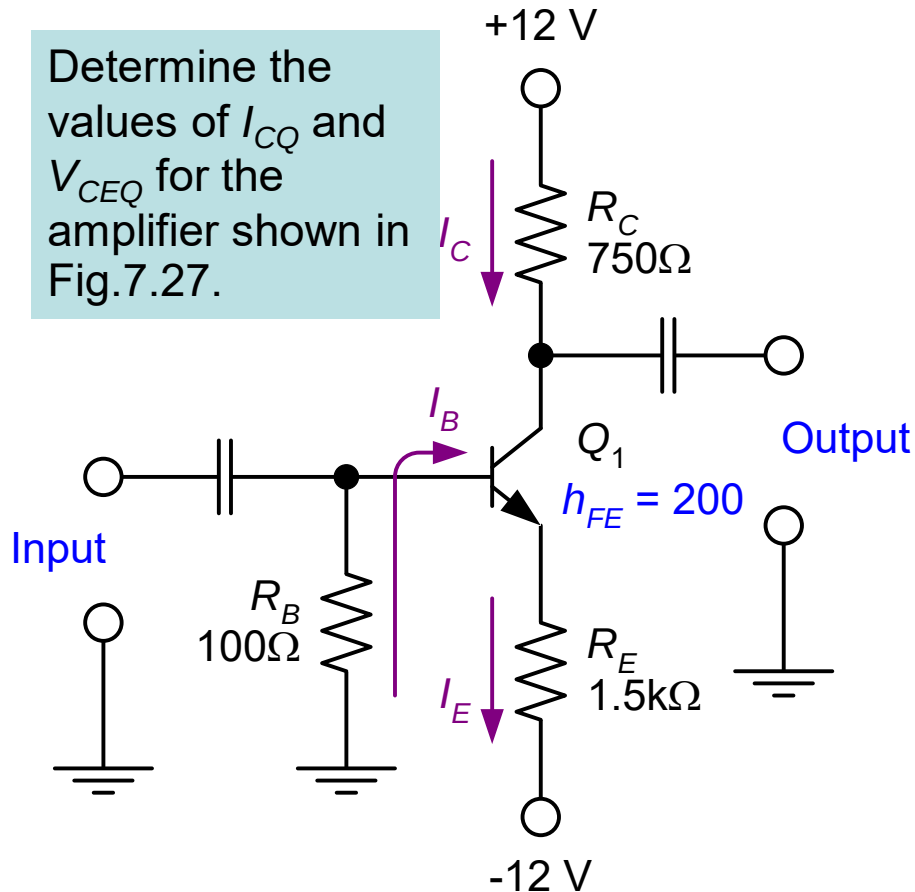
$$V_{CE} = V_{CC} - I_C R_C - I_E R_E + V_{EE}$$

Assume that $h_{FE} \gg 1$.

$$V_{CE} \cong V_{CC} - I_C (R_C + R_E) + V_{EE}$$

Fig 7.27 Example 7.12.

Determine the values of I_{CQ} and V_{CEQ} for the amplifier shown in Fig.7.27.



$$I_B = \frac{12V - 0.7V}{R_B + (h_{FE} + 1)R_E}$$

$$= \frac{11.3V}{100\Omega + 201 \times 1.5k\Omega} = 37.47\mu A$$

$$I_{CQ} = h_{FE} I_B = 200 \times 37.47\mu A$$

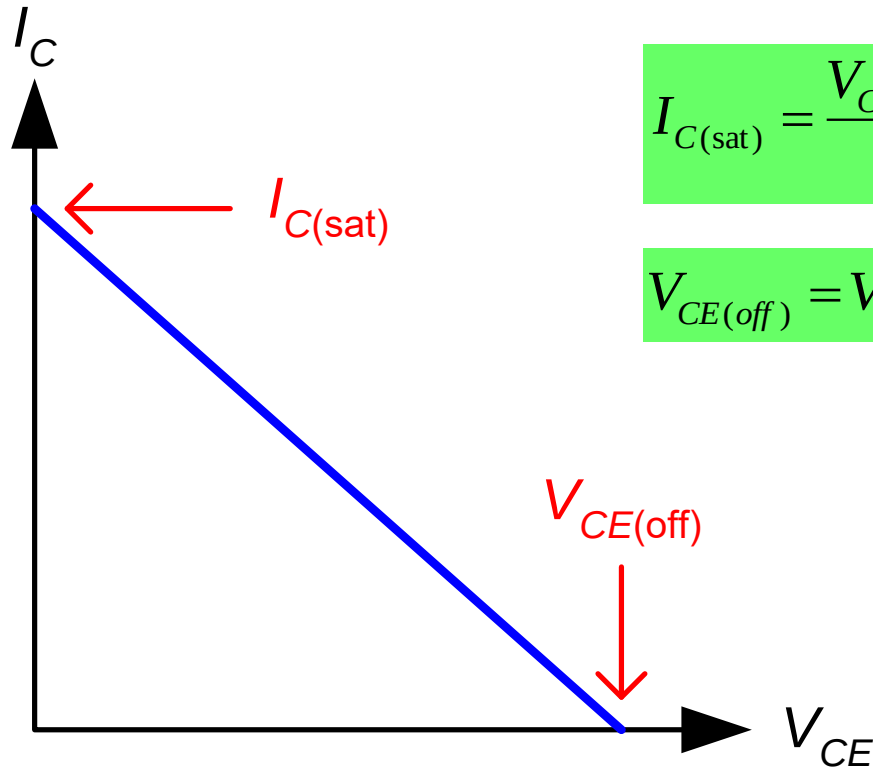
$$= 7.49mA$$

$$V_{CEQ} \cong V_{CC} - I_C (R_C + R_E) - (-V_{EE})$$

$$= 24V - 7.49mA (750\Omega + 1.5k\Omega)$$

$$= 7.14V$$

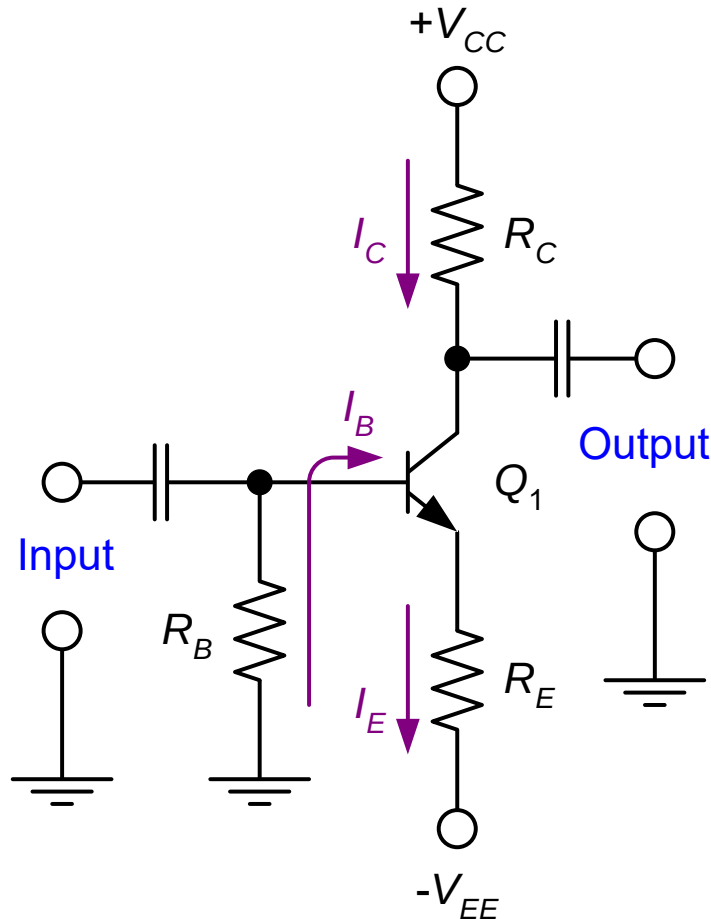
Load Line for Emitter-Bias Circuit



$$I_{C(sat)} = \frac{V_{CC} - (-V_{EE})}{R_C + R_E} = \frac{V_{CC} + V_{EE}}{R_C + R_E}$$

$$V_{CE(off)} = V_{CC} - (-V_{EE}) = V_{CC} + V_{EE}$$

Fig 7.28 Emitter-bias characteristics. (1)



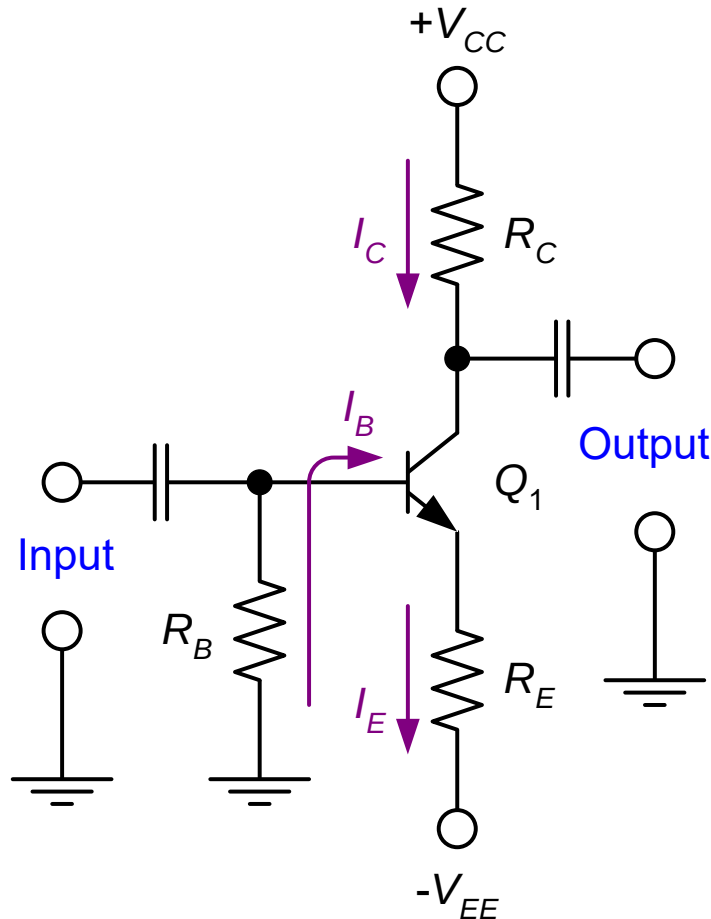
Circuit recognition: A split (dual-polarity) power supply and the base resistor is connected to ground.

Advantage: The circuit Q-point values are stable against changes in h_{FE} .

Disadvantage: Requires the use of dual-polarity power supply.

Applications: Used primarily to bias linear amplifiers.

Fig 7.28 Emitter-bias characteristics. (2)



Load line equations:

$$I_{C(sat)} = \frac{V_{CC} + V_{EE}}{R_C + R_E}$$

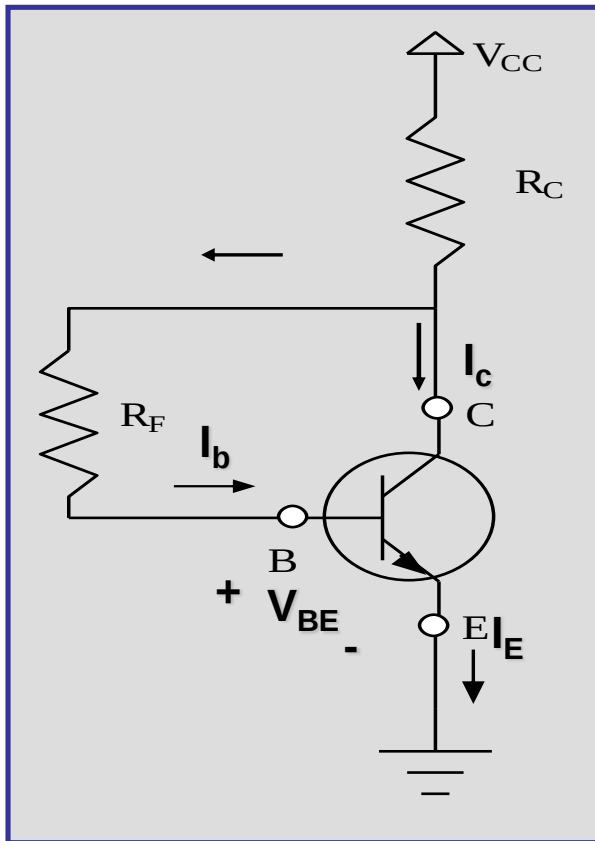
$$V_{CE(off)} = V_{CC} + V_{EE}$$

Q-point equations:

$$I_{CQ} = (h_{FE}) \frac{-V_{BE} + V_{EE}}{R_B + (h_{FE} + 1) R_E}$$

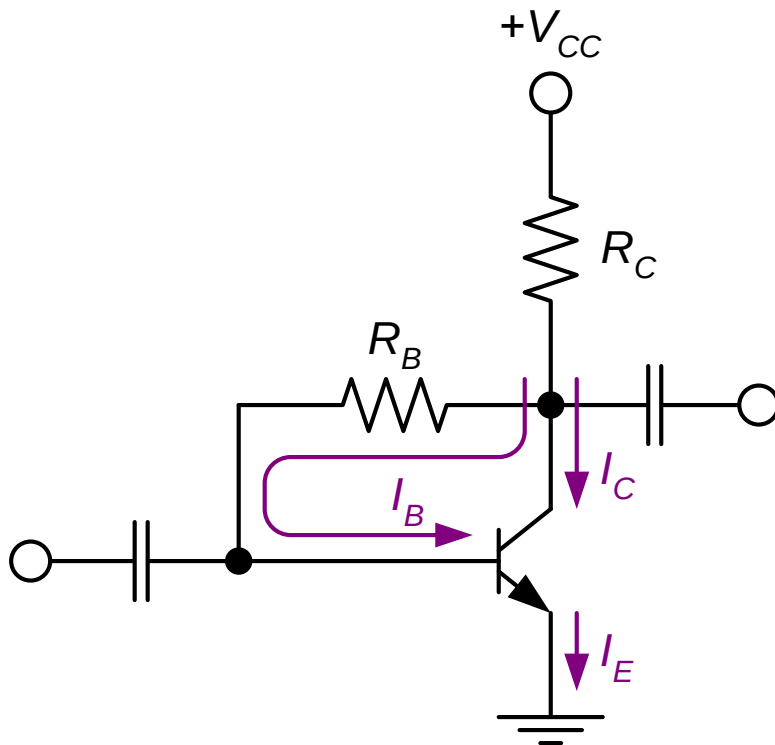
$$V_{CEQ} \cong V_{CC} - I_{CQ} (R_C + R_E) + V_{EE}$$

The Collector to Base Bias Circuit



This configuration employs negative feedback to prevent thermal runaway and stabilize the operating point. In this form of biasing, the base resistor R_F is connected to the collector instead of connecting it to the DC source V_{CC} . So any thermal runaway will induce a voltage drop across the R_C resistor that will throttle the transistor's base current.

Fig 7.29 Collector-feedback bias.



$$V_{CC} = (I_C + I_B)R_C + I_B R_B + V_{BE}$$

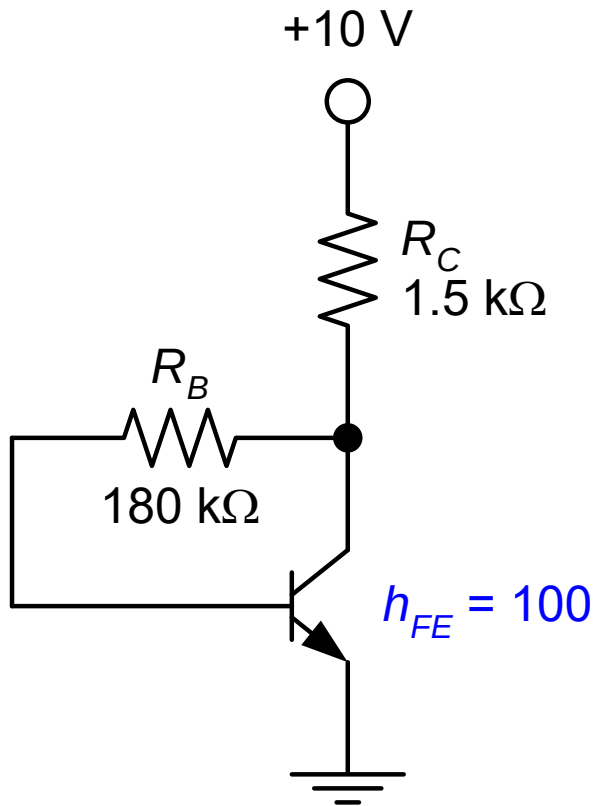
$$I_B = \frac{V_{CC} - V_{BE}}{(h_{FE} + 1)R_C + R_B}$$

$$I_{CQ} = h_{FE} I_B$$

$$V_{CEQ} = V_{CC} - (h_{FE} + 1)I_B R_C$$

$$\cong V_{CC} - I_{CQ} R_C$$

Fig 7.30 Example 7.14.



Determine the values of I_{CQ} and V_{CEQ} for the amplifier shown in Fig. 7.30.

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (h_{FE} + 1)R_C}$$

$$= \frac{10V - 0.7V}{180k\Omega + 101 \times 1.5k\Omega} = 28.05\mu A$$

$$I_{CQ} = h_{FE} I_B = 100 \times 28.05\mu A$$

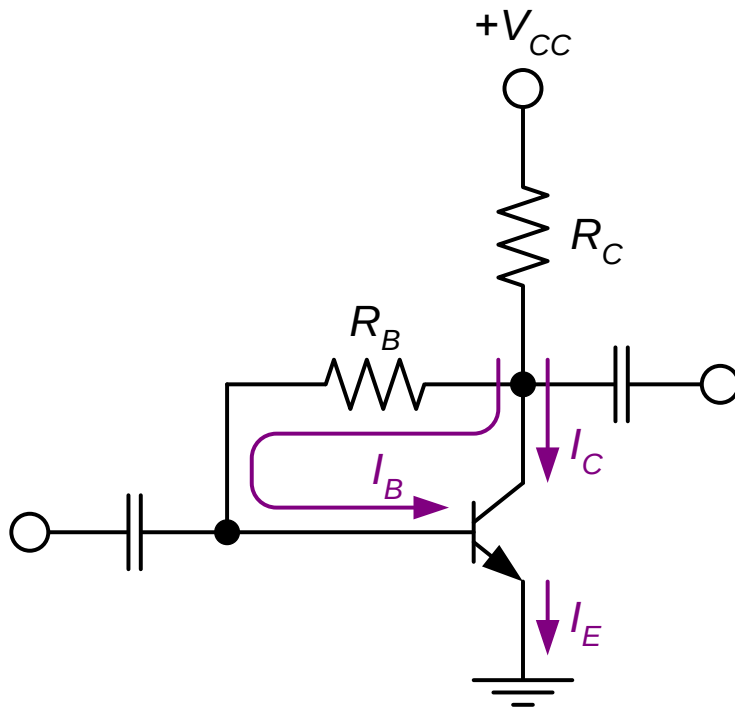
$$= 2.805mA$$

$$V_{CEQ} = V_{CC} - (h_{FE} + 1)I_B R_C$$

$$= 10V - 101 \times 28.05\mu A \times 1.5k\Omega$$

$$= 5.75V$$

Circuit Stability of Collector-Feedback Bias



h_{FE} increases



I_C increases (if I_B is the same)



V_{CE} decreases



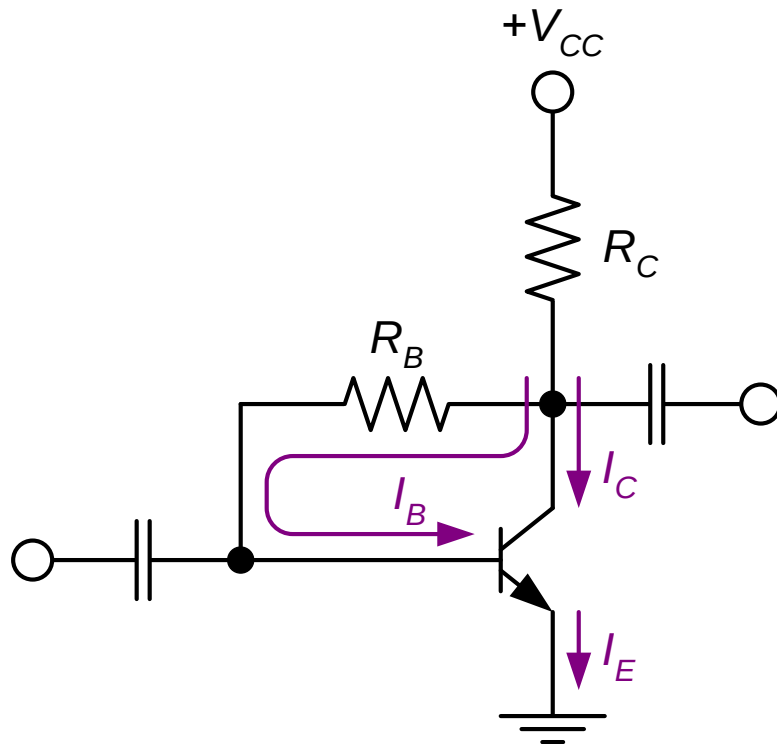
I_B decreases



I_C does not increase that much.

Good Stability. Less dependent on h_{FE} and temperature.

Collector-Feedback Characteristics (1)



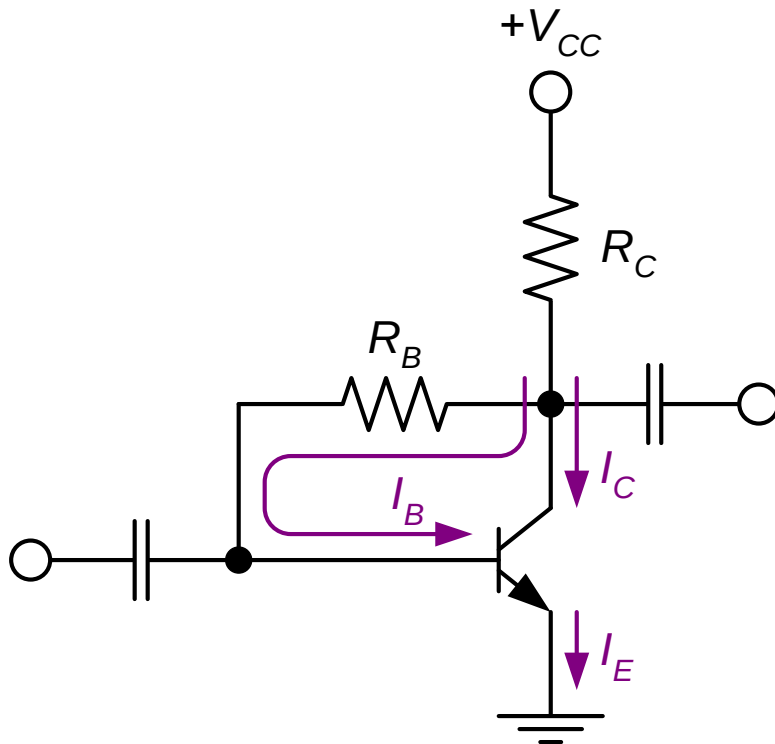
Circuit recognition: The base resistor is connected between the base and the collector terminals of the transistor.

Advantage: A simple circuit with relatively stable Q-point.

Disadvantage: Relatively poor ac characteristics.

Applications: Used primarily to bias linear amplifiers.

Collector-Feedback Characteristics (2)



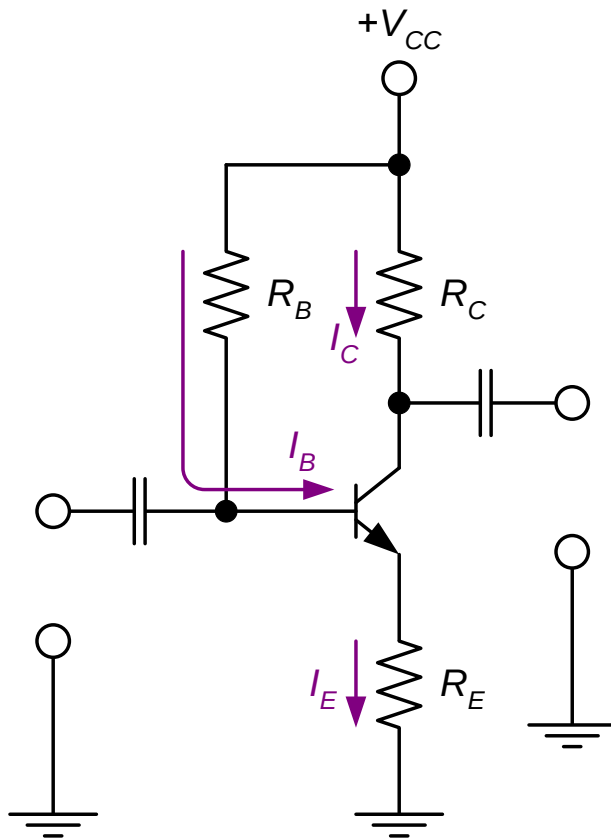
Q-point relationships:

$$I_B = \frac{V_{CC} - V_{BE}}{(h_{FE} + 1)R_C + R_B}$$

$$I_{CQ} = h_{FE} I_B$$

$$V_{CEQ} \cong V_{CC} - I_{CQ} R_C$$

Fig 7.31 Emitter-feedback bias.



$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (h_{FE} + 1)R_E}$$

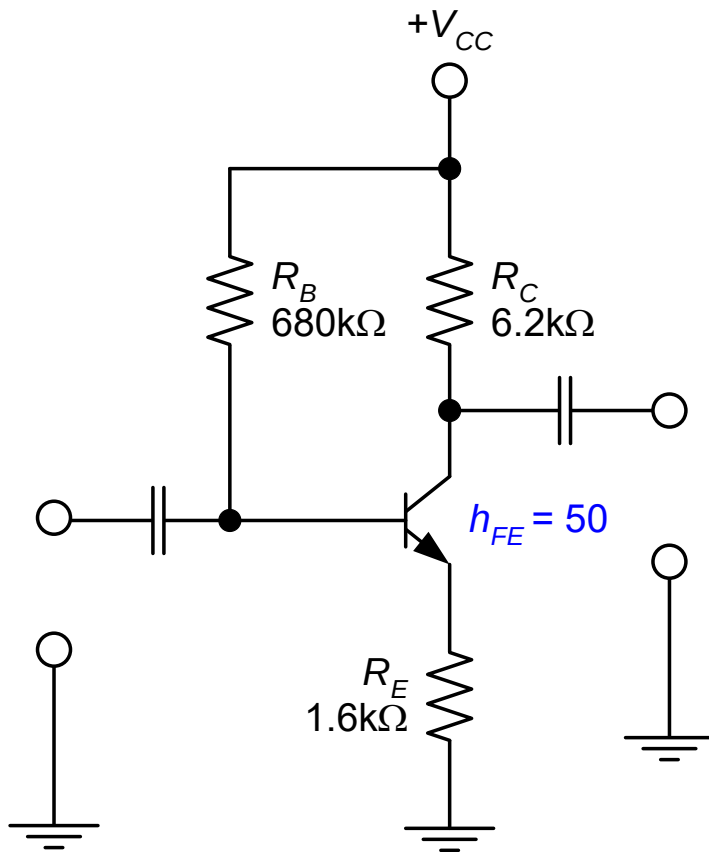
$$I_{CQ} = h_{FE} I_B$$

$$I_E = (h_{FE} + 1) I_B$$

$$V_{CEQ} = V_{CC} - I_C R_C - I_E R_E$$

$$\cong V_{CC} - I_{CQ} (R_C + R_E)$$

Fig 7.32 Example 7.15.

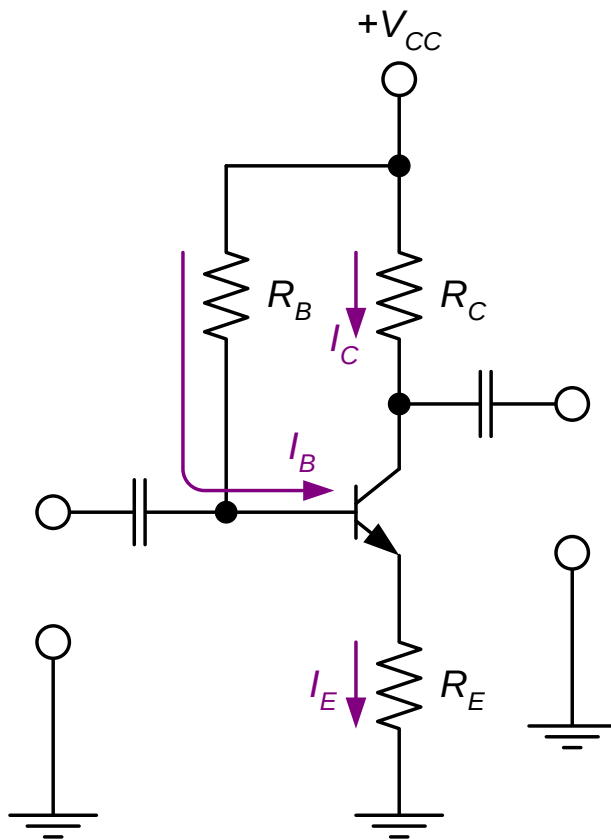


$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (h_{FE} + 1)R_E} = \frac{16\text{V} - 0.7\text{V}}{680\text{k}\Omega + 51 \times 1.6\text{k}\Omega} = 20.09\mu\text{A}$$

$$I_{CQ} = h_{FE} I_B = 50 \times 20.09\mu\text{A} = 1\text{mA}$$

$$V_{CEQ} \cong V_{CC} - I_{CQ}(R_C + R_E) = 16\text{V} - (1\text{mA})(7.8\text{k}\Omega) = 8.2\text{V}$$

Circuit Stability of Emitter-Feedback Bias



h_{FE} increases



I_C increases (if I_B is the same)



V_E increases



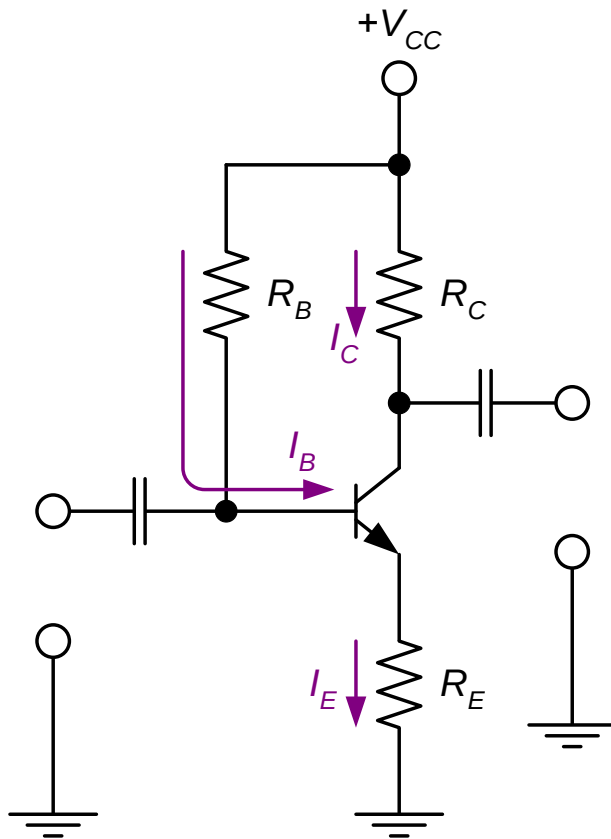
I_B decreases



I_C does not increase that much.

I_C is less dependent on h_{FE} and temperature.

Emitter-Feedback Characteristics (1)



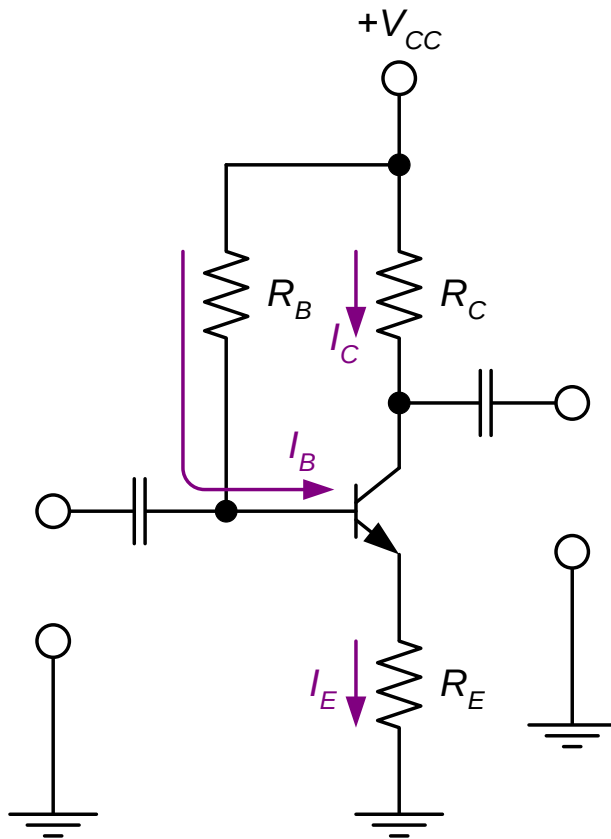
Circuit recognition: Similar to voltage divider bias with R_2 missing (or base bias with R_E added).

Advantage: A simple circuit with relatively stable Q-point.

Disadvantage: Requires more components than collector-feedback bias.

Applications: Used primarily to bias linear amplifiers.

Emitter-Feedback Characteristics (2)



Q-point relationships:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (h_{FE} + 1)R_E}$$

$$I_{CQ} = h_{FE} I_B$$

$$V_{CEQ} \cong V_{CC} - I_{CQ} (R_C + R_E)$$

Summary

- DC Biasing and the dc load line
- Base bias circuits
- Voltage-divider bias circuits
- Emitter-bias circuits
- Feedback-bias circuits
 - Collector-feedback bias circuits
 - Emitter-feedback bias circuits

Digital Systems

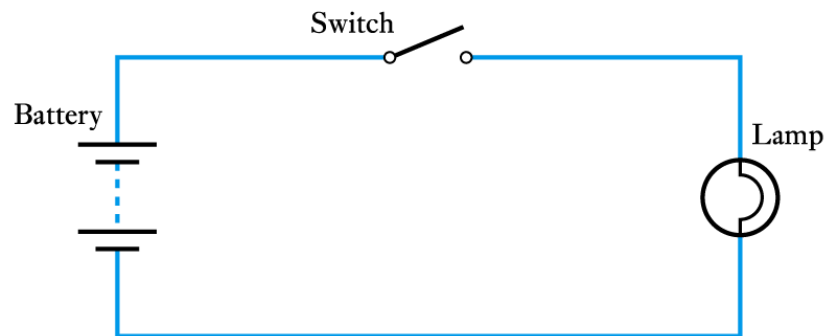
- Introduction
- Binary Quantities and Variables
- Logic Gates
- Boolean Algebra
- Combinational Logic
- Number Systems and Binary Arithmetic
- Numeric and Alphabetic Codes

Introduction

- Digital systems are concerned with digital signals
- Digital signals can take many forms
- Here we will concentrate on **binary signals** since these are the most common form of digital signals
 - can be used individually
 - perhaps to represent a single binary quantity or the state of a single switch
 - can be used in combination
 - to represent more complex quantities

Binary Quantities and Variables

- A **binary quantity** is one that can take only 2 states



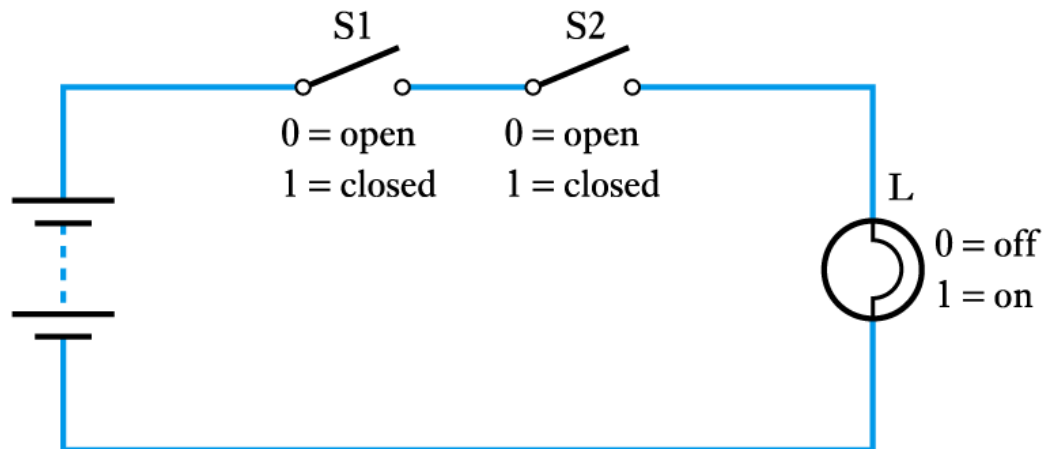
A simple binary arrangement

<i>S</i>	<i>L</i>
OPEN	OFF
CLOSED	ON

<i>S</i>	<i>L</i>
0	0
1	1

A truth table

- A binary arrangement with two switches in series



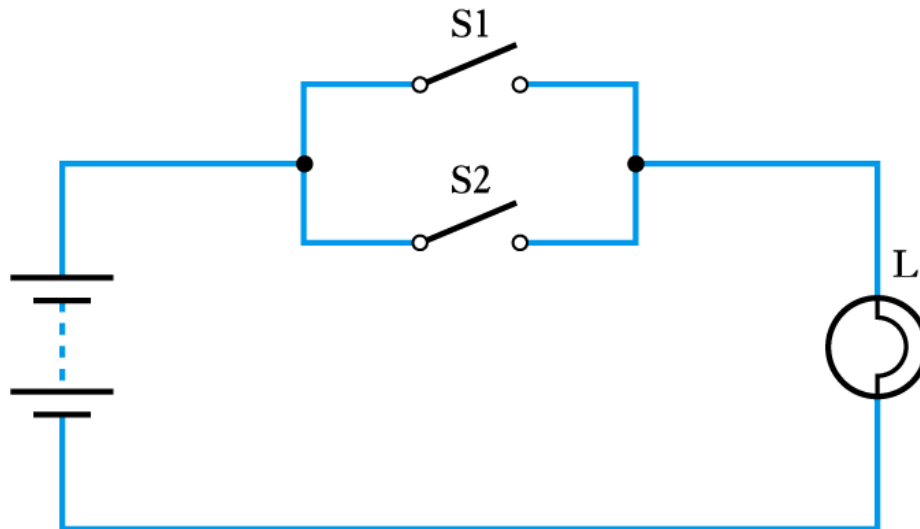
(a) Circuit

S1	S2	L
0	0	0
0	1	0
1	0	0
1	1	1

(b) Truth table

$$L = S1 \text{ AND } S2$$

- A binary arrangement with two switches in parallel



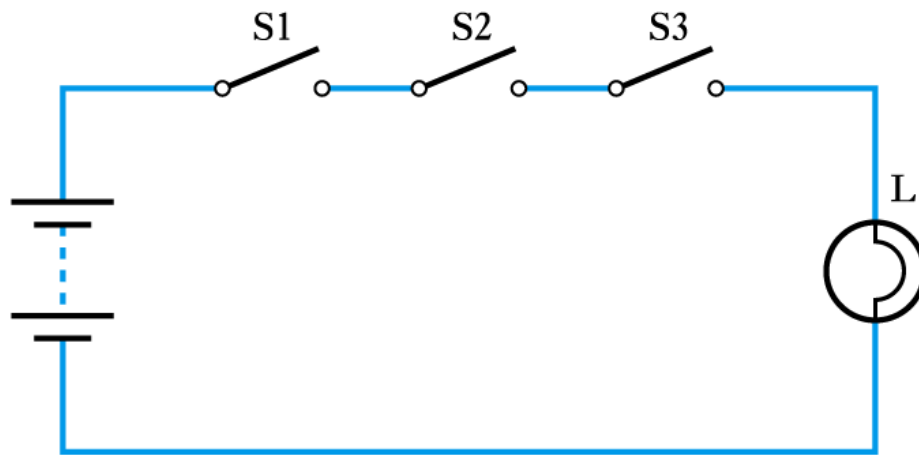
(a) Circuit

S1	S2	L
0	0	0
0	1	1
1	0	1
1	1	1

(b) Truth table

$$L = S1 \text{ OR } S2$$

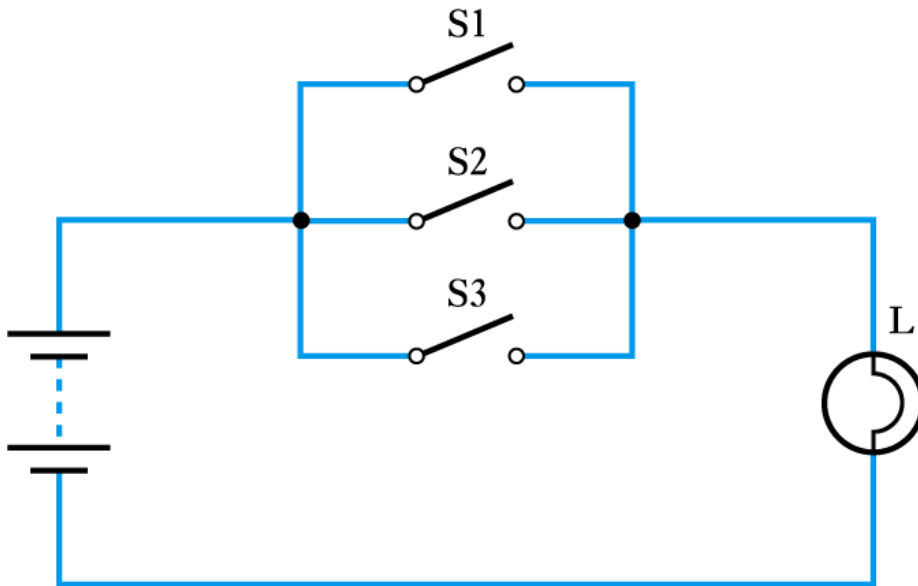
- Three switches in series



S1	S2	S3	L
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

$$L = S1 \text{ AND } S2 \text{ AND } S3$$

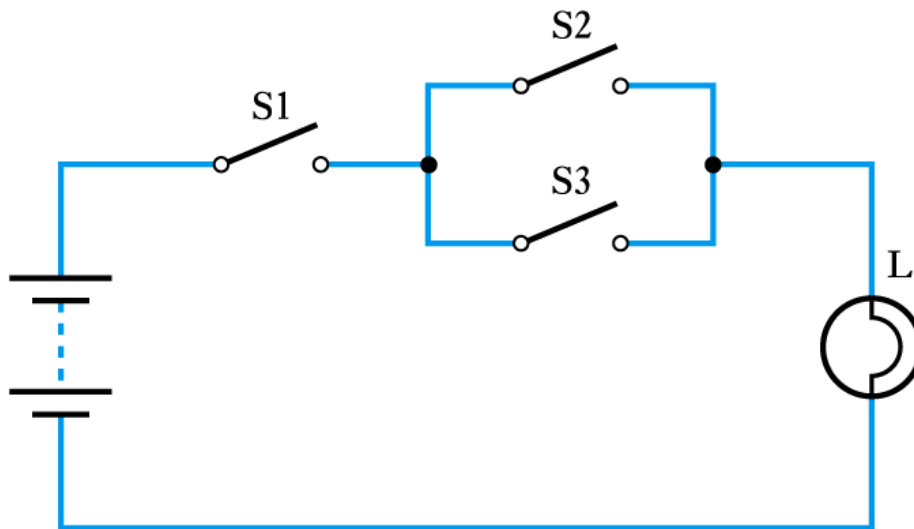
- Three switches in parallel



$$L = S1 \text{ OR } S2 \text{ OR } S3$$

S1	S2	S3	L
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

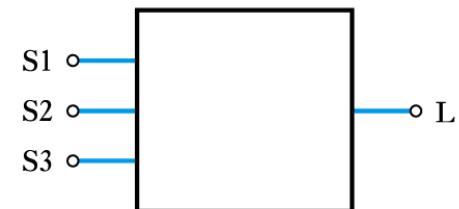
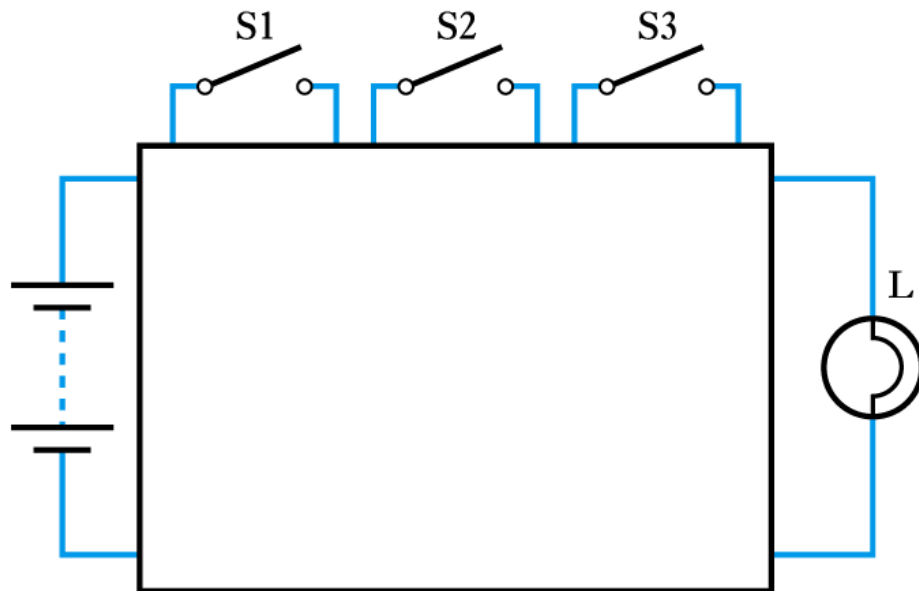
- A series/parallel arrangement



S1	S2	S3	L
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$L = S1 \text{ AND } (S2 \text{ OR } S3)$$

- Representing an unknown network



Logic Gates

- The building blocks used to create digital circuits are **logic gates**
- There are three elementary logic gates and a range of other simple gates
- Each gate has its own **logic symbol** which allows complex functions to be represented by a logic diagram
- The function of each gate can be represented by a **truth table** or using **Boolean notation**

■ The AND gate



(a) Circuit symbol

A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

(b) Truth table

$$C = A \cdot B$$

(c) Boolean expression

■ The OR gate



(a) Circuit symbol

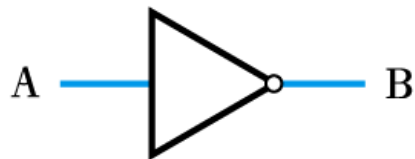
A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

(b) Truth table

$$C = A + B$$

(c) Boolean expression

■ The NOT gate (or inverter)



(a) Circuit symbol

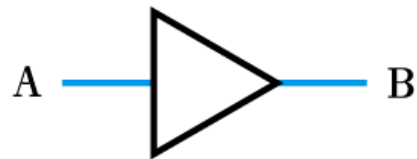
A	B
0	1
1	0

(b) Truth table

$$B = \bar{A}$$

(c) Boolean expression

■ A logic buffer gate



(a) Circuit symbol

A	B
0	0
1	1

(b) Truth table

$$B = A$$

(c) Boolean expression

■ The NAND gate



(a) Circuit symbol

A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

(b) Truth table

$$C = \overline{A \cdot B}$$

(c) Boolean expression

■ The NOR gate



(a) Circuit symbol

A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

(b) Truth table

$$C = \overline{A + B}$$

(c) Boolean expression

■ The Exclusive OR gate



(a) Circuit symbol

A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

(b) Truth table

$$C = A \oplus B$$

(c) Boolean expression

■ The Exclusive NOR gate



(a) Circuit symbol

A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

(b) Truth table

$$C = \overline{A \oplus B}$$

(c) Boolean expression

Boolean Algebra

- **Boolean Constants**

- these are '0' (false) and '1' (true)

- **Boolean Variables**

- variables that can only take the vales '0' or '1'

- **Boolean Functions**

- each of the logic functions (such as AND, OR and NOT) are represented by symbols as described above

- **Boolean Theorems**

- a set of **identities** and **laws** – see text for details

■ Boolean identities

AND Function	OR Function	NOT function
$0 \bullet 0 = 0$	$0 + 0 = 0$	$\overline{0} = 1$
$0 \bullet 1 = 0$	$0 + 1 = 1$	$\overline{1} = 0$
$1 \bullet 0 = 0$	$1 + 0 = 1$	$\overline{\overline{A}} = A$
$1 \bullet 1 = 1$	$1 + 1 = 1$	
$A \bullet 0 = 0$	$A + 0 = A$	
$0 \bullet A = 0$	$0 + A = A$	
$A \bullet 1 = A$	$A + 1 = 1$	
$1 \bullet A = A$	$1 + A = 1$	
$A \bullet A = A$	$A + A = A$	
$A \bullet \overline{A} = 0$	$A + \overline{A} = 1$	

■ Boolean laws

Commutative law $AB = BA$ $A + B = B + A$	Absorption law $A + AB = A$ $A(A + B) = A$
Distributive law $A(B + C) = AB + BC$ $A + BC = (A + B)(A + C)$	De Morgan's law $\overline{A + B} = \overline{A} \bullet \overline{B}$ $\overline{A \bullet B} = \overline{A} + \overline{B}$
Associative law $A(BC) = (AB)C$ $A + (B + C) = (A + B) + C$	Note also $A + \overline{A}B = A + B$ $A(\overline{A} + B) = AB$

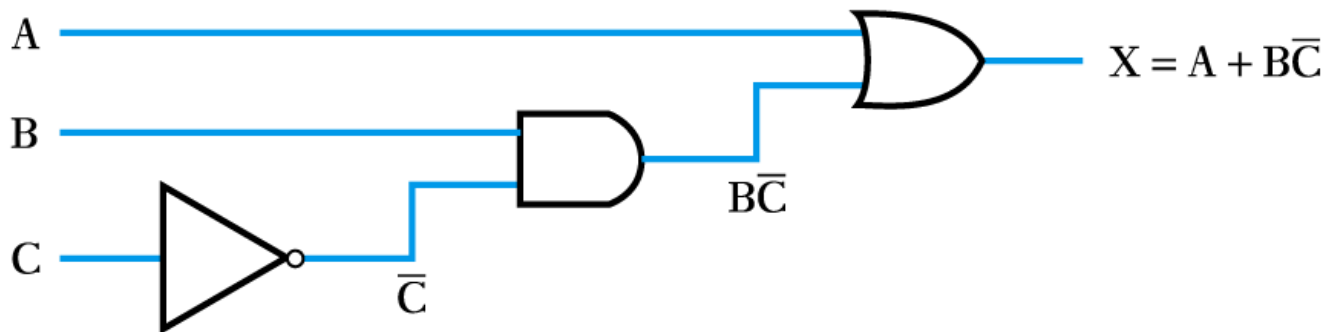
Combinational Logic

- Digital systems may be divided into two broad categories:
 - **combinational logic**
 - where the outputs are determined solely by the current states of the inputs
 - **sequential logic**
 - where the outputs are determined not only by the current inputs but also by the sequence of inputs that led to the current state
- In this lecture we will look at combination logic

■ Implementing a function from a Boolean expression

Example

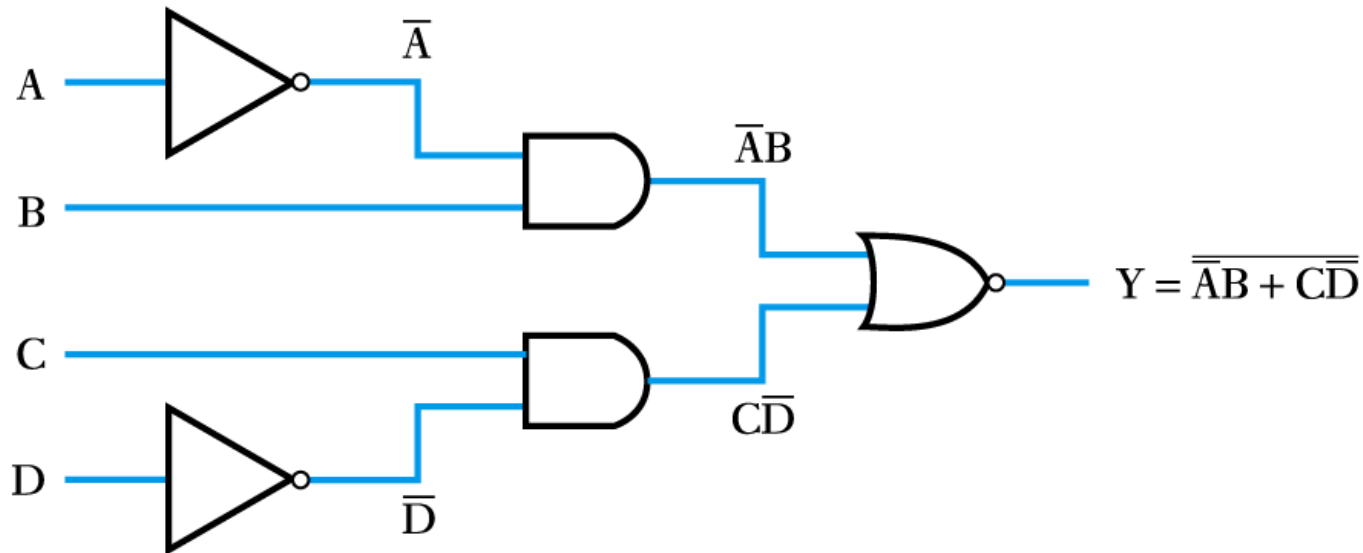
Implement the function $X = A + B\bar{C}$



■ Implementing a function from a Boolean expression

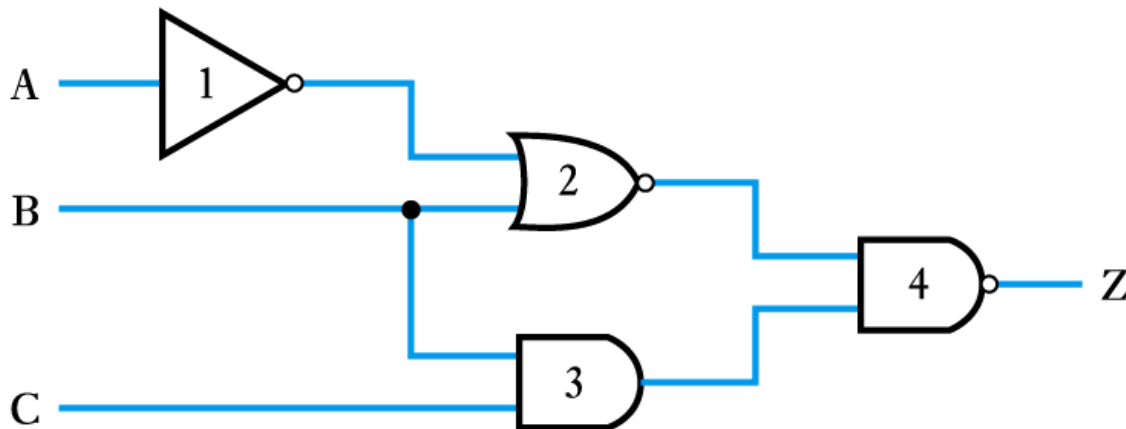
Example

Implement the function $Y = \overline{\overline{A}B + C\overline{D}}$



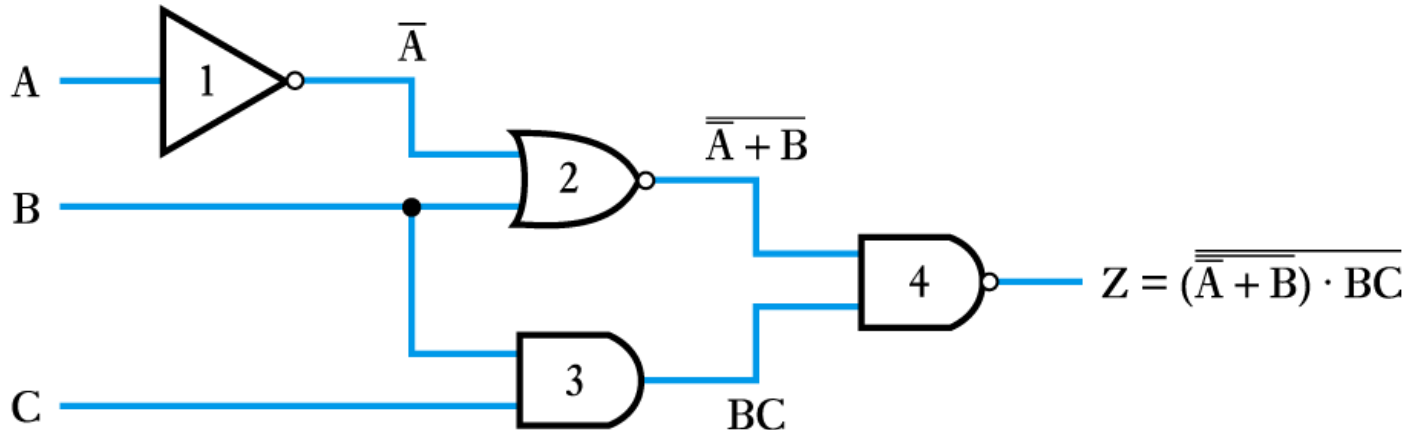
- **Generating a Boolean expression from a logic diagram**

Example



Example (continued)

- work progressively from the inputs to the output adding logic expressions to the output of each gate in turn



■ Implementing a logic function from a description

Example

The operation of the Exclusive OR gate can be stated as:

“The output should be true if either of its inputs are true, but not if both inputs are true.”

This can be rephrased as:

“The output is true if A OR B is true, AND if A AND B are NOT true.”

We can write this in Boolean notation as

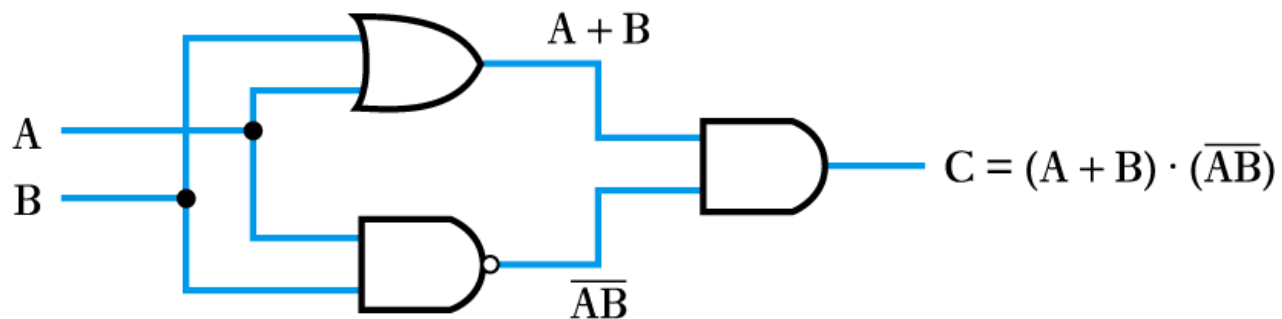
$$X = (A + B) \bullet \overline{(AB)}$$

Example (continued)

The logic function

$$X = (A + B) \cdot (\overline{AB})$$

can then be implemented as before



■ Implementing a logic function from a truth table

Example

Implement the function of the following truth table

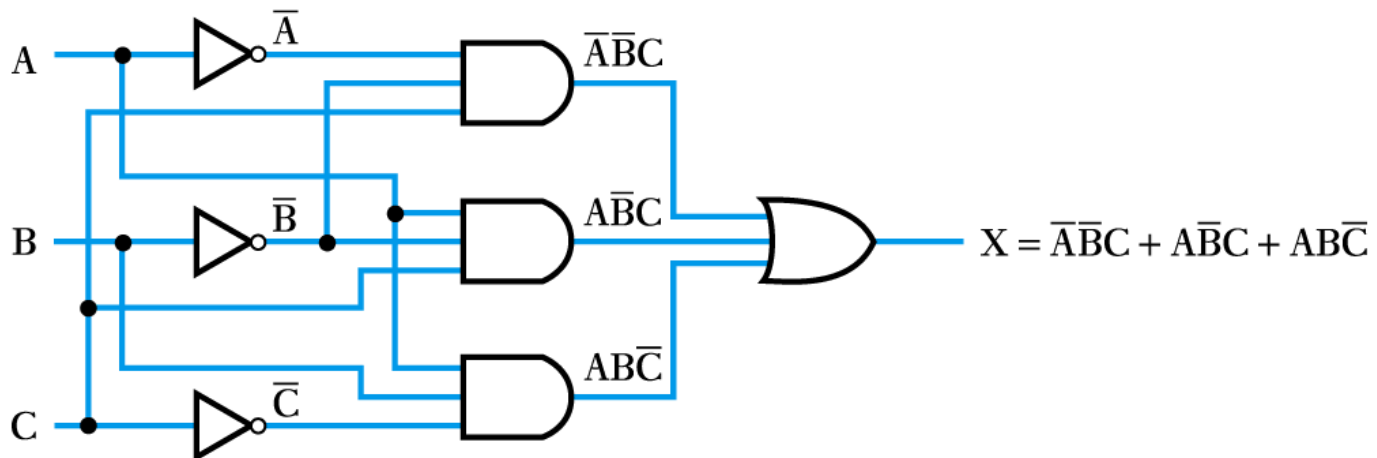
A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

- first write down a Boolean expression for the output
- then implement as before
- in this case

$$X = \bar{A} \bar{B} C + A \bar{B} C + A B \bar{C}$$

Example (continued)

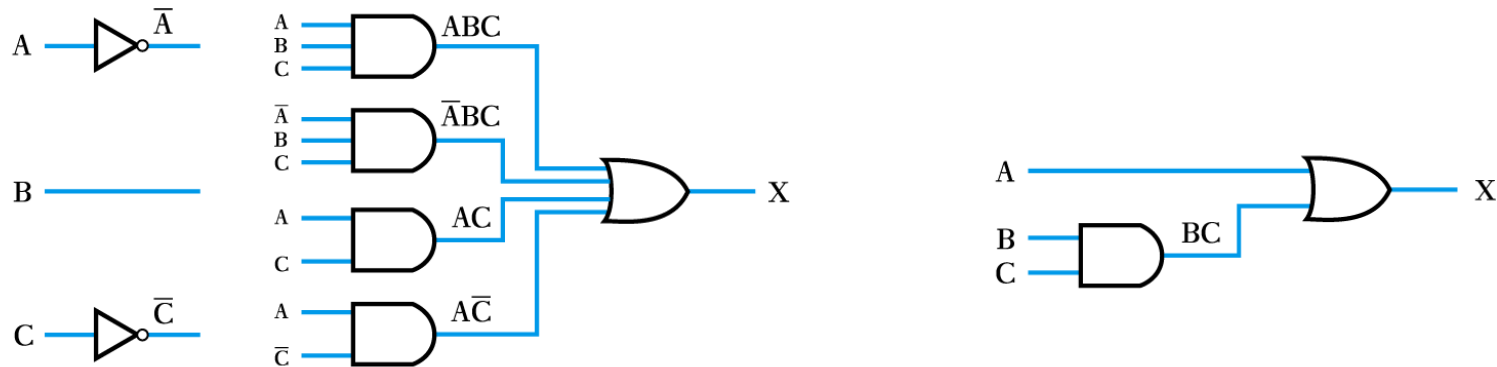
The logic function $X = \bar{A}\bar{B}C + A\bar{B}C + AB\bar{C}$ can then be implemented as before



- In some cases it is possible to *simplify* logic expressions using the rules of Boolean algebra

Example

$X = ABC + \bar{A}BC + AC + A\bar{C}$ can be simplified to $X = BC + A$
hence the following circuits are equivalent



Number Systems and Binary Arithmetic

- Most number systems are order dependent

- **Decimal**

$$1234_{10} = (1 \times 10^3) + (2 \times 10^2) + (3 \times 10^1) + (4 \times 10^0)$$

- **Binary**

$$1101_2 = (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$$

- **Octal**

$$123_8 = (1 \times 8^2) + (2 \times 8^1) + (3 \times 8^0)$$

- **Hexadecimal**

$$123_{16} = (1 \times 16^2) + (2 \times 16^1) + (3 \times 16^0)$$

here we need 16 characters – 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F

■ Number conversion

– conversion to decimal

- add up decimal equivalent of individual digits

Example

Convert 11010_2 to decimal

$$\begin{aligned} 11010_2 &= (1 \times 2^4) + (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (0 \times 2^0) \\ &= 16 + 8 + 0 + 2 + 0 \\ &= 26_{10} \end{aligned}$$

■ Number conversion

– conversion from decimal

- repeatedly divide by the base and remember the remainder

Example

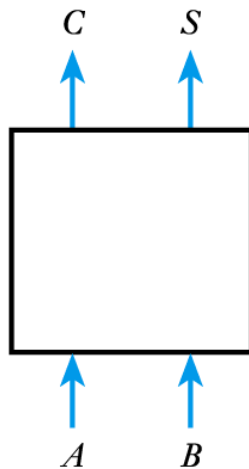
Convert 26_{10} to binary

	Number	Remainder
Starting point	26	
$\div 2$	13	0
$\div 2$	6	1
$\div 2$	3	0
$\div 2$	1	1
$\div 2$	0	1

↑ read number from this end
=11010

■ Binary arithmetic

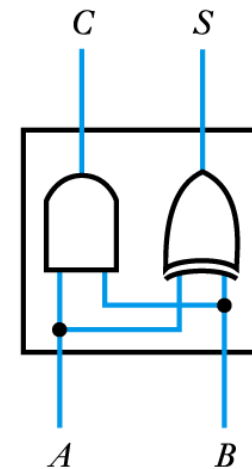
- much simpler than decimal arithmetic
- can be performed by simple circuits, e.g. half adder



(a) Block diagram

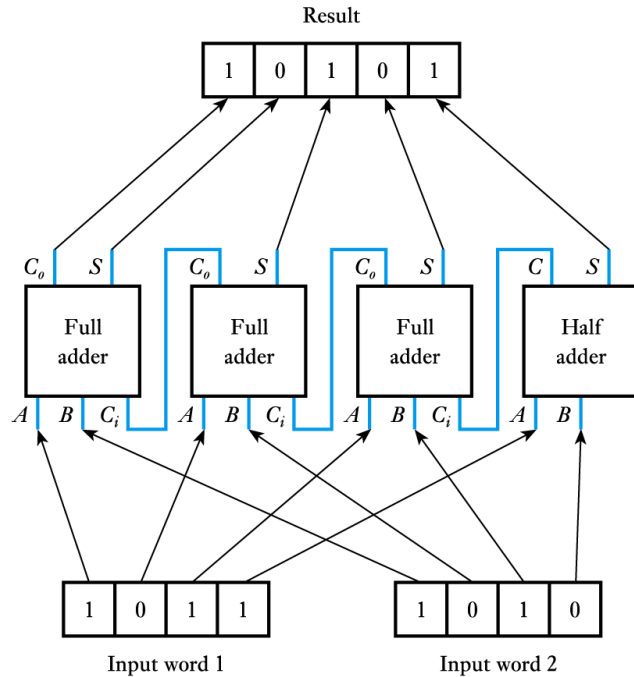
A	B	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

(b) Truth table



(c) Circuit diagram

- More complex circuits can add digital words



- Similar circuits can be constructed to perform subtraction – see text
- More complex arithmetic (such as multiplication and division) *can* be done by dedicated hardware but is more often performed using a microcomputer or complex logic device

Numeric and Alphabetic Codes

■ Binary code

- by far the most common way of representing numeric information
- has advantages of simplicity and efficiency of storage

Decimal	Binary
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000
9	1001
10	1010
11	1011
12	1100
etc.	etc.

Numeric and Alphabetic Codes

■ Binary-coded decimal code

- formed by converting each digit of a decimal number individually into binary
- requires more digits than conventional binary
- has advantage of very easy conversion to/from decimal
- used where input and output are in decimal form

Decimal	Binary
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000
9	1001
10	10000
11	10001
12	10010
etc.	etc.

Numeric and Alphabetic Codes

■ ASCII code

- American Standard Code for Information Interchange
- an alphanumeric code
- each character represented by a 7-bit code
 - gives 128 possible characters
 - codes defined for upper and lower-case alphabetic characters, digits 0 – 9, punctuation marks and various non-printing control characters (such as carriage-return and backspace)

Numeric and Alphabetic Codes

- **Error detecting and correcting codes**
 - adding redundant information into codes allows the *detection* of transmission errors
 - examples include the use of parity bits and checksums
 - adding additional redundancy allows errors to be not only detected but also *corrected*
 - such techniques are used in CDs, mobile phones and computer disks

Key Points

- It is common to represent the two states of a binary variable by '0' and '1'
- Logic circuits are usually implemented using logic gates
- Circuits in which the output is determined solely by the current inputs are termed combinational logic circuits
- Logic functions can be described by truth tables or using Boolean algebraic notation
- Binary digits may be combined to form digital words
- Digital words can be processed using binary arithmetic
- Several codes can be used to represent different forms of information